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KEYNOTE PRESENTATIONS

Miller Puckette

Abstract

When we listen to a well-played music, we hear it as if the player were making the music, and the instrument were a transparent conduit through which the music flows. When listening to a bad performer, our attention often is attracted to the instrument itself. When making music with a computer, it can be hard to avoid calling attention to what the computer is doing; and yet it would seem desirable, at least by analogy, to have the computer be as transparent a part of the music-making process as an instrument can be. In this talk I'll try to identify some patterns of computer usage that either call attention or avoid calling attention to the medium (the computer); many useful and entertaining examples are available from teh standard computer music repertory. I'll try to make some useful generalizations as to what a musician might wish to try to do or avoid doing to evade some of the many artistic traps the computer can so easily lure us into.

Biography

Miller Puckette obtained a B.S. in Mathematics from MIT (1980) and Ph. D. in Mathematics from Harvard (1986). He was a member of MIT's Media Lab from its inception until 1987, and then a researcher at IRCAM (l'Institut de Recherche et de Coordination Musique/Acoustique), founded by composer and conductor Pierre Boulez. There he wrote the Max program for MacIntosh computers, which was first distributed commercially by Opcode Systems in 1990 and is now available from Cycling74.com. In 1989 Puckette joined IRCAM's musical workstation team and wrote an extended version of Max, called Max/FTS, for the Ircam Signal Processing Workstation.

Puckette joined the Music department of the University of California, San Diego in 1994, where he is now chairman. From 2000 to 2011 he was Associate Director of Center for Research in Computing and the Arts (CRCA) at UCSD.

He is currently working on Pure Data ("Pd"), a leading open-source real-time multimedia arts programming environment, with software contributions by many others worldwide. Puckette has collaborated with many artists and musicians, including Philipe Manoury (whose Sonus ex Machina cycle was the first major work to use Max), and Rand Steiger, and Vibeke Sorensen (as part of the Global Visual Music project). Since 2004 he has performed with the Convolution Brothers.

*****

Michael Young

Abstract

Computational systems able to collaborate with human improvisers are *live algorithms:* able to cooperate proactively, on an equal basis, with musicians in performance. This is an ideal that raises fundamental questions about creativity and group interaction, how these might be computationally modelled. Can musicians and computers relate to one another, just as human musicians do? Can an audience recognize and appraise this relationship? Live algorithms offer the prospect of a new understanding of real-time creative practice that differs from the established paradigms in live electronic music: *computer as instrument* and *computer as proxy.* Drawing upon ideas from social psychology and pragmatics, collective music-making can be viewed as a special case of social cooperation, evidenced primarily through sound. To attempt a functional description of live algorithms is to model modes of cooperation and causal attribution that occur between proactive agents in a shared *sonic* environment. The challenge of live algorithms is to find genuinely original ways for humans and computers to work together - an original way to make music. The ideal live algorithm paradigm is *computer as partner.*

Biography

Dr.Young is a composer with interests in real-time computer music, interdiscipinary practice and performance. He was born in South Shields in 1968. He studied music at Oxford University and composition with John Casken & Peter Manning at Durham University, completing a PhD in 1995. He is a Senior Lecturer in Music at Goldsmiths, University of London in the Electronic Music Studios (EMS). Previously, he lectured at Oxford Brookes University and University of Wales, Bangor. His recent music has focussed on interactive/generative systems, exploring the interface between composed and improvised music-making. He is co-founder of the Live Algorithms for Music Network. He has worked with jazz musicians and improvisers as pianist, laptop musician & composer.
[source: http://homepage.mac.com/oobop/myounginfo/about.html]

PEER REVIEWED PAPERS

QUANTITATIVE CATEGORIZATION OF TONAL MUSIC STYLES

Hector Bellmann
Queensland Conservatorium
Griffith University

Ray Duplock
High Performance Computing Centre
Queensland University of Technology

ABSTRACT

Tonal music style is a substantial quality of the music material that can be recognized to belong to individual composers or their epoch. We were interested in the aspects of style that are detectable in notation. The objective of this study was to identify the main variables that allow for effective classification of tonal music styles. This was a quantitative study based only on measurable features giving rise to numerical continuous variables resulting from actual measurements carried out by software. As many features as practicable were operationalized and software routines devised to carry out the measurements on a purposely built database that stood for the whole range of styles of the period of common practice. The resulting dataset was investigated by machine learning algorithms. Key and rhythm measures and frequencies of use of scale degrees turned out to be the variables of highest predictive validity. The three key indicators were then used to effectively classify pieces by composer.

1. INTRODUCTION

Paisley [1] is credited with the concept that style is best accounted for by ``minor encoding habits''. He quoted Bernard Berenson's statement [1:221]: ``Obviously what distinguishes one artist from another are the characteristics he does not share with others'' leading to the notion of style as `deviation from the norms'. This idea opens the door for quantitative analyses of music. Although there is a long tradition of these, going back to Gabura [2][3], they continue to be in the minority. Recently, Clarke & Cook [4] have provided support for this approach which ``embodies a principled awareness of both the potential to engage with large bodies of data, and the appropriate methods for achieving this''. This converges with David Huron's [5] statement that ``quantitative methods allow us to observe patterns that might otherwise be difficult or impossible to decipher''.

Within this trend, our own interest points specifically to style, a concept whose use is ubiquitous but a phenomenon that is neglected in itself. For example, Klaus Döge, discussing Dvorák [6] characterizes his American period by the use of ``pentatonism in the melodic line, a flattened leading note, plagal cadences, drone accompaniment, rhythmic ostinato and strongly syncopated rhythms''. Such list may make us feel impressed by the acumen of the analyst, but it may also bring about a lingering uneasiness caused by the lack of a conceptual frame of reference for style. The reader could wonder whether such tally is really exhaustive, or is it perhaps biased, and what is the possible range of the list of features from which the analyst picked these items. Naturally, these doubts would not have been raised if there was an established conceptual structure for style, which makes one wonder how it is possible that a fully developed theoretical characterization of tonal music style does not yet exist. It is likely that the frequent encounter with multiple works of a certain composer leads our minds to the formation of a certain `prototype' of the composer's style, to the point that we can roughly classify those works in relation to how representative or idiomatic they are. But this process is entirely unsystematic, and its limitations become apparent when one is faced with the discovery of a similar work of unknown composer. There are no tools to measure how close the new work is to others of the probable composer to justify attribution.

The notion of developing tools for attribution based on quantitative measures led our interest to the aspects of style that are detectable in notation, which is all we have when a score is discovered. If a system of fundamental measurable variables could be found that characterized style on the basis of notation alone, the basis for the development of attribution tools would have been set. The quantitative approach is ideally suited to this pursuit. This study is based on the notion that a comprehensive enough set of features that allows for stylistic classification of musical works should lead to the identification of the main conceptual dimensions of style. Measuring as many meaningful parameters of the music as possible would result in a set of values amenable to quantitative analyses. The goal involved avoiding subjective judgments, and for this purpose it would be ideal to preferably rely on continuous variables resulting from actual measurements carried out by purposely-designed algorithms implemented as software.

2. METHOD AND MATERIALS

An overview of the method comprises operationalizing as many features as practicable; then devising algorithms to measure them and implementing them into software; using these programs on a suitable sample of pieces standing for the period of common practice; and finally analyzing the resulting data. In this process, three stages can be recognized:

2.1.1. Corpus collection and digitization}

The need of a suitable sample of the whole range of styles of the period of common practice on one side, and the time limits imposed by practical constraints on the other, led to the compromise of creating a database of 297 piano or harpsichord pieces, i.e., 11 pieces each from from 27 leading composers from Handel to Shostakovich. The pieces were selected by size and assumed representativeness, scanned and OCR'ed into musicXML files.

2.1.2. Feature selection and measurement

Observable features were extracted from the main areas of music, namely key, rhythm, harmony, and melody. Each feature was operationalized and software algorithms were developed to measure them preferably into continuous numerical variables. The algorithms were applied to the database, thus yielding a set of numerical results for each piece.

2.1.3. Data analysis and interpretation

These results were collected in spreadsheets to make them available to analytical algorithmic methods in order to identify the main dimensions. Once these were identified, they were used to classify the database by composer.

2.2. Key determination

All the features, except for the rhythm variables, are pitch-related. Therefore, an essential initial step was the automatic determination of key as a point function, i.e. at every point of the score.

This task was carried out according to a method described in [7], which is essentially based on finding the maximum dot-product between the pitch content of the piece and a set of numbers statistically obtained from a large set of works which represent the ideal frequencies of the scale degrees. The fact that, in major and minor scales based on the same tonic, the so-called modal degrees fall on different notes, made necessary to process everything separately by mode. The set of numbers used is shown in Table 1.

Table 1 - Key set for both modes

2.3. Sliding window

Scale degree	Major	Scale degree	minor
Tonic	16.80	Tonic	18.16
Raised Tonic	0.86	Lowered Supt.	0.69
Supertonic	12.95	Supertonic	12.99
Raised Supt.	1.41	Mediant	13.34
Mediant	13.49	Lowered Subd.	1.07
Subdominant	11.93	Subdominant	11.15
Raised Subd.	1.25	Lowered Dom.	1.38
Dominant	20.28	Dominant	21.07
Raised Dom.	1.80	Submediant	7.49
Submediant	8.04	Lowered Subt.	1.53
Raised Subm.	0.62	Subtonic	0.92
Subtonic	10.57	Lowered Tonic	10.21

This method determines the key of a fragment of the piece that can stretch to the whole of the piece, giving in that case the global key. But practically every piece modulates at some point, which introduces a confounding variable if the initial key is assumed throughout. The way to obtain the key for each section as well as the points of modulations is to calculate it during a window wide enough to be stable and short enough not to bypass modulating passages and then slide the window along the score, repeating the calculation as new notes are being added and the old ones dropped.

While the key remains the same, successive calculations yield results that vary slightly. If the piece modulates, the key can be seen as a piecewise function of time affected by noise. The window is a low-pass filter whose resolution and stability depend on its width. The solution is a compromise.

2.4. Measured features

The following are the features that were measured in each of the music areas:

2.4.1. Key

In order to assess key parameters, three scalar variables were measured:

(a) A previous study [8] had suggested that the tonal weakness of a piece correlates with the rate of decay of the dot products, that is, the less clearly tonal a piece is, the more slowly falls the value of the successive dot products. This led to the calculation of the average percental distance between the top scalar dot product for each slice and each of the five runners-up. The mean of those averages for the whole of the piece was labeled *Global average dot product*. It was found that historically, its value is relatively stable before 1800 but declines gradually from then on until the eve of the Great War. Its absolute value, which is affected by parameters such as the width of the measuring window, has not been normalized but with the parameters used in this study, its values for our database varied from a minimum of

560 (Prokofiev: March Op.3 No.3) to a maximum of 2022 (Satie: Gnossienne No.1).

(b) An index that indicates the propensity of the piece to modulate chromatically was called *Modulation index*, and its value calculated as the percental ratio between the sum of the tonal distances for each key transition and the normalized total duration of the piece. For our database, the values of this index varied between a minimum of 0 and a maximum of 30.662 (Prokofiev: Visions fugitives Op.22 No.5)

(c) Likewise, a similar index for the propensity of the piece to modulate diatonically was called *Intermodal index* and calculated as the percental ratio between the sum of the inter-mode changes and the normalized total duration of the piece. For our database, the values of this index ranged between a minimum of 0 and a maximum of 7.033 (Shostakovich Prelude Op.34 No.5).

2.4.2. Rhythm

As rhythm is a multi-dimensional phenomenon, a rhythm-measuring scheme was devised comprising four practically independent scalar variables:

(a) *Beat rate*, measured as beats per minute. For our database, the beat rate varied between a minimum of 24 (Satie: Gnossienne No.5) and a maximum of 510 (Scriabin: Etude Op.8 No.3);

(b) Number of *onsets per measure*. For our database the value of this variable ranged between 5.29 (Liszt: Nuages gris) and 52.245 (Ravel: Ondine);

plus two indexes of rhythmic variability that refer to the pattern of onsets for the piece:

(c) A way to measure changes in the longitudinal profile of the piece would be to consider the variations in the number of onsets taking place at each subdivision of the bar. Thus, the Index of longitudinal or *Between-measures* variability, is equal to the ratio of the sum of the variances for the number of onsets occurring at each subdivision of the measure and the total number of onsets. For our database, the value of the Between-measure variability varied between a minimum of 0.144 (J.S.Bach: Gigue from French Suite No.3) and a maximum of 24.953 (Debussy: Prelude No.1); And

(d) Likewise, a way to capture the variability occurring within each measure would entail considering the variations across all the subdivisions of each bar. Thus, the Index of transversal or *Within-measures* variability, equal to the ratio of the sum of the variances for the number of onsets occurring in each measure and the total number of onsets. For our database, the value of the Within-measure variability ranged between a minimum of 0.239 (Chopin: Etude Op.25 No.12) and a maximum of 24.953 (Ravel: Menuet from Sonatine).

2.4.3. Scale degrees

One of the simplest measurements that can be effected, the frequencies of use of the individual twelve scale degrees - in both modes - was carried out on each piece. Since each piece yields 24 values, these can be considered a set of 24 scalar variables, or more properly a single 24-component vectorial variable.

2.4.4. Harmony

Using the concept of sonority [9] rather than chord to refer to each simultaneous combination of pitch classes, what was initially measured - separately for major and minor modes - was the frequency of use of each of the possible sonorities made of up to four different notes, i.e. the single monad, the six dyads, the 19 triads and the 43 tetrads. These values, however, constitute a sparse matrix because the majority of the sonorities are seldom used. A double set of 69 scalar variables, the majority of which is zero, was not very convenient for analysis. The consideration that only a few of the sonorities receive a name in harmonic theory - the six dyads, five triads and seven tetrads - suggested that it was better to limit the measurements to this 18-variable reduced set.

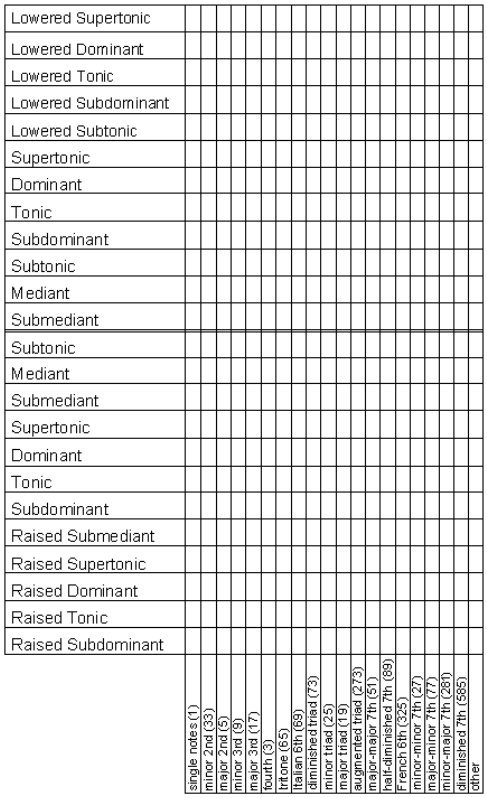


Fig.1 - Grid for representation of harmonies

Considering, for example, that each sonority, such as major and minor triads and major-minor sevenths, can play different harmonic functions such as tonic, dominant, and so on, it is arguable that the frequency of use of each combination of sonority and function will have stylistic significance. The concept can be generalized to the set of all of the sonorities and all the

functions, resulting in the measurement of the frequency of use of each of the sonorities playing each of the harmonic functions. Fig.1 illustrates the grid comprising the combinations of sonorities and functions, and Fig.2 shows an example, the frequency of the sonority-function combinations for Handel's Sarabande from Suite 16.

This set turns out to comprise 432 scalar variables, although it can be more properly viewed as a single second order tensorial variable for each piece in the database. This compares unfavorably with the size of the database, which comprises only 297 cases, and is also a sparse matrix, as Fig.2 illustrates - for most of the squares the frequency is zero. Consequently, we decided to collapse the matrix in two directions, thus replacing the 432-component second order tensor by two vectorial variables, one an 18-component measure of harmonic frequencies (irrespective of function), and the other a 24-component measure of the frequencies of harmonic functions (irrespective of harmony).

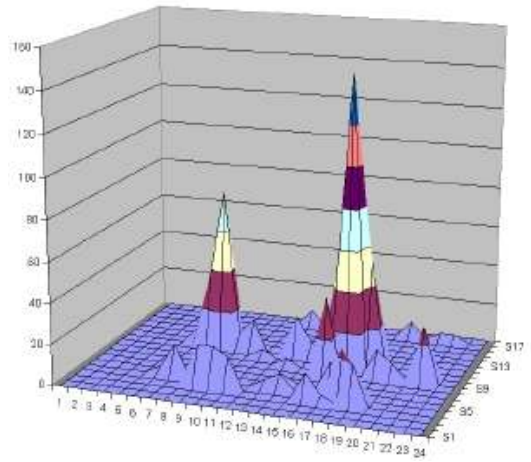


Fig.2 - Harmony frequencies for Handel's Sarabande from Suite 16

2.4.5. Melody

In keyboard music, in general, voices cannot be automatically individualized. Particularly after the end of the Classical period, `voices' exist only in the mind of the composer and they tend to appear and vanish as a matter of convenience. As counterpoint lines are beyond disentangling, it was thought necessary to approach the materials as if they were basically melodies accompanied by a bass line. Unfortunately, the perception of melody is linked to psychoacoustic principles and there does not seem to be any valid strategy to objectively identify it. This led to the conclusion that the only practical way to operationalize a melody-like variable was to consider the frequency of use of scale degrees occupying the highest pitch in the piece. This operationalization in many cases will coincide with the melody, in others it will not, but it has the advantage of measuring an objective variable.

Consideration of the two-part principle suggested also the convenience of carrying out a similar measure for the bottom note of the music, which would agree with Schoenberg's [10] referring to the bass as ``the second melody'' in his theoretical writings. Therefore, the melody-related variables were two 24-component vectorial variables measuring the scale degree frequencies of the notes occupying the highest and lowest positions at each point in the piece.

3. RESULTS

3.1. Data

The preceding feature measurements were carried out by software routines applied to each of the pieces in the database. The result was a set of 139 numerical continuous variables which constitute the dataset for the analyses. As the expectation was to use statistical techniques for the analyses, data were subjected to the customary screening for distribution and subsequent transformation of variables when they did not comply with acceptable levels of normality and skewness.

It has to be remembered that the main purpose of the study was to unveil the main conceptual dimensions of style, and the expectation, that measuring a comprehensive set of features would capture the style of the pieces. According to this goal, it is necessary to analyze the whole of the dataset together. For example, if the most important dimensions were to be identified, they would have to emerge as the principal components of the complete set of variables.

Nevertheless, the dataset proved impregnable for statistics. The number of variables was too large for any statistical technique except for Principal Component analysis. This would have been the appropriate technique for the stated purpose, but Principal Component Analysis can be applied only when there are clusters of variables in the data. This is assessed by mens of the the values of the covariance matrix as well as the Kaiser-Meyer-Olkin measure of sampling adequacy. Both were found to be below acceptable levels, meaning that Principal Components was not an appropriate way to analyze the data because there were no significant factors to be extracted.

3.2. Data mining algorithms

Machine learning algorithms, being assumption-free, non-linear and non-parametric and some - like Classification and Regression trees (CART) and Random Forests - being able to handle highly complex data structures with a large number of variables such as this dataset, constitute a more powerful alternative to statistics, and were the tool of choice for the analyses.

3.2.1. CART

The CART algorithm was developed by Breiman, Friedman, Olshen and Stone [11]. A complete

mathematical description of the algorithm is given in Breiman et.al, (1984). Readable introductions are available in Hastie, Tibshirani and Friedman (2009) and Steinberg (2009). The CART algorithm represented a major milestone in the evolution of artificial intelligence [12]. The CART unsupervised machine learning procedure, used when there are no target variables, is an ideal alternative to traditional clustering algorithms [13]. CART unsupervised learning was used to identify two or three important variables that would be responsible for partitioning the data into homogeneous segments or clusters. This procedure lets the whole set of predictor values determine the model without need for variable transformations. Starting on node 1 containing all the cases, the first binary split is done using *modulationindex*. If the value of this variable is < 4.01, cases are separated into the left child node; if the value is > 4.01 they are separated into the right child node. Further binary splitting then occurs on the children nodes. The variable used as the binary splitting variable on both of these child nodes *intermodalindex*. Unsupervised CART, unlike a traditional CART model, does not require an optimal sized tree to be developed. The tree model in unsupervised CART can be pruned back as required to reveal data groupings of interest or significance. In our analysis, pruning the tree back to the first two binary splits provides an important insight about the two variables that are the most important in the initial tree splits.

3.2.2. Random Forests

CART introduced the concept of a tree model. Random Forests extended this concept by developing a collection of individual trees or a ``tree ensemble''. Although each tree is independent of any other, the results of each of these trees is combined via a voting process (for categorical models) or by averaging (for regression models). Since each tree is independent, adding trees to the model does not create overfitting. In this way, this algorithm tends to washout any artifacts and model instabilities that can occur with single tree models, yielding results that are robust and stable [14]. A complete mathematical description of this algorithm is also given in this source.

Random Forests was run to classify the database, with composers being the target, or *Y*, variable. After the run, the algorithm provides the list of overall variables of importance for the classification. The list shows that:

- the most important three variables are the three key indicators:
  - *modulation index*
  - *intermodal index*
  - *Global average dot product*
- of the next five variables, four are the rhythm indicators;
- with the exception of two functions and one harmony, the remaining 24 variables are the frequencies of use of all the scale degrees.

Thus, from a conceptual point of view, Random Forests classifies fundamentally on the basis of the key, rhythm and scale degree variables.

3.2.3. Gene Expression Programming (GEP)

The previous two algorithms agreed about the most important variables that can be used for a classification model (*Y* = the composer). In view of those results, the three key variables found to be of the greatest importance from the CART and Random Forests models were included within a new evolutionary algorithm called Gene Expression Programming (GEP). GEP was developed by C. Ferreira [15] and can be used to construct binary classification models or regression models. *Y* takes the value of 1 for a particular composer, 0 otherwise; there being 27 different composers, 27 binary GEP classification models were run. Using standard GEP binary classification settings gave an overall classification accuracy of 86.6% with a minimum of 74.1% for Scriabin and a maximum of 95.6% for Schubert.

To remove any suspicion that GEP could have overfitted the classification solutions, a second trial was carried out, splitting the 297 dataset records into a training set of 189 records, and an independent test set of 108 records. This trial was run on six composers chosen at random. In this trial the GEP models gave a classification accuracy for the training set of 80.9%. The average classification accuracy for the test set was 72.8%. As expected there is a deterioration for the training values due to the reduction in size of the training dataset. Likewise, the test values are lower since validation is done using only 36% of the original dataset records; however, being able to achieve a predictive accuracy across the six randomly selected composers of over 72% on the basis of completely new pieces, is a remarkable result.

4. DISCUSION

The limitations of statistics have been made explicit by their inability to deal with the dataset of this study. Two independent data mining algorithms, however, have shown their effectiveness and the agreement of their results. The question may arise whether, once the main dimensions for classification have been identified, would it not be possible to do statistical studies based on these variables? In our view, the central problem is that the dataset of this study turned out to be very complex in the sense of comprising non-linear variables with complex interactions. Data mining algorithms can deal with a large number of variables without imposing conditions on their distributions, linearity of mutual interactions, but statistic methods cannot. Nevertheless, two attempts were carried out to see if it was possible to cluster the

data through statistical methods using the found most important variables as predictors. The SPSS Two-step cluster is a method supposed to automatically find the optimum number of clusters. Two-step was tried using the three key variables as predictors, but the result was that the procedure identified a single cluster comprising the whole dataset. A second attempt used as predictors the best 13 variables as identified by Random Forests. But for a second time, Two-step cluster put all the cases in a single cluster.

Therefore, the knowledge of the most important variables of the dataset does not bring the classification of tonal music within the reach of statistics. But since powerful data mining algorithms are available for the home computer, studies of this kind could become more widespread to the extent that the data mining software becomes affordable for the public.

5. CONCLUSIONS

This study has identified, among the measured features, the most important dimensions of tonal music style. These are measures of key and mode changes, rhythm measures and the relative frequency of use of scale degrees. These could be the basis of a conceptual model and a basic taxonomy of tonal music styles. Naturally, it is not possible to exclude the existence of other unmeasured variables that could turn out to be of comparable importance. The first candidates could be the frequencies of transitions between scale degrees and between harmonies. But notice that 24 chromatic scale degrees give rise to 552 possible transitions between them, a set of variables that far exceeds the total number of variables used in this study. Considering transitions probably requires a different approach.

The study has also proven it is possible to automatically identify a composer's style from the score alone, and created a system for the identification, which is a valuable contribution to authorship studies.

We believe this study opens new grounds for musical research both in philosophical and methodological senses. It shows it is possible to take a quantitative approach, generating quantitative measures for music variables and carrying out precise measurements, excluding as much as possible the influence of subjective judgments. It also shows the value of classifying and clustering music scores (and composers) by data mining methods as opposed to outdated statistical procedures.

Also, on the methodological side, the understanding that the tonalness of music is based on highly consistent proportions of use of the scale degrees makes indefensible for musical studies to be based on notes instead of scale degrees, and suggests the necessity of adopting a reliable method to determine key as a point function.

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RETHINKING INTERACTIVITY: FROM MATERIAL TO ORGANIC MODELING.

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ABSTRACT

This paper examines the limitations that a materialist approach to interactivity has imposed on the genre, creating a disjunction between the richness and complexity of life experiences and the linear, representational nature of interactive art. The author proposes that a shift to an organic philosophy provides a method of accounting for the fluid complexity of relational forces composing an art event, and examines drift and noise as two strategies by which the artist might begin to structure a distributed creative agency within the event.

1. INTRODUCTION

'The change from materialism to "organic realism"… is the displacement of the notion of static stuff by the notion of fluent energy. Such energy has its structure of action and flow and is inconceivable apart from such a structure.' Alfred North Whitehead [22].

Imagine for a moment that you are out walking in the street. To go for such a walk is to create through the endless flow of interaction. Bodily and spatially, each step – and indeed within each step – perceptual, sensorial and social possibilities are opened up, assemblages of forces gathered, altered and reconnected, complexities multiplied, memories activated. The moment is saturated with affectual relations and intensities. With the fall of the same step, previous possibilities perish, simultaneously propelling the endless opening of fresh possibilities of connection [7]. Try to map all the relations that go to make up one instant – one occasion: within your body, between body and world, mind and body, object and object and so on. You will have to consider subatomic, atomic and molecular forces with their general disregard for what we view as discreet bodies. You will want to account for the way texture and gradient of the terrain shapes movement, rhythm and posture and how sensory perception - vision and hearing and so on – begin to ready the body for the next step; how the force of physical habits – body memory – shape patterns of movement in the present moment. Present also will be all the events of relation that have gone into making each tree, stone, person or sound you are interacting with, effecting your body more or less forcefully. And so on. Then there are the mental forces – 'inextricably intertwined' [22] with the physical - memories, anticipations, random associations made and forgotten, affects that will subtly or bluntly alter you, the myriad mental processes that sit below conscious perception yet nevertheless shape and reshape your body. Beyond that instant, in the next occasion, the concrescence of all these forces creates anew this simple act of walking the street. It is a constant, complexly enmeshed act of creativity: when we look honestly all things, as Whitehead says, are vectors of relations [22].

Such an everyday act is saturated with complexity and invention, and rich with potential. But now imagine you are in a gallery, in some interactive installation. Things happen as you move around – sounds, lights, video or whatever. Perhaps once triggered by your presence the work pretty much does its own thing, perhaps it continues to develop as you engage. Either way, it so often lacks the complexity, intertwined-ness of body and work, the perceptual nuance, the fluidity, surprising originality of connection and thickness of experience of a simple walk outside. While at best the interactive experience might seek to expand awareness of the processes of perception and relation, in fact it too often remains programmatic, lacking in subtle and surprising combinations of memories, affects, sensations and prehensions. This is not to suggest that interactive art's role is to mimic life, but rather that many such works display a paucity of life's rich, heightened experience of connection and possibility.

To me, this is a continuing disappointment that cannot be simply explained away by accusing any particular artist of a lack of imagination. Thus in this paper while I intend to briefly cover some of the common criticisms of the limitations – experientially and ethically – of interactive art, I wish to move beyond such critique to what I would argue is the underlying philosophical problem: the denial of the fundamentally fluid 'vector' nature of interaction. It is this issue that I believe underlies the narrowness of both the invention and critique of interactive art – a 'narrowness in the selection of evidence' as Whitehead might argue [22] - which in its attempts to reduce the field of discussion to a manageable stability, succeeds only in denying the actual nature of the event. My discussion draws heavily on Massumi and Manning's thinking on the shift from

interaction to relationality[1], which I have attempted to extend through concepts of drift and noise in order to extrapolate practical methods by which relation might continue its drive towards novelty in the interactive artwork.

2. THE MYTH(S) OF INTERACTIVITY.

What are our expectations of interactive art forms? Perhaps that it expands the range of art experiences available to the audience, offering levels of 'free choice' and embodied experience seemingly lacking in more traditional art forms; that it will be participatory on some level unavailable in the supposedly more passive enjoyment of traditional forms; and that it will be experiential rather than representational. The existence of some essential qualitative - and indeed moral - judgment of difference between the 'interactive' and 'non-interactive' forms by proponents of interactivity has often been promoted. Simone Osthoff's argument might be cited, for example, that Lygia Clark's work utilizes the viewer's own energy, synthesizes mind and body and explores the sensorial, and replaces the object with the experience – all in ways that painting and sculpture cannot [14] Stoichita's statement that in Lozano-Hemmer's work '(w)e are no longer before the (interactive) work, we are in the work' [4], or Roy Ascott's claim of 'moving beyond the object', from observed effect to participation, and elsewhere that participatory art is the opposite of 'traditional' forms that distance one from the process [1], all can imply some kind of moral superiority in the interactive.

Is there really open-ended decision making in generative or interactive art? In her essay entitled '*Interactivity means interpassivity*', Mona Sarkis argues that the participant in interactive art remains a passive 'user', assembling the artist's vision without any real free choice [17]. De Mèridieu likewise warns that 'we should not delude ourselves: interactivity can conceal programmed actions and predetermined pseudo-choices' [13]. Massumi also cautions that interactive works can dictate an involvement and proscribe possibilities [11], and Lev Manovich describes the rise of interactive art as a shift from representation to manipulation [5]. Here the question of (free) choice seems to be, as Peacock argues, one on which the success and failure of interactivity commonly balances, stating that '(d)ecision making of some kind is a necessary condition of the interactive' [15].

Can interactivity really offer more – or even as much choice as, for example, a painting - can it offer as many options to the viewer either in the way they assimilate content or in the choices of levels of involvement in the work? Even an exhibition of paintings might offer the viewer fairly free reign in their manner of experiencing the space: the choice to skim over some works, view them in any order, dip in and out of concentration and so on – all fairly banal choices that one would take for granted. Interactive works, on the other hand, often require a high level of participation to achieve any presence. As Massumi notes, 'it is important to remind ourselves that there can be a kind of tyranny to interaction' [11]. That is, it is not necessarily liberating to dictate a level of involvement, in itself a kind of operation of power over the viewer that the artist exercises that needs to be recognized. We should not forget, as Manning warns, that the sensory technologies at the base of many interactive works have 'problematic pasts, both as displacers of the corporeal body and in assemblages of control' [8], and, while many artists who utilize surveillance technologies claim to be displacing the power dynamics by making art with such tools of control, they may retain such potential in the artist's hands - hardly a paradigm shift.

Perhaps participatory works might claim a certain freedom from representational content - aiming more for a visceral experience that narration, contemplation or reflection. In one sense it is true that a painting's content is all constructed prior to encounter with the viewer, dictated by the artist and waiting to be read. However even in the most didactic, narrative driven image there presents the possibility – one could argue even inevitability – for a freedom of association, that is, for myself as a viewer to link elements I see to memories. Personal (and cultural) associations are inevitable, whether a colour that reminds me of a flag, facial features I associate with a friend, lighting effects that somewhere in my brain trigger memories of a half forgotten film, muscle memory or a prehension of movement made conscious through an association with a figure's awkward pose. This is not simply a reviving of old memories, but actualization of virtual memory that creates new thought within the event, exactly the kind of 'interaction' that fits with Manovich's argument that the notion of interactivity must become inclusive of notions of psychological processes, mental as well as physical or temporal connections [6].

Any artwork, in this psychological sense, might be read as loosely 'generative', in that on some levels an individual experience still emerges from the combination of viewer and work that in its singularity inevitably begins to escape the confines of the artist's control. Interactivity can struggle to allow such excessive layering and complicating of dialogues. That is, the interaction is often necessarily productive - lacking the multitude of virtual potentials, interactions struggle to become excessive, to outstrip function and destabilize orderly systems of exchange. It is perhaps no wonder that interactive technologies and displays sit so comfortably in didactic museum displays. Ironically the very participation that in art is intended to free the viewer from constraints operates effectively to direct and lecture them. As Massumi argues, to utilize such technologies in a becoming, emergent fashion they need

[1] In particular Massumi's *Semblance and event* [10], and Massumi and Manning's *Propositions for an expanded gallery* [13].

to be freed from 'use-value' or 'exchange-value', move beyond 'prodding a participant to gain a response', and take on a more speculative nature [11].

What space for contemplation does the interactive installation allow? The curse of interactivity, like video art, is that often I must either abandon midway through boredom, endure to a set endpoint or at least move through in a set order. Here the 'empowerment' that Ascott claims for interactivity, allowing the 'individual to participate fully in the workings of the system' [1] can be rather like the participation in riding a train: certainly I am bodily involved in the machinations of travel, but with limited entrance and exit points and heading inexorably in a prescribed direction. It's a kind of roller-coaster ride experience that that contains a certain level of visceral thrill but that can also, as Poissant notes, 'enclose one into a schema of manipulation rather than propose a real space for dialogue' [16]. The risk is that my movements loose their incipient, 'becoming' qualities, and there is the possibility in participatory works such of merely 'performing the software' [7].

Ascott contends that interaction can become trivial, in a closed, linear system with finite data [1] – a flicking of an 'on' switch with my presence, a prompting of a software program to jump to the next prearranged scene, as in a video game. This is often in sad contrast to everyday lived experience with its endless emergent qualities. These qualities, Manning argues, are a building elasticity of virtual potential, such as the mental and physical prehensions and pre-accelerations that shape the actuality of a movement [7], resolving such tensions - a 'satisfaction' of the prehension [23]. This physical/psychological connection seems to me the kind of visceral involvement that is often lacking in programmatic interactive art. The excess of the virtual is replaced by the probable, open endedness by specific purposes. The art that announces itself as interactive too often limits itself through its dedication to the representation or demonstration of its interactivity, and its insistence on demarcating a stable zone of relation between two discrete bodies – the viewer and their environment.

3. ART AS EVENT: A RELATIONAL MODEL.

It might seem apt here to demand more from the interactive artist (and critic): more complexity, more imagination and more inventive. Of course more creativity will always have its place, but the problem underlying interactive artworks' limitations lies primarily, I would argue, in the philosophical conception of an object, a subject and a work of art. Whitehead's methodology is to re-invent the field of the problem in order to find a new solution. Following this I want to ask here how might we think (and construct) a field of interactive art so as to encourage the fluidity and layered inventiveness that might now be lacking?

Massumi argues that interactivity describes a simple back and forth between two elements that remain discrete and stable [3]. That is, in this context, it is a material view of the world in which I am a stable subject and the art work a stable object, both transcendent of the event of our encounter. But in the 'Whiteheadian' universe of organic philosophy the scenario is very different. Any notion of an 'enduring substance sustaining persistent qualities' [22] is replaced by an actual entity as a 'process…not describable in terms of the morphology of "stuff".'[2] These actualized entities are atomic. That is, they do not change in themselves; rather they exist only in the instance of their becoming, perishing in actualization to be replaced by new actualizations (which we may or may not chose to view from certain perspectives as being ostensibly the same): an endless advance towards intensity and invention. Viewed thus, 'objects' are endpoints in processes of concrescence of complex events of relation, and a 'subject' (or 'superject', as Whitehead prefers) arises out of experience rather than interacting with a world that it somehow remains transcendent of [22].

As in life in general, the artwork here *is* the encounter: art is an event of relations. This notion of relationality, Massumi says, addresses objects and bodies from the point of view of their ability to change and respond – 'a coming together in a fusional event…a telescoping into a potential becoming' [3]. That is, rather than operating as 'establishment and reproduction of an ordered relay' [3], the relational is an immediate, 'emergent process' where something new occurs out of the relations [12]. Thus Lozano-Hemmer's insistence that his work is not interactive but 'relational' [2] - focused not on the fixed or mechanical elements of interaction, but on the potential for establishing relations, that always have an immanent, virtual quality to them. Relations are always improvisational, fluid and emergent, Manning states [7], a 'becoming' connectivity. So I might now seek to address the 'event' of the connections rather than the work as a stable object – the way relations develop between – or create - my body and the work, a 'mutual incipiency', a process of change and response [3]. Manning and Massumi use the term 'co-causal' [12] to describe this mutual emergence of the new through the flux of the forces of relations. Varela's use of 'enaction' similarly describes such events of relations between the world and body as events of mutual creation, neither wholly internal nor external. This is a 'co-determined' world rather than a stable environment that one is 'parachuted into' [20].

This might be thought of as self evident, useless information. After all, if all things are composed from events of relation, are not all artworks thus composed, regardless of the artist's intentions? I would argue, however, that the way many interactive works in fact

[2] An 'entity' is anything that is actualized. Whitehead also uses the term 'occasion' as interchangeable with entity, and this perhaps expresses the eventness of things more overtly. [22]

operate is to attempt to stabilize such unfoldings, erase the connections to the virtual – the future potential for immergence – and establish enduring actualized connections. With the shift in emphasis proposed I am no longer thinking of creating a stable artwork, but as an emergent or potential (that is, virtual) event that may occur or is occurring. What exists beforehand might be thought of as a proposition for an event, but it exists *as* an event only in a temporal relationship (or rather as a nexus of relationships) with the viewer, enfolded and unfolded through interaction, each nexus creating a singular event. Embodied enaction is always directed towards the 'next', the continuing evolution of the event [20], and therefore towards the virtual. Such events create body-artwork assemblages – contingent networks of interconnections, with multiple, unplanned, potentially contradictory variables of relation.

So in articulating the experience – the event of my body's temporary and temporal co-causal relations with a work I might view the artwork as an 'technique of relation' [12] and be assessing its ability to act as an inventive catalyst for affective co-causal relations to arise. Now we have changed the creative emphasis to the enabling of change, and abandoned control of outcomes in favour of propositional launching points. In doing so whole series of relational forces that resist any practical control might be brought into play to complicate and activate invention within the work. My concentration is now on the buildup of energy and rhythm between and within body and work; how the event moves beyond a mapping of simple cause and effect (move this way, a particular sound occurs), into something that has a 'self-tendency, (a) life movement' [12]. Now complex multiple actions and potential relations might catalyze a singular experience, perhaps moving beyond being able to be mapped or articulated. What is felt/perceived here in the moment might be more intensities of pure sensation, a building of energies expressed through combinations of movement, sound, image, posture and so on – but also potentially contradictory affectual relations that push and pull at the body; a kind of psychologically topological experience, being manipulated into different tonalities by the ever reconfiguring connections.

The excessive nature of such relations outstrips function and destabilizes orderly systems of exchange [2] - affect and sensation can never be fully incorporated into the productive perception, there is always an inarticulate remainder to complicate consciousness. Thinking relationally here begins then, to open artistic design to at the very least the inclusion of a wider range of relation in the functioning of the work. With the emphasis on process not outcomes, slippery, hard to define conjunctive and disjunctive forces – affects, inarticulate sensations, micro-perceptions, emotional tonalities – all might have the potential to be thought of as a fuzzy palette of options to be encouraged.

4. STRUCTURING ACTION AND FLOW.

But for the practicing artist, engaged with an interactive, generative or what we might now term 'relational' art practice, questions of how to structure or enable fluidity and maximize open-ended potentiality in practical terms is still problematic. How to structure so as allow for multiple, surprising outcomes, and how to create organic movement – the complex flow of prehension, synthesis and perishing pursued endlessly by further such creation – remains a question. For we cannot ignore structure - chaos in itself does not seem to me an answer, neither is mimicry of the everyday. Rather the special thickness of experience and the surprise of unusual connection and revelation that the art event can offer needs to be retained without losing the kind of underlying complexity and entanglement of everyday experiential involvement in the world.

Any such discussion of practicalities is necessarily propositional. Indeed, speculation – a drive towards an unknown conclusion – might be seen here as an advantage: inherently part of any technique interested more in establishing lines of flight than destinations. Think of it more as a kind of meta-modeling – an attempt to create potential tools for multiple usages, with the understanding that different combinations of techniques must be assembled to address each singular artistic problem [21]. Meta-modeling, Guattari says, 'abandon(s) all universalizing pretensions' involves a disentangling of oneself from systems of modeling that 'pollute our ways of thinking' [21], creating instead a contingent critical 'bricolage' of possible approaches to be utilized for the particular analysis at hand.

The two questions I then want to ask of any interactive work are: 'how can an art event generate its own satisfaction?' and 'how can the drive towards novelty be maximized in the event?' Without any intent of providing definitive answers, the two concepts I wish to consider in relation to these questions are those of drift and noise.

4.1. Drift.

Think first of the making of a relational artwork as the building up of propositions. These propositions might be multiple, possibly contradictory. If sound 'A' can happen, or sound 'B', but not both sounds, the sound not actualized still has, as Whitehead says, a creative role to play – both as a 'giveness' that shapes paths of potentiality, and as a continuing link to the virtual. The negated proposition remains a link to what might of happened, to unrealized potential that 'vibrate(s) against the conformal' [22]. Propositions then provide ongoing links towards the potentialities of the event, to a 'second phase' of the virtual: its 'real' potentiality rather than the 'general' potentiality [22], conditioning the potential by inclusion of the circumstances of the emergent event: those selected by the artist (layout, software, sounds, images, shapes and so on), plus what the participants

bring (physical capabilities, tastes, moods), plus then, to varying and probably lesser extents, the worldly circumstances surrounding the art event (culture, politics, geography, art histories, weather or what ever), which then create its virtual milieu.

An entity, Whitehead states, 'feels as it does feel in order to be the actual entity it is' [22]. The propositions composed within the art event, are launching points, 'lures towards feelings'. These feelings are none other than the prehensions [22] -internal again - whose drive toward satisfaction is the realization of some potentiality of the entity. 'Feelings' in the sense of prehensions are not necessarily anything to do with conscious thought. Rather they are a drive towards completion of an occasion. Thus an inanimate entity might be seen as being just as capable of a feeling, as driven towards its own satisfaction, as a sentient one. A sensor, for example, in itself might have the proposition of a tendency to notice movement. This may not happen, it is a potentiality, constrained by the given – its position, the mechanics of its construction, and so on. It has 'sensitivity' towards searching for this movement, the incoming sense data that drive its completion, its satisfaction in that instance as a movement sensor – whether it actually senses movement or not (exclusive potentials that in that in any occasion only one can be actualized while the other remains virtual).

And, although we might chose for conveniences sake to think of the art event as a single entity, it is perhaps better viewed as a 'society'[3] of entities, divisible into multiple, overlapping and simultaneous events or entities, each seeking – and competing – for its own satisfaction. That is, during events of concrescence it is always at a point of unfolding, facing multiple potential paths towards the various satisfactions. Multiple, fluid assemblages of eyes/brain/image, ears/noise/speakers/current, software/ sensor/movement data and so on, are each in themselves divisible again, each seeking resolution of their feelings. Such art events might begin with multiple proposals, luring even greater multitudes of prehensions, held in both inclusive and exclusive relation to each other, seeking – competing even – for their satisfaction, driven that is by the creative urge to turn potential into actual. This philosophical stance emphasizes most forcefully that art events are composed from ground up, with an understanding that the concrescence of forces builds towards an endpoint of an actual event, discovered and motivated within the occasion itself by complexities of virtual and actual forces.

How does the art event 'chose' which prehensions it follows through to satisfaction, which entities will actualize? Having set itself into motion through its propositional structuring and gained through feeling its own agency, it is not beholden to any external intentions or drive – it must sort itself out internally. But it does not strive to be the best event it can – the most efficient, original or surprising. That would again imply some kind of transcendent motivation. Rather, we could say, it drifts. 'Drift', as Varela says, implies a system that makes do – it seeks the 'viable' rather than the 'optimal' [20], it is 'pragmatic', its motivation is to find *a* satisfaction, not *the* satisfaction, it makes do with what it has, cobbles together a solution. Enabling a process of drift 'takes the place of task oriented design', Varela argues [20], it implies a system which is truly interactive – both within itself and its given circumstances – composed through that activity rather than representative of determined function or outcome.

A relational artwork capable of such drift might take many forms, creating many differing events. This does not mean that it drives towards making events necessarily different - it is indifferent to the quality or quantity of difference it generates, indifferent to the *demonstration* of change and relation that haunts so many interactive works. It settles where it settles. Perhaps some days the events generated would be markedly variable, on others it might seem to settle around the same outcomes. Change to one parameter, one tiny force operating within the assemblage might result in a complete turn around, little noticeable change or be negated by some other factor and perish unnoticed. But the artist relinquishes control over this, leaving it – encouraging it – to work itself out: it does what it does, whether disappointing one occasion, surprising the next.

4.2. Noise.

The 'society' that is the artwork assemblage can endure because new entities emerging within the art-assemblage conform to common feelings – their emergence is shaped in part by their relation to the society [19]. That is, although they retain potential to change, drift towards some semblance of stability is still possible. How then might a continuous and vigorous drive towards reinvention be structured into the event/s? Rather than just concentrate on the agency of the event to establish layers of relation, how can their perishing and replacement also be driven internally? To become an event that gains the power of continual self-invention of the everyday experience rather than one that drifts to a point of stasis requires a system that is able to include not just connectivity, but disconnections, failed, disruptive, competing and destructive relations and account for interference in the communications. Michel Serres proposes that 'noise' – that is, the interference in a relation – is a necessary condition of its existence, stating that 'if a relationship succeeds, if it is perfect, optimum and immediate; it disappears as a relation' [18]. That is, relations are a condition of difference in a system or assemblage rather than arising out of harmony or equilibrium. Relations are full of 'losses, flights, wear

[3] While entities themselves continually perish and are replaced, the things we experience as enduring actualities such as art objects or people are in Whitehead's terms 'societies' [19].

and tear, errors, accidents, opacity' that are their creativity, Serres states, and a system of 'pure rationality', without such excesses, interferences and disruptions is in essence a fascist system [18].

Serres' concept of the parasite presents a potential mechanism to complicate and expand the idea of co-causality. The parasite here has multiple meanings, being both a literal parasite, feeding off the energy (physical or social) of another, but also more importantly meaning the noise in the system of relations[4]. Now too there is a 'noise' in the system – the parasite – to propose. The parasite is essentially creative in that it forces into existence new logic, new combinations, and new orders of exchange [18]. It turns one type of energy into another, turns body movement into electrical current, software computations into sound, image into psychological discomfort. It disrupts communications, but produces something else (excessive) through its (mis)translation of relations. This third position in the system is itself unstable; the roles are interchangeable, fluid - each position is potentially noise for the other two - they lie in between any absolute or fixed position, always fuzzy and multiple; contradictory and irresolvable [18].

Thus it is a system of internally organizing and foregrounding the roles of instability and of difference in creation, embracing the possibilities of disruption as well as connection. It is a (self-organizing) multiplier of relations – it bifurcates any stable exchange, as a derivation from equilibrium it has an 'abuse-value' rather than exchange-value [18] to create new relationships through the eruption of difference; difference that 'recharges the activity of relating from which all experience emerges', Massumi states, to be seen not as deconstruction, but 'continued construction. Reconstruction on the fly. Not interruption: recharging, resaturation with potential' [9]. This implies creating a propositional structure where relations not only layer, but have also the inbuilt potential to interrupt each other. Whether such interruptions actualize is not the issue – even as virtual noise they create open-endedness: potential disruptions that nevertheless can create a tension, act on any actualized relation to keep it provisional, on the point of change or collapse. Each virtual configuration of the parasite introduces excess, something interesting and complicated to the relationship, keeps it on the verge of the event of expansion or transformation, and multiplies its virtual qualities rhizomically. As Massumi says, 'the virtual is the mode of reality implicated in the emergence of new potentials…its reality is the reality of change: the event' [10].

[4] Whitehead's 'non-conformal propositions' might be seen as something of an equivalent to the parasite, as their inclusion, Stengers argues, introduces the possibility of the disruption of social continuity, that is, the destruction of order and the introduction of novelty [19].

On a practical design level, the implications might involve firstly the acknowledgment and encouragement of a wider range of disruptive relations, and secondly the construction of software and generative events with the inbuilt potential to interrupt and distort each other. The first of these factors involves coming to an understanding of ways in which sensorial, affective and social relations affect the actual individuated experience in any event. That is, for example, understanding how the emotional tonality I bring to an event will colour my experience, magnifying some aspects, minimizing or negating others, connecting one experience to memories; or the disruption my movements bring to any stability of software/sensor relations; or how the vibrations of sounds felt through the floor will complicate the sense information gained through the ears; or how the affective tonality of the room alters with the arrival of another body, creating a hyper awareness or 'transparency' of temporality and my body in relation to the event, making me hyperconscious of my posture, disrupting my image of myself. And so on, and so on... While on one level banal or obvious, the exploitation and enhancement of these naturally slippery relations brings to the event a natural unpredictability to any planned interaction – continual subtle re-tunings of relations that modulate and invent. Secondly, as explored in the next section, in the more overtly concrete design of software or sensor technologies for interactive works, factoring in potential perishings or negations becomes as important as the establishment of possible relation.

4.3. Thinking Software (Noisily).

As a tentative step to thinking these ideas in some practical relation to computer music and software, I want to briefly consider Ableton Live, a DJ program much used by sound and interactive artists. The program, while quite capable of operating in a linear, highly structured manner, has also interactive potential – to be controlled by sensors registering a performer or larger environment or data stream via MIDI, or through programs such as Max or Isadora, and for events to follow a more flexible, topological process. Of course in conceiving of the software within a larger assemblage of an art event composed also of parts more readily able to operate provisionally – bodies, sensors and so on – we can to some extent 'diffuse' the pre-designed nature of software responses, but there is a need to address the potential for drift and disruption within the software itself, its 'minor' potentials - its capabilities to operate outside of its original design brief. I want to propose some tactics for thinking this conception of the program.

Live is designed make multiple loops conform to a set tempo – an external stable structuring - so as a first step we might consider the possibility of allowing instead a rhythmic drifting. Live organizes sound clips both inclusively and exclusively: multiple tracks arranged horizontally can play simultaneously, while each track contains multiple clips arranged vertically that are exclusive of each other. Thus simultaneously sounds

operate in both a compossible ('and, and, and…') manner horizontally and an incompossible ('or, or, or…') manner vertically - this is already the beginning of a potential network structuring. A dispersed rhythmic structuring can be initiated by turning off the quantization function for each clip so that they begin playing as soon as triggered. Now externally driven, incidental rhythms arise out of the internally generated tension of multiple loops – an outcome of multiple relational events, a contingent structuring rather than a centrally organizing rhythm as an architect of sounds; rhythm as an expression of difference, an elasticity 'stretching towards' an unknown next [7].

Live's midi connections are highly flexible and most operational parameters can be linked to MIDI. Link a clip to a movement trigger, and you initiate a simple cause and effect relation. This can be complicated through building into the design multiple potential relations. For example, other sensor events might potentially turn off the clip, and/or swap it for a different sound, and/or modulate its volume so that it might be inaudible or dominant, and so on. In such relatively simple ways we move from a linear causation of relation of movement = sound (realization of the possible), to multiple complex potential events intermeshed within a nexus of relations, the 'noise' of disruption a continual force moving the process into reconfiguration. Nexūs of relations here can be seen to begin to operate not just as independently self-satisfying, but also as complexly and fluidly inter-related through disjunctive events of emergence, as potential noise within relations, constructing through disrupting. This enriched connection to the virtual proposes relation as more than just complex vector relations of physical interdependence.

Multiply this exponentially for each sound event and its virtual potentials added to the equation. Think too of proposing these individual actualizations of sound as micro-perceptions - that is, not necessarily individually consciously recognized, but layering in combinations (of tones, timbres, overtones, rhythms, textures), to produce a 'society', the perceived sound, while retaining difference and their atomic nature.

This then is a topological web of relations, self modulating and designed as much for potential perishing as connections, primed for necessary injections of chaos through its multiply entangled chains of cause and effect. Again, such tinkering represents both small, seemingly inconsequential moderations, and at the same time, I would argue, a paradigm shift: the death of the (software) author to be replaced by the propositional event. The artist proposes sound events, in excess of possible actuality. With causality dispersed, notions of an artist or software 'agent' are replaced by a co-causal 'agency' (that is, a subject replaced by process). Within such simple tactics we begin to see sounds within the system becoming free floating events inhabiting a virtual soundscape - sounds as societies vibrating internally and externally with the tensions of relation - they begin to hum with difference and potential.

These are at best starting points for freeing a software program from its use-value, from productive function and stability. Beyond this 'tweaking' there must somehow be a larger picture in which we can conceive of a 'performative' software itself as process, as real time events, intensifications of creative relation. A material conception of a software positions it as a stable structure, pre-existing and determining in its interactions, a framework in which relation plays out – transcendent of the surrounding assemblage. But how can the sound proposed by the software ever escape the reality of assemblages of projection and capture – speakers and ears – through which its eventness is actualized? Can we perhaps more radically begin to understand it as co-produced? That is, as a software potentiality that is itself (re) constructed out of the event of its actualization. In this organic approach we might begin to see each event as a gathering or holding together of its surrounding constraints. In each event there is an atomic re-enaction of software relations and potentiality – a re-creation, with inherent opportunity for variation and invention, movement and novelty. Thinking through and exploiting this kind of actualization, capable of an internal recreation of its potentiality, is a re-saturating of software with a virtual that is in itself relationally eventful: composed eventfully, always at a point of unfolding or bifurcating. This, perhaps, is a path towards a minor conception of software, a flight path from the software hegemony, that artists might begin exploit to create a relational, dynamic art event inclusive of computer programming.

5. CONCLUSION

The point, I might argue, to a shift from material to organic modeling of the interactive event, is in a sense to not have a point: to rescue such art-events from purposefulness, to encourage growth, mutation and destruction without predetermination, to enable an event to generate its own forces of concrescence and find its own satisfaction. This implies not an absence of artistic input in any negative sense, but a shift towards propositional, speculative structuring with an emphasis on the intensification of experienced relation.

Whitehead's philosophy offers the vocabulary to describe these dynamic, emergent and complex events of relation, which, he argues, cannot be conceived as static, stratified 'stuff', they can only exist in the unfolding of their eventness. The task for the artist, from this standpoint, is to steer interactivity back from an artificially stable plain towards the propositional, and to invent ways to keep the event, the temporal experience of participation, unstable, keep assemblages fluidly creative. Here perhaps discourse can move beyond questions of chance and freedom that hound interactivity

to death, and focus on the 'eventness' of participation rather than demonstration of its mechanics, in ways that are capable of opening up to excess and disorder, embracing the complexity of movement, sensation and affect that is the nature of embodied experience.

For me this is a positive shift that represents an expansion of the potentiality of interactivity, an opportunity to absorb the lessons of the Conceptual art movement's radical move from the representational to the propositional without losing the immersive, visceral strengths of the interactive experience.

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MORPHING AURAL SPACES IN 7.1 MUSIC

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ABSTRACT

This paper outlines and discusses a method for improving interactivity in using space (aural and musical) as a compositional device in surround composition. The relative merits of 7.1 over 5.1 in this context are summarised. The question is asked if 'space' were granted equal compositional importance with the sounds themselves, what forms might this take, and how might one negotiate technical limitations that are imposed by digital audio workstations. Examples of technical and creative approaches are given.

1. INTRODUCTION

As a composer and music technologist I have an interest in creating music[1] in surround for broader dissemination than just for those people fortunate enough to experience immersive works at 'one-off' special events with advanced technical catering. Once such events are completed, so is the opportunity to experience these works. Consequently, effective broader dissemination of music mixed for surround sound requires utilisation of consumer technologies.

Further, the level of detail required to localise sound has been the subject of psychophysical studies [1, 12, 18, 22, 36] and perceptual studies [6, 9, 24, 28, 29, 31, 33]. It has been proposed that, "*at least* six loudspeakers (not including LFE) are needed to reproduce the spatial impression of a diffuse sound field" [12]. My experiences in working with surround support these findings, most particularly with respect to the importance of phantom images created between pairwise speakers [8, 20, 21, 23, 25, 29, 32], commonly identified as an issue with 5.1 [6, 9, 12, 24, 28, 29, 31, 33].

There is a wide array of consumer accessible formats and configurations [14, 15, 16, 19], that comprise more than 6 speakers, however the HD 7.1 surround configuration utilised by *Dolby® TrueHD* and *DTS-HD Master Audio™* is the default for the Blu-ray DVD format. Consequently I suggest that this configuration is most likely to be a 'default' configuration and have accordingly chosen this as a surround platform for my research.

The Dolby® TrueHD and DTS-HD Master Audio™ 7.1 surround formats (Blu-ray) attempt to address the recognised weaknesses of 5.1 and 3/4.1 [6, 9, 12, 24, 28, 29, 31, 33] by using 6 equidistant speakers each separated by 60º (in conjunction with the centre and LFE channels). This configuration therefore allows a consistent set of speaker pairings to occur around the entire 360º arc.

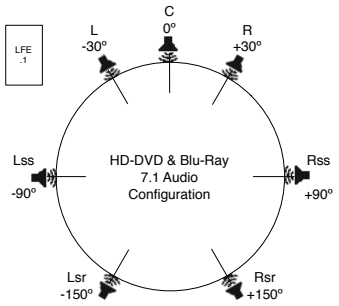


Figure 1 - 7.1 Blu-ray specification

My current research focuses on utilising a series of pairwise phantom images available within the HD 7.1 configuration that exploit psychoacoustic elements of sound localisation and physical attributes (generalised) of HRTF. I am treating the surround speaker array as isotropic for compositional purposes and exploring the manipulation of multiple aural spaces that can themselves be moved around the panoramic field (360º). I employ techniques to move space in conjunction with or separately to the sounds that excite space and (partly inspired by the idea of quantum mechanics) that move sound *across* space (without moving sound or space). Of particular interest in my work then is the ability to morph sounds and spaces fluidly, whether in tandem or separately, and to assess levels of cognitive dissonance or consonance that occur.

2. BACKGROUND AND CONTEXT.

Working variously as a musician and composer, sound designer and engineer, I am deeply engaged by the concepts of sonic energy and narrative and the musical elements of timbre and dimension[2].

I started working with surround sound in 2002. I experimented with Quadraphonics, Ambisonics and 5.1 before predominantly settling on 5.1. At this time 5.1 was gaining momentum as a publicly accessible (driven by home theatres, later supported by HDTV surround broadcasts) and, therefore, viable platform for the wide

[1] While there is no single and intercultural universal concept defining what music might be, within the context of this paper I define music as an art form whose medium is sound.

[2] Herein, dimension includes: [1] the ambient field in which the sounds sit (A piece can have multiple ambient fields); [2] the spectral height from low to high frequencies; [3] the perceived distance from the listener; and [4] the panoramic width of each sound itself.

dissemination of surround music and "auralisation" [36]. Commercial releases of musical works were on the rise. Available repertoire included: classical works which were relatively easy to produce in surround, albeit the surround channels commonly for room reflections; re-issues of older albums such as Pink Floyd's *Dark side of the moon*, Eric Clapton's *Slowhand*, Billy Joel's *Piano Man*, Queen's *A night at the Opera*, Beach Boys' *Pet Sounds,* etc.; and works re-envisioned by the artist or created specifically for surround such as Sting's *Ten Summoner's Tales*, Chick Corea's *Remembering Bud Powell*, Studio Voodoo's *Club Voodoo*, Dave Grusin's *Two for the road,* etc. At the same time, experimental and art music artists were exploring new ground musically and technically [2, 10, 11, 13, 17].

Two fundamentally different aims of rendering spatial audio were apparent: (I) to achieve physiological realism (to emulate or document self-created, biological, architectural or natural acoustic environments); and (II) to create a sonic canvas for surreal (conflicting but apparently real[7]) or completely imaginary forms of creative expression[3].

This paper focuses on the creation and manipulation of surreal or imaginary spaces for compositional purposes and makes no attempt to reconcile the difficulties inherent in documenting reality or physiological realism. The context being scrutinised is therefore one where the artist is either the architect of the aural space(s), or one where a work uses aural space as part of the musical structure – ergo an aural architect [3].

3. MUSICAL AND AURAL SPACE

The term 'space' in music is most commonly used in reference to two concepts, motion or environment. With reference to motion of sound (e.g. melodic shapes), the term space signifies action and non-action. I refer to this usage herein as musical space. The second usage of the term space refers to the sonic characteristics of the location with which the sound interacts, herein referred to as the aural space. Aural space has 4 key domains - width, depth, height and temporality (length, width, depth and height over time[4]).

With reference to aural space, Emmerson [7] proposes the idea of a frame (a defined area of interest), applied progressively from the largest to the smallest scale. The frame comprises a landscape bounded by the acoustic horizon (distance limit from which a listener can hear sonic events) [3, 30, 34]. The landscape then comprises an arena, within which we find a stage, upon which we frame an event as depicted in figure 2 [7].

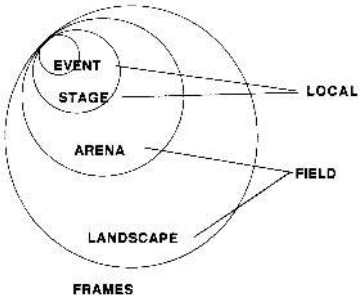


Figure 2 - Soundfield Frames , after Emmerson [7]

The aural architect, during both recording and post production stages, has control over the creation of landscapes, arenas, stages and events from the foreground out to the acoustic horizon. Further, multiple frames can co-exist, within which multiple and different landscapes, stages and arenas can be created in cognitive dissonance or consonance.

3.1. Sound as Objects in Space

An increasingly broad array of sounds—abstract, generated, permutated or extracted from other sounds, often unidentifiable or other-worldly—have become considered common constituent elements of complex textures in music from art through to contemporary commercial music, soundtracks and installations. This is evident in works from composers such as Pratella, Russolo, Varese, Schaeffer, Boulez, Cage, Schafer, Xenakis, Stockhausen, Brecht, Horner, Zimmer, DJ Spooky, Cabaret Voltaire, Scanner, Newton Howard, and Elfman to name a few.

Similarly, advances in technology and parallel changes in aesthetic sensibilities have lead to an expanded use of ambience and aural spaces—abstract, generated, permutated or extracted, sometimes unidentifiable and other-worldly—within which these sounds are heard to exist. A large range of sounds now exist that do not have a pre-established connection to a particular aural space. Not only can we place sounds within as many different types of 'space' as imagination allows, but as Trevor Wishart points out "*we can combine real (that is, recognisable as being possible in the real world) or imaginary, objects or spaces in any combination"* [36:146].

This creates a (wonderful) paradox between sound and aural space highlighting one of the paradoxes of compositional process; does a composer think more about using sounds to build a composition or using composition as a means to organize sounds? In designing a sonic work, this chicken and egg question can equally be reframed as to what comes first, the 'space' or the 'objects' in it? There may be no definitive answer, however this paper of course prioritises discussion around Emmerson's concept of framing sounds (events) within arenas and landscapes.

[3] It is important to note that these approaches are not mutually exclusive.

[4] A fifth of course can be seen in changes (delta) to each of the 4 key domains.

4. MAKING THE CONNECTIONS

Mixing music is a performative art form. The same real-time reactive and collaborative sensibilities that are central to performing with other musicians are present when mixing. As a performer I typically engage with one instrument at a time, however as a composer or mix engineer I engage with all instruments simultaneously. This demands a heightened sense of flow, speed and dexterity in decision-making, and an ability to creatively stand back from the 'whole' while at the same time delving deeply into the minutiae of all the sound elements. No different to any instrumental performance, a mixing performance requires technologies that enable interactivity and control-efficiency such as a performance controller (e.g. Monome) or mix control surface rather than the limited control of a computer mouse or track pad. As will be shown, surround functionality in Digital Audio Workstations (DAWs) is currently greatly limiting in terms of accessibility and functionality, technically and creatively.

Of particular interest herein therefore are two aspects of the surround panner: accessibility and functionality.

4.1.1. Accessibility

Surround panning functionality is typically only available to mono or stereo tracks. Multichannel tracks with more than two (e.g. stereo) channels presume hardcoded outputs and as a result do not present full featured surround pan controls[5]. Figure 3 shows two tracks in Protools for example, the left track being a 7.1 track with no pan controls and the right being a stereo track with surround panning controls. In other DAWs (e.g. Logic) a single locator is enabled for all outputs. This technical limitation brings significant creative limitations, not the least of which is a lack of control over creating phantom images.



Figure 3 - Surround panner availability

This handicap can however be circumvented to a degree. For example, one single 7.1 event can be broken down into 5 tracks comprising 3 stereo pairs (L-R, Ls-Rs, Lss-Rss) and 2 mono outputs (C and LFE). This however increases the performance challenges substantially. For example, a work may easily have at least 4 different blanket spaces. These may include early reflection 'ambience' to provide extreme depth control (commonly a convolution reverb), a short room-styled space to provide a bloom or halo to a sound, a longer reverb (typically with minimal early reflections), and a delay (to simulate early reflections without smearing or clouding the mix). Each track (5) of the single 7.1 'instrument' requires an aux send for each space (4) resulting in 20 aux send faders (each requiring control) for each single 'instrument'. The pieces I typically work on have between 40 and 140 tracks. For this reason, the routing limitations inherent in surround-sound DAWs are understood to be significant.

4.1.2. Functionality

A surround panner is a device for controlling the location of a sound source using a positional reference on an x-y linear grid. Various points on the grid equate to discrete outputs, with points in between these using amplitude panning to determine levels to multiple outputs. While surround panners are typically used to locate sound within a surround speaker system, the method proposed herein utilises surround panner functionality to route signal to different auxiliary busses for the purpose of applying different digital signal processing (DSP) to discrete paths.

This method requires the creation of multichannel (e.g. 7.0) auxiliaries, which are then separated into stereo or mono busses in the DAW. At the time of writing this, the only DAWs that provide this capability are Avid's Protools[5] (using either Protools HD or Protools LE with surround tools) or MOTU's Digital Performer. Avid's Protools has been chosen for the purposes of this paper.



Figure 4 - 7.1 Bus (& sub-paths) for morphing pairs

Figure 4 shows the bus setup for a single 7.0 surround auxiliary send (Morph-1). This is separated into 3 stereo sub-pairs (Morph 1a – 1c) and a mono send (Morph 1d). The matrix view to the right of image shows which panner outputs are used for each pair. With reference to figure 5 (superimposed numerals), Morph 1a is then using outputs 1 and 2, Morph 1b is using outputs 3 and 4, Morph 1c is using outputs 5 and 6, and Morph 1d is using output 7.

In this scenario then, a single 7.0 auxiliary is acting as the morphing controller for up to four discretely configurable spaces, therefore can be considered as a morphing-panner. The stereo returns (with associated DSPs) have their inputs assigned to Morph 1a, Morph 1b and so on. The outputs of the DSP channels are assigned a position in the surround field using an output surround panner.

[5] Different DAWs have been tested for features, including Apple's Logic Pro (v6-9), Avid's Protools (v7-10), Steinberg's Cubase (v5), Cockos' Reaper and MOTU's Digital Performer (v6-7).

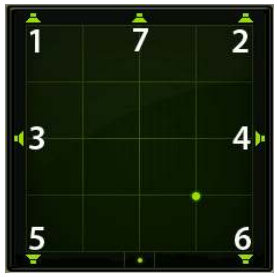


Figure 5 – Surround panner (numerals superimposed)

The positional reference point on the morphing-panner therefore no longer correlates with the perceived location of the event, instead representing send levels to multiple DSP effects. Multiple outputs operating simultaneously from a single controller results in the ability to morph multi-various spaces using simple gestures.

Given that the panner also has 3 dimmable locations (Center{7} and Sides{3,4}), the complexity of results achievable from simple movements on a single controller (x-y positional reference) can be quite profound. Adding 3D divergence (front, rear and front-rear) provides further routing possibilities with the potential to send to all (four in this example) arenas from a single controller if desired. Figure 6 shows the signal path of a single stereo channel and two of its associated stereo returns (Morph 1a and Morph 2d).

In this example (figure 5), a mono event is being sent (ignoring pan law adjustments) at 100% level to outputs 4 and 6, and 50% level to outputs 3 and 5. If 3 and 4 were disengaged completely, the controller would effectively become quad (+ centre), meaning that the same controller placement results in 100% level to output 6, and 50% level to outputs 2 and 5.

Figure 6 shows a setup in Protools (4 mono audio tracks and 4 stereo pairwise spaces) with the auxiliary surround panners (morphing-panners) configured as morphing controllers. These are nested above the output surround panners that then determine the location of each event within the surround speaker setup.



Figure 6 – Morphing sends (4) & stereo returns (4)

Figure 7 depicts the signal flow for 2 of the 8 available (in this example) spaces when using two surround sends.[6]

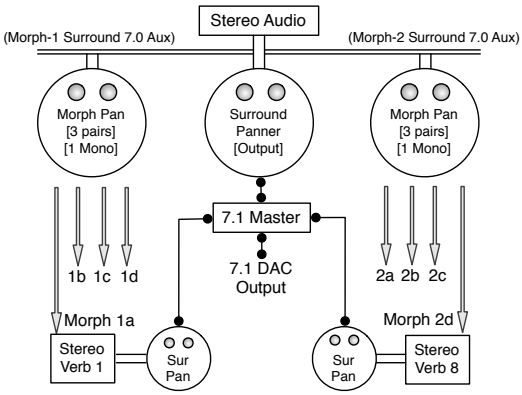


Figure 7 – signal flow for morphing spaces

5. CREATIVE APPLICATION

The following example[7] describes the framing of an event where the only change is to the extent and nature of the landscape. The event is a 13 second extract from a whale song recorded using a hydrophone. I have subjected the sample to a forensic cleansing to remove extraneous noises so the result also contains no depth information. Subsequently it is not possible to gauge proximity with the result that the sound appears close. The sample is repeated 3 times over a 2-minute interval.

The event is routed via a single surround panner to 5 separate stereo arenas with a combination of stereo and mono inputs with all having stereo outputs)

- L routed to input Left distant halo (stereo input ambience)
- R routed to input Right distant halo (stereo input ambience)
- Ls routed to input bandpassed ping-pong delays 1 (mono-in, stereo-out in series with the stereo halo)
- Lss routed to input bandpassed ping-pong delays 2 (mono-in, stereo-out in series with the stereo halo)
- Rs routed to input long reverberant space (stereo input reverb)
- Rss routed to input long reverberant space (stereo input reverb)
- C routed to nowhere (allows option for source signal only)

Figure 8 depicts the (approximate) location of each space (positioned using output surround panners) during the time of the event. The whale song event remains in the same position for the 2-minute timespan, located between Left and Left Surround speakers. With the morphing-panner at 0º (position 7) the overall event is framed in a strong early-reflection 'halo' (aural space) that places the sound into the distance. As the event

[6] The example used throughout this paper would result in 8 configurable spaces, however up to 14 would be available across two sends if all mono inputs were used (or 16 if LFE controls were included).

[7] This example is taken from an original musical work (unpublished) titled Viscosity.

continues, the morphing panner is brought back and left toward the 270º position that keeps the event in the distance, but further introduces some bandpassed (distant) long delays (aural/musical space) to support the impression of spatial distance. The delays are located toward the rear left of a front-facing listener and are therefore distinctly locatable but also appear (in series) within the halo and therefore enhance the overall sense of size. From there the morphing panner is brought toward 90º, which introduces a long reverberant (musical) space that is located in an opposing location to both the delay and the original distance halo. The soundfield starts by seeming real, but morphs into the surreal via the introduction of surreal and imaginary arenas. As a call and answer, another short whale song sample is rendered through a middle distance room sound (110º), which provides a combination of halo (aural) as well as temporal (aural/musical) space. The overall effect is still relatively subtle, stopping well short of auditory sensory saturation [4] that could be achieved with more extravagant choices.

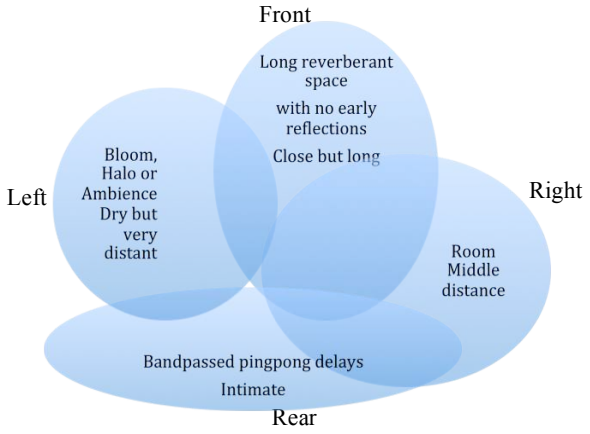


Figure 8 – Example multi-space placement

In contrast to placing a mono sound (the event) in front-left, this event can now be framed on multiple stages in multiple locations. This creates an expanded landscape that occupies the complete surround field without ever shifting the event from front left. Importantly, this whole scene has been made to happen with a simple movement of a single controller

6. CONCLUSION

In contrast to the rather obvious effect of moving sound events around the listener, moving sound in associated aural spaces through pairwise planes utilising surround panning functionality produces a sophisticated fluidity and seamlessness. Space can now take on a more significant role in composition.

Being able to route sounds into multiple spaces allows a greater sophistication in defining each sound's (and subsequently the overall) timbre, depth, width, proximity and perceived energy. The physical constraints on doing this may typically make the approach seem unfeasible, however I have found that reducing the interactions to a single controller means that creative flow, speed and dexterity in decision-making have all been enhanced.

The equidistant speaker array used in 7.1, and the isotropic nature of it, greatly enhance (in my experience, not yet empirically validated) the ability to use multi-various musical and aural spaces in pairwise planes and from any direction. Certainly, I have found the ability to achieve smooth morphing of space to be ineffective in 5.1, but very effective in 7.1. Importantly, being able to create, manipulate and mediate sophisticated spaces and cause them to interact in their own right (e.g. series vs parallel) has proven to be a rich source of compositional inspiration.

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New digital interactions with John Cage's *Variations IV, V* and *VI.*

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ABSTRACT

To celebrate the centenary of John Cage's birth in 1912, Western Australian new music ensemble Decibel undertook the realization of the American composer John Cage's (1912 – 1992) complete Variations I – VIII. *The works offer a unique insight into the development of Cage's approach to composition practice, aleatoric approaches, spatial arrangements and the use of electronics. Entitled the "John Cage Complete Variations Project", Decibel created a performance of the eight pieces in around an hour. The preparation and reading of the scores that make use of transparent sheets (*Variations I, II, III, IV and VI*) has been adapted using digital score creators and readers. This permits real time generation of measurements and graphics, as well as the assemblage of performance symbols, that can occur during the actual performance of the works. This paper examines the approach to the* Variations *whose instructions result in the employment or creation of maps:* Variations IV *(1963),* V *(1965) and* VI *(1966).*

1. INTRODUCTION

John Cage's eight *Variations* composed between 1958 and 1967 are a varied collection of compositions prepared in very different ways. They take score forms that range from very precise instructions (*Variations I, II, III, IV, VI*) to reflections on early performances (*Variations V*), handwritten sketches (*Variations VIII*) or only few words (*Variations VII*). The works encapsulate Cage's interest in maps, astronomy, system design, spatial sound production and multimedia. They at times incorporate similar orchestrations to his *Imaginary Landscapes* series (five works composed between 1939 and 1952) such as radios, tapes and oscillators. The *Variations* introduce new equipment to Cage's composition toolbox – antennas, light sensitive resistors and telephony. As David P. Miller suggests, the *Variations* offer "a trajectory away from self-contained concert pieces (*Variations I* and *II* in particular) and culminating in theatrically ambient works that draw on an increasingly broad range of source material" [1]. In addition, the approach to notating and communicating Cage's ideas undergoes a dramatic development throughout the *Variations.*

Despite a number of performances of individual *Variations* (though surprisingly few it would seem), *Variations IV* and *VI* have been performed the least. This may be due to a number of factors; the complex and rather convoluted instructions provided for *Variations VI* and the possibility of unperformable arrangements or outcomes in *Variations IV*.

Transparent plastic sheets are provided with the scores in *Variations I, II, III, IV* and *VI. Variations V* and *VII* and *VIII* use text only, and appear more of a record of a past performance than as an instruction per se[1]. Text features heavily in all the *Variations* scores: however, even with the transparent sheets provided in the score folders, detailed instructions as to what is to be done with these sheets are provided and are often difficult to interpret[2]. Yet the key distinguishing qualities of *Variations IV* and *VI* is the requirement to generate specifications for the placement and directions of sounds in the space. More information is provided about the where the sound should be produced, than how to produce it. As such, they have offered up different challenges for real time digital score generation when producing what Miller has called 'performance scores'[1]. *Variations IV, V* and *VI* create performance 'maps' rather than scores, and performing them involves a process of exploration and discovery.

Decibel has been working with graphic, mobile and real time screen scores for some time, and the *Complete Variations Project* has provided a possibility to apply experience and expertise gleaned from the creation of new works to the realization of historic pieces [23]. Each of the *Variations* has a performance score produced digitally, and realized in real time, where the computer performs cut ups, random placements, measurements, joining up of figures and realisation of the resulting score onto a computer screen or projection to be read by performers[3].

As part of its performance strategy, Decibel distributes scores in MaxMSP devised players to multiple Apple MAC laptop computers over a wireless network, driven by a master computer. This made it necessary to synchronize score events by linking all the required data for sending across the network. As certain networking protocols, such as User Datagram Protocol (UDP) transmission do not ensure that information is received in the same order as it is sent, concatenating the information into a single matrix data structure ensured that the order of information remain intact. To facilitate this networking and allow troubleshooting, a network utility has been developed by James to facilitate and monitor network traffic in MaxMSP, the interface of which can be seen in Figure 1.

[1] Cage subtitles *Variations V* on the cover page with "thirty seven remarks re and audio visual performance" and footnotes the scores with references to a certain performance of it (the premiere)[14].

[2] An admission shared by David P. Miller in his paper on the *Variations* [1].

[3] *Variations V, VII* and *VIII* do not require scores at all, as performance notes are all that are required [13,16,17].

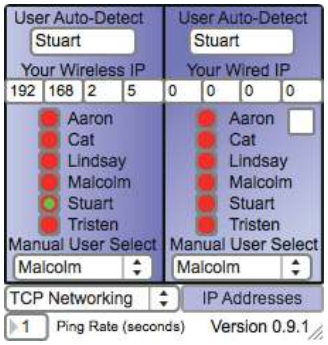


Figure 1: The interface for the Decibel network utility has the ability to be used for TCP and UDP networking, on wireless and wired networks.

This has meant that all the scores created for the *Variations* project are visible to each performer at all times, as required. This also means that any projections of the scores may come from any performers computer. The generation of data required for creating and displaying the scores of *Variations IV* and *VI* was carried out using a combination of Max/MSP and Java programming by Decibel members Stuart James (MaxMSP), Lindsay Vickery (MaxMSP) and Aaron Wyatt (Java). The premiere of the Variations project was held in was a projection room at the Goethe Institut in Palermo, Italy in early 2012 [21].

2. VARIATIONS IV (1963)

Variations IV has been dubbed the pivotal work in the *Variations* series of works, because "it takes the distribution of sound sources within and outside a given space as its primary point of interest, laying the ground for the commitment to total environments that marks *Variations V* and *VII* (and arguably *VI*)"[1]. Cage's ongoing interest in new technology seems to have been directed towards an environment that would allow him to map sonic space by "allowing access to the full range of all the specific parameters of sound"[2]. Here Cage refers not only to parameters such as timbre, dynamic, pitch and so on, but also spatialisation.

Variations IV is the second part of a group of three works of which *Atlas Eclipticalis* (1961) is the first and 0'00" (1962) – also know as 4'33" No.2 - is the third. The sequence relates to HideKazu Yoshida's interpretations of Japanese Haiku poetry, and aligns lines of the poetry to different psychological states: *Atlas Eclipticalis* represents 'nirvana', *Variations IV* represents 'samsara', (the turmoil of everyday life) and 0'00" 'individual action' [3]. This 'turmoil of everyday life' could be considered realised twofold in *Variations IV*. The placement of performers around and outside the space, away from the stage or concert hall brings the musicians into the world, away from the theatrical presentation of the stage. In addition, the reluctance of Cage to give specific instructions as to what sounds be produced, going as far as to suggest that "a performer need not confine himself to a performance of this piece. At any time he may do something else"[13]. Both these parameters offer up risk and uncontrollability, much more than in previous *Variations*. Here, Cage is moving closer to a different paradigm for music altogether, where performers are removed almost completely.

The actual location of the performers - playing or 'doing something else' - is clearly marked by way of the score. Continuing the use of transparencies as in the first three *Variations,* Cage provides a sheet with seven points and two circles to be cut into separate parts. One circle is placed on a map of the performance venue, to be provided by the performers. The other eight remaining points and circle are dropped "inside or outside" the map and lines then are drawn from the fixed circle to the points [13]. The second circle only becomes operative when a line intersects with it. Cage then instructs that "sounds to be produced at any point on the lines outside the theatre space"[13]. An intersection with the dropped circle indicates a sound producing system inside the space. The instructions are tailored according to a theatre space, a multi floor building, apartment, cave or outdoor space. Decibel worked in a theatre for the first performance. In *Variations IV*, the provided symbols are not to be 'performed' as in previous *Variations*, or even given musical parameters at all. Rather, the performance score provides arrangements for the placement of the sounds in particular space, providing another major shift in Cage's compositional approach and the beginnings of his interest in spatialisation.

A recording of *Variations IV* recorded by America's KPFA Radio for the birthday of Cage's long-term collaborator David Tudor in 1965 produced a work some thirtyone minutes long. It features loudspeakers placed in hallways outside the performance space that interacted with speakers inside the space [5]. The performance features a range of radio announcements, radio static noise and music, which come to the listener from different areas of the space, and heard as different timbres and dynamics on the recording. An earlier 1964 performance in Los Angeles' Feigen-Palmer Gallery went for some 6 hours, and featured two rooms fitted out with complete sound systems that included recording and mixing equipment, numerous radios, tape players and record decks [7]. In this performance, microphones were placed strategically inside and outside the building (one was suspended above the bar, another out in the street to catch the passing traffic)[6].

Decibel developed upon earlier performance score engines created for *Variations I – III* to compile *Variations IV*[4]. Due to low processing overheads *Variation IV* was implemented solely in MaxMSP. All scored elements were drawn using the quickdraw primitives found in the jit.lcd object, permitting the

[4] For a detailed examination of what was involved in the creation of the first three *Variations*, see Vickery, Hope, James. (2012). "Digital adaptions of the scores for Cage *Variations I*, II and III" [4]

creation of all necessary objects referred to in Cage's score: the circles, lines, points, and the lines to be drawn between them. A map of the venue was attained and inserted into the program, which then complied the data and superimposed it over the map[5] (see figure 2).

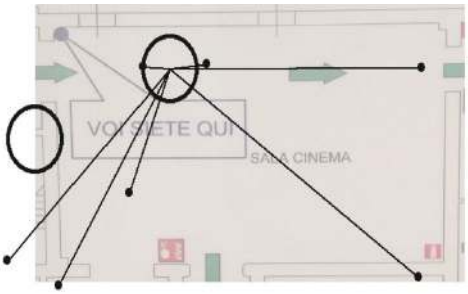


Figure 2: The score generated for the first Decibel performance of *Variations IV*. In this version, all points and lines were accessible to the performers.

A series of random coordinates were generated for determining the location of the two circles, seven points, and resulting lines. These random numbers are then used to instruct jit.lcd to draw these objects to the screen, emulating the 'dropping' of the cut up transparency. There are three layers of jit.lcd with a differing transparency assigned to each, allowing the superimposition of the graphics over the venue map. A threshold mechanism was employed to control the amount of transparency applied to each layer, allowing an organic control of the way visuals appear and disappear in the real time realization of the score, fading in and out.

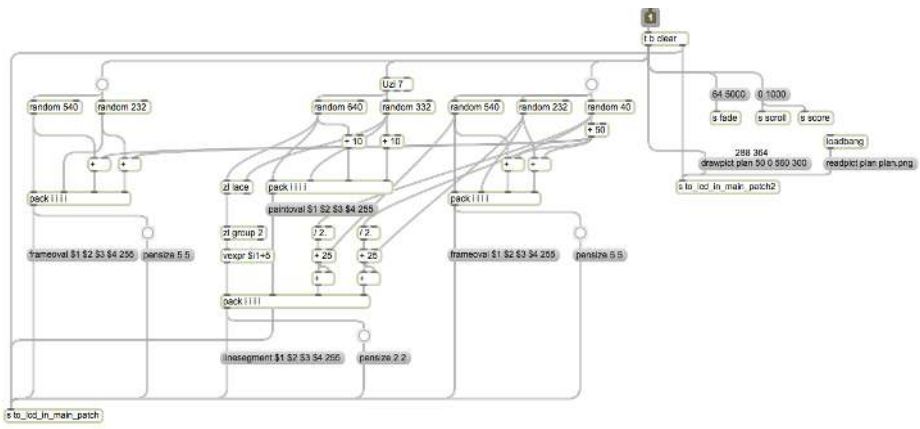


Figure 3: The MaxMSP patch used to generate the score for Variations IV.

As the graphics are 'dropped' randomly over the map, it is possible for certain outcomes to be less successful than others, depending on the venue and the portability of equipment. For example, some venues have inaccessible rooms encircling them, meaning it is difficult to get outside the building and still be audible, though I'm sure Cage would approve all the same. Often certain venues have areas that could be deemed interesting to use and this systems enables some flexibility in including them in the performance. The realtime generation of the lines on the map enables a quick turnover of different performance score options if desired, until a satisfactory result is achieved (figure 4).

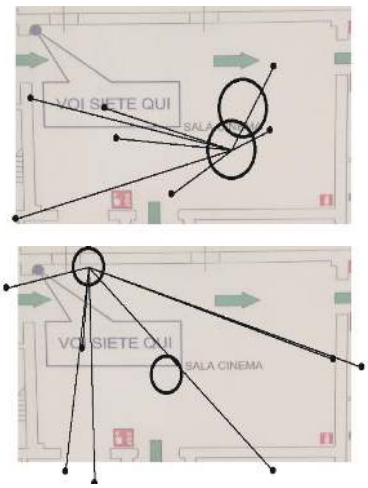


Figure 4: Two other iterations of a possible score for *Variations IV* at the premiere performance. Here, certain impossible arrangements have resulted, such as behind walls that are not accessible.

Decibel generated some five or six arrangements before settling on one thought to be workable, as shown in Figure 2. Figure 4 shows two other possible options. Six performers situated themselves on the map. Unlike early performances using radio and tapes, Decibel engaged viola, electric cello, spoken word, bass flute (which ended up in a bathroom) and one performer dedicated to 'doing something else'. In addition to these roles, the performers walked about, talked to each other, opened and closed doors, flushed toilets and other took part in other activities. As can be seen in Figure 2, the second circle was not engaged and as such a 'sound system' was not employed. Unlike some early performances of *Variations IV* by Cage and Tudor, dance did not make part of this performance[6][13]. The score was projected onto the empty stage, and a series of

[5] For the premiere, it was a photograph taken of the fire exit map beside the stage.

[6] Merce Cunningham's choreography for *Field Dances* (1963) and *Cross Currents* (1964) was featured [5, 20].

versions were generated until one that was achievable given the resources available was attained[7].

3. VARIATIONS V (1965)

Cage does not provide a score for *Variations V*, instead he offers the possibility for the performers to obtain a star chart and to use it "as though there were a drawing of the controls available and – on transparency – transcription from astronomical atlas which (if it were superimposed) would give suggestions for use of controls" [14].

Decibel chose the chart shown in Figure 5 (which resembles Cage's own score for his *Fontana Mix* (1958)). During the performance the chart is continuously repositioned, moving smoothly in the vertical and horizontal dimensions, jumping to particular new positions and expanding and contracting. The performers realize the work by interpreting the components of the score that are framed by a circle, colour-coded to correspond to each player, as "suggestions for use of the controls". Each performer's circle also moves freely around the screen[8]. As in Variations IV, all programming was completed in MaxMSP.

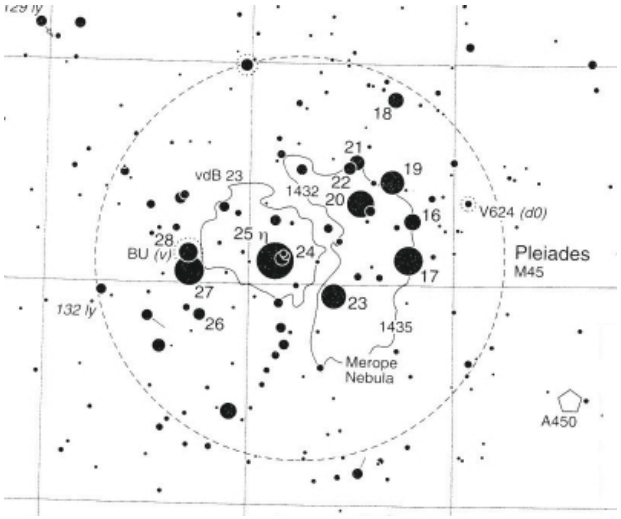


Figure 5: The Astronomical chart used for Decibel's version of *Variations V*.

The trajectories of the circles are programmed so that their movements are indeterminate, but permutate between a range of contrasting behaviors: wander – in which the movement of the circles are completely independent; follow – in which the trajectory of the one circle is mirrored by the remaining ones; and converge in which the (moving) position of the one circle is the circles converged upon by the others. In addition to these behaviours, each circle is programmed to grow in size at indeterminate points in the performance, indicating that the corresponding performer should take a more soloistic role. These behaviours are illustrated in Figure 7.

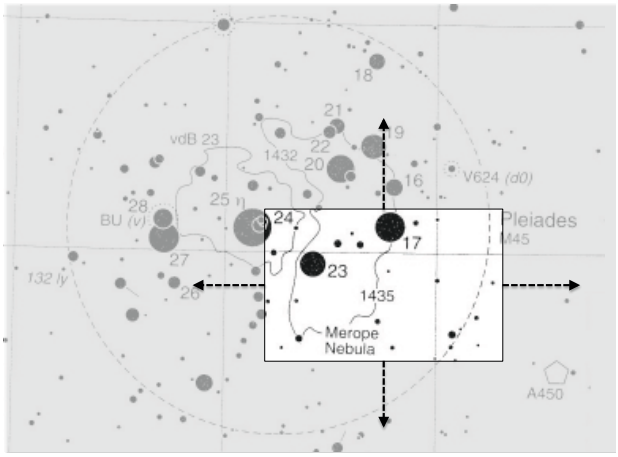


Figure 6: A representation of the relationship between the mobile score and the visible score that appears on the performer's screen (shown here as the opaque rectangle in the lower left-hand corner). The visible portion of the mobile screen is continuously repositioned during the performance.

The movement of the score and the behaviours of the circles provide a focal point of the performers reading the score. The performers are left to interpret the meaning of the symbols in relation to the electronic control parameters at their disposal. Therefore the structural outcome of any particular instantiation of the work is extremely indeterminate, relying as it does upon indeterminate trajectories both of the score and the circles as well as the performer's interpretations.

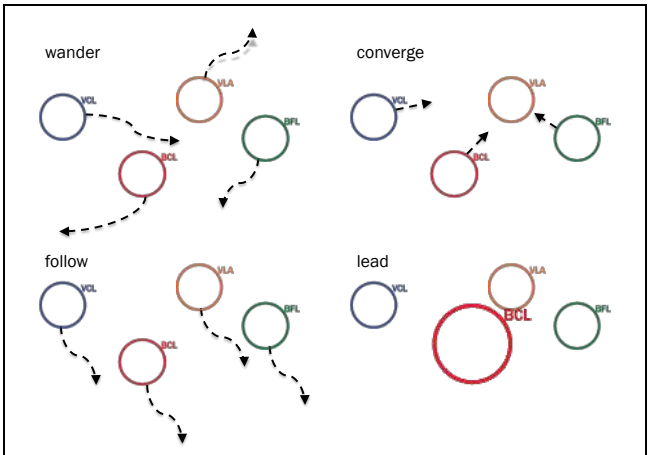


Figure 7: A graphical representation of the behaviours of circles in *The Talking Board.*

[7] The premiere performance had many restrictions on accessing outside space, and the speakers were not easily moved, and had very short cables. As such a few different versions were generated as part of the performance.

[8] A similar arrangement was employed by Decibel in Cat Hope and Lindsay Vickery's work *The Talking Board* (2011), with a collaged image created by the composers.

4. VARIATIONS VI (1966)

Like all the *Variations* after *V*, *Variations VI* is characterized by "the unpredictable interactions of multiple simultaneous systems"[8]. After the very open nature of *Variations V*, *Variations IV* returns to the methodology of transparencies, yet now generating a different range of instructions for the performance score. Much of Cage's writing on music in the 1950s focused on the identification of frequency, amplitude, timbre and duration, yet here in *Variations VI*, he has a different range of sonic territory to explore [8]. Terms such as 'varied' and 'unvaried operations' are the only 'musical' instructions provided [15]. All other information pertains to arrangements of various elements that are available be used in the performance.

The period between *Variations IV* and *VI* was not prolific for Cage, but did feature works where electronics were becoming prominent[9]. As Miller points out a key to the iteration of *Variations VI* is the requirement to decide what is what Cage calls a 'sound system', and how one may articulate and arrange components within that [1]. One of the most liberating elements about *Variations VI* is the way the score works to fit what is available, rather than the need to provide what is demanded. The

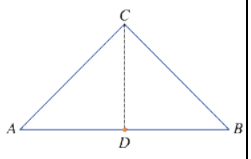
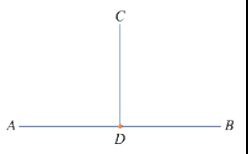
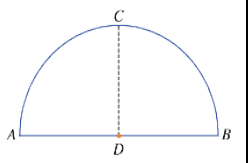
Cage's parameter name	Cage's symbol	Decibel's assignment	Decibels generated symbol, showing the co-ordinates measured
loudspeaker	Hand drawn triangle	amplifier	
component	Hand drawn Bisected short lines	Effects, effect pedals	
Sound sources	Hand Drawn Half circle	performer	
Sound system	Straight line	Total combination of the above	

Table 1: A list of symbols and assignments used in Decibels performance of Variations VI.

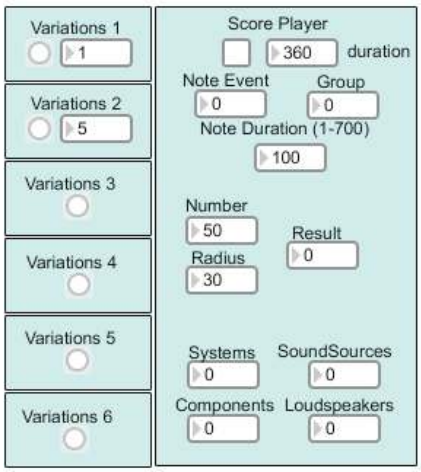


Figure 8: The master control panel for Decibel's realisation of the Cage Variations. The selections for the number of each parameter required in Variations VI are inputted bottom right.

interface that controls all the electronic score generation for each of the *Variations* provides easy input of the parameters required to generate the performance score – sound sources, components, loudspeakers and the aforementioned sound systems (figure 8)[10]. Where the performance score for *Variations IV* provides only spatial arrangement information, *Variations VI* dictates equipment arrangements and some minimal musical direction. Cage's goal of composing "notations that circumscribed a field of musical possibility out of which an unrepeatable stream of unique sounds and actions could emerge" had become a reality[10].

In terms of curating a varied and engaging *Variations* program, the orchestration of *Variations VI* is pivitol. It is framed by the two most chaotic and large scale works (*Variations V* and *VII*) and as such requires careful consideration of pace and orchestration to maintain audience engagement. After the first four variations being presented as acoustic in nature, and *Variations V* using a range of electronic sound producing tools, it was decided to make the instrumentation of *Variations VI* electronic, reflecting the direction Cage was taking with

his compositions at that time. Decibel members Cat Hope (noise bass guitar) and Malcolm Riddoch's (noise guitar) instrumental and effect set ups were nominated the 'sound systems'. In this way, the experienced improvising performance styles of these musicians offered a contrast to the more chaotic and 'style-free' approaches in the neighbouring *Variations*, and offered a framework in which to read *Variations VI*.

[9] This period saw the composition of Electronic Music for Piano (1964), Rozart Mix and Variations V (1965).

[10] Variations VII and VIII were not included, as they have no score to generate. For Variations V we used an astronomical atlas as suggested, and developed a mechanism to 'read' the map in realtime. Thus it was included in the master control panel.

Provided with the score instructions for *Variations VI*, are symbols on a transparent sheet intended to be cut out and dropped onto a sheet of paper with a vertical line drawn vertically down the middle. Each of these symbols signifies what we will call a different 'arrangement parameter' as allocated by Cage. These arrangement parameters instruct how to group parts that make up a unit. They differ from the instructions provided on how to read the resulting performance score. The random placement of the cut out symbols on the lined sheet signifies how different arrangement parameters are to be allocated and interpreted. These parameters are listed in Table 1, showing what Cage provides as material, and how Decibel has re generated the meanings and the actual symbols digitally.

Decibel programmed an environment in MaxMSP where it was possible to virtually 'drop' the symbols onto the computer screen (being the 'non transparent sheet' described in the score) in realtime, a similar process that was used to realise *Variations IV*. The final column in the table above shows how the symbols were measured and categorized to enable their digital rendition and behavior. Each of these symbols were derivative of the same structure and dimensions, and a geometric transformation was applied to the coordinates A, B, C and D, giving a random distribution of rotation and translation to each symbol when they 'drop' onto the screen. The distance from each of these symbols to the straight line is measured and grouped differently depending on their proximity. The perpendicular distances are measured according to the vector equation:

$$d = \frac{\left|\left(x_2 - x_1\right)\left(y_1 - y_3\right) - \left(x_1 - x_3\right)\left(y_2 - y_1\right)\right|}{\sqrt{\left(x_2 - x_1\right)^2 + \left(y_2 - y_1\right)^2}}$$

The digital replication of the cut out symbols provided on the transparency was achieved using Java. The symbols are grouped based on the extent of their proximity to each straight line or system with the following code:

```
private void assignGroups()
{
  double minDists[] = new double[shapes.length];
  for (int i = 0; i < minDists.length; i++) {
    minDists[i] = inlets[0] + inlets[1];
  }
  for (int i = 0; i < lines.length; i++) {
    for (int j = 0; j < shapes.length; j++) {
      double dx = lines[i].getX(1) - lines[i].getX(0);
      double dy = lines[i].getY(1) - lines[i].getY(0);
      double m = dy / dx;
      double c = lines[i].getY(0) - (m * lines[i].getX(0));
      double px = ((dx * shapes[j].getX(3)) + (dy *
      shapes[j].getY(3)) - (dy * c)) / (dx + (dy * m));
      double py = (m * px) + c;
      double distSquared = pow(shapes[j].getX(3) - px, 2) +
      pow(shapes[j].getY(3) - py, 2);
      double dist = sqrt(distSquared);
      if (dist < minDists[j]) {
        minDists[j] = dist;
        shapes[j].setGroup(i);
      }
    }
  }
}
```

The program also included the ability to extend the lines, as requested by Cage, and seen in Figure 9; "Drop each reserved symbol on the non- transparent sheet. Two adjacent straight lines which converge (or would if extended) or cross symbolize a sound system or systems" [15].

There were a number of reasons for adopting Java. There was need for the fast creation of both recursive structures and management of arrays or lists to enable the drawing of the symbols. A lower level language is more adept at intensive number crunching and list management tasks than a higher level graphical language such as MaxMSP which is based on a scheduled paradigm, where particular care must be taken in relation to ordering of events. MaxMSP provides considerable documentation on procedural management and message ordering, including the right-to-left and bottom-to-top approach to the arrangement of objects in a patcher window. However, when combining this with the construction, deconstruction and permutation of lists recursively, it became apparent that these are better implemented using a programming language such as Java or C. This had been tested in the score generation for Variations II, a score that requires many measurements to be taken into consideration. In realtime score generation, time is key. The real time generation of the data processing for *Variations II* took 6.27 milliseconds computational time in Java as opposed to 1860 milliseconds computational time in MaxMSP, a significant difference. This involved sequential tasks such as list construction, permutation, deconstruction, geometric algebra, and storage of relevant results into a Jitter matrix. The Jitter Application Programming Interface (API) is accessed directly from the main patcher window.

The information enabling the rotation, placement and gouping of the symbols is then sent over the network to any computers for displaying the score. The score information is formatted in a specific way, and consists of a single list of numbers starting with a header describing the variation number, display information such as transparency, the number of systems, components, sound sources, and loudspeakers that appear, and an interleaved list of x and y cartesian coordinates for generating all of the required symbols followed by a group number. The group numbers are used to color each symbol in the score differently, making it clear which objects relate to which system. Figure 9 shows the score generated for Decibel's premiere performance of *Variations VI,* where two elipses were added manually after the symbols were 'dropped'. These indicate the grouping of symbols relating to the each of the sound systems avialable, rather than the colouring of the symbols, as the

performance took place before the colouring of symbols has become available.

The rotation of each object was then interpreted by the performers live. The vertical or horizontal tendencies of each symbol represented how much a particular system is varied or not varied, respectively. The musicians interpreted these in their performance style, reading the score from a project in any direction they chose, following the symbols as indications to different degrees of change in a performance that lasted some eight minutes.

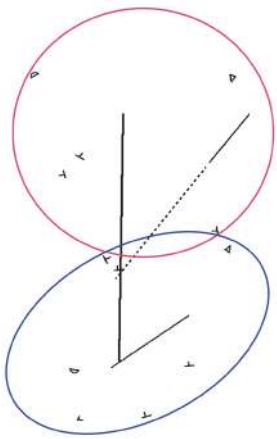


Figure 9: the first iteration of Variations VI with manual groupings indicated by a red (upper) and blue (lower) ellipse.

Cage notes that the distribution of sound in space is indicated by the relationship of the 'dropped lines' to the fixed line of the screen by noting "the orientation of the converging straight lines with respect to the non transparent (vertical) line may suggest distribution of sound in space' [15]. As a response to this instruction, the amplifiers were directed to project the trajectory as specified in the score, and shown in Figure 10.



Figure 10. Decibel members Cat Hope (bass) and Malcolm Riddoch (guitar) performing Variations VI in Palermo. Photo by Davide Carozza.

5. CONCLUSION

DeLio points out that Cage has provided "a significant attempt to introduce various notions of multiplicity into musical discourse" [18]. The *Variations* epitomize this intention. Yet the impetus behind Decibel's realisation of these works has been principally performative: to create practical tools for the realisation of the works, that retain both indeterminacy and the precision of the Cage's specification yet allow an open performative involvement characterized by the groups skills. The digital rendering of these works aims to provide performance scores where accurate performances are more likely, and avoid the problem of so called 'faking it').

The digital rendering of the performance scores enables multiple precise realtime instantiations that can become an efficient part of the very performance of the works. It enables performers to choose an almost indefinite range of possibilities for performance, but also provides the ability for performers to choose the best of a series of possible performance scores, particularly in regard to *Variations IV*. The network utility enables trouble shooting where score coordination is an important consideration.

This *Cage Variations Project* is a work in progress where performance plays an important part of the research methodology [22]. Each performance opens up new possibilities, ideas and problems to solve. The current phase of development heading towards the second performance involves making the score players function on iPads, replacing the quickdraw protocol with Quartz 2D drawing tools, and the adoption of OSC for network facilitation. This will lead to the development of a 'Cage Variations Performance Score Generator' as a stand alone application that will combine the score generators, players, network facility and management in a single application. And then, Cage's important compositional traits as exemplified in the *Variations* will become available and easier to use than ever, through a wide distribution for use on the latest consumer end technology.

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FROM AUTONOMOUS TO PERFORMATIVE CONTROL OF TIMBRAL SPATIALISATION

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ABSTRACT

Timbral spatialisation is one such process that requires the independent control of potentially thousands of parameters (Torchia, et al., 2003). Current research on controlling timbral spatialisation has focussed either on automated generative systems, or suggested that to design trajectories in software is to write every movement line by line (Normandeau, 2009). This research proposes that Wave Terrain Synthesis may be used as an effective bridging control structure for timbral spatialisation, enabling the performative control of large numbers of parameter sets associated with software. This methodology also allows for compact interactive mapping possibilities for a physical controller, and may also be effectively mapped gesturally.

1. INTRODUCTION

Timbral spatialization techniques are ways of dispersing a sound in space by placing each component frequency sub-band in a different localised position across a multichannel speaker array.[1] The technique has its origins established not only in spatialisation theory, but also in FFT[2] techniques and granular synthesis. In an abstract sense both FFT and granular techniques use a windowing technique to reduce sound into smaller constituent components. These components or *particles*[3] of sound are broken down either by their relative frequency or position in time. They may be described as spectra in the frequency domain, and as grains or quanta of sound in the time domain. Timbral spatialisation effectively extends this process to allow for the independent panning of individual spectra, allowing a performer and/or software to "re-sculpt" a sound both timbrally and spatially. This technique can be applied to the re-synthesis of spectra derived from recorded sounds[4], and may also be applied to tones generated synthetically.

Timbral spatialisation opens up the possibility of re-composing sounds in terms of timbre and space, and exploring this technique in a performative sense may allow the performer/composer to explore concepts such as Ligeti's concept of *permeability* where the individuality of timbre and interval give way to a more abstracted and chaotic, and increasingly difficult texture.[5] The underlying intent here in exploring timbral spatialisation in a performative sense, is to explore modifications on live sampled input, and to explore the timbral and spatial transformation that occurs through this interaction. This allows for a realtime acousmatic re-presentation of these familiar sounds. As Smalley has written on spectromorphology:

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1 Schmele, T. 2011. Exploring 3D Audio as a New Musical Language. Masters Thesis. *Music Technology Group, Department of Information and Communication Technologies, Universitat Pompeu Fabra*

2 FFT being the Fast Fourier Transform, a term more often used in realtime frequency domain synthesis, bearing in mind this is an efficient computation of the discrete fourier transform or DFT.

3 The "quantum of sound" was first written about by Gabor in 1947. He proposed some early theories that were to be later coined granular synthesis by Xenakis in his text Formalized Music (Xenakis, 1971). Curtis Roads later described the compositional practice centered around the use of grains *Microsound*. Whilst Granular Synthesis should not be confused with frequency domain synthesis, there is a parallel in the sense that both techniques employ windowing techniques and the opportunity for sounds to be reconstructed either in the frequency domain or the time domain.

4 Torchia R, and Lippe, C. 2003. "Techniques for Multi-Channel Real-Time Spatial Distribution Using Frequency-Domain Processing." *Proceedings of the International Computer Music Conference, Singapore:* 41-44.

5 Ligeti, G. 1965. "Metamorphoses of Musical Form," *Die Reihe, (*ed.) K. Stockhausen and H. Eimert, (trans.) by C. Cardew, Bryn Mawr: Theodore Presser: 5-11.

> The wide-open sonic world of electroacoustic music encourages imaginative and imagined extrinsic connections because of the variety and ambiguity of its materials, because of its reliance on the motion of colourful spectral energies, its emphasis on the acousmatic, and not least through its exploration of spatial perspective (Smalley, 1997).

Such techniques of spectral synthesis and decomposition within the performance space evoke notions of Schaeffer's analytical (or 'reduced') listening mode, yet here they are realised through the process of sound diffusion.[6] Careful selection of sound sources can make spatial percepts a compositional area to explore.[7] With natural sound sources, whose timbre's are dependent on features such as attack transients, the spatial re-distribution of a sound's spectral content can be an especially interesting musical effect.

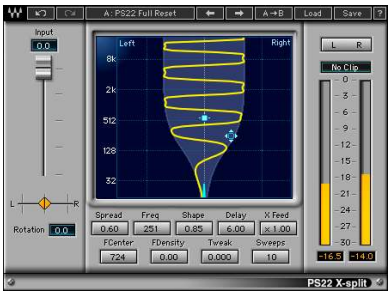


Figure 1. The Waves PS22 Mono-to-Stereo Plugin

Out of the many approaches to spatialisation, including developments in ambisonics and wavefield technology, Dolby and DTS surround encoding and decoding processes, binaural and head-related transfer function methodologies in 3d-audio, and rapid panning modulation synthesis, timbral spatialisation focuses largely on the localization of spectra with respect to azimuth in the median plane.[8] Early implementations of such a concept arose in some mono-to-stereo tools such as the Waves PS22 plugin allowing the engineer to shift the relative position of various frequency bands between two speakers resulting in a pseudostereo[9] effect.

Timbral spatialisation methodology largely draws on the fundamental theories of sound localization established by Blauert, namely the perception of localization based on a sounds *Interaural Time Delay* and *Interaural Level difference*.[10] Recent interest in spatialisation systems has also seen new research into the gestural control of spatialisation techniques.[11] However, in the case of timbral spatialisation Normandeau states:

> At the moment, the only way to design trajectories in... [software]... is to write every movement line by line, which is not adequate for complex movements. (Normandeau, 2009)

Currently implementations of control for timbral spatialisation have been limited as compared to many other techniques.[12] Previous implementations have employed the use of automated or pre-composed systems such as drawing the spatialization or using feature analysis from other sound sources creating a spatial cross-synthesis.[13] Building on the work of Torchia and Lippe in 2004, Kim-Boyle also used the boids algorithm to determine the spatial location of each frequency band, and later also used simulations of clouds of smoke.[14] More recently Barreiro has simplified the control of such a system by reducing the system to 8 frequency bands for adopting a more static and simplified method of control.[15]

Interestingly Barreiro reports a strong dependence on the input sound for the success of the technique:

> In general, sounds with a broad spectral content tend to sound diffused, providing an enveloping sonic image. Sounds with energy concentrated on specific regions of the spectrum, on the other hand, usually sound more localised." (Barreiro 2010)

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6 Smalley, D. 1997. "Spectromorphology: Explaining Sound Shapes." *Organised Sound: Vol. 2, no. 2. Cambridge: Cambridge University Press:* 107-126. and Lauke, K. and E. Stefani. 2010. "Music, Space, & Theatre: Site-specific approaches to multichannel spatialization." *Organised Sound*.

7 Kendall, G. 2007. "The Artistic Play of Spectral Organization: Spatial Attributes, Scene Analysis and Auditory Spatial Schemata" *Proceedings of the International Computer Music Conference*, Copenhagen: 63-68.

8 Normandeau, R. 2009. "Timbre Spatialisation: The Medium is the Space." *Organised Sound* 14, no. 3.

9 Such pseudostereo methods are discussed in Streicher, R. and F. A. Everest. 2006. *The New Stereo Soundbook: Third Edition*. TAB Books: Audio Engineering Associates, California

10 Blauert, J. 1969. "Sound localization in the median plane,"*Acoustica* 22: 205-213.

11 Marshall, M. T., J. Malloch, and M. M. Wanderley. 2009. "Gesture Control of Sound Spatialization for Live Musical Performance." *Gesture-Based Human-Computer Interaction and Simulation: 7th International Gesture Workshop*, GW 2007, Lisbon, Portugal (2007), Revised Selected Papers, Springer-Verlag, Berlin, Heidelberg.

12 Kim-Boyle, D. 2008. "Spectral Spatialization: An Overview." *Proceedings of the International Computer Music Conference*, Belfast.

13 Torchia R, and Lippe, C. 2004. "Techniques for Multi-Channel Real-Time Spatial Distribution Using Frequency-Domain Processing." *Conference on New Interfaces for Musical Expression (NIME)*, Hamamatsu, Japan: 116-119.

14 Such as jit.p.vishnu, jit.p.shiva and jit.p.bounds; Refer to Kim-Boyle, D. 2008. "Spectral Spatialization: An Overview." *Proceedings of the International Computer Music Conference*, Belfast.

15 Barreiro, D. L. 2010. "Considerations on the Handling of Space in Multichannel Electroacoustic Works", *Organised Sound 15(03)*, 290–296.

In 2011 James suggested that Wave Terrain Synthesis may be used as a means of controlling timbral spatialisation, rendering such a complex processing system more manageable in live performance.[16] It is arguable that both the multi-dimensionality, transformational, and morphological nature of Wave Terrain Synthesis is synonymous to the kinds of transformations needed for controlling timbral spatialisation. Wave Terrain Synthesis is also simple and effective to control, especially when mapped to gestural controller input.[17] Using Wave Terrain Synthesis as a method of controlling timbral spatialisation allows a performer to sculpt the localisation of many individual frequency bands with a comparatively small number of control parameters.

2. REALTIME AUTONOMOUS CONTROL

Whilst the motivations of this project have extended to many different areas in the field including techniques involving sound spatialisation, table lookup, FIR filtration, convolution synthesis, as well as concepts such as multi-dimensionality, timbre (i.e. tone color and spectrum), phase quadrature, evolution, trajectory, morphology, transformation, and gesture, the concept originally in mind was to give a performer a simple tool from which they could push and pull frequencies from one location in space to another. The most obvious of these that initially evolved was an idea of biasing high frequency in one direction and low frequency in another. If we were to replicate this on a 2-speaker system using two 30-band equalizer's for each speaker, we would achieve this bias by inverting one of the filter curves across the horizontal axis as we see in Figure 2. In a way this is synonymous with the concept of a linear crossfade, as effectively all we are doing is shifting a frequency band out of one signal and boosting it proportionally in another. When this process occurs, psychoacoustically we perceive a frequency band moving from the "centre" of a stereo image, and off to the side.

In software these kinds of filters are often implemented in the frequency domain, and referred to as FIR filters. Instead of a global linear crossfade controlling the overall amplitude of the resulting sound, we have many virtual faders that are each independently determining the amplitudes of each respective frequency band:

$$A_{amp} = \frac{\theta}{\theta_{MAX}}$$

$$B_{amp} = 1 - \frac{\theta}{\theta_{MAX}}$$

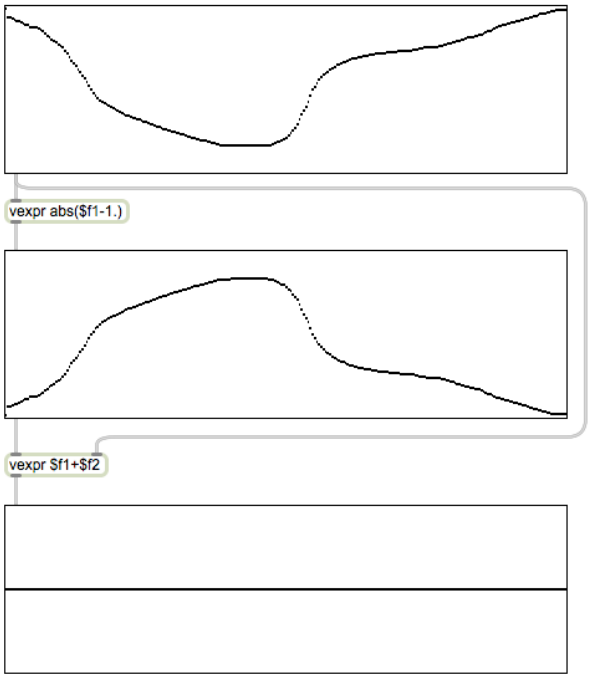


Figure 2. A multi-band filter curve, it's inverse curve, and the two curves added together showing equal distribution of energy

The importance of "phase" and "inverted phase" are important here, as the intended idea is that if each fader level for both channels is added together, and compared to all other fader levels, the accumulated energy should be constant for all frequencies as we see in the final graph in Figure 2.[18]

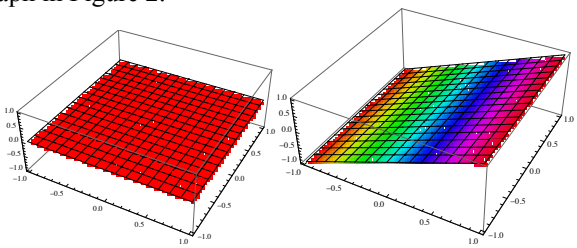


Figure 3a. Signal equally spectrally distributed out to all loudspeakers
Figure 3b. Signal is distributed out to the loudspeakers in such a way that the sound spectrally shifts from low to high frequency across the room.

By extending this system to use 2-dimensional wave surfaces, we are merely controlling the way frequencies are distributed by manipulating a virtual plane. So how do we result in two signals that are effectively opposite "polarity" or 180 degrees out-of-phase? We can achieve this using a circular orbit, reading values off the 2-dimensional plane while following this circular orbit. In figure 4a we can see our circular orbit. One trajectory effectively begins from point A, and the second

16 James, S. 2011. "Multidimensional Data Sets: Traversing Sound Synthesis, Sound Sculpture, and Scored Composition." *Proceedings of the Australasian Computer Music Conference, Auckland.*

17 Mills, A. and R. C. De Souza. 1999. "Gestural Sounds by Means of Wave Terrain Synthesis." *Congresso Nacional da Sociedade Brasileira de Computação* XIX. http://gsd.ime.usp.br/sbcm/1999/papers/Anderson_Mills.html

18 We in fact require an equal power panning curve to ensure that different frequency bands move between the speakers with an equal power relationship maintained. This may be easily remedied by reshaping the amplitude settings for all frequency bins using a transfer function $w(t) = \sin\left(\frac{\pi}{2}t\right)$

trajectory begins at point B. Both move in an anti-clockwise direction. The advantage of this system is we can rotate the terrain or the trajectory any number of times, and the phase relationship of A and B remain the same.

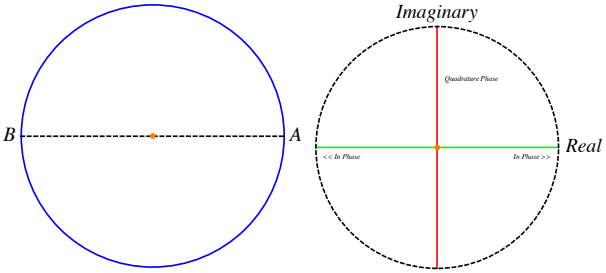


Figure 4a. A circular trajectory structure used to lookup values over a 2-dimensional plane. Whilst only one trajectory is generated, a phase displaced version of this trajectory is used for looking up values over the 2-dimensional plane for the second loudspeaker.
Figure 4b. A graphical representation of the Hilbert Transform.

Generating a circular trajectory orbit is achieved by a well-known process described scientifically as the Hilbert transform. This is achieved by mapping our original "real" signal against the same signal displaced in quadrature phase or a 90 degree phase shift.

Visualising the trajectory has been achieved by writing the trajectory to a jitter matrix allowing the performer to track the trajectory on a video display. Whilst the output is broken periodically with a clear message, the changes over time are smoothed with the use of video feedback; this smooths the transitions over a period of time, creating changes that naturally decay over time. This is visible in Figure 5.

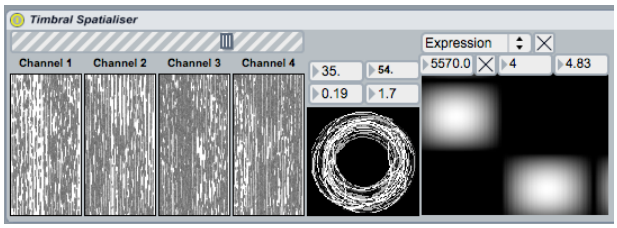


Figure 5. A version of the timbral spatialiser created for Max4Live. The trajectory is plotted in the centre of the interface. To the right of this is the terrain contour, and the left-most part of the patch shows the changing spectrum for all 4 output channels.

In a four speaker system, we phase shift the second, third, and fourth signals by 90, 180, and 270 degrees respectively. We can view this phase shift in Figure 6. As we know the exact number of samples in one single periodic revolution of our trajectory, we are able to specify the phase shift as an exact number of samples delay. If our FFT window size is 2048 samples, this gives us a revolution of exactly 1024 samples which, if our sampling rate is 44100Hz then our fundamental frequency is approximately 43Hz. This also translates to a signal delay of 256 samples, 512, or 768 respectively.

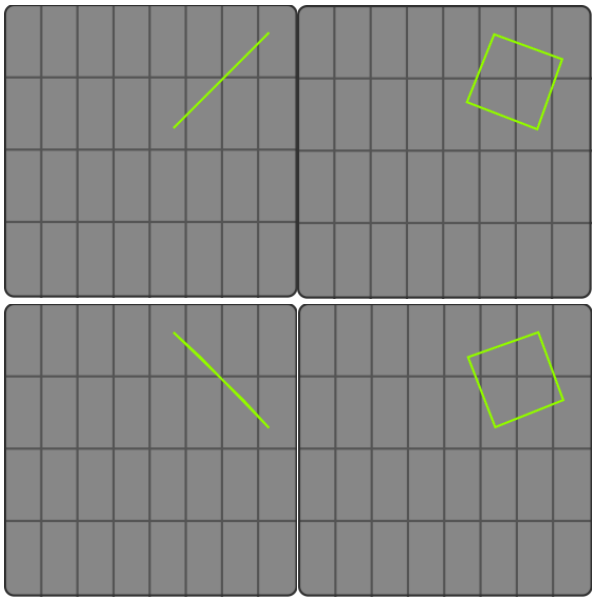


Figure 6. Plots showing the phase divergence of each signal. Top-left is in phase (0 degrees), top-right is 90 degree's out-of-phase. Bottom-left is 180 degree's out-of-phase. And bottom-right is 270 degree's out-of-phase.

Similarly in an eight speaker system, we extend the phase shift to 1/8, 1/4, 3/8, 1/2, 5/8, 3/4, and 7/8 to create 8 discreet outputs. We can use a series of delays at 128, 256, 384, 512, 640, 768, and 896. Figure 7 shows the phase distribution of 4- and 8-channel speaker configurations used.

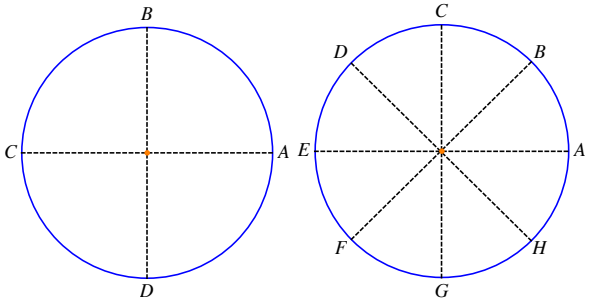


Figure 7. Plots showing phase distribution of speaker channels. To the left we have a quadraphonic configuration A, B, C, and D where A and B are front right and left, and C and D are rear left and right channels. To the right we have a plot showing phase distribution for 8 channels. Again these follow in series in an anti-clockwise direction.

Advantages for this mapping strategy include the ability to tilting the terrain structure resulting in a spectral divergence, rotation of the terrain and/or trajectory resulting in a rotation of a specific frequency distribution in a circular speaker array, scaling adjustment of the trajectory signal correlates to spatial separation, geometric contortions of the terrain and trajectory structures result in effective translations spatially, haptic rate dynamics can be introduced using dynamic terrain surfaces (i.e. motion video), and audio rate timbral spatialisation can be achieved by modulating the trajectory signal.

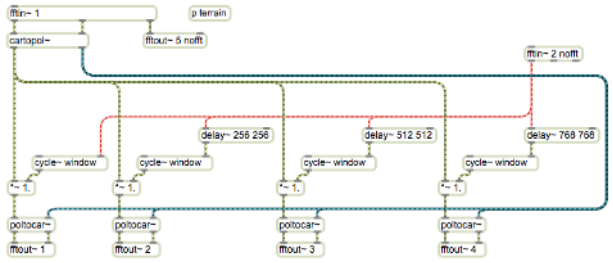


Figure 8. The internals of the FFT sub-patcher; this patch manages the phase shift applied to the filter contour, and calculates the component frequencies to be sent out to each speaker.

To some extent this method does create a significant amount of timbral "spill" between all speakers, so it was also necessary to devise ways in which to make the spectral bands between each speakers narrower and more precise. This was achieved by implementing spectral gates, a method where the performer is able to specify a threshold through which frequencies can be allowed through. At sensitive settings the system lets through everything, but at more insensitive threshold settings the system only allows frequencies through at higher energies or amplitudes.

Another method was to create window functions for each speaker that "narrow" the spectral window being read from the terrain or plane. This way the curve is determined by a smaller arc rather than a full circumference of the circle.

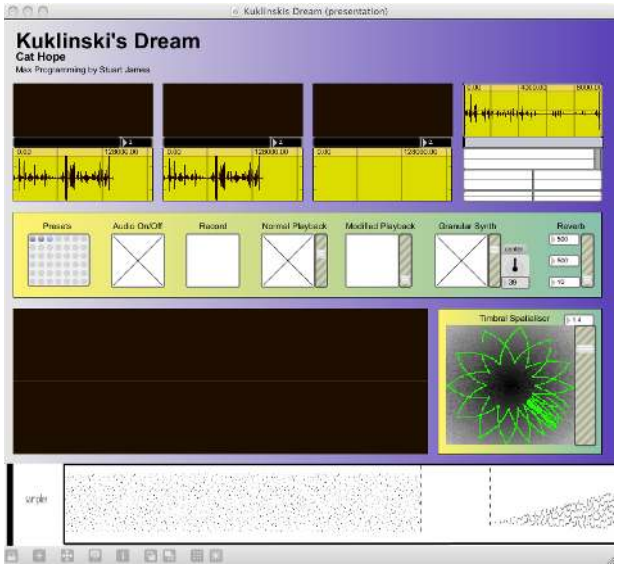


Figure 9. The interface for "Kuklinski's Dream", a composition by Cat Hope. This patcher designed by Stuart James uses automated timbral spatialisation as a means of dispersing sound about a quadraphonic system.

3. SPATIAL TRANSFORMATION VERSUS GEOMETRIC TRANSFORMATION

Previous research has also focused on transformational processes applied to both the terrain and trajectory structures.[19] The potential for using terrain surfaces that exhibit random and "noisy" topographies translate to a resulting control contour that exhibit these same sorts of characteristics. Further transformational processes have included video feedback and spatial remapping, as well as geometric transformation of the trajectory structure using processes such as affine transformation, filtration, bit-rate reduction, and delay lines. In a way audio filters (in the IIR sense) create a sculpting tool for audio not tool dissimilar to the blur and sharpen tools we find in graphics. We can use these filters to change the shape of our control signal.

It is these geometric transformational processes that can extend the scope of this methodology in performance. By using Wave Terrain Synthesis as a bridging control in this way we have the opportunity to morph and contort the behavior of another audio process; for example these geometric transformations may be used to twist, push and pull the timbre spatially. In terms of gesture to sound mapping, Mills and De Souza explored the slow translation of these trajectory orbits over a terrain surface, and found this to be effective in creating expressive control.[20] However with the additional extensive range of transformational parameters here, there is further scope and increased flexibility in how gesture may influence sound via such a model. Exploration of this mapping flexibility will be a major focus of investigation in this research project.



Figure 10. In the top-right we see a geometrically contorted version of the image found in the top-left. The bottom-left and bottom-right show a virtual 3-dimensional plot of both images above.

[19] Roads, C., et al. 1996. *The Computer Music Tutorial*. Cambridge, Massachusetts: MIT Press and James, S. 2005. "Developing a Flexible and Expressive Realtime Polyphonic Wave Terrain Synthesis Instrument based on a Visual and multidimensional methodology." (Masters Diss., Edith Cowan University).

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4. PERFORMATIVE CONTROL

The physicality of the human body in electronic music performance is still often seen to be a fundamental missing element in electronic music.[21] In comparison to other performance practices, audiences of electronic music performances often find it hard to relate the physical gestures they see with the auditory outcome they hear.[22] Machines do not interpret movement in the same way human beings observe and experience it. As Camurri and Moeslund state:

> Human beings seem to have little or no problem with perceiving and understanding the expression of gestures of musical performers and dancers on a scene. Even when we are not able to see all details of the performers' movements and/or bodies, ...we also perceive parts of the body that may be momentarily occluded because of an unfavourable viewing angle, i.e. we will in most cases correctly assume that a person continues to move the whole body even though we actually only see some parts of the body moving. Our abilities to sense quite accurately both the actual movements and their expressive and emotive features become even more remarkable when we try to replicate these abilities with machines. What's easy for us may be very difficult or even impossible for machine-based systems of vision. Yet developing technologies for machine-based vision and gesture recognition has attracted considerable effort, because such artificial systems of vision and gesture recognition may have many applications in human–computer interaction (HCI) (Godøy, et al., 2009, p238).

The process of gesture to sound mapping has many possibilities, but in performance practice there are additional aesthetic considerations. The performers physical engagement with software relates to the theatre of live performance. There has been much recent work involved with the analysis of musician's posture and gestural movement during performance.[23] This has involved not only the scientific analysis of gesture to sound but also the psychology of the performers physical experience, and how this translates to musical expression.

Recent developments in computer hardware have started to redefine how we physically interact and engage with technology. With the introduction of multi-touch devices like the Jazzmutant Lemur, Apple iPhone and iPad, it is possible for a performer to access different functions directly via a customizable graphical user interface. Other sensory technology such as the Vicon motion camera can track the form and movement of the physical body in 3-dimensional space. The XBox Kinect is another motion capture device that uses infrared light with depth perception allowing for the capture of 3-dimensional movement and acceleration of the human body in space. Many of these technologies allow for a more diverse interpretations of body movement and gesture.

With the use of tactile multi-touch technologies such as the iPad, the intention here is to extract control information from hand gestures and movement which will then in turn be used to geometrically contort terrain and trajectory structures used by Wave Terrain Synthesis to then in turn generate control information for timbral spatialisation. In this way Wave Terrain Synthesis serves as a software control bridge. Refer to Figure 11 for a schematic. This process will effectively modulate the parameter sets generated by Wave Terrain Synthesis allowing to expressively control many different parameter streams simultaneously. As noted in the abstract, timbral spatialisation requires potentially thousands of control parameters, and to make this process ideally responsive, these control signals must operate at audio rate.

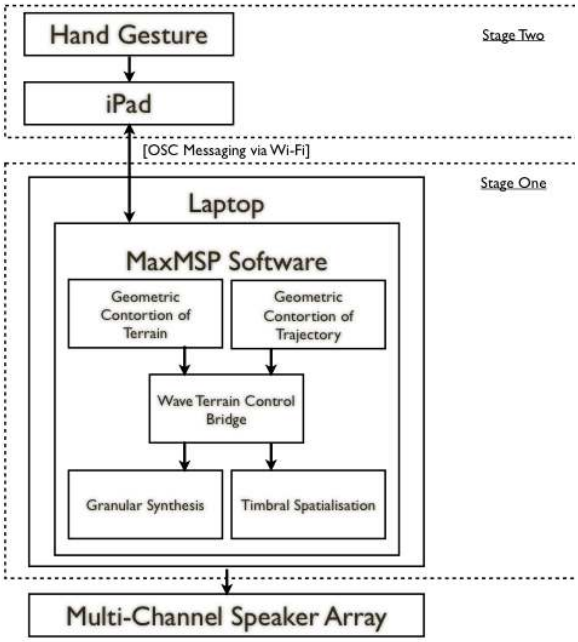


Figure 11. A schematic outlining how both the iPad controller and software are bridged, bearing in mind that Wave Terrain Synthesis bridges the iPad to the timbral spatialisation.

An effective physical controller for these kinds of transformations requires a multi-sensory device. Hsu has explored the use of the Wacom Tablet for the purposes of drawing trajectory structures[24], but new technologies are emerging that promise new directions through customisable multi-sensory and tactile control such as the Arduino, as well as mobile and programmable multi-sensory devices such as the Apple iPhone and iPad. Finding effective ways and means of gesture mapping for trajectory motion and how to effectively generate

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21 Turner, T. "The resonance of the cubicle: Laptop Performance in Post-Digital Musics." *Contemporary Music Review* 22, no. 4 (2003): 81-92.

22 Stuart, C. "The object of performance: Aural Performativity in Contemporary Laptop Music." *Contemporary Music Review* 22, no. 4 (2003): 59-65.

23 Kapur, A., G. Tzanetakis, N. Virji-Babul, G. Wang, and P. R. Cook. "A Framework for Sonification of Vicon Motion Capture Data" *Proceedings of the International Conference on Digital Audio Effects*, Madrid, Spain (2005).

24 Hsu, W. 2002. "A Flexible Interface for Wave Terrain Synthesis." *PERformance & NETworking Colloquia*, San Fransisco State University, Department of Computer Science. http://cs.sfsu.edu/news/pernet/02/04-24-02.html and http://userwww.sfsu.edu/~whsu/TERRAIN/

and/or manipulate a terrain surface strikes right at the heart of effectively performing using this method. The choice of physical controller or sensor is critical in being able to effectively map gestural information from a performer. A Polhemus Stylus or similar device provides position and orientation information for a single point in space via a stylus tip, whilst a gesture interface can input many positions since the system tracks multiple features simultaneously. Further investigation will involve experimentation with two dimensional and three dimensional motion cameras such as the Vicon and Kinect interfaces to test their complementarity with the traversal of multi-dimensionality. This is a topic for further discussion at a later stage.

The way in which these processes are mapped will determine exactly what auditory process the movement determines. For example, hand and arm gestures may influence the shape and course of the trajectory. In this way body movement will effect the multichannel imaging and distribution of frequencies, pulling them around in a more abstract sense. As a means of clarification, the term sound sculpture could be used here loosely to describe the physical act of sculpting a terrain surface with physical input much like Dan Overholt explored with the MATRIX interface.[25] Since the terrain largely affects the resulting sound, the perceived effect here is such that the performer is "sculpting" the sound gesturally via such a process.

Advantages of the iPad as a controller device include the fact that it is multi-touch, wireless, it has a software development kit available, a proliferation of different apps intended for controlling music software remotely,[26] it uses OSC (open sound control) as a protocol. There are also music controller apps where the user interface is completely programmable such as TouchOSC[27], MrMr, C74, and Ardumote HD.

One major distinction to make here is the difference between direct controller-to-parameter mapping strategies, as opposed to gestural recognition and mapping strategies. This project is in the process of evaluating both options given the multi-control and multi-tactile nature of the iPad. In this way research involves the systematic categorization of possible hand gestures, bearing in mind that in the case of the iPad, movement is commonly tracked at the fingertips. In this way, gestures will be separated by the number of fingers used (i.e. that is between 1 and 10.) From here it is possible to correlate those gestures commonly used for Apple OS level control and functions, and develop new strategies for control.

Whilst the iPad is currently being used to export coordinates of multi-finger movements over Wi-Fi, the way in which this data is managed is currently performed by MaxMSP. Current implementations firstly determine the number of fingers used in the gesture. A second step involves the filtering and formatting of this information to a specific destination based on the number of fingers used, and step three involves either processing of this information, or analysis for the purposes of pattern recognition or manipulating the data into a useful format for mapping. The two examples in Figure 12 include a patch used to recognise a physical oscillation pattern, like what string players use for vibrato, and extracting information from a "pinch" gesture.

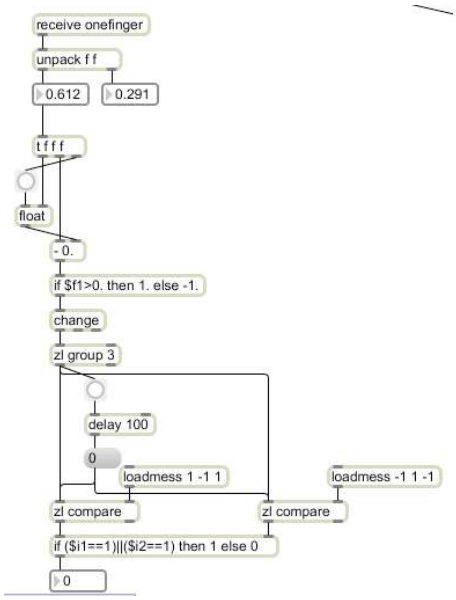


Figure 12a. A patch designed to recognise a vibrato movement.

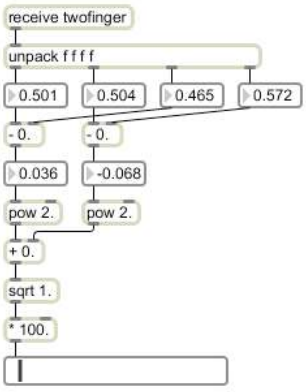


Figure 12b. A patch designed to output the distance between two fingers, ideal for when one creates a "pinch"-like gesture

Advantages for iPad integration include the option for tilt sensing, which can be measured using the

25 Overholt, D. 2002. "New Musical Mappings for the MATRIX Interface." *Proceedings of the 2002 International Computer Music Conference*. http://www.create.ucsb.edu/~dano/matrix/ICMC2002.pdf

26 Applications currently include ProRemote, ProRemote Light, ProTransport, TrixMix, Pro-XY, MrMr, TouchOSC, ITM MidiLab, ITM Pad, c74, iOSC, iTM Tilt, Remokon for OSC, OSCemote, rain., Griid Pro, Runxt Life, Breath OSC Interface, Hex OSC S, Control, eyoControl, OSC Physics, Griid Pro, SonicLife, Ardumote HD, [v] Remote for iPad, GyrOSC, touchAble, expressionPad, Kapture Pad, DrawJong, Live Music Coder M^2 OSC, OraisonLight, dot E++, HexaChrom, and Runxt Life Plus

27 TouchOSC has an editor utility downloadable from the hexler.net website allowing the user to custom design their own multi-touch control work surfaces.

accelerometer sensor built into the iPad. In this way we may be able to control the tilting of the terrain structure, resulting in spectral divergence. A two finger rotation can be mapped to the rotation of the terrain or trajectory, which then correlates with the rotating of frequencies around a multichannel array. The two-finger "pinch" can be mapped to scaling adjustments in the trajectory signal, correlating to spatial separation. Multifinger movements, that is those using between 5 and 10 fingers, can be used as geometric contortions of the terrain and trajectory structures result in effective translations spatially.

5. CONCLUSIONS

The focus of this paper has been to outline a new methodology for controlling timbral spatialisation, and to discuss how a gestural controller, in this case an iPad, may be used to govern the way in which Wave Terrain Synthesis controls timbral spatialisation gesturally. Whilst there is still much work to be done in regards to the gestural development and its implications on the performability of such a technique as timbral spatialisation, what is established at this point is a clear trajectory and course of action penultimately leading towards a more thorough investigation of these pertinent issues.

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EZITHER: EXTENDED TECHNIQUES FOR CUSTOMISED DIGITAL BOWED STRING INSTRUMENT

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Victoria University of Wellington,
New Zealand

ABSTRACT

The EZither is a custom designed musical instrument that affords extended musical control through the use of electronic sensors and a microcontroller. This extended control allows the possibility of new forms of interaction with the instrument and new approaches to sound. This paper outlines the custom hardware and software design of the instrument. It also discusses how the relationship between musical instruments and technology can provide new possibilities and approaches to composition.

1. INTRODUCTION

Music has a rich history of creating new practices by integrating new technology. One way in which technology has been integrated into music practice is by taking traditional designs and modernising them with new technology. This is often advantageous as users can bring years of experience to a device, and technology can have the role of complementing and extending an already existing device. This has been particularly true in designing new interfaces for musical expression.

The Hyperbow [1] is an instance of how technology is enhancing the possibilities of expression with the addition of sensor technology. The Hyperbow provides in-depth data about the subtleties of how the performer is using the bow, which is then used to affect the sonic outcome through digital signal processing. The Hyperbow is coupled with the Hyperviolin [2], which is designed in a way that, through multi-channel audio analysis software and embedded wireless bow hardware technology, allows the performer to have greater control over the sonic output.

Other instruments have combined traditional acoustic instruments with electronic sensors to extend the functionality and capabilities of the instrument. The ESitar [3] is a modified 19-stringed, pumpkin shelled, traditional North Indian instrument. Through the information received from multiple sensors placed on the instrument, the user is able to capture data about fret number, pluck time, thumb pressure, as well as pluck direction and three axes of the performer's head tilt. This allows the performer to have more control over the sound output of the performance through effects and looping mechanisms as well as deduce important data from the performance for analytical and documentation purposes.

Dan Overholt has also worked with expanding the range of possibilities for string instruments with his Overtone Violin [4]. He utilises data from the movement of the bow and the bow hand, as well as the sound information produced by the player. This information is then used to sculpt the sound and the sound is then sent back to the sounding body of the fiddle through an internally mounted tactile sound transducer, which is controlled via DSP running on an attached iPod Touch. This creates a two-way communication system and the body of the fiddle is resonated, giving the performer haptic feedback.

The EZither's design is influenced by these modern hyperinstruments discussed above. It has been named the EZither as it can be loosely attributed to the zither family of instruments. However, its influences range from the guitar to the ancient Chinese Guzheng and the Japanese Koto. Traditionally, a zither is a string instrument, usually found in Eastern or Asian countries [5]. They have no neck and the strings are connected directly to the sound box. The design of the zither allows it to be played either on the lap or on a flat surface. There are fretted and fretless varieties and the number of strings varies widely throughout the family. Usually Zithers are plucked however, some, like the Hammered Dulcimer, are struck and some are bowed. Most Zithers have many strings that are fixed in pitch to allow the performer to be able to play melodies in various keys.[1] The EZither is a ten-string zither that is designed to be played on a flat surface. It is designed to be bowed, however it can be struck and plucked as well. It features a dual-bridge system, which allows for a wide range of tunings and the possibility of unique techniques.
This paper contains six sections: Section 2 describes the build of the physical body of the EZither. Section 3 describes the playing techniques that are possible on the instrument. Section 4 gives an overview of the electronics that are used and Section 6 describes how the instrument has been implemented. The paper concludes with a discussion on how the interaction between the performer and instrument is enhanced due to the electronic components in Section 7.

[1] http://en.wikipedia.org/wiki/Zither

2. DESIGN OF PHYSICAL BODY

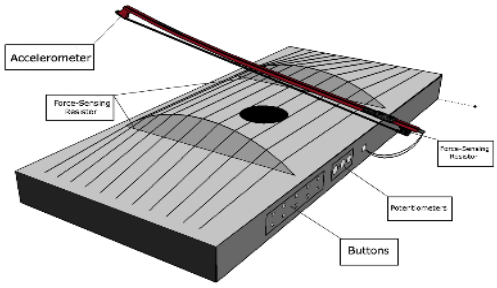


Fig. 1. This figure shows an overview of the design of the EZither.

2.1. Body

The EZither's body is made from a simple box design. Plywood was used for affordability and was glued and nailed together to create the box. A hole was drilled into the top of the box to act as a sound hole to promote greater acoustic volume. The choice of using a box shape as the body of the instrument was integral to the intended use of the instrument. The instrument was designed originally to create drone textures and the box shape is an acoustic structure which reinforces frequencies and promotes the sustain of the instrument. Aluminum sheet metal was screwed into the wooden body to act as reinforcement to the base. Due to the tension that the strings put on the body of the instrument, as well as the structure being weakened by drilling holes through its frame to mount sensors, the metal base was necessary to ensure the body was structurally sound. Aluminum sheet metal was also used to mount buttons, lights and potentiometers to the body and the material for the bridges.

2.2. Bridges

The bridges were an important aspect in the design of the instrument, as we wanted to be able to isolate a single string when bowing. This was achieved by creating a bridge with a circular segment, based on traditional string instruments. The two bridges also divide each string into three segments, each with their own pitch. This allows for up to 30 pitches to be tuned. The tuning of each segment is determined by the tension and the length of the section, meaning that proportional relationships can be used to create a system of pitches, which are in tune with each other. The bridge is made out of aluminum sheet metal, which produces a strong metallic timbre and is a very malleable material. The bridges are not fixed and instead are secured by the downward force of the strings and small tabs, which help to anchor the bridge.

2.3. Strings

The strings used are metal wound strings, designed for acoustic guitars. They were chosen because of the metallic quality of their sound and for pragmatic reasons like cost and accessibility. They are secured into the body of the instrument by guitar bridge pins, which stop the string into the base of the body.

3. PLAYING TECHINIQUES

The EZither has been designed so that the user can easily bow or pluck the strings of the instrument. It allows for a wide array of playing techniques that are mirrored in other instruments as well as some unique techniques, made possible by the design of the instrument.

3.1. Bowing

The strings can be bowed and the full range of traditional techniques used with other string instruments are applicable to this instrument. Sul ponticello bowing is especially pronounced on the instrument as the metal bridges severely alter the sound, as the bow gets closer to them. The metallic timbre is most pronounced when the strings are bowed in the sul ponticello position or when they are bowed with much pressure. The resulting sound is similar to the grating sound of a string instrument being bowed directly on the bridge or a scratch tone being produced.

3.2. Pizzicato

The strings can be plucked and, due to the choice of strings, the resulting sound is more similar to a guitar than a classical string instrument. Each string is easily isolated as the distance between strings is much greater then other string instruments.

3.3. Vibrato and Pitch Bending

The dual bridge system allows for the free hand of the user to manipulate the pitch of the string. Since the bridges split the string into three sections, the user can change the tension of a string in one section and bow or pluck in another section. This is very similar to the Chinese Guzgheng and Japanese Koto, which can be manipulated in the same way. This also allows the user to be able to play more pitches than the ten pitches the strings are tuned to, expanding the pitch functionality and capability of the instrument.

3.4. Harmonics

By lightly touching the string with a fingertip, harmonics are easily played on the EZither. It is very responsive to harmonics and numerous harmonic nodes can be obtained.

4. ELECTRONICS

The sensors used on the EZither are an accelerometer, three force-sensing resistors, three potentiometers and

five buttons each with an LED. They all communicate with a computer via an Arduino Mega 2560.[2]

4.1. Arduino Mega 2560

The Arduino Mega 2560 is the microcontroller that forms the core of the electronics in the EZither. It allows for the sensor data to be communicated to the host computer. The Arduino Mega was chosen because it features several parallel on-board analog to digital converters allowing for the extraction of data from multiple analog and digital sensors, as well as, controlling visual feedback outputs.

4.2. Accelerometer

The accelerometer used is the ADXL 335 Triple Axis accelerometer with a breakout board. It allows for three axes of movement and sends continuous analog data. It can measure the static acceleration resulting from motion, shock or vibration. The accelerometer is positioned on the tip of the bow.

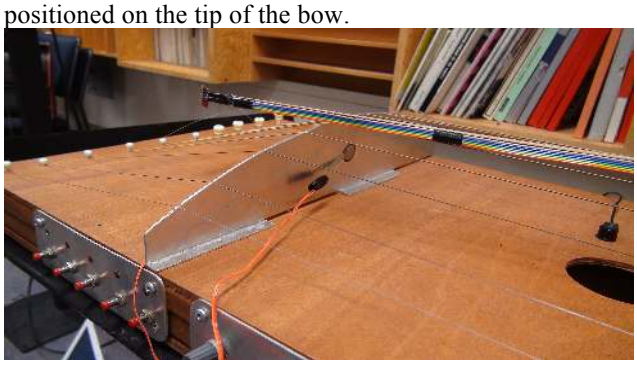


Fig. 2. This figure shows the position of the accelerometer at the tip of the bow.

4.3. Force Sensing Resistor

FSR's will vary their resistance depending on how much pressure is being applied to the sensing area. The harder the force, the lower the resistance. These FSR's can sense applied force anywhere in the range of 100g-10kg. Three FSR's are used on the instrument. Two 0.5 inch FSR's sit on the left bridge of the instrument, which makes them easily accessible to the left hand of the performer and one smaller 0.16 inch FSR is fixed onto the bow near the frog. This is placed so the performer can access it easily with their right hand thumb.

4.4. Buttons and LEDs

The buttons used are simple momentary panel-mounting buttons. They function as a toggle button and a LED represents the state of each button. The buttons and LEDs are mounted onto the instrument with a panel made from aluminum sheet metal.

4.5. Potentiometers

Three 10k ohms, linear rotary potentiometers are used on the EZither. They are simple components that allow a great deal of control and have been used to control the master volume of the output and the wet/dry level's of effect chains.

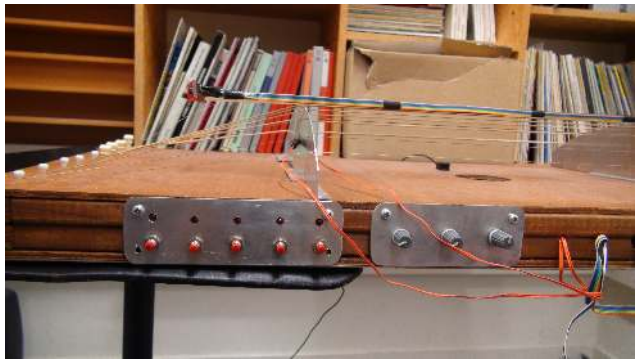


Fig. 3. Shows the position of the buttons, LEDs and potentiometers.

4.6. Hiduino Firmware

The Arduino was modified using Dimitri Diakopoulos' 'hiduino' custom built firmware, which converts the Arduino into a class-compliant, driverless MIDI/HID device [6]. Normally, the Arduino microcontroller communicates by implementing a virtual COM port via USB for basic serial I/O. This provides a problem because most music programs do not recognise serial data natively, and thus the data needs to be converted by an intermediary program so that the data can be implemented. The hiduino firmware bypasses this extraneous step, and allows the Arduino to communicate directly to musical applications via MIDI. This makes the data chain more streamlined, universal and stable.

5. IMPLEMENTATION

5.1. Data Implementation

Multiple parameters of the bow movement can be extracted from the three-axis data of the accelerometer. The raw data gives the static position of the bow. By storing a history of the data, the velocity and acceleration can be deduced as well. This gives nine, continuous streams of analog data that can be mapped to sound parameters just from the accelerometer.

Often, during live performance, the accelerometer data may want to be triggered on and off. The force-sensing resistor that is positioned on the bow allows the performer to have control over when the accelerometer data is being received and updated by the user. While the FSR is held down, the accelerometer data will be continuously read and updated. Otherwise, the accelerometer data is ignored and the values are locked to the last received message. This is helpful as it allows the performer to change a parameter of the sound with

[2] http://www.arduino.cc/

the accelerometer and when desired, hold that parameter at a constant point while being free to move the bow.

A layer system has been implemented with the buttons to maximize their functionality. Each layer has one instance of the five buttons which all send out an on and off message on a unique channel. By pushing the two outer buttons simultaneously, the layer is changed and the buttons now send on and off messages on a new unique id. This can be repeated endlessly, and to return to the original layer, the first and third buttons can be pushed simultaneously.

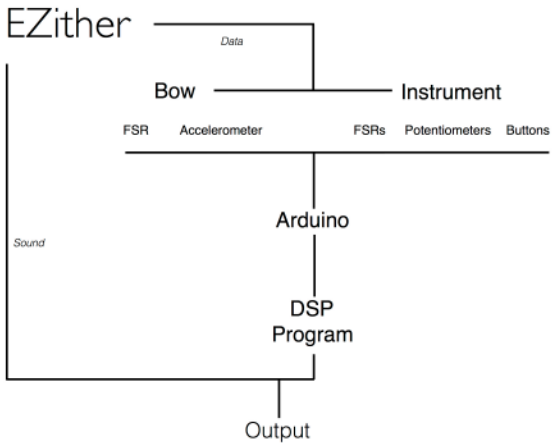


Fig. 5. Shows the signal flow of data.

5.2. Musical Implementation

The control data sent from the EZither has been implemented in multiple ways to extend the EZither. The electronics can serve as an on board sampler which can trigger sounds. In this case, the function of the control data is to accompany the acoustic sound of the instrument.

These sounds have been triggered using two methods so far. The buttons can be used to simply launch a sound, leaving the FSR's, potentiometers and accelerometer free to affect the samples. The acceleration data from the bow can also be used to trigger sounds. Certain gestures can be captured, allowing for a gestural creation of the sound. By using the bow to create samples, the connection between the physical gestures of the performer and the sound created is easily apparent for the audience.



Fig. 4. Insomnia being performed.

The function of the electronics can also be to extend the acoustic sound of the instrument by controlling digital signal processing. This method has been used in two compositions that have been written for the EZither and other new musical interfaces. In these pieces, a contact microphone is placed on the face of the instrument and the incoming audio is then processed in Ableton Live[3]. Using a contact microphone has proven to be the most suitable as it isolates the sound coming from the EZither, which lowers the potential for feedback, and it also picks up the resonance of the instrument. Other programs could have been used but Ableton was chosen as it easily allows for MIDI mapping and custom-built Reacktor[4] patches can be used in Ableton with the Reaktor FX plugin.

In both compositions, the incoming audio was received on one channel and then sent out via busses to auxiliary channels. On each channel, a different effects chain was implemented and the buttons controlled which channels were armed. Various plug-in effects were used including the built-in Ableton effects, as well as, third party plug-ins and custom built plug-ins made in Reaktor. Within these effects chains, parameters of the effects were controlled by the control data received from the accelerometer, FSR's and potentiometers.

Other ways that the EZither has been implemented is by using the buttons to control a looping module. This allowed for sounds to be recorded and layered on top of each other. This is quite an effective implementation for the EZither as it can create mass chords and dense textures. This implementation allowed the EZither to function as a supporting instrument within an ensemble.

The data available from the performers interaction with the bow has been the most expressive and interesting out of the sensors. By using the bow to control the processing of the acoustic sound, the performer can control the sound by using physical gestures. This allowed for a more expressive performance, and for the audience to understand the link between the performers interaction with the instrument, and the resulting sound.

6. CONCLUSION

In all forms of music using acoustic instruments, gesture plays an important role in the production of sound and the interaction of the performer with the audience. In the case of an acoustic instrument, the performers interaction with the instrument and the resulting sound is an easily discernable causal chain. When a performer pushes down a piano key, the audience hears the result of this physical gesture. This is not always the case in live electronic music. Performers may use a range of

[3] http://www.ableton.com/

[4] http://www.native-instruments.com/#/en/products/producer/reaktor-5-player/

interfaces and instruments that are new and whose methods of sound production are unknown to the audience. This creates a disjunction in the audience's understanding of the performers interaction with an interface or instrument and the resulting sounds.

By using electronic sensors to capture gestural activity, the causal connection of the performers physical gestures and the resulting sound output can be mirrored and even enhanced. The resulting sound output can take various forms but the causal connection can still be apparent to the audience. Instruments like the EZither that utilise electronic sensors to capture the performers gestural movement can mirror the causal chain of acoustic sound production. The accelerometer is used in this way on EZither as gestural movement is translated into a sonic outcome. As the bow is rotated, the sound of a reverb effect can swell and then be quickly muted by a flick of the bow. When using the acceleration data of the bow, a forceful flick of the bow can trigger a new sound or amplify the acoustic sound of a gesture. In this case, the bow can function in its traditional role of producing sound by vibrating a string, as well as, controlling the sound manipulation over the resulting sound.

With more development, these types of interactions can expand the gestural control of a performer, and because the sound is not limited to the acoustic restraints of the instrument, the interaction can be more expressive then previously possible.

7. ACKNOWLEDGEMENTS

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IMPLEMENTATION OF A REAL-TIME MUSICAL DECISION-MAKER

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ABSTRACT

In this paper, we present a novel implementation of a real-time musical decision-maker. In our scenario, the behaviour of a musical decision-making agent is specified as a constraint satisfaction problem. Generally, constraint satisfaction problems are solved using methods which are not suitable for real-time musical performance, because the amount of time they will take to arrive at a solution is unpredictable. We have developed a musical decision-making agent which can solve some musical constraint problems (i) in a predictable length of time and (ii) usually much more quickly than alternative methods. It works by using an efficient data structure called a binary decision diagram to represent the constraint satisfaction problem.

1. INTRODUCTION

We use the term *musical agent* to refer to a computational entity intended to play music either autonomously or alongside a human performer. The computer music literature contains many reports on musical agents (see, e.g. [6] for a review) and a great variety of algorithms and architectures have been used in their implementation [10, 12, 20]. Less attention has been paid to the methodology of designing musical agents—that is, to the approaches to the design of musical agents which are not specific to particular music systems or situations.

We have previously studied a number of approaches to the creation of tools with which musicians can design the musical decision-making behaviour of a musical agent for use in their creative work. We have studied the use of *partially observable Markov decision processes* (POMDPs), whereby a musician can design aspects of an agent's behaviour by tuning the *reward function* of a POMDP model [13]. In addition, we have studied a *design-by-example* approach in which a musician can specify the behaviour of an agent by supplying examples of the desired behaviour [15].

In these studies, the focus was on design methods with two particular characteristics. The first is that they do not require musicians to be experts at computer programming or algorithm implementation or development (though, of course, they may be). The second is that the musician's aim is to design agents which make relatively high-level musical decisions, for example, the sequencing of musical elements such as sound samples and short MIDI sequences; or controlling the parameters of digital audio effects or processes which generate low-level musical material.

Recently we introduced a toolkit for the design of musical agents, which was also created with the aforementioned characteristics in mind [14]. The *Agent Design Toolkit* allows a musician to design the behaviour of a musical agent by first recording some examples of the desired behaviour and then following an iterative design process in which the musician can add to, or edit the examples, as well as adjust the parameters of a set of machine learning algorithms which learn the musical patterns in the example data. At each design iteration, an agent is created, which the musician can audition in order to decide what modifications to make to the examples and/or the parameters of the machine learning algorithms. This paradigm, in which a user interacts with machine learning algorithms iteratively to arrive at a satisfactory result, is known as *interactive machine learning* [7] and it was first used in the context of computer music by Fiebrink et al [8].

The Agent Design Toolkit is comprised primarily of two separate pieces of software. The first is the *Agent Designer*, which is used to record examples and run the machine learning algorithms. Its output is an *agent behaviour specification*; a file which specifies the behaviour of a musical agent. This file is then loaded into the *performer module* for real-time use in a musical performance.

In this paper, we report an a completely new implementation of the performer module, which is much more computationally efficient and is capable of executing far more complex behaviours than was previously possible with the toolkit. It relies on the use of a data structure called a *binary decision diagram* [4], which can be used to solve certain types of computational problems much more efficiently than with alternative methods.

The remainder of this paper is structured as follows. In Section 2, we give an overview of the Agent Design Toolkit and how it is used. In Section 3, we describe the performer module and our new implementation using binary decision diagrams. In Section 4, we discuss the performance improvements using the new performer module. In Section 5, we discuss the Agent Design Toolkit and its application in the context of other research, before concluding in Section 6.

2. OVERVIEW OF THE AGENT DESIGN TOOLKIT

In [14] we describe the Agent Design Toolkit and discuss a preliminary study of its use. In this section, we give a detailed overview of the toolkit. This will provide the context for understanding the issues involved in the implementation of the performer module.

2.1. Design paradigm

Our goal is to enable the creation of musical decision-making agents that can be adapted by music practitioners to their specific creative contexts. In a typical scenario, a musician has assembled a *music system* comprising an ensemble of virtual instruments, digital audio effects and algorithmic processes, all of which can be parametrically controlled (i.e. there is a set of variables whose values determine the output of the music system). An example of such a system is an Ableton Live project (www.ableton.com). In this context, the project is the music system which the musical decision-making agent will control. The Agent Design Toolkit provides a means for a music practitioner who is not an expert in algorithms or computer programming to design the behaviour of such an agent. The agent's decision making process can involve the use of external musical variables, e.g. incoming MIDI data and therefore the Agent Design Toolkit can be used to create both generative and interactive music systems.

2.2. Machine Learning Algorithms

Here, we give an overview of the machine learning algorithms used in the Agent Design Toolkit to learn the musical patterns in the example data. Separate machine learning algorithms are used to model (i) the -dependencies between the variables which control the music system and (ii), the manner in which they change over time. For (i), we use *association rule learning* (ARL) algorithms (see, e.g. [9]). These algorithms can be used to find deterministic rules describing dependencies between the variables. The rules are in the form of *implies* rules. For example, a music system may have separate variables which control the activity of the bass drum, the snare drum and the hi-hat, and the following rule may be discovered:

Bass drum = On, Snare drum = On ➔ Hi-hat = On

which can be read as "when the bass drum and the snare drum are sounding, then the hi-hat is sounding too". The musician can configure various aspects of the rule learning process.

For (ii)—learning the manner in which variables change over time—we use *variable order Markov models* (VMMs) [19]. Given the current value of a variable and its history of previous values, a VMM can be used to efficiently calculate the set of new values that the variable may take next, and their associated probabilities. The musician can choose which variables are modelled by VMMs and the maximum *order* of the VMMs (i.e. the maximum number of historical values to take into account when calculating the probability distribution for the next value that the variable will take). Higher order models produce sequences closer to those found in the set of examples provided by the musician.

As part of the design process, the musician must choose an integer-valued *priority* for each variable with which a VMM is associated. This is necessary for the following reason. At performance time, the agent uses the VMMs to choose values for the variables with which they are associated. As this is done for each variable independently, the VMM-chosen values can be inconsistent with the ARL-derived rules. When a conflict of this kind arises, the VMM-chosen value associated with one or more of the variables must be ignored so that no rules are broken. The agent uses the variables' priority values to choose which ones will retain their VMM-chosen values and which ones will not. Variables with high priorities are more likely to retain their VMM-chosen values in preference over those with low priorities.

2.3. Agent behaviour specification

To summarise, we list the different elements of an agent behaviour specification output by the Agent Designer and used by the performer module:

- **Music system variables:** The attributes of each variable associated with the music system. These include the *domain* (the values which the variable may take); a flag indicating whether the variable is *ordinal* or *categorical* (i.e. whether or not the values in the domain have an intrinsic numerical ordering); a flag indicating whether the variable is *controllable* or not (i.e. whether it is to be controlled by the agent, or just used in the decision-making process); the order of the VMM, if necessary; the *priority* with which the VMM is used; and a flag indicating if the value of the variable should be output at performance time.
- **Custom variables:** The attributes of the custom variables, which are additional, musician-specified descriptors that are intended to help the machine learning algorithms to find salient musical patterns (see [14, 16]).
- **Variable order Markov models:** The VMMs associated with certain variables and custom variables, as chosen by the musician.
- **Rules:** The set of rules describing the dependencies between variables (including custom variables), as discovered by the ARL algorithms.

For the purposes of illustration, Figure 1 shows a screen-shot of the "Variables" panel from the Agent Designer. The panel is used for choosing which variables will have VMMs associated with them, and for defining custom variables.

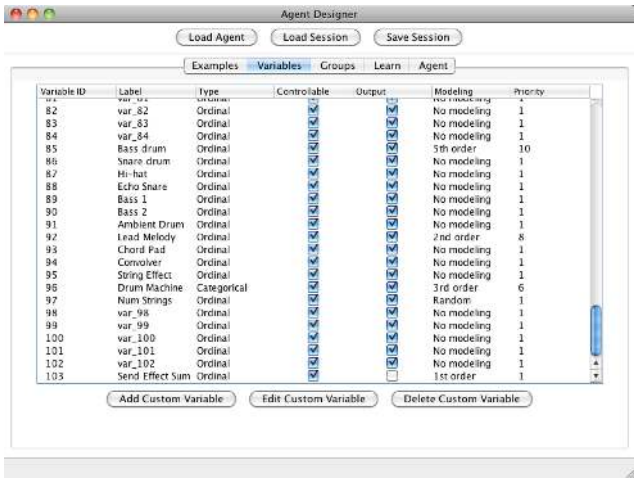


Figure 1. A screenshot of the Agent Designer.

3. THE PERFORMER MODULE

In this section we give details of the performer module. We start by describing the procedure which must be executed by the module. We then describe our original implementation. Finally, we describe its successor, which is the new implementation based on binary decision diagrams.

3.1. Algorithm for choosing new parameter values

The performer module is responsible for choosing new values for the music system variables in real-time, during a musical performance. In the following, we use the term *configuration* to refer to a particular set of values taken on by the music system variables and custom variables; and we refer to the set of configurations allowed by the rules as the *set of allowed configurations*. When an update of the variables is required, the following steps are taken:

I. Values of the uncontrollable variables (those not used to control the music system, but which describe external musical processes) are read in. Then the set of allowed configurations is reduced so that it includes only those configurations in which the uncontrollable variables have these values.
II. The variables which have associated VMMs are iterated over in order of their user-defined priorities. For each one (i) a value is chosen from the VMM, and (ii) the set of allowed configurations is reduced so that it only includes those configurations in which the variable has the newly chosen value. If there are no allowed configurations in which the variable has this value, the set of allowed configurations is not changed (i.e. the VMM-chosen value is ignored).
III. A random configuration is drawn from the remaining set of allowed configurations, and the values of the controllable music system variables are output.

3.2. First implementation

In the following, we describe how these steps were implemented in the performer module used in the first version of the Agent Design Toolkit. This will serve to introduce some key concepts and motivate the new implementation described below.

In the first version of the toolkit, the set of allowed configurations at the beginning of step (I) was represented by the set of rules output by the association rule learning algorithms (see previous section). Together, these rules form a *constraint satisfaction problem* (CSP; see e.g. [1]), which is a problem defined by a set of constraints on the values of a set of variables. It is possible to use many types of constraints to define a CSP including the *implies* rules discovered by ARL algorithms and other types required to define custom variables (see previous section). In step (I), when the values of the uncontrollable variables were read, they were encoded as additional constraints in the CSP.

During step (II) values are chosen for the parameters with which temporal models have been associated. Each time a value is drawn from a temporal model, it is encoded as a constraint and added to the CSP. Before continuing, the CSP is tested to see if it is feasible (i.e. at least one solution exists), since it is possible that a value has been drawn for a parameter, which does not satisfy one or more constraints (thus constraining the parameter to have this value would make the CSP infeasible). If the CSP has become infeasible, the new constraint is removed and no constraint is added for that parameter. If not, the new constraint is allowed to remain. A CSP can be tested for feasibility—and solved, if a solution exists—using a *constraint solver*: a piece of software designed to solve CSPs. The first version of the performer module was implemented in Java and a constraint solver Java library called *Choco* (www.emn.fr/z-info/choco-solver) was used.

Finally, in step (III) a random solution is drawn from the CSP comprising the constraints corresponding to the rules found by the ARL algorithms, as well as those corresponding to the values of the uncontrollable variables, and those corresponding to the values drawn from the temporal models. In the first version of the performer module, this was done by using the *Choco* library to find *all* of the solutions to the CSP and then choosing one randomly.

There were two reasons for our use of a constraint solver for the first implementation of the performer module. First, the rules discovered by ARL algorithms are standard constraints which can be directly used in a constraint solver with low development overhead. Second, general purpose constraint solvers, including *Choco,* support a wide variety of constraint types. The custom variables are implemented using different constraints, and in the early development of the Agent Design Toolkit, it was unknown which custom variables would be included. Thus, it was advantageous to allow for many different possibilities by using a general purpose constraint solver.

We have found that the implementation based on the *Choco* constraint solver generally works well with agents designed to control a small number of variables, or a set of variables with small domains. However, it has a serious flaw which arises from the fact that it is generally impossible to predict in advance (i) how long it will take to solve a CSP and (ii) how many solutions it will have (and how much computer memory will be required to store them). This means that at any time during a musical performance, a CSP may arise for which a random solution cannot be found within the time limits set by the real-time requirement. This is more likely with an agent designed to control many variables, or variables with large domains, since the number of possible variable configurations grows combinatorially. Nevertheless, the key issue is that it is generally impossible to predict when a problematic CSP will arise, and this makes the constraint-based solution unsuitable for live music performance.

In the next subsection, we describe an alternative way to implement steps (I)-(III) above. It involves representing the CSP as a binary decision diagram. This is a data structure which makes it possible to find random solutions to CSPs in a predictable length of time, without requiring all of the solutions to be stored.

3.3. Binary Decision Diagrams

Binary decision diagrams[1] (BDDs) are representations of Boolean functions [11] and in this section, we show how they can be used for real-time musical decision-making. To make use of a BDD, we first transform our CSP into a *Boolean satisfiability* problem (referred to as a *SAT problem*). This can be understood as a CSP in which all of the variables are Boolean; they can only take on values of *true* or *false*. A SAT problem is equivalent to a Boolean function (i.e. a function of Boolean variables) that gives an output of *true* if all the constraints are satisfied and *false* otherwise.

Before continuing, and with reference to our reasons for using a general purpose constraint solver in the first implementation of the performer module (see Section 3.2), we note that not all constraints supported by a general purpose constraint solver can be readily translated to SAT. However, at the time of implementing the BDD-based version of the performer module, the set of custom variables had been defined so it was known that the required constraints could be translated to SAT by available tools.

Once the Boolean function representing the original CSP has been created, it can be transformed into a BDD, which represents the Boolean function as a *directed-acyclic graph* [4]. This two-step process to compute the BDD (CSP to SAT, SAT to BDD), can be performed offline (i.e. before step (I) above). Once it has been completed, it is possible to perform steps (I)-(III) very efficiently and in a predictable length of time. This is because the BDD has special properties which make it a very attractive representation in the context of real-time musical decision-making. Using a BDD, the following operations are possible [5, 11]:

Check if there are any solutions: This is required for step (II) above. Using a BDD, it can be performed in constant time.

Count the number of solutions: This is required for choosing a random solution. It can be done in *O(nB)* time, where *n* is the number of variables in the Boolean function and *B* is the number of nodes in the BDD.

Choose solutions randomly, with all solutions being equally likely: This is required for step (III) and it can be performed in *O(n)* time or less.

Projection: This is the operation whereby the solution space can be restricted to one in which extra constraints are true, such as when the values of uncontrollable variables are read in step (I) and when values are chosen for temporally modelled variables in step (II). It can be performed in *O(B)* time.

All of these operations require the BDD to be constructed to begin with. While the process of transforming a Boolean function into a BDD can be done offline, and therefore is not time-critical, it has two pitfalls which we mention here. First, for a given Boolean function, the size of the BDD (i.e. the number of nodes, *B*) is very sensitive to the way in which the Boolean variables are ordered. A sub-optimal variable ordering can lead to a BDD which is much greater in size than that which would result from the optimal variable ordering. The problem of finding the optimal variable ordering is very hard to solve (it is coNP-complete, in computer science terminology) [11]. However, heuristic algorithms exist which can generally find reasonably good variable orderings. The second pitfall is that some Boolean functions simply cannot be compactly represented using a BDD, even if the optimal variable ordering is known [11]. However, these problems have not prevented BDDs from being successfully used in many different problem domains.

3.4. BDD-based implementation

We have implemented a BDD-based performer module as a plugin for the Max interactive platform (www.cycling74.com). It is written in C++. In this section, we present the implementation details and details of the third party software libraries on which our implementation is built. As described in Section 2.4, the output of the Agent Design process comprises three parts: descriptions of the variables, including the custom variables; VMMs for certain variables; and a set of rules describing the dependencies between variables. The core functionality of performer module is to performs steps (I)-(III) above. To do this it requires a BDD representing the CSP corresponding to the rules.

The construction of the BDD from the CSP is a two-stage process, as described in Section 3.3. The first stage of this process (CSP-SAT) is performed by the Agent Designer, using the Java-based *Sugar* CSP library (http://bach.istc.kobe-u.ac.jp/sugar/) [22]. The SAT

[1] We use this term as it is generally used in the literature to mean *reduced, ordered* binary decision diagrams [11].

representation, along with the agent behaviour specification, are loaded into the performer module. The performer module is based on the *CUDD* package (http://vlsi.colorado.edu/~fabio/CUDD/) for creating and manipulating BDDs (as well as other decision diagrams). This is used to convert the SAT representation into a BDD at load-time, and to perform the BDD projection operations and find random solutions at performance time. Though the conversion from SAT to BDD can in theory take a long time—and so might better be done as a separate, offline process so as not to delay the loading of an agent—we have not found it to take very long in practise (see next Section).

4. PERFORMANCE OF THE BINARY DECISION DIAGRAM

In the previous section, we argued that a performer module based on a constraint solver was unsuitable for live music performance due to its unpredictability. We proposed that an implementation based on BDDs is superior because the resources (time and computer memory) required during performance are more predictable. The uncertainty associated with the BDD-based implementation lies in the conversion from a CSP to a BDD which can result in a BDD which is too large to solve in real-time. However, it will be known at design time (rather than performance time) if an agent cannot be used, and if this is the case, steps can be taken to alter the agent or find another course of action. Furthermore, since BDDs have been used in many other applications to greatly improve the speed at which constraint problems can be solved, we have reason to believe that not only will the BDD-based performer will be more predictable, but it will often be much more efficient as well. In this section we present results which support these arguments.

To compare the performance of the BDD-based performer module with the constraint solver based one, repeated measurements were made of the time taken by each implementation to update the values of sets music system variables. Three different agents were used: they were (i) an agent for performing electronic music, described in [14]; (ii) an agent for performing improvised electroacoustic music, described in [16]; and (iii) an agent for performing drum and bass music, also described in [16]. The number of variables and rules associated with the agents are given in the second and third columns of Table 1. Each performer module performed with each agent for 1000 variable updates, and the times taken to perform the update calculations were recorded. The computer used for all measurements was a 2007 model Apple Macbook Pro with a 2.4 GHz CPU and 4 GB of RAM.

For each set of 1000 measurements, we calculated the normalised standard deviation (i.e. the standard deviation divided by the mean) to show the spread of times taken in a way that is independent of the time values themselves. This gives a good indication of the predictability of the time required to perform a variable update. In addition, we calculated the mean time taken across all 1000 measurements, to compare the efficiency of the two implementations.

The results for each agent as well as the mean across all agents, are shown in Table 1, columns 4-7. Clearly, the time taken for parameter updates is more predictable using the BDD-based performance module for which the mean normalised standard deviation of the time per update was 0.13 compared 1.33 for the constraint solver based implementation. To further illustrate this, we include Figure 2, which shows the normalised update times for 100 of the 1000 updates performed by each of the two performer modules using the electronic music agent. In addition, the results show that for each of the agents used, the BDD-based performance module was much faster than the constraint solver based one (a mean time per update of 0.08 ms compared to 75 ms).

We have not encountered a CSP which resulted in a BDD too large for real-time performance. The BDD sizes (numbers of binary variables and numbers of nodes) are given in Table 1, for each of the three agents used for this experiment. Also given in the time taken to create the BDD (averaged over 10 trials). The most complex agent was the electronic music agent which was used to control 102 music system variables (the number of variables given includes custom variables). It resulted in a BDD with 8262 nodes which took approximately 14.7 seconds to create from the SAT representation. Our results do not show the average time per update growing with the BDD size and number of variables. This can be attributed to the other variations between agents, such as the number of VMM-modelled variables which affects the number of BDD operations that must be performed.

Agent	# Vars	# Rules	Normalised Std. Dev.		Mean time per update (ms)		# Binary Variables	# BDD Nodes	BDD Creation Time (s)
			Con	BDD	Con	BDD			
Electronic [14]	103	500	1.5	0.1	175	0.06	226	8262	14.76
Electroacoustic [16]	13	26	1.3	0.2	25	0.08	36	1040	0.002
Drum and Bass [16]	14	18	1.2	0.1	25	0.1	32	291	0.001
		Mean	**1.33**	**0.13**	**75**	**0.08**			

Table 1. Comparison between the BDD-based performer module (BDD) and the constraint solver based performer module (Con) for three autonomous agents. See text for details.

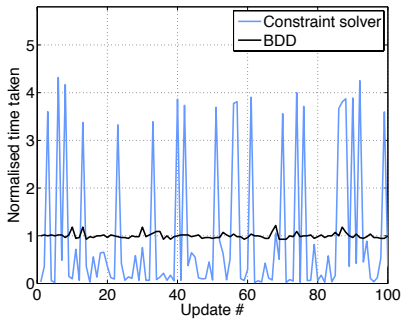


Figure 2. Normalised time taken to update variable values for the constraint solver and BDD-based implementations.

Finally, we note an additional problem encountered using the constraint solver based performer module when running the electronic music agent. On occasion, the number of solutions to the CSP was too large and they could not be stored in memory. This meant that the software could not perform the update. For the timing measurements shown above, restrictions were added to the agent specification to prevent this occurring. That the BDD-based implementation allows random solutions to be found without calculating and storing all solutions is an important advantage.

These results are consistent with BDD-theory, which says the update time required by the performer module only depends on the BDD size and the number of variables, which do not change during a musical performance. This contrasts with the constraint solver based implementation for which the update time depends on the number of CSP solutions, which can change from one update to the next. The results also show that for the agents used—which are representative of the agents produced by the Agent Designer Toolkit to date—our BDD-based performer module is much more efficient than the constraint solver based one. Some of this increased performance can be attributed to the inherent speed difference between Java and C++. In addition, there are constraint solvers which are faster than *Choco*. However neither of these factors accounts for the size of the performance improvement, particularly for the electronic music agent.

5. DISCUSSION

Considerable research has been done into the application of constraint programming (i.e. the use of CSPs and constraint solvers) to modelling music theories, both traditional and novel (see [2] for a survey). Much of this work has focussed on ways to represent music (i.e. a musical score) so that constraints of sufficient sophistication and musicality can be used. In comparison, the block-based representation of musical data used in the Agent Design Toolkit is very simple. However, in our case constraints are learnt whereas in the work surveyed in [2], they are specified by the user. Thus, in the Agent Design Toolkit, any constraint-related capability must be paired with a corresponding machine learning capability and the machine learning problem of discovering patterns as complex as those found in the theory of classical harmony, for example, is an extremely difficult one. Our use of custom variables (see Section 2.3, [14] and [16]) is one way in which a musician can increase the sophistication and musicality of the representation and thereby enhance the ability of the machine learning algorithms to find musically salient patterns and relationships.

There has been some research into the use of Constraints in real-time, interactive applications [3, 17], but only with manually specified constraints on musical data. As previously mentioned, it is impossible to know in advance, how long a constraint solver will take to find a solution. Thus, in [3] where a constraint solver is used in a real-time scenario, the workaround is to use a timeout which stops the search if a solution cannot be found within a specified duration. In contrast, the predictability of our BDD-based solver is a great advantage in real-time applications.

5.1. Future work

To improve the performer module, we plan to investigate alternative ways to deal with the possibility that a VMM-chosen value might not be consistent with the association rules. For example, a method has been proposed for adjusting the probabilities in a Markov model to account for external constraints while still remaining consistent with the model [18].

In the context of the Agent Designer Toolkit as a whole, we plan to carry out a thorough evaluation in an upcoming series of user studies. In addition to usability

testing, we will evaluate the toolkit in its capacity as a *creativity support tool* [21] and the extent to which it is useful to computer music practitioners in their creative work relating to agent design and automation. To this end, we have already integrated the toolkit into the Ableton Live music production software [16].

6. CONCLUSION

Constraint based systems are a powerful way for musicians to make musical decision-making agents. Constraints provide a way of formally specifying a set of relations between musical elements that can accurately capture a musician's conceptualization of their music. However, CSPs are not generative models, meaning that although they provide an accurate representation of a set of musical constraints, they cannot be used to derive musical decisions. In addition, they can be difficult to solve in a real-time context. The BDD provides a real-time generative representation of a CSP an a way that is practically useable by musicians. To our knowledge, this is the first report on the use of binary decision diagrams in a real-time music application.

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OSYNC: CROSS-APPLICATION PERFORMANCE SYNCHRONISATION OF BEAT-DRIVEN MUSICS

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ABSTRACT

OSynC is a protocol consisting of standardised messages for communicating metrical information and rhythmic descriptors for use in networked computer-based music performance. It uses the Open Sound Control (OSC) encoding to send descriptors from a host machine to any number of receivers, while receivers can in turn send changes to the host. The OSC encoding is a de facto standard for musical control, and is an extensible, low-bandwidth method for communicating information independent of platform and application. OSynC places particular emphasis on communicating rhythmic musical information at a range of musical timescales, providing details on musical context that are desirable for performance of beat-driven musics. It also aims to provide flexibility for musical improvisation, as well as intelligibility for performers. The authors describe OSynC's implementation as a Max patch, VST plug-in, and Max For Live device. These implementations are available from http://x37v.com/x37v/osync/.

1. INTRODUCTION

Collaborative computer-based music performance encompasses a wide variety of musical styles and configurations. From the laptop orchestra to smaller ensembles, a common requirement is the synchronisation of participants for precisely timed output. This is particularly crucial in the performance of beat-driven musics, including contemporary electronic dance music, such as IDM, dubstep, and so on. A system for networked performance of beat-driven musics must necessarily be robust and low-latency. Additionally, the system should provide the ability for real-time flexibility and improvisation in performance. The issue of synchronisation in performance is thus not purely computational, but also needs to take into account musical aspects that are most useful to performers.

The authors acknowledge that the conceptual framework behind networked computer and performer synchronisation is not a new one, however this paper serves as a formal framing of a synchronisation protocol based rhythmic descriptors on multiple time-scales, as opposed to metronomic pulse. The authors additionally acknowledge that there is a body of documented, undocumented, uncredited contributions to, and research into netjamming and synchronised computer performance.[1]

Existing models for synchronisation may rely on non-computational means (i.e. initiating a simultaneous performance/clock start via visual or aural cues between performers), however different applications, computers, and hardware may suffer from varying latencies and jitter [1], causing eventual desynchronisation. Metronome-based timing information may also be sent across a network from one user to another, in the form of pulses or 'ticks', while timecode (or SMPTE) information can also be used as a means to indicate temporal position. The system proposed here (OSynC) aims to extend these existing paradigms. It describes a method for transmitting musical timing features that are important to performance, while maintaining flexibility and cross-application functionality. Given the continued growth of the laptop as performance instrument, networked synchronisation is not a new issue by any means, but remains an area for improvement.

2. EXISTING MODELS

To date, several methods have been used for synchronising two or more performers across a network. Early examples include MIDI clock and word clock, which are subject to comparatively significant limitations of bandwidth, distance, and frequency, and are prone to time drift. Other MIDI-based synchronisation methods include DIN sync, used for linking hardware synthesisers. MIDI, however, possesses a comparatively small bandwidth of 31.25 kilobits/sec, compared with 10+ megabits/sec for networking technologies used by Open Sound Control (OSC) [2]. Meanwhile, the frame resolution of MIDI timecode implies suitability for video rather than music, making it unsuitable for control-rate musical interactions. A further drawback to each of these methods is their tailoring towards hardware (rather than software) synchronisation.

Synchronisation using the computer's audio capabilities has been proposed as a solution to the

[1] Existing examples include Hajdu's "Quintet.net" Internet based performance system [http://quintet.net], Wessel, Wright and Momeni's collaboration with Shafqat Ali Khan [http://www.loopers-delight.com/LDarchive/200106/msg00131.html], and synchronisation strategies of laptop orchestras [http://soundlab.cs.princeton.edu/publications/plork_icmc2006.pdf].

limitations of MIDI. Max's *sync~* object, for instance, is designed to receive MIDI real-time pulses and fill in the 'blanks' to create an audio-rate phasor wave synchronised to a pulse. Streaming audio-rate information across a network, however, raises its own problems with regard to different machines possessing different clocks and sample rates. Although a number of ways for transmitting audio over a network have been proposed (such as Olaf Matthes' largely abandoned *netsend~* and *netreceive~* objects for Max and Pd [3], and Plasq's Wormhole2 VST plug-in [4]), latency and synchronisation issues are not necessarily resolved with higher bandwidth. Furthermore, both of the above synchronisation systems lack the ability to transmit descriptive tags useful for musical performance.

There are a number of industry-supported methods for inter-application (and inter-machine) synchronisation. Both ReWire and VST SystemLink transmit playback-oriented information, including play state, bar, beat, and time signature information. ReWire, for instance, allows one application to drive another (client-server model), though this is not intended for use as a collaborative synchronisation tool for performers, as the assumption is that both programs are running locally. Meanwhile, Steinberg's VST SystemLink allows separate machines to be synchronised, but focuses primarily on sharing processor power between separate computers and additionally is developer-specific (i.e. both computers must be running Steinberg software). In 2008 Cycling '74 implemented a *transport* object for Max, designed to generate ReWire-like timeline-based transport information, but this is similarly intended for local machine communication within Max. The ability for inter-application communication is embedded in Max For Live, however communication is limited to Max and Ableton Live on a local machine.

OSynC differs from the systems described above, in that it is designed as a cross-application, control-rate method for synchronising sound objects in networked performance, which is platform- and application-independent. It focuses specifically on the synchronisation of musical attributes and their application in performance.

3. OPEN SOUND CONTROL

OSynC uses the Open Sound Control (OSC) encoding, an open system for network communication between computers and musical devices developed by the Center for New Music and Audio Technology (CNMAT), UC Berkeley [5]. The positives of an open communication protocol for musical control have been discussed in detail [6], in regard to applications such as communication of audio features [7], time-stamped clock synchronisation [8][11] and controlling VST plug-ins [9]. OSC's openness makes it a valuable tool for cross-application communication; as Wright [6] states, "Part of what makes OSC 'open' is that it comes with no standard set of messages every synthesiser must implement, no preconceptions of what parameters should be available or how they should be organised […] This form of openness has led to great creativity among OSC implementations, supporting idiosyncratic, creative software and hardware" [6, p. 194]. The OSynC standard aims to leverage the openness of OSC while proposing a base set of descriptors useful to musical performance and synchronisation.

OSynC proposes a set of standardised namespaces for synchronisation of rhythmic network-based performance. It also defines a priority of order for these messages. Similar OSC namespace standardisations have been proposed, such as SYN for synthesiser communication [10]. Wright [6] suggests that standardising namespaces is desirable for reasons of compatibility between applications and implementations.

OSC sends a continuous stream of 'packets' (or datagrams) describing the instantaneous state of a musical interface [11]. OSynC implements this idea, and reports information on the instantaneous state of position within rhythmic phrase on several timescales, as opposed to a metronomic pulse. Because OSC sends low-bandwidth packages via UDP (without an acknowledgement package sent in return from the receiving computer/s), OSynC messages can saturate the available bandwidth[2], or be sent at a user-specified rate. As OSynC sends a complete set of messages in each packet, dropped packets are not problematic, because the ensuing packets will contain the most up-to-date state information.

While research has been carried out on precise synchronisation using timestamps to allow systems to adjust for latency [8], OSynC does not timestamp messages for the sake of latency adjustment, although this could be provided in future implementations. It does however timestamp messages to ensure that message order is maintained. Clock synchronisation is not a concern in this research, as accuracy beyond the perception of the performer is not required[3]. Sample accuracy is therefore not necessary in OSynC, but a simple latency compensation can be implemented by timing a round-trip of packets and adjusting for latency on the host machine.

4. OSYNC

4.1. Description of OSynC

While OSC is typically used to send instrument and gestural control parameters and hardware control messages, OSC's openness as indicated by Wright [5] means that any musical information from the host can be

[2] Tests show that between 13,500 and 17,000 datagrams per second could be sent between applications on a local machine, or between 3600 and 20,000 per second between performers over computer-to-computer network, depending on Max's Overdrive setting (benchmark tests carried out by authors). Speeds above this are possible, but may result in dropped packets.

[3] Wright [6, p. 195] addresses network latency, concluding that latency of up to 10ms is acceptable in a musical performance.

treated as a transferable piece of information via OSC. Here, OSynC is used to transmit transport information between performers using a one-to-many host-receiver system. OSynC information is sent to receivers' IP addresses by the host via UDP. The host is responsible for transmitting OSynC messages, however any receiver of the system can alter the output of the host machine if desired (e.g. increasing master tempo). Although OSynC shares a number of similarities with ReWire's transport information, it possesses the advantage that the application does not need to support ReWire, only OSC. Furthermore, OSynC's descriptors are extendable to provide more detailed rhythmic information useful to performance.

4.2. OSynC information

OSynC transmits packets which define the following aspects of rhythmic state, in order:

0. Timestamp (for message order): float
: In the alpha implementation by the authors, an enumerated packet-stamp is included, not for latency adjustment or synchronisation, but to ensure that messages are received in the correct order. This safeguards against messages being re-ordered due to network load. If a message's timestamp is earlier than the current message's timestamp, it can be disregarded. In time, high-resolution timestamps are likely to replace this packet-stamp, hence its name.

1. Play state (on/off): integer
: Indicates whether the OSynC host transport is currently playing.

2. BPM/tempo: float
: Indicates beats per minute as specified by the host.

3. Time signature: integer array
: Numerator/denominator pair signifying number and type of beats.

4. Time elapsed between fractions (outlined below) as millisecond value: float
: Used for synchronising time-based effects between users (such as delay times), or generating events above fraction-level resolution (e.g. at 120 BPM [4/4 time signature] the milliseconds between 32nd-note fractions is 62.5).

5. Total bar count from initialisation: integer
: Indicates time (in bars) elapsed from the start of performance; can be used to schedule future events between performers.

6. Phrase-level bar count (1–4): integer
: Allows performers to work within shared phrasings comprising multiple bars.

7. Beat within the bar (1–4 per bar, or configurable by the user based on time signature): integer
: Allows performers to remain synchronised within the bar.

8. Fraction within bar (0–31 per bar, or configurable by the user based on time signature): integer
: Indicates basic temporal framework for note event occurrence, with the assumption of 4/4 timing. Calculated based on time signature (32 × numerator ÷ denominator, e.g. 4/4 equates to 32 demisemiquaver events per bar; 3/4 equates to 24 per bar; 7/8 equates to 28 per bar). This representation has been used by the authors as a simple method of driving a step sequencer.

9. Ramp (0.–1. across the course of a bar): float
: Indicates partial position within a bar, useful for creating arbitrary timing alterations on the receiver side.

A typical OSynC message packet will appear as follows:

```
/osync/timestamp 773495082.
/osync/play 1
/osync/bpm 125.7
/osync/timesig 4 4
/osync/fractime 59.665871
/osync/barcount 12
/osync/bar 4
/osync/beat 3
/osync/fraction 18
/osync/ramp 0.578125
```

Early implementations of OSynC transmitted 'incomplete' packets (without timestamps) that only described changed attributes, hence only *ramp* messages were sent between *fraction* messages. It was therefore important that messages from the host machine were sent in the above order and that packets were not dropped. The current model for OSynC suggests that 'complete' messages should be sent to circumvent any ambiguity created by lost packets. Because of this single packet design, message order is less of an issue, but it is still recommended that the host packs the message in this top-down order to ensure that higher-level musical context-related information (e.g. *bar*) arrives before note-level instruction (e.g. *fraction*). Lastly, it should be emphasised that OSynC is particularly dedicated to performance use; its current implementation forgoes sample accuracy in favour of flexibility and ease of use in a performance situation. Nonetheless, it is capable of synchronisation with no audible latency between performers.

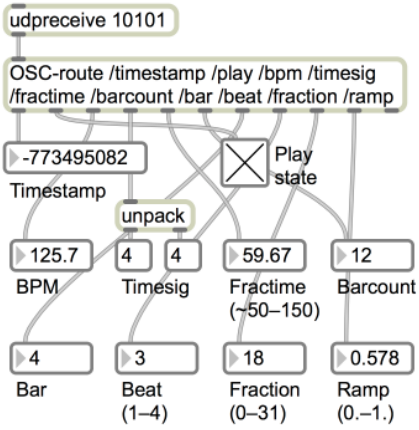


Figure 1. Received OSynC messages within Max

4.3. Benefits of OSynC

OSynC therefore does not rely on the strict regularity of pulse, but rather the regular update of information. Jitter in regularity, or dropped packets, are not a major concern, given the high rate of messages sent. OSynC differs from existing synchronisation protocols such as MIDI real-time messages in that if a connection is lost, positional synchronisation with the host is re-established as soon as the next package is received, instead of continuing from the last dropped packet.

OSynC focuses on sharing useful musical information between performers on a variety of timescales, from the macro-level (*play* and *barcount*), to meso-level (*bar*), sound object level (*beat* and *fraction*), down to micro-level (*ramp*) (after Roads [12]). This allows performers to work with different levels of rhythmic and structural form, in a way that is intelligible to all participants. OSynC additionally allows bidirectional communication between host and receiver; although all OSynC messages originate from the host, the receiver can alter higher-level controls such as tempo, time signature, and play state, forming a discontinuous feedback loop (Figure 2).

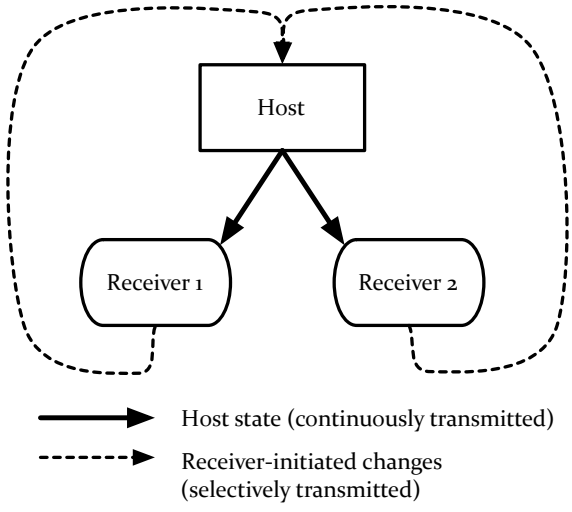


Figure 2. Host/receiver network flow using OSynC

A receiver can use OSynC in a prescriptive sense, or re-interpret transmitted values to their own ends (e.g. adding *ramp* value to *barcount* in order to provide continuous representation of position[4]). The *ramp* value, in particular, is a powerful musical descriptor. A receiver can use the *ramp* value as an arbitrary method of counting a number of beats within a bar by multiplying it by an integer or float, to create polyrhythmic interplay against the host to any degree of complexity. Additionally, *fractime* could be multiplied or divided to convey tempo-synchronous timing events beyond the fraction level.

Another benefit of OSynC is that host and receiver machines can run at different sampling rates and require no specialised hardware, whereas synchronisation systems such as VST SystemLink require sample rates to be the same in both systems. Additionally, OSynC is not software- or hardware-dependent, and can be implemented as a standalone within a development environment (Max, Pd, SuperCollider), or as a plug-in if non-development software is used as the host (e.g. multi-track audio software).

4.4. Current implementation

The first version of OSynC was built in 2007–2008 [13]. Its initial goal was to improve communication between the multi-track recording environment Cubase and MaxMSP, using Cubase to both maintain tempo and capture events generated in Max, while maintaining a uniform position between both programs to make subsequent editing easier. To do this, an OSynC host VST plug-in was built using the Pluggo SDK, which leveraged *plugsync~*'s ability to obtain ReWire-like transport information from a host environment. Here, Cubase generated timing events that were interpreted by MaxMSP, allowing synchronisation between the two applications on the local machine. One of the drawbacks of this implementation was that OSynC transmitted the host's state *once* per signal vector. The vector size used by plug-ins was the same as the I/O size, meaning that a VS of 512 samples produced up to 11ms waver, a limitation of this particular host environment. There was therefore a tradeoff between temporal accuracy and efficiency. This initial implementation also used a simplified descriptor system, which only updated *changed* values (i.e. *bar* was only reported at the start of a new bar). This contradicted OSC's remit as a stateless message system.

Since the introduction of Max For Live (2010), an implementation of OSynC within this multi-track recording environment is a simple task[5]. Similarly, the

[4] This could be useful in controlling elements that should evolve over the course of a musical performance, and could foreseeably be used to control non-musical aspects of a performance, such as lighting.

[5] The re-integration of many Pluggo objects into the Max For Live package means that a network synchronisation system like

introduction of the *transport* object for Max enables Max to act as a host system by wrapping OSC descriptors around the *transport* object's output. Although these programs enable synchronisation, they do not in themselves suggest a solution to cross-application (beyond Max and Live) or networked synchronisation. A wider implementation of OSynC as a VST plug-in, for example, would allow any VST-compatible host to act as an OSynC host, and any implementation of OSynC within an audio development environment like SuperCollider, Max, or Pd to act as a host, or as a receiver. The current implementation of OSynC as used by the authors is as a means of synchronising two performers using Max as a generative performance environment, however there are myriad potential uses for the OSynC system. Implementations of OSynC for Max, Ableton Live, and Pluggo are available from http://x37v.com/x37v/osync/.

4.4.1. Example 1: Networked cross-application performance using a variable number of participants

A host running Ableton Live transmits OSynC information, via Max For Live, to two receivers across a network. One receiver is using Max, and another is using Pd. The Ableton Live host is triggering scenes at a variety of BPM and continually addressing both receivers using OSynC. As the scene is changed, each receiver is updated with current information in order to generate note events in time with the host. The receivers' time-synchronous effects (e.g. delay) are updated via *fractime*. Both receivers can additionally slow down or speed up the BPM of the Max For Live host. A third receiver, running Processing, creates time-synchronous visuals alongside the performers.

4.4.2. Example 2: Working with flexible timings in OSynC

A host, using an implementation of OSynC in SuperCollider, transmits timing information to a receiver running Max. Using transitions in the *ramp* value to generate note events, the Max receiver can work at a different time signature to the host in order to produce polyrhythmic output, while still staying synchronised with the host at the bar- and phrase- level. Secondly, the Max receiver can use a lookup function to create temporal variation [14] from the events output by the host at the bar- or beat- level, such as dynamic temporal phrasing and swing (Figure 3). Additionally, the Max receiver uses a combination of *bar* and *ramp* values to drive a dynamic filter effect over the course of a four-bar phrase.

OSynC can be built in Ableton Live, using Live as the host and *plugsync~* to report on host state.

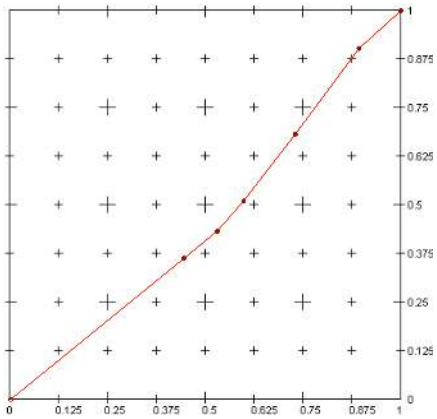


Figure 3. Altering linearity of playback using *ramp* value in conjunction with a lookup function

4.5. Future implementations

The current implementation is limited to receivers known to the host, i.e. the host explicitly sends OSynC messages to fixed IP addresses. An area to investigate in future is the ability for a new receiver to join a multicasted performance by requesting OSynC packets from a host, without prior explication of the receiver's address. Further research into the scalability of performer numbers and network load would also be required. Additional musical metadata such as rhythmic phrasing, root note, and musical mode could also be implemented to allow performers to synchronise modal changes during a performance. Extended benchmarking of OSynC to increase efficiency is also a possibility. A more formalised rule for timestamping based on OSC timetags, to provide high-resolution timestamping of OSynC packets, is also an area for future inquiry.

5. CONCLUSION

Although OSynC has been used as a performance synchronisation system by the authors for a number of years, this paper proposes OSynC as a method for standardising beat-driven performance, rather than as a definitive implementation of this idea. Attention has been paid to prioritising useful performance information, while maintaining intelligibility and controllability. One of the benefits of OSynC as proposed here is that descriptors can be used, ignored, or extended, according to performer needs. All musical factors needed by the authors are described here, however musicians may also wish to communicate additional information such as pitch spaces, timbral settings, dynamics, rhythmic groupings, light controls, text, and other information not specifically related to synchronization. With this alpha implementation these additional ways of participant synchronisation have been considered but have been left up to the end user. The authors welcome other implementations for other software.

The standardisation of an open synchronisation protocol encourages compatibility between applications

and third-party plug-ins, as well as performers. Initial results from the use of OSC as a means for synchronisation show promise for wider application. Because OSC is an open encoding requiring no special hardware, and the number of programs supporting this standard continues to increase, OSC (and OSynC) may be fertile ground for future development.

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RETHINKING INTERACTIVITY: FROM MATERIAL TO ORGANIC MODELING.

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ABSTRACT

Solenoid-based drum beaters are the drum-striking method of choice in mechatronic percussion systems. While solenoids are simple both electronically and mechanically, they produce a highly nonlinear dynamic response: volume levels created by the beaters do not correspond with input values in a linear manner. The goal of this paper is to quantify this nonlinearity and discuss methods by which the composer and performer who works with such systems can, in the face of such nonlinearity, still create musical works with expressive dynamic range. Additionally, avenues for future work are discussed, as are methods for calibrating and retrofitting existing systems to behave in a more linear manner.

1. INTRODUCTION

Among instrument families, percussion instruments are perhaps the most popular candidates for conversion to automatic and mechatronically-augmented systems. The primary reasons for the relative popularity of such instruments are largely due to the comparative simplicity of drums and other percussion objects when compared to more complex families such as string and woodwind instruments.

In spite of their apparent simplicity, mechatronic drums are not without their share of problems and challenges to the musician attempting to interact with such systems through composition and performance. In order to reliably work with existing systems, the artist needs to consciously compensate for three primary physical shortcomings in the currently available open-loop robotic percussion systems: the problem of latency between the output event and the mechatronic system's physical response, the problem of physical limitations of the speeds at which notes can be repeated by mechatronic drummers, and the problem of nonlinear output dynamics in response to a linear input value. This paper will focus on the last issue, that of the nonlinear dynamic response and how to work with existing musical robots which use open-loop solenoid-based percussion systems.

After a brief history of significant work done in the field of musical robotics and automatic musical instruments in Section 2, Section 3 will introduce the systems as tested and will more thoroughly explore the problem of nonlinear dynamic response and why composers and performers find it important to take into account during composition and performance. Section 4 attempts to quantify the dynamic response of the TrimpTron, a rotary solenoid-based drum beater. Finally, Sections 5 and 6 discuss integrating solutions proposed in prior section and overcoming the problem altogether via closed-loop approaches. Section 6 discusses such closed-loop approaches and introduces current and future work in the field of solenoid-based mechatronic percussion systems.

2. RELATED WORK

Composers and instrument builders have been fascinated by automatic musical instruments for centuries. [2] and [7] describe many of these early systems, most of which used punched or pinned rolls and barrels to store musical information. With the advent of the loudspeaker, automatic instruments faded into the background.

The later decades of the 20th Century saw a resurgence in such systems, with the emergence of several notable workers. Sound artist Trimpin, whose works are described in [1], has done a great amount of personal research and work on solenoid-based percussion systems, going so far as to experiment with feedback techniques and drum beaters utilizing multiple solenoid coils. In addition to Trimpin, fellow early musical roboticist Godfried-Willem Raes, founder of the kinetic arts collective Logos Foundation, has created many open-loop solenoid-based percussion instruments [6].

The last two decades have musical robotics arise as an academic discipline. [3] describes many academic artists who conduct and share research toward artistic ends. Several current researchers have focused on mechatronic percussion systems: Eric Singer's LEMUR collective consists of an array of open-loop solenoid-based systems used in an installation context as well as tools for live performance [8]. Researcher Gil Weinberg has in recent years focused on a variety of percussive systems.

Weinberg's work has revolutionized many aspects of human/computer interaction [10].

Second author Ajay Kapur's research (exemplified in [4] and [5]) has paved the way for this paper, laying down a foundation of quantitative study of solenoid-based percussion systems. Kapur's more recent work with collaborator Michael Darling, which provided the authors with a motivation for this paper, is described in Section 3.1.

3. SOLENOID-PERCUSSION SYSTEMS

Solenoids are mechanically simple, providing instrument builders with a straightforward means of converting electrical signals into motion. Due to this simplicity, solenoids are often used as the basis for mechatronic drum systems. The following subsections will describe the authors' solenoid-based systems as well as the challenges in creating dynamically predictable performances and compositions with such systems.

3.1. The Machine Orchestra

The Machine Orchestra, developed by second author Ajay Kapur and collaborator Michael Darling, is the performance-oriented outlet for the first two authors' musical robot compositions. Within the framework of the Machine Orchestra, new mechatronic designs are built and tested in a real-world interactive performance scenario. [11] details the ensemble. The necessities of performing with solenoid drummers in The Machine Orchestra provided the impetus for the work done in this paper: creating a workflow for interacting with open-loop solenoid percussion systems.

3.1.1. The KarmetiK NotomotoN: TrimpTron and KalTron

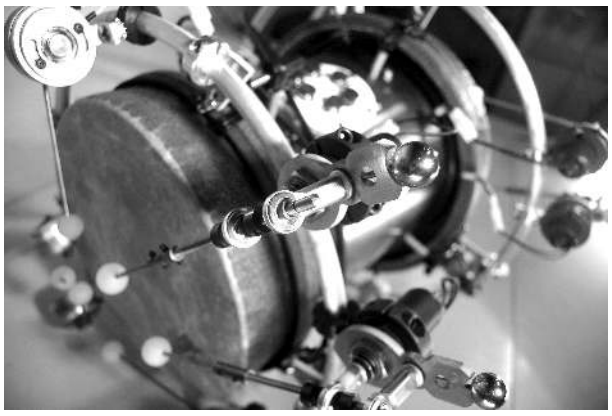


Figure 1. The KarmetiK NotomotoN, with KalTron and TrimpTron beaters visible.

The KarmetiK Notomoton, described in [12], is a robotic drummer which represents the most advanced integrated solenoid-based drummer in the Machine Orchestra. There are two types of drum beaters used in the NotomotoN: the TrimpTron, which utilizes a rotary solenoid, and the more mechanically intricate KalTron, which utilizes a linear pull-type solenoid. Both designs are by Michael Darling, closely inspired by work done by Trimpin. Figures 2 and 3 illustrate the key components of both types. The NotomotoN has become a central player in the Machine Orchestra; with its integration in the orchestra, the problem of the highly nonlinear dynamic response of its beaters has become quite obvious.

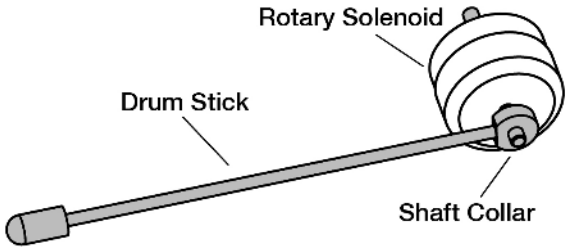


Figure 2. The components of a rotary-style solenoid percussion beater, similar to the TrimpTron beater.

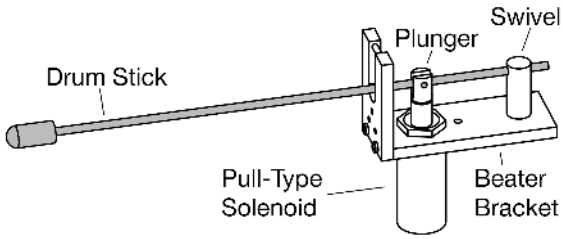


Figure 3. The components of a pull-type solenoid percussion beater, similar to the KalTron beater.

3.2. Motivation: The Problem of Dynamic Response

As mentioned in this paper's introduction, solenoid drum beaters do not produce linear output loudness in response to linear input values. This nonlinear dynamic response proves highly troublesome when corresponding with composers and performers who do not have access to the instrument: while the output dynamics are specified as MIDI CC values ranging from 0-127, the nonlinearity means that those without access to the instrument can not know how the solenoids will behave in real-world performance scenarios.

As an example, visiting composers have found that pre-sequenced compositions behave unpredictably when played on the NotomotoN's TrimpTron beaters. A perceived need to more closely quantify the actual behaviour of the beaters on the NotomotoN has led to the work undertaken in the following sections. It is the authors' hope that this paper can provide background and reference to future collaborators and guest composers when preparing to work with the myriad of

open-loop solenoid drum systems of the Machine Orchestra.

4. EXPERIMENTS

4.1. Experimental Setup

The remainder of the work shown in this paper will concentrate on our attempts to understand the nonlinear dynamic response of the TrimpTron rotary solenoid beater. To determine the degree to which the TrimpTron is nonlinear, experiments were undertaken to compare the input values with actual output values.

4.1.1. Hardware

The hardware used in the dynamic range tests is divided into two sections: the solenoid driver hardware and the datalogging hardware.

The solenoid driving hardware is similar to that used by the Machine Orchestra. It consists of a microcontroller using the HIDUINO firmware suite [13] to convert MIDI HID values to logic-level control voltages. The logic level signals are then amplified using power MOSFET chips capable of driving the 24V DC solenoids. Dynamics are controlled using a standard chopper drive method of solenoid control.

The datalogging section consists of hardware which records the output power created by the drumstick striking a surface. To test the dynamic response of the TrimpTron rotary solenoid beater, a piezoelectric pickup was used. The piezoelectric pickup output data to an Alesis D4 16 bit drum module. The D4 drum module converted the piezo's pickup to MIDI CC values, which were transmitted to datalogging software on a PC.

4.1.2. Software

To properly analyse the values picked up by the Alesis D4 drum module, the D4's output range had to correspond with the maximum value output by the TrimpTron: to normalize the MIDI values output by the D4 against the velocity commands input to the TrimpTron, correlation software was written using the ChucK programming language [9]. The correlation software sent a series of values to the TrimpTron and compared those with the values returned by the D4.

To gather dynamic response data, a linear series of output values were transmitted to a HIDUINO-equipped AVR microcontroller. For each of the linearly increasing output messages sent to the solenoid via the microcontroller, the corresponding value received by the piezo pickup was recorded. The following subsection discusses the results obtained by the tests.

4.2. Results

The data obtained in the tests is shown in Figures 4, 5, and 6. As was suspected prior to the quantitative testing, the results were shown to be very nonlinear. Figure 4 displays this nonlinearity: in a linear system, each measured value on the Y-axis would correspond to the same input value on the X-axis. Instead, the majority of the dynamic range is obtained with input values between 0 and 10. While this range varies with the beater's proximity to the drumhead, the characteristic rapid saturation holds true regardless of proximity.

In an effort to see the trend of the system, nine trials were taken for each value. These were averaged and are shown in Figure 5. A histogram of these values is shown in Figure 6. Again, the system's preference for returning values above 100 (out of a possible 128 values) is shown.

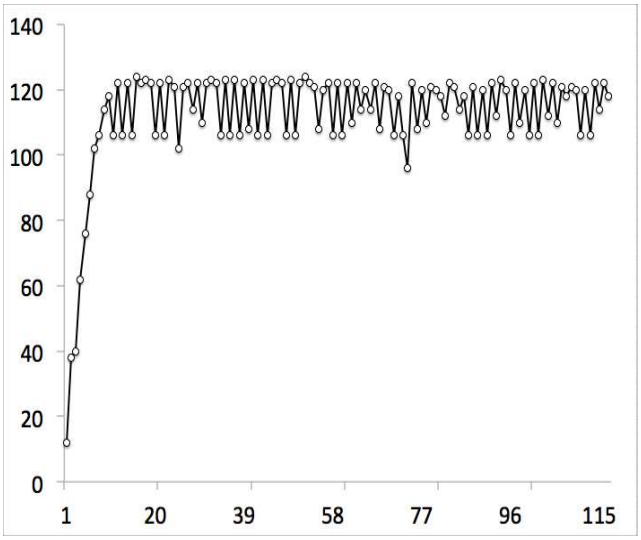


Figure 4. The dynamic response of a 128 linear output events (X Axis), compared to the recorded, calibrated output value (Y-Axis).

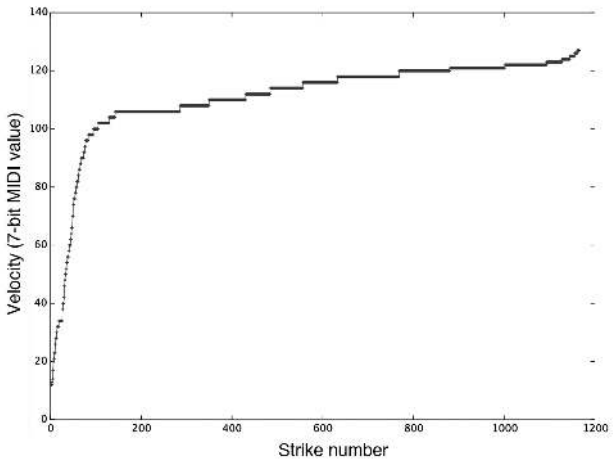


Figure 5. The dynamic response of a TrimpTron rotary solenoid drum beater averaged over nine trials. The nonlinearity of the response is clearly visible, with velocities from 0-100 underrepresented and higher velocities favoured.

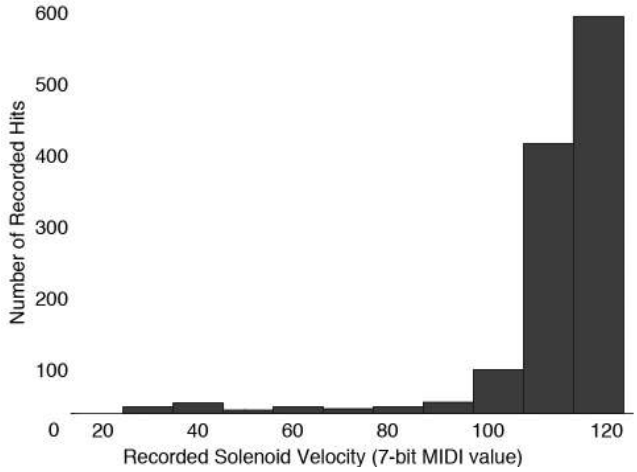


Figure 6. A histogram of the dynamic response of 1152 recorded strikes of a TrimpTron. The tests consisted of nine linear iterations of output velocities from 0-127, with results favouring values above 100

4.3. Experimental Conclusions

Based upon the results of the quantitative analyses described in Section 4.1, we can conclude that an uncalibrated, open-loop solenoid percussion beater similar to the rotary-style TrimpTron beater produces usable variances in its dynamic response in only the first 10 values of a possible 128 values. Thus, in order to vary the beater's volume, values between 0 and 10 should be sent to the beater. Any greater values will simply result in the TrimpTron-style beater outputting its maximum possible value. The following section discusses methods by which the limitations of such a system can be addressed.

5. INTEGRATION

As discussed in the previous section, the nonlinear output characteristics shown in Figures 4, 5, and 6 present composers and performers with a challenging situation: how to create expressive dynamics with such nonlinear drum beaters. The solution can be as simple as working within the constraints of the nonlinearity: values between 0 and the saturation point can be used, with the remaining resolution offered by the 7-bit MIDI CC values ignored. The disadvantage of such an approach is that it requires composers to be familiar with the system. Other approaches affording greater user-friendliness have been undertaken and are described below.

5.1. Linearisation Through Calibration

One approach to making use of the limited dynamic range of the TrimpTron drum beater is to stretch its useful values across the entire 7-bit range of possible input values. To do this, a calibration routine is utilized. Figure 8 depicts the signal flow in such a routine: firstly, the maximum output value of the solenoid is determined in a manner similar to that discussed in Section 4.1.2. Such normalization allows for inconsistent placement of piezo pickups: the pickup can be placed at various points on a drumhead and will still return a value equal to the maximum output level of the TrimpTron beater. Next, the dynamic range of the TrimpTron is determined via a variety of search techniques: with the successful linear search technique, output values are linearly iterated until the output value matches the normalized linear value. In this way, a linear response curve, similar to that shown in Figure 7, can be created.

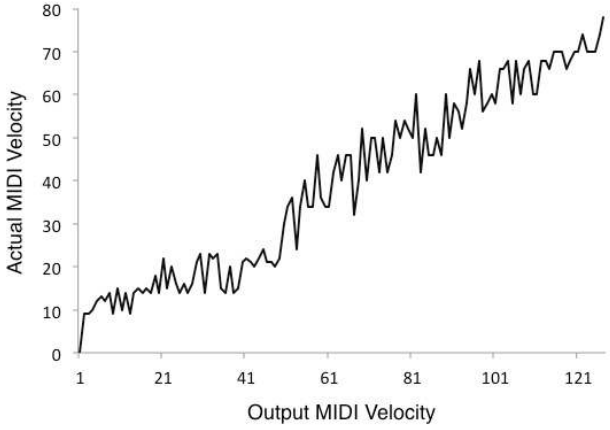


Figure 7. The linearised output of a calibrated TrimpTron beater. With the calibrated system, useful dynamic values can be obtained with all seven bits of the input value.

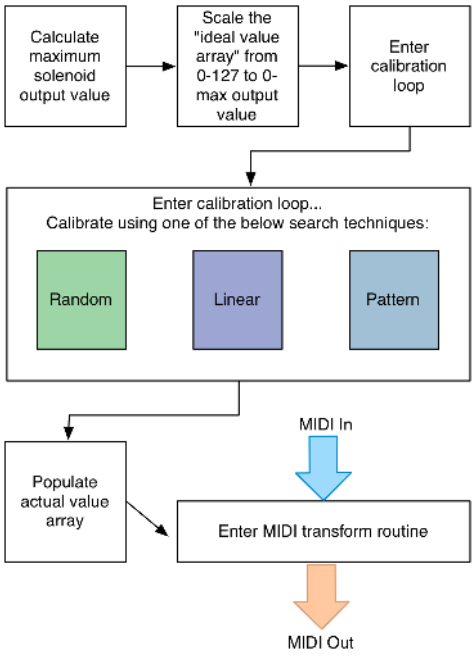


Figure 8. The signal flow for a pre-performance calibration routine for solenoid drum beaters.

5.2. Lookup Tables Using Dynamic Markings

A second approach to working within the dynamic range constraints of nonlinear open-loop solenoid percussion systems lies in the use of lookup tables. Figure 4 shows that for MIDI messages between 0 and 20, there are approximately eight useful distinct output levels. A lookup table was made which divided the 128 possible MIDI output levels into eight steps, each of which corresponded with a traditional dynamic marking (*PPP* through *FFF*). Composers and performers were informed of this dynamic range and could therefore compose within the constraints of real-world dynamic markings. This approach is simpler than the aforementioned calibration routines. However, it is more passive, lacking the ability to compensate for inconsistent beater placement.

6. CONCLUSIONS

The open-loop nature of existing solenoid percussion systems invites nonlinear behaviour as well as a myriad of other inconsistencies in performance. The clear solution to such a problem is to employ closed loop systems. By closing the loop and using an active feedback-based control system, a much greater degree of consistency can be obtained. The upshot of the work done in this paper is to convince us to continue to pursue working toward employing closed-loop systems in the Machine Orchestra. Such systems are currently under development, some of which build upon the semi-closed-loop approach used in the pre-performance calibration routine described in brief in Section 5.1.

Until such closed-loop systems have fully replaced the current open-loop robotic drummers, though, the results obtained in this study, as well as the calibration and lookup table-based solutions, will prove useful to composers interested in interacting with those robotic percussion systems which utilize solenoid-based drum beaters.

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AN APPROACH TO FEATURE EXTRACTION OF HUMAN MOVEMENT QUALITIES AND ITS APPLICATION TO SOUND SYNTHESIS

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ABSTRACT

This study investigates qualities of human movement that relate to the artistic organisation of sound. Identification of human movement qualities can assist in making expressive computer music interfaces and sound art installations which respond appropriately to expressive human movement.

In this paper, we present an empirical approach to exploring the relationship between sound and movement, in which participants are asked to move in response to sounds selected randomly from a palette of samples based on a mapping between Laban movement qualities and spectromorphological properties. Participants' movement is recorded using a depth-sensing computer vision system and later analysed with respect to the movement qualities exhibited. Participants are also asked to verbalise their thoughts and responses in a follow up interview.

The collected data, will be used in the design of an interactive installation based on the principles of the relationships between sound, movement and cognition established through our investigation.

1. INTRODUCTION

The aim of the research is to establish an understanding of common associations between organised sounds and movement qualities in order to facilitate the creation of expressive computer music interfaces, responsive art installations, and soundscapes for contemporary dance theatre.

The significance of this research is evidenced by the growing interest in computer music interfaces, seen in the New Interfaces for Musical Expression conference and in the burgeoning commercial market for computer music instruments and broader movement-focused computer game interfaces such as the Nintendo Wii and the Microsoft Kinect.

Our hypothesis is that, within a given local culture, a common set of associations between sound qualities and movement qualities may exist and such an association is bi-directional and symmetric (i.e. that specific sound qualities will evoke specific movement qualities in response, and vice versa). There is an underlying assumption that specific movement qualities are expressive of a person's expressive intent.

We aim to benefit the creative community by informing the design of both artistic works and commercial products which depend upon the automatic interpretation of user intent through physical movement, particularly where that interpretation results in the production of sound or music, or where sound or music is used to shape human responses to an environment. As an example, Maes et. al [1] have recently published work on a gestural control interface for vocalists.

2. RELATED WORK

This work builds upon a strong body of literature, well summarised in Godøy and Leman [2]. For example, Hevner's pioneering work in establishing empirical connections between emotion and music [3] [4] [5] helped to establish an experimental framework for investigating the psychological basis for human responses to music, and this approach has underpinned the majority of subsequent studies. In collecting a statistically significant set of response data Schubert et al. [6] shows that there are common emotional responses to musical features, and that the subjects' felt response is more significant than the emotion that subjects perceived was intended to be expressed by the music [7] [8].

For this reason, capturing subjective responses to sound and movement form an important part of this research. Inclusion of these subjective responses will also form an important part of the final installation as a bridge between observable stimuli and internal response, and to allow participants in the final artwork to gain insight into the experience of others at a cognitive and linguistic level.

Like Maes [1], we approach this work on the basis of the Embodied Music Cognition paradigm [9]. This perspective provides both empirical evidence and an explanatory theory which allows for a basic level of "objective" meaning to inhere in sound due to humanity's common cognitive strategies, while still allowing for individual and cultural differences. On this basis, we consider an investigation into associations between movement qualities and sound qualities will make a useful contribution to the overall understanding of embodied cognition of sound, as well as having practical application to the design of expressive music performance interfaces.

3. APPROACH

In examining the relationship between movement and sound, we have taken an empirical approach similar to that used by Maes [1], in which we observe spontaneous movement in response to sound stimuli. The following sections describe the design and selection of the sound stimuli, the movement capture system, and the analysis and results.

3.1. Movement Qualities and Sound Design

Central to our hypothesis is the idea of a bidirectional relationship between sound and movement qualities, arising from the premise that, as embodied beings, our sound processing capacity is related to the way we generate and perceive movement at the neurological level.

Drawing upon Laban Movement Analysis (LMA) [10] we sought to identify correlations between Laban Effort Shapes, as a canonical set of movement qualities, and the spectromorphological attributes of sound developed by Smalley[11]. Marshall[12] takes a similar approach in the application of Smalley's spectromorphology to the design of an interactive sensor-based sonification system. Smalley's theory of source-bonding is applied to the design of sounds to exploit participant's natural tendency to associate sounds with a (potential) source. Principles of Auditory Scene Analysis are also applied to ensure that individual auditory streams remain distinct and recognisable.

In our approach, the concept of source bonding is explored through empirical observation of the participants' spontaneous movement in response to a variety of sound stimuli drawn from Smalley's spectromorphological typology. The association between physical gesture and spectromorphological type are inherent in that Smalley's typology and LMA effort shapes share several identical or highly similar categories, such as floating, pushing/pressing, flinging/flicking. Leman's work on embodied musical cognition suggest that these associations are grounded in the very cognitive processes used for the perception of sound. Godøy[17] further proposes an explicit linkage between Schaeffer/Smalley's typology/morphology and sound producing/accompanying gesture.

Other associations between Smalley's typology and LMA are not so obvious at the category level, and we need to examine common combinations of qualities in order to identify a mapping.

First, we consider the components of effort in LMA, which are Space, Weight, Time and Flow. Each of these forms an axis in a four dimensional model of movement effort, as shown in Figure 1.

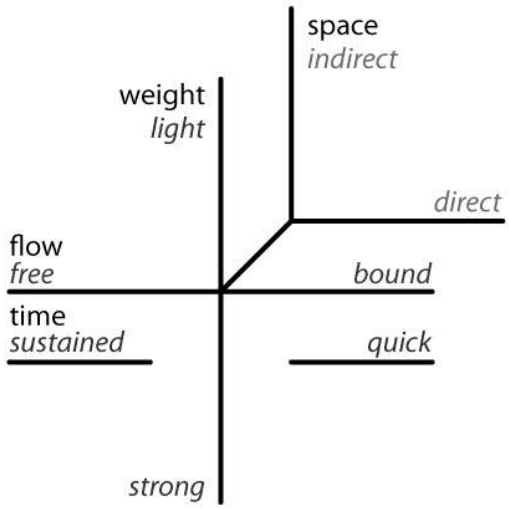


Figure 1. Laban Effort Graph [16]

The dimensions of the Laban Effort Graph are well summarized by Zhao[13], as shown in Table 1. The Space, Time and Weight factors of the effort model produce eight basic effort types, summarised in Table 2. Flow is considered separately, as explained by Laban[14]: "Movements performed with a high degree of *bound flow* reveal the readiness of the moving person to stop at any moment in order to readjust the effort if it proves to be wrong, or endangers success. In movements done with *fluent flow*, a total lack of control or abandon becomes visible, in which the ability to stop is considered inessential.

Motion factor	**Dimensions**	**Examples**
Space attention to the surroundings	**Indirect (flexible)**: spiraling, deviating, flexible, wandering, multiple focus **Direct**: straight, undeviating, channeled, single focus	Waving away bugs, surveying a crowd of people, scanning a room for misplaced keys Threading a needle, pointing to a particular spot, describing the exact outline of an object

Weight attitude to the movement impact	**Light**: buoyant, weightless, easily overcoming gravity, marked by decreasing pressure **Strong**: powerful, forceful, vigorous, having an impact, increasing pressure into the movement	Dabbing paint on a canvas, pulling out a splinter, describing the movement of a feather Punching, pushing a heavy object, wringing a towel, expressing a firmly held opinion
Time lack or sense of urgency	**Sustained**: leisurely, lingering, indulging in time **Sudden (quick)**: hurried, urgent, quick, fleeting	Stretching to yawn, striking a pet Swatting a fly, lunging to catch a ball, grabbing a child form the path of danger, making a snap move
Flow amount of control and bodily tension	**Free (fluent)**: uncontrolled, abandoned, unable to stop in the course of the movement **Bound**:	Waving wildly, shaking off water, flinging a rock into a pond Moving in slow motion, tai chi, fighting back tears, carrying a cup of hot tea

Table 1. Motion Factors and Effort Elements (from Zhao 2001, p. 41)

In terms of spectromorphology, Smalley[11] describes seven characteristic motions: fly, float, drift, throw/fling, rise, flow, push/drag, all of relate to dimensions of *motion rootedness*, *motion launching (attack), contour energy and inflection, and internal texturing.* Note that these are motions in terms of the spectra of sound, rather than motions in spatialisation of sound. The relationship between these dimensions and the characteristic motions is not fully explained by Smalley, however indications are given in the text and summarized in Table 3.

Effort Type	**Space**	**Weight**	**Time**
Slashing	Flexible	Strong	Quick
Gliding	Direct	Light	Sustained
Pressing	Direct	Strong	Sustained
Flicking	Flexible	Light	Quick
Wringing	Flexible	Strong	Sustained
Dabbing	Direct	Light	Quick
Punching	Direct	Strong	Quick
Floating	Flexible	Light	Sustained

Table 2. Basic Laban Effort Types

Motion type	**Rooted - ness**	**Attack**	**Contour**	**Texture**
Fly	rootless	emerging	slow / evolving	sustained
Float	rootless	emerging	slow / evolving	sustained
Drift	rootless	emerging	slow / evolving	sustained
Throw/ Fling	partial	pressured	rapid	granular
Rise	partial	emerging / pressured	rapid	streaming / turbulent
Flow	rooted	emerging / pressured	rapid	sustained / granular
Push/ Drag	rooted	pressured	slow	turbulent

Table 3. Spectromorphological motion qualities

In comparing Tables 2 and 3, it is possible to derive some correlations between the two models: *time* in LMA relates to both the *motion launching (*or *attack)* and the *texture* quality of spectromorphology; *weight* relates to the *rootedness* quality; and *space* relates to *contour*. This mapping is imperfect, but is sufficient for us to design the sounds which form the basis of our empirical approach to examining the relationship between movement and sound. Expressing Smalley's spectromorphological movement qualities in terms of more common synthesis parameters, we have taken the following approach to designing/selecting sounds which we expect to correspond to various LMA movement qualities. These are summarized in Table 4.

Motion Factor	**Movement Quality**	**Synthesis properties**
Time	Quick	short percussive sounds (short onset, short continuation)
	Sustained	Sustained sounds (longer onset, long continuation, long decay)
Weight	Light	low density sounds with harmonic content and greater reverb, higher pitch
	Strong	high density sounds with greater noise, less reverb, lower pitch
Space	Direct	steady pitch and spectra, less reverb
	Flexible	varying pitch and spectra, more reverb

Table 4. LMA Motion Factors and sound synthesis properties

4. DATA COLLECTION METHOD

At the time of writing, data collection and analysis is on-going, with 5 participants so far. Participants we recruited by referral and were screened for basic aptitude with respect to the tasks of the study. Participants come from a mixture of backgrounds, with 60% having greater than 5 years music performance training and experience. All participants reported that they listened to music for recreation and to alter their mood.

Sounds were designed or selected from existing corpora according to the qualities described above and assembled into a playback system that allowed for random selection and playback, one sample at a time. Participants were asked to respond to the sounds they heard by moving in whatever way they felt appropriate. The nature of the study does not lend itself to the use of a control group, however participants were given a trial task to confirm their understanding of the task and resolve any questions or concerns.

The order of the triggered sounds was recorded in a sequencer for later alignment with movement data. Movement data was collected using a Microsoft Kinect sensor and software from OpenNI [15] which allows RGB and depth-sensing camera data to be recorded. Participants were then interviewed and audio from the interview session was recorded[1].

5. ANALYSIS

5.1. Movement Analysis

Initial investigations into the data was conducted by manual tagging of the audio/video stream, labeling sections of video with instances of the recognition methods shown in Table 5.

Labelled audio/video data was then segmented according to the sound stimulus, and aligned across all subjects, allowing us to identify common movement qualities that emerged for any given stimulus (e.g. sustained movement for sustained sound, e.g. floating).

Basic Effort Type	Recognition Method
Gliding (sustained, direct, light movement)	long duration, low velocity movement without changes in direction, maintaining distance between hands and torso.
Slashing (quick, flexible, strong)	long duration, high velocity movement with changes in direction
Punching (quick, direct, strong)	short duration, high velocity movement with suddent acceleration and deceleration involving whole arm (elbow moves from torso)
Floating (sustained, flexible, light)	long duration, low velocity movement with changes in direction
Dabbing (quick, direct, light)	short duration, low velocity move-ment
Pressing (sustained, direct, strong)	long duration, low velocity movement without changes in direction, hands moving away from the body
Flicking (quick, flexible, light)	short duration, high velocity movement localised to the hands (elbow doesn't move far from torso)
Wringing (sustained, flexible, strong)	long duration, low veloc- ity movement with hands constrained body space (hands and arms close to torso)

Table 5. Effort type recognition methods

Table 6 shows the results for data collected to date. Shaded cells indicate the intended quality criteria for a given effort type, while number values give the normalised number of actual observations of that quality criteria for the given sound stimuli category.

Some sound stimuli were consistently associated with the expected movement criteria. Sounds in the floating and wringing categories produced expected behaviour; gliding also produced fairly consistent results, with some variation velocity and constancy of hand-torso position. Dabbing and flicking responses were crossed over in their observed velocity, but tended in the correct directions (lower for dabbing, higher for flicking): quantitative analysis of the motion capture data will be used to more closely investigate these responses to determine whether a clear boundary can be observed in terms of velocity. Some stimuli, such as wringing, tend to be variable in the movement responses they generate. Others, such as slashing and punching, generate visibly different movements, which are not strongly distinguished by the recognition criteria, and do not align with the expected results.

5.2. Analysis of interviews

Post-activity interviews with subjects revealed a number of interesting threads. A common theme was that subjects moved in response to visual imagery, i.e. their imagination of how the sound looked, rather than how they felt in response to the sound. Subjects reported that they had little emotional response to the sounds, and moved in a way that mimic'd the sound, rather than expressing personal intent.
Some subjects reported that complex or ambiguous sounds offered more room for interpretation and

[1] This research is being conducted under ethics approval for Minimal Risk Application 1136360.1, University of Melbourne.

personal expression, compared to sounds which were coherently of one type (e.g. percussive sounds or floating sounds).

Other common observations were the relevance of certain sounds to parts of the body. Subjects commonly related higher pitched or short duration sounds to the extremities, while lower pitched sounds with rich harmonic content were related to the whole body.

6. CONCLUSIONS AND FUTURE WORK

Our objective in examining the sound-movement mapping is to ultimately reverse mapping and use movement quality recognition to control various sound synthesis parameters, and in this way, build a gestural control system which is expressive of the users intent.

The stimulus-response data collected and the subject observations collected so far are encouraging in terms of supporting the underlying embodied music cognition approach that we have taken. The specific movement quality attributes selected are generally robust in terms of categorising the observed overall movement quality, and support the idea that there is an association between sound qualities and movement qualities that manifest in observably common ways. Refinement of the selected quality criteria for certain sound stimuli is needed to capture the wider variety of movement observed.

The most unexpected result observed so far is that our assumption that this association is bi-directional and symmetric appears to be false. Subjects report that, while their movements related to shape of the sounds, they were, in general, not expressing any intent. Even though subjects were clearly in control of their own bodies, their comments indicate a general sense of lacking agency.

This was particularly evident in one case, where in response to sounds designed to relate to the 'punching' effort type, the subject moved *as if being punched*. If the subject were to be moving with the intention of generating a punching sound, it would be more natural to be moving as if punching (in line with the general embodied music cognition theory that sound is often processed in terms of what motion would produce that sound).

In retrospect, the asymmetry of the agent-patient relationship is obvious, but it does raise interesting questions about what issues will arise when attempting reverse the sound-motion mapping developed during this phase of our research and apply it to synthesis control. Even more interesting, however, is the subjects' reported sense of lack of intent in responding to sound stimuli. This raises many questions in terms of the way in which interactive synthesis systems may be used to mediate performer-audience, performer-instrument, and performer-performer relationships, and is an area which we intend to explore further.

In terms of practical future work, additional data collection and analysis is required to gain greater confidence in the results we have presented here, as well as gain greater insight into the most effective movement quality criteria for effort type recognition.

The manual motion quality analysis conducted in this study will be used to build an automated motion quality recognition system utilising the OpenNI skeleton tracking system to extract 3D coordinates of key joints (hands, elbows, head, torso, hips, knees, ankles) and analyse the data according to the motion quality recognition methods presented here.

In addition, a supporting study into classification of the designed sounds is also planned. Subjects will be requested to apply their own freely associated labels to sounds they have grouped together, as well as applying the fixed set of effort type labels in order to validate the sound-movement association at a cognitive labelling level.

The understanding gained through completing these studies will be used to drive the development of a gestural control system for synthesis, to be used in an interactive audio-visual installation. We also expect to be able to apply this deeper understanding of sound-movement

Movement qualities	Pressing	Wringing	Dabbing	Punching	Floating	Gliding	Slashing	Flicking
Long duration	1.00	1.00	0.00	0.50	1.00	1.00	0.38	0.00
Short duration	0.00	0.00	1.00	0.50	0.00	0.00	0.63	1.00
Low velocity	1.00	0.80	0.25	0.75	1.00	0.67	0.13	0.00
Med velocity	0.00	0.20	0.75	0.25	0.00	0.17	0.38	0.67
High velocity	0.00	0.00	0.00	0.00	0.00	0.17	0.50	0.33
Steady direction	0.00	0.00	0.75	0.25	0.00	0.33	0.75	1.00
Changing direction	1.00	1.00	0.25	0.75	1.00	0.67	0.25	0.00
Sudden accel/decel	1.00	0.40	0.75	1.00	0.00	0.17	1.00	1.00
Gradual accel/decel	0.00	0.60	0.25	0.00	1.00	0.83	0.00	0.00
Hand-torso distance constant	0.00	0.00	0.00	0.25	0.25	0.50	0.00	0.00
Hands-torso distance increasing	0.00	0.00	0.50	0.25	0.63	0.50	0.63	1.00
Hand-torso distance variable	1.00	1.00	0.50	0.50	0.13	0.00	0.13	0.00
Elbow-torso distance constant	0.00	0.20	0.50	0.25	0.25	0.50	0.00	0.00
Elbow-torso distance increasing	0.00	0.00	0.25	0.25	0.63	0.33	0.75	1.00
Elbow-torso distance variable	1.00	0.60	0.25	0.50	0.13	0.17	0.25	0.00

Table 6. Observed movement quality criteria for various sound stimuli (normalized average number of instances).

association to sound design for dance performance, and computer game audio in way that delivers inherently meaningful audio responses to performer/user gesture while going beyond simple mappings which lack nuance and lasting appeal.

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Adapting John Cage's *Radio Music* for a digital score player

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ABSTRACT

This paper discusses the creation of a digital score player and installation version of John Cage's *Radio Music*. The context surrounding the work's composition is explored as well as the changing context of the work in light of the evolution of radio broadcasting since the 1950s. The score for the work is explored in detail particularly in regard to the issues its determinate and indeterminate aspects, and their implications upon the performance of the work. The concept of "available indeterminacy" is introduced to describe the real-world limitations that exist as a result of Cage's specifications for the work. The presence of form-bearing structural features and consequently the potential for the emergence of an indeterminate and nonlinear formal structure from the performance of Radio Music is investigated.

1. INTRODUCTION

Cage's works including radio as a sound source[1] emerged during a period when radio was exercising an increasingly important cultural role in American society. Although radio broadcasting had existed since the 1920s [3 pp. x-xi], it wasn't until the 1930s that "radio's aesthetic possibilities, it's potential to reach vast audiences, and its wealth of institutional resources attracted composer's from a broad stylistic spectrum" [18 p. 106]. The Second World War propaganda successes the consequent consolidation of radio services the also brought greater focus to the ideological influence that this medium was bringing to bear.

In 1945 Adorno directed his attention to radio, declaring:

> Music under present radio auspices serves to keep listeners from criticizing social realities; in short, it has a soporific effect upon social consciousness. (…) While radio marks a tremendous technical advance, it has proved an impetus to progress neither in music itself nor in musical listening. [2 p. 232]

By the end of the 1960s Marshall McLuhan was proclaiming a very different point of view.

> But while radio contracts the world to village dimensions, it hasn't the effect of homogenizing the village quarters" [16 p. 408].

While Cage was ideologically much closer to McLuhan's vision [15 p. 145][12 p. 564] than Adorno's, his recasting of the radio as a autonomous musical instrument does respond to one important question that Adorno posed in his article: "Does a symphony played on the air remain a symphony? Are the changes it undergoes by wireless transmission merely slight and negligible modifications or do those changes effect the very essence of the music?" [2 p. 230]. In Cage's radio works, the decontextualisation of the sonic contents of the radio broadcast is laid bare. Cage discussed his view of the radio as an abstraction in an interview with Kostelanetz:

> I didn't myself turn on a radio to listen to it; but when I was going through the streets or when a neighbor was playing the radio, and so forth, I listened as though I were listening to a musical instrument. (…) It made it possible for me to listen to radio with great interest, no matter what it was doing. [14 p. 220]

Cage's stance, listening to the sonic emanations from a radio in the same manner that he would listen to a piano or a thunderstorm, remained contentious: for example in more than 10 years later musicologist Nicholas Ruwet used *Radio Music* as an example of "non-music" [17]. Radio broadcasts (amongst other media) have, today, long been accepted as a simulacrum of the "real thing". What might be called "Cagean Listening" in which "sounds are just sounds" [9 p. 134], and consequently his *Radio Music,* is still a challenge for the audience.

Interestingly Cage wrote this work in semi-seclusion, living with only four others at the Stony Point artist's colony of Paul Williams at the edge of Harrimann state forest in upstate New York. Nicholls has noted that Cage's move to the countryside in 1954 paradoxically brought "increasing interaction with electronic media" [17 p. 105].

2.THE SCORE

Cage's score for *Radio Music* is deceptively simple. Each of the eight parts bears a similar set of instructions:

> Part A of Radio Music to be played alone or in a combination with parts B-H. In 4 sections (I-IV) to be programmed by the player with or without silences between the sections, the 4 to take place

[1] The most prominent of these works include *Credo in US* (1942) for radio or phonograph, muted gongs, tin cans, electric buzzer and tom-toms and prepared piano [8]; *Imaginary Landscape No. 4* (1951) for 12 radios [4]; *Speech 1955* (1955) for 5 radios and newsreader [5]; *Radio Music* (1956) 1-8 radios [6]; and *Music Walk* (1958) [7] for 1 or more pianists who also play radios and produce auxiliary sounds by singing or other means.

within a total time-length of 6 minutes. Duration of the individual tunings free. Each tuning to be expressed by maximum amplitude. A _____ indicates 'silence' obtained by reducing amplitude approximately to zero. Before beginning to play, turn radio on with amplitude near zero. [7 Part A].

The remainder of the score consists of columns of frequencies separated by roman numerals, signifying the work's four sections. The score is a form of tablature, signifying physical actions, changing the selected carrier frequency manually with the radio dial, but not sonic outcomes [13 p. 386].

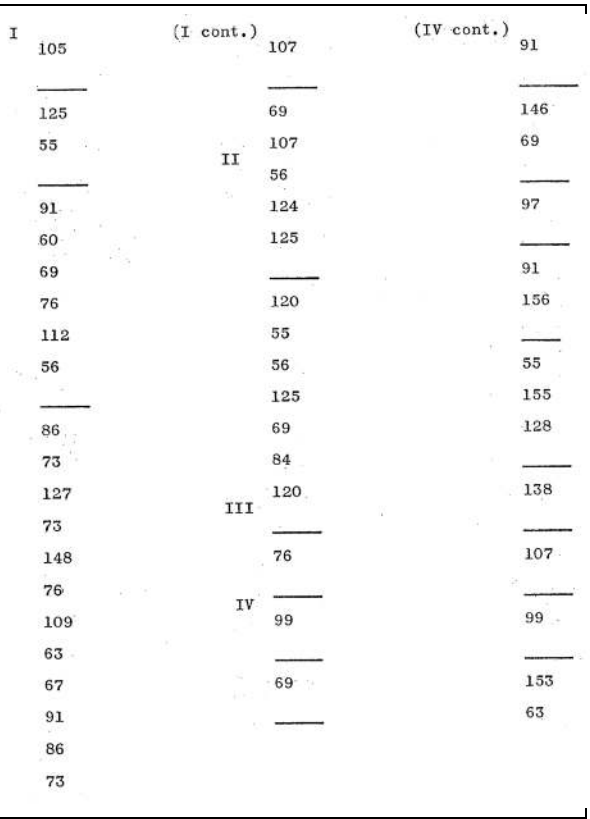


Figure 1. Radio frequencies arranged in columns in the score of Part A of *Radio Music*.

Although the "authentic instruments" for performing *Radio Music* (RCA Victor "Golden Throat" Radios) similar to the one shown in Figure 2) are still available, "the contemporary radio landscape is completely different from the one in 1956" [11]. In effect the indeterminacy of the score has brought "authentic performance" of the piece to the brink of silence, through the "looming switch-off of the analogue radio signal in favour of the DAB type digital signal" [20] throughout the world.

The carrier frequencies listed by that Cage fall between 55 and 155. This has been something of a puzzle for performers of the work as the AM band is generally labeled in kHz, and there is very little activity in this range. The most likely solution to this conundrum is that Cage was referring to frequencies in hectohertz (hHz). It was common to use the range 55 to 160 to represent the typical AM band of 550 to 1600 kHz, on radios of that era to save space on the dial (as shown in Figure 2) [11].



Figure 2. A 1950s RCA "Golden Throat" radio similar to the type John Cage conceived Radio Music for. Note the AM dial shows carrier frequencies displayed in hectohertz (hHz).

Although many have commented on the indeterminate nature of this work, there are a number of elements of *Radio Music* that are quite precisely prescribed. The impact of these determinate elements upon what might be called the "available indeterminacy" - that is the range of choices[2] that may be made without generating outcomes that fall outside of the determined elements of the instructions – will be discussed in the next section.

In Cage's instructions he states that the four sections of *Radio Music* should "take place within a total time-length of 6 minutes" [7], but that the "duration of the individual tunings (is) free". Each of the eight parts has a different number of carrier frequencies to be performed (see Table 1.). The part with the largest number of carrier frequencies is C with 86. In order to perform all 86 in the space of 360 seconds (within a total time of 6 minutes) the performer must retune 86 times at an average rate of just over one every four seconds, not leaving a great deal of leeway for freedom of duration.

Indeed, if the work is to be performed in four sections as specified by Cage, the calculation of the average time allowed for each retuning is dictated not by the part with the greatest number of carrier frequencies, but by whichever part has the largest number of frequencies in any given section. In section I the largest number of frequencies is in part G with 44 changes; II - C with 30; III - H with 38; and IV - F 40 changes. This means that for each section, the average length of each retuning, in the part with the largest number of cariier frequencies, is calculated by dividing the maximum duration (6 minutes) by the sum of these numbers (44+30+38+40) i.e. 360/152.

[2] The consequence of an indeterminate element in a score is always a performative choice.

Determinate

	A	B	C	D	E	F	G	H
Total no. of Frequencies per part	68	51	86	42	36	70	71	46

Number of frequencies per part per section	A	B	C	D	E	F	G	H
I	28	15	37	8	8	7	**44**	*3*
II	11	14	**30**	2	10	16	6	*1*
III	3	2	11	23	2	7	2	**38**
IV	26	20	8	9	16	**40**	19	4

Largest number of frequencies per section	I	II	III	IV
	44	30	38	40

Indeterminate

Duration	152-360s
Number of parts	1-8 parts

Maximum and Minimum length of sections (s)	I	II	III	IV
	min-max	min-max	min-max	min-max
	44-196	30-182	38-190	40-192

Maximum and Minimum length of frequencies (s)	A	B	C	D	E	F	G	H
	min-max	min-max	min-max	min-max	min-max	min-max	min-max	min-max
I	2-7	3-13	1-5	6-25	6-25	6-28	1-4	15-65
II	3-17	2-13	1-6	15-91	3-18	2-11	5-30	30-182
III	13-63	19-95	3-17	2-8	19-95	5-27	19-95	1-5
IV	2-7	2-10	5-24	4-21	3-12	1-5	2-10	10-48

Table 1. Determinate and Indeterminate characteristics of *Radio Music*.

Therefore, the parts with the largest number of frequencies in each section must change tuning at an average every 2.37 seconds. This extremely rapid rate of change means that, although indeterminate, the "available indeterminacy" in duration of the individual tunings is, in fact, always quite limited for some of the performers.

However, since Cage allows for *Radio* Music to be performed with between one and eight parts, these speculations about average durations vary depending on the number and combination of parts[3].

Another variable is the length of the sections themselves, which is only determined by all of the performers completing the prescribed set of frequencies and of the total duration, (the sum of the part with the longest durations). Although Cage specifies that the total duration of the work should not exceed six minutes, the performer is not prohibited from attempting shorter durations.

These factors mean that although it is not possible to get an exact indication of the rate at which each player must realize the frequencies in their set, limits can be proposed upon the average duration that a player will need to maintain in order to avoid a situation in which there is not enough time remaining to complete the prescribed frequencies in their set with six minutes. Table 1 proposes these ranges for each part and section in the work (on the basis that all eight parts are being performed).

Cage's directions for "each tuning to be expressed by maximum amplitude" (Cage), implies that the spaces between the tunings should be lower in amplitude, perhaps even silent. To achieve this, the performer must operate the tuning and volume dials on the radio

[3] For this reason the performance of the work by the Western Australian new music group *Decibel* (of which the author is a member) on 20 June 2011 used only parts A, B, D, E and H: the five shortest parts.

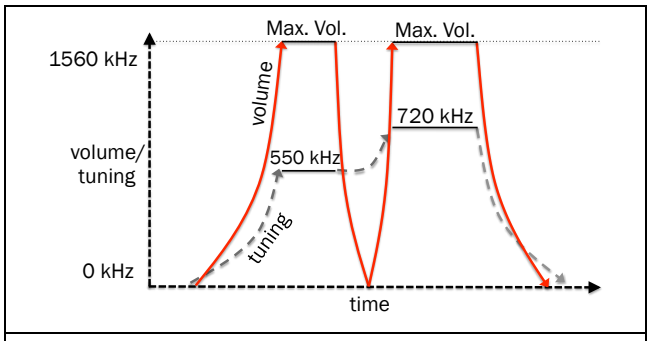


Figure 3. Manipulation of volume and Frequency controls necessary for the performance of *Radio Music*.

simultaneously and somewhat independently[4]. Figure 3 illustrates these actions on a graph: showing the volume dial always moving between silence and the maximum volume and the tuning dial moving smoothly between each frequency.

There is a both a limit upon the speed at which the performer can accomplish this task and presumably a limit upon how much time is needed for the audience to perceive of the particular frequency that has been reached. It is suggested here that one second is the lower boundary of this limit.

In performance, the physical difficulties of shifting between the score and the dial of the radio in order to reach the correct frequency are very real. Although it might be argued that a professional musician should not need to look at their instrument, the number of professional radio players is of course rather limited.

The issue of communication between performers is also a factor in *Radio Music*. Since the sonic output of the radios is indeterminate, it is difficult to take auditory cues from the other performers. As each performer is freely choosing the duration of each frequency it is not possible to take temporal cues from the other performers. These combined factors make a precise reading of *Radio Music* a serious challenge.

3.DIGITAL PERFORMANCE VERSION

The digital version of *Radio* Music alleviates some of the issues discussed in the previous section. The score is projected or networked on multiple computer screens. It consists of graphical representations of the radio dial for each of the parts being performed as shown in Figure 4. A red vertical line indicates the destination carrier frequency to be reached and a black vertical line moves continuously to indicate the suggested current position of the tuning indicator. The destination frequency is displayed numerically on the left hand side of the screen and a notional current frequency is displayed on the right side of the moving black line. This allows the performer to follow the changes in frequency in a more intuitive kinetic manner.

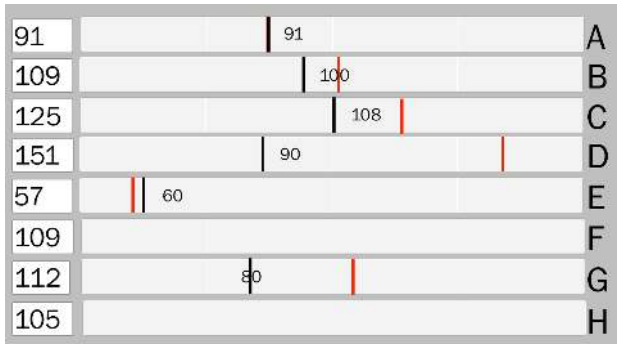


Figure 4. The Real-time Score for the digital version of *Radio Music*.

The issue of the duration of each frequency is calculated through a "hierarchy of indeterminacy", taking into account that the available choices diminish hierarchically from formal levels with longer durations (such as the total duration of the work) to shorter formal levels (such as the duration of sections and individual frequencies). Once the total duration of the work is set, duration of the sections cannot exceed that length and similarly once the length of the sections has been chosen, the total duration of the frequencies cannot exceed that length.

The control panel for the digital version of *Radio* Music, provides the option to:

- choose which parts will be included in the performance;
- set the total duration of the work;
- set the average duration of pauses between the four movements;
- choose the degree of allowable deviation from the notional average duration of the four movements
- choose the minimum duration assignable to each frequency.

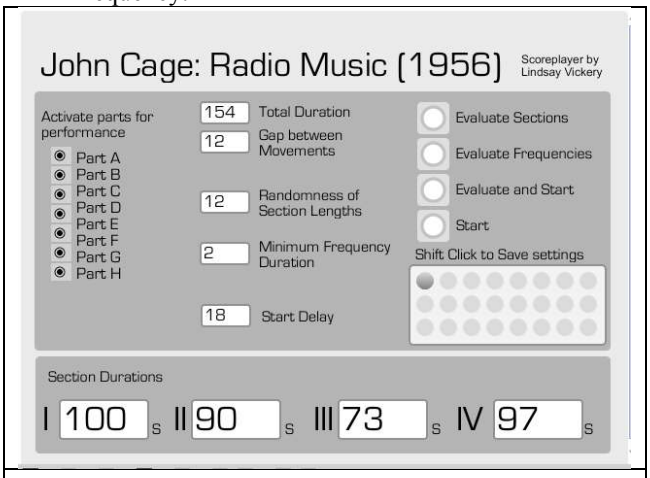


Figure 5. The control panel for the digital verson of *Radio Music*.

The performer(s) may choose to evaluate and re-evaluate the section durations before the performance,

[4] Unlike *Imaginary Landscapes No. 4* in which two performers are used for each of the 12 radios.

total duration (D)	calculation	360 s			
largest no. of frequencies (f) performed by any part in sections I+II+III+IV		f^{i}=44	f^{ii} =30	f^{iii}=38	f^{iv}=40
sum of the largest frequencies (F)	$f^{i} + f^{ii} + f^{ii} + f^{iv}$	152			
relative proportions (p) of each section	$f^{(i-iv)}/F$	p^{i}=0.29	p^{ii}=0.20	p^{iii}=0.25	p^{iv}=0.26
average duration in seconds of each section (S) ($D*p^{i-iv}$)	S^{i}, S^{ii}, S^{iii}, S^{iv}	104	71	90	95
randomness of section lengths	$S^{(i-iv)}$(+/- r)	104(+/-8) 100 (-4)	71(+/-8) 63(+1)	90(+/-8) 83 (-7)	95(+/-8) 91(-4)
Scaled	$(S^{(i-)}(+/- r))$ / $S^{(i-iv)}(+/- r)* S^{i}$	107	67	89	97

Table 2. Calculation of the indeterminate section lengths of *Radio Music.*

allowing proportions that are deemed unsuitable to be discarded.

Performers are given a range of options for selecting both the determinate factors in Radio Music (such as the number of parts performing and total duration) and the "available indeterminacy" of indeterminate factors (such as the amount of variability in section and frequency duration) on the control panel shown in Figure 5. They may also choose the average duration of the gaps between movements and the delay prior to the score activating.

The duration of the work must fall between 360 seconds (the longest duration specified by Cage) and the sum of the largest number of frequencies that must be performed in each section. It should be noted that because the choices diminish with shorter total and sectional durations, the work is necessarily less diverse, in terms of durations, as its total duration decreases. For example when the duration is set at its minimum, the performer with the largest number of frequencies in each section will need to maintain an average of one second per frequency change.

Indeterminacy in the durations of the sections is obtained in a number of stages. Initially, a notional average length for each section is calculated. This is equal to the largest number of frequencies in a section[5], divided by the sum of the largest numbers of frequencies for *all* four sections, multiplied by the total duration of the work. On the control panel, the performers may choose the degree of "available indeterminacy" to be used when choosing how much deviation each section can have from the notional average length. This process is outlined in Table 2.

The final duration of the frequencies is calculated in a slightly different manner. Again the performers may choose the degree of indeterminacy available when calculating the lengths of frequencies, however the final duration is chosen by randomly distributing the number of frequencies in each section over the total duration of each section. For example, when calculating the durations of frequencies of Part G in Section I using the greatest degree of indeterminacy, the 44 frequencies are randomly distributed over the 107 possible seconds of Section I (giving a range of between 1 and 63 seconds for each frequency). When the degree of indeterminacy is set at the minimum level, the 44 frequencies are equally distributed every 2.43 seconds of the total 107 second duration.

The digital performance version of *Radio Music* has three possible modes, score only, automated volume and installation mode. In the score only mode, a realtime screen score is generated for the performers to follow, changing frequencies as indicated by the score and adjusting the volume to match the changes in frequency. Automated Volume mode transfers control of the volume to the computer, allowing the performers to focus only upon changing frequencies.

In the Installation version (which has not yet been fully implemented), the computer operates both the volume and frequency changes via Ardiuno. The radios used in the Installation version would be modified to allow direct electrical control of volume and the control of the frequency via a stepper motor. Each radio must be calibrated to its zero point (the lowest frequency it can reach) and the range of frequencies employed in *Radio Music* (55hHz to 156 hHz)[6].

4. FORMAL IMPLICATIONS

The problems associated with achieving a precise performance of *Radio Music* have already been discussed. The question remains, however, why it would be necessary to "precisely" perform an indeterminate work. The answer lies in the determinate aspects of the work, which arguably infer a consistent if variable formal structure. Adherence to Cage's specifications and consequently to the structure that emerges as a result of these specifications, was the primary aim of this project. Although the "spirit of the work" is

[5] Which in turn is dependent upon the Parts that have been chosen by the performers.

[6] This version remains currently unrealised because of issues arising from obtaining stepper motors of sufficient precision.

important, *Radio Music* and many other works by Cage, have been approached by many performers in a way that foregrounds the freedom and indeterminacy of his concepts without sufficient attention to the determinate elements and their implications.

When considering the formal implications of *Radio Music* it is necessary to consider the sonic material from which it is comprised. Radio broadcasts are strongest at a specific frequency, but are spread, increasingly weakly over a band of frequencies about 10 kHz on either side, so a specific carrier frequency number may be "in the middle" of a broadcast frequency or in the "detuned" penumbra of the broadcast in which the broadcast is still present, but often mixed with static or spill from other broadcasts. In regard to the density of the AM band, it might be expected for a radio to pick up local stations quite strongly, but also to detect more or less strongly, regional (or at least non-local) stations.

Unlike the score for a conventional musical instrument, the carrier frequencies listed do not specify pitch. However broadcast frequencies are precise and predictable and contents of radio broadcasts are usually roughly predictable - Classical Music, Rock, Jazz, talkback and so on – and in cases with detailed program listings quite precise. Radio Stations often exhibit strong consistency over a short period of time (such as found in *Radio Music)*, generating what we might consider a linear "narrative", that emerges as a consequence of the timbre of a presenter's voice the continuity of a piece of music, the subject matter under discussion and so on. Many broadcasters also have a distinctive timbre and dynamic, resulting from varied studio equipment, production approaches and broadcast strength.

Although *Radio Music* can not rely on the elements typically found in musical discourse, it does potentially draw on recognizable elements that could be expected to give rise to some of the experiences typical of musical listening, such as expectation, surprise a sense of departure and return, conjunction and disjunction of the sonic materials and so on.

Forty-four different carrier frequencies are used in the 150 frequency changes specified amongst the eight part of *Radio Music*. These are graphed in ascending order in Figure 6.

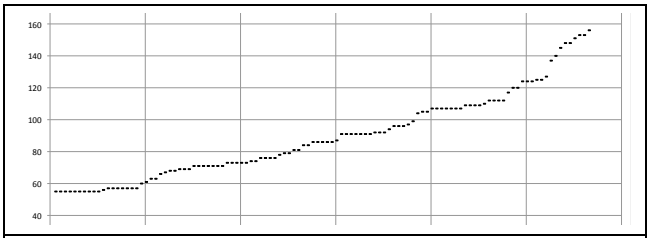


Figure 6. The occurences of carrier frequencies in *Radio Music* graphed in ascending order.

The number of repetitions of each frequency is shown in Figure 7, illustrating that although 19 frequencies are only specified once, many others are repeated (between two and ten times) as shown in Figure 7.

incidence	no.	frequencies
1	19	56, 60, 61, 66, 67, 78, 87, 94, 97, 99, 104, 110, 117, 127, 137, 140, 145, 151, 156
2	11	63, 68, 74, 79, 81, 84, 105, 120, 125, 148, 153
3	4	69, 92, 96, 124
4	3	76, 109, 112
5	2	73, 86
7	4	57, 71, 91, 107
10	1	55
blank	37	

Table 3. The incidence of repetition of carrier frequencies in *Radio Music*.

Although it is not possible to predict exactly when each carrier frequency will be performed, they are performed in a strict order, and due to the limitations upon available indeterminacy due to the narrow time constraints required by Cage, as discussed previously, it is possible to create a probabilistic picture of when particular frequencies will occur in relation to one another. It should be noted that along with broadcasts from radio stations, static which often has specific colours in different parts of the radio band, is also struck as performers tune from frequency to frequency.

Figure 8 is graphs the carrier frequency changes in Section I of *Radio Music*, spacing them equally over the same duration. It gives an approximation of how active the eight parts are in relation to one another.

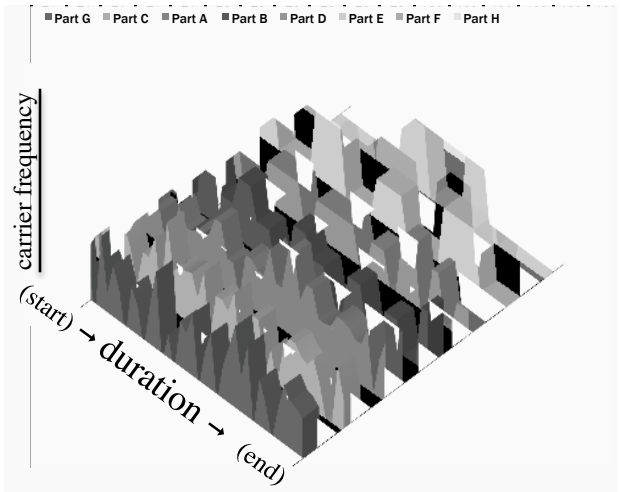


Figure 8: A representation of the carrier frequency changes in the eight parts of Section I of *Radio Music*, spacing the changes equally over the same duration.

If this data is viewed in cross-section, with each part outlined as a "wire-frame", as shown in Figure 9 it is possible to see certain structures emerging from this "indeterminate" work: repetitions of the same frequency reoccurring at different points during the section and gradually rising frequencies that may more toward or away from a radio station (as shown in Figure 6 some frequencies are "clustered" together, while others are more isolated form one another). These inherent structures suggest at least the possibility of structures being detected.

Recurring carrier frequencies and clusters of adjacent frequencies occur across the different parts during the work as illustrated in Figure 6. All of the parts are, of course heard simultaneously, resulting in an audible structure relying upon musical features including the materials of the broadcasts themselves, the timbral and dynamic variation between the broadcasters, the varied nature of interstitial static in the AM band and finally the spacial location of the radios themselves. Listening to *Radio Music* performed precisely reveals a complex interplay and dialogue between these features.

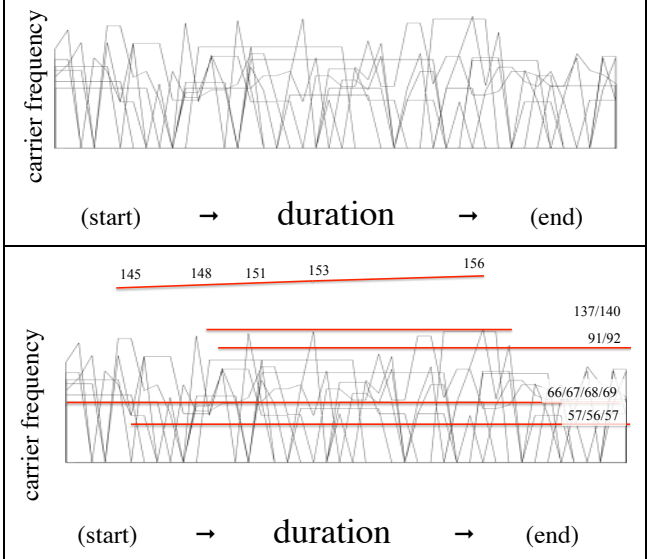


Figure 9. A "wire-frame" cross section of the representation of carrier frequency changes shown in Figure 8. The lower graph shows the incidence of recurring and gradually shifting carrier frequencies with potentially form-bearing characteristics.

Cage stated that his radio works make "available something which you're already in. You are bathed in radio waves (…) radio simply makes audible something that you thought was inaudible" [10 p. 19)].

Had he intended only to make this point, he might have permitted the performers of *Radio Music* to freely explore the AM band or even have provided specific timing[7] and non-specific frequencies. That fact that Cage specifies frequency rather than duration implies that the precise structure – the proportions of different frequencies and the possibilities for their temporal relationships with one another was both intended and crucial to the composer.

[7] Many of Cage's works from this period (for example *4′ 33″* (1952), *59 1/2″ for a String Player* (1953), *34′ 46.776″ for a Pianist* (1954), *31′ 57.9864″ for a Pianist* (1954) and *26′ 1.1499″ for a String* (1955)) explore this territory.

5. PERFORMANCE OF *RADIO MUSIC*

To explore how Cage's specifications for *Radio Music* match up to the realities of AM radio in the early 21st century, Figure 10 shows the 44 frequencies specified in the work plotted against the actual existing local and regional radio stations that might be detected from Perth, Western Australia in 2011. As can be seen, only four of the eleven local radio stations coincide with frequencies specified by Cage. In regional stations are included the number goes up to 17 out of 44 stations.

Although the Australian Communications and Media Authority's 2006 report on AM Radio Issues claims that the "AM sector is not in terminal decline" [1], since the introduction of commercial FM licenses in the 1980s audience share of AM radio stations has fallen dramatically [1 p. 14].

The result of the ever-diminishing number of AM band radio stations is that *Radio Music* will be increasingly comprised of static rather than signal. There are several possible responses to this issue:

1. to embrace the realties of evolving technology and realize the work principally with the "noise" of static;
2. to "transpose" the work into a frequency band such as FM which is still relatively populated with Radio stations;
3. to "narrowcast[8]" internet radio stations in the vicinity of the performance.

The first approach is certainly defensible, but will undoubtedly result in outcomes in which the recognisable structural features potentially present in the work will be greatly diminished. The second approach might be likened to performing Baroque music on modern instruments. FM is less noisy than AM, but would still provide many of the outcomes found in an AM performance: such as noise between stations and tuning that is off-station.

The final approach is complex, requiring multiple digital receivers and transmitters. Interestingly it could afford the possibility for reconstructing an AM environment similar to the one for which the work was originally envisaged or even an "authentic" using recordings of broadcasts from 1956. The legality of this approach would be an issue dependent upon varied broadcasting regulations in different jurisdictions.

6. CONCLUSION

Douglas Kahn describes Cage's *Radio Music* as a work pitting "disintegration against disintegration, noise against noise" [12 p. 578]. It is certainly true that this is a work that takes a very commonplace medium and rends its sense of continuity and familiarity. However, it can also be said that the continuity and familiarity of radio broadcasting is also the essential material of *Radio Music*, material which provides structural features that are decontextualised and recontextualised in relation to one another in the work.

This paper has shown the potential for the emergence of structure, albeit an indeterminate nonlinear structure in *Radio Music* and the benefit of attempting to realize the work in the most precise manner possible to maximize its inherent characteristics.

The problem with the issue of "available indeterminacy" in regard to this particular work is that when left to make durational choices "freely", performers are painted into a corner in which they must curtail their freedom, having to either slow or accelerate their durations in order to in order to finish each of the four sections together, and in order to complete the work in the designated maximum length of six minutes.

The digital score player for *Radio Music* provides the opportunity to circumvent these issues and to generate numerous varied instantiations of the work. It would be possible to allow performers to save the parameters of an "ideal" performance that for example generates a very clear articulation of the work's inherent formal structure. This feature has not been added as it might arguably contravene the composer's clear desire for indeterminacy to play a role in the performance. Like many of Cage's works *Radio Music* prescribes a process

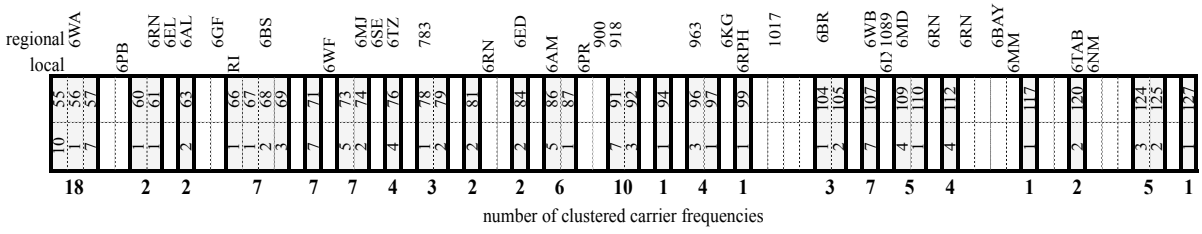


Figure 10. The carrier frequencies, the number of occurrence of each frequency and the number of clustered frequencies specified in *Radio Music* plotted against the actual broadcasting regional (6WA-6BAY) and local (6PB-6NM) AM radio stations in Perth, Western Australia in 2012.

[8] The term "narrowcasting" describes a radio or television service whose reception is limited in some way, such as being targeted to special interest groups, being intended for limited locations (for example, arenas or business premises), being provided during a limited period to cover a special event, to provide programs of limited appeal, or for some other reason. [1].

for exploration of sound, and the digital score player is a tool for aiding this process.

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APPROXIMATING OBJECTIVE AESTHETIC ASSESSMENT WITH ARTIFICIAL NEURAL NETWORK

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ABSTRACT

This article discusses the possibility of using an Artificial Neural Network to provide aesthetic assessments of automatically generated counterpoints. It presents in detail several different architectures of such networks and analyses training strategies. In particular, it demonstrates how to train such a network for providing objective assessments (such as those prescribed by course materials on counterpoint) and why this approach is not entirely practical. It also observes that Artificial Neural Networks may, in fact, exhibit some form of machine subjectivity.

INTRODUCTION

Human perception of beauty is subjective. Each individual perceives the same piece of art differently. Moreover, the same individual can provide different aesthetic assessments of exactly the same piece of art at different times. Human impressions are biased by various criteria, not necessarily related to art. Culture, time of the day, mood, politics, religion, relationship with the author, all this can affect the perception of beauty.

Nevertheless, it is possible for a number of humans to agree on some universally accepted attributes of a piece of art responsible for beauty. For example, the rules of counterpoint can objectively determine if a musical composition (i.e. a counterpoint) is *beautiful* or *ugly*. In particular, the number of repetitions or the number of parallel consonances can be used as a numerically expressed aesthetic score.

The objective aesthetic assessment, based on clearly defined rules, can be used as a basis of a computer program. Such an artificial judge is then able to assess pieces of art automatically. Typically, it will provide exactly the same judgements of the same piece of art every time without any bias. It may also assess several different art pieces as equally beautiful, while human observers will value them differently. In other words, when computer says that these two melodies are equally beautiful, I may say that I like this one better than the other, but I don't know why I like it better. Perhaps it resembles something I have heard before, which I liked a lot?

A computer program is usually nothing more than a series of clearly defined steps, a prescription how to achieve something, solve an equation and in general, is usually ultra objective. From the programmer's perspective, computer programs always do exactly what they were instructed to do, mercilessly. Within this context, is it possible to create an artificial aesthetic judge, which will have some subjectivity? In particular, is it possible to train a computer program in aesthetics to the extent that we, as humans, will explain why two pieces of art are equally beautiful, but the program will still say that it likes one of them better then the other? Is it possible to create a computer program, which will calculate a different aesthetic assessment of the same melody at different times and will be biased by its previous experiences?

The human brain is totally different to an electronic computer. While we don't know exactly how the brain works, we know it is composed of neurons. The hypothesis is that an artificial neural network may in fact exhibit some form of machine subjectivity.

Neural networks described in this paper were trained to reproduce the objective aesthetic assessment of automatically generated counterpoints. It has been observed that during the experiments neural networks exhibited some level of machine subjectivity, which are further analysed and explained.

BACKGROUND INFORMATION

Artificial Neural Networks constitute well established technology. They are particularly good for detection and classification of patterns, picture recognition (such as handwriting), signal processing and filtering [8]. They have already been used in the field of music and music research [5]. In particular they were used for automatic generation of musical compositions [2, 3, 4] and recognising musical genre [6].

Artificial Neural Networks were also used as Fitness Functions in conjunction with Genetic Algorithms [2, 3]. However, this approach has proven to be problematic [1].

AESTHETIC ASSESSMENTS

For the purpose of the experiments described in this paper, aesthetic assessments were clearly divided into two distinctively different categories: objective and subjective.

Objective Aesthetic Assessment

In the context of this paper, the objective assessment can be defined as a numeric score (or a combination of scores), which is universally accepted and agreed upon by certain group of people, such as composers or teachers. It is also independent of individual preferences of each human. For example, the number of parallel imperfect consonances in a counterpoint or number of repeated notes can be considered as an objective assessment (or part of it). Such precisely defined criteria can be used to described two different counterpoints as equally beautiful, because they contain exactly the same number of repetitions and parallel imperfect consonances.

The aesthetic guidelines were extracted from the course material for the undergraduate counterpoint classes at the University of Adelaide, that are taken by composers [7].

Subjective Aesthetic Assessment

This objective aesthetic assessment differs substantially from a subjective impression of each human listener. The main reason for it is that the musical notation is merely a representation of a piece of music rather than the music itself. Hence, we do not assess the music notation, but the music that the notation describes or represents. Moreover, each individual human listener will most likely assess music differently than his fellow listeners. The reasons may also include personal bias toward or against the creator of the composition, current mood and whatever the listener did or experienced recently. In other words, if there are two counterpoints with exactly identical objective aesthetic scores, one of them may be perceived as more pleasing than the other. And we, as humans, will have difficulty to formulate a believable justification for our preference, apart from saying that it just "sounds better".

Assessment Calculation

The rules of objective aesthetic assessment can be easily turned into a computer program (as long as they are clearly described). Such program will be able to identify counterpoints, which are identically pleasing. We may be able to say that we like them differently. However, we may not be able to provide any justification as to why we like one better than the other. This kind of subjective assessment will also be beyond machine comprehension.

SUBJECT OF THE EXPERIMENT

All the counterpoints used in the experiments were automatically generated by an expert system written in Prolog [9]. In particular, the system calculated all the valid counterpoints for a given cantus firmus. The number of counterpoints was frequently very high (typically exceeding one million for a 12-note cantus firmus). To decrease the computational requirements and speed up the experiments, the number of all valid counterpoints was further reduced by limiting the cantus firmus to 4 notes only and locking in one particular rhythm. This way it was possible to reduce the number of all valid counterpoints to less than 100.



Figure 1. Cantus Firmus used in the experiments.

Each of the valid counterpoints was assigned an objective score consisting of 3 values: the highest number of repeated notes, number of parallel imperfect consonances and the total number of parallel/contrary motions (where each oblique movement was scored at 1 and each parallel movement was scored at 2). Values closer to ZERO were considered *nice*, while higher values represented increasing levels of *ugliness*. These counterpoint attributes were selected based on the coursework material used at the University of Adelaide for teaching counterpoint.

The aforementioned scores were calculated automatically by the program. From the machine point of view, the selection of counterpoint attributes used for calculating the objective aesthetic score is arbitrary. As long as each counterpoint can be assigned a value calculated in any way, it is good enough for the purpose of the experiment.

INITIAL EXPERIMENTS

The purpose of the first set of experiments was to connect counterpoints to the input of a neural network to see how well it could learn to approximate objective aesthetic scores. Each counterpoint together with its cantus firmus, was encoded numerically to represent note pitches ranging from 0 to 128. Rhythmical values were broken into a series of crotchets played legato. Hence, a semi-breve became 4 crotchets with the same pitch value.

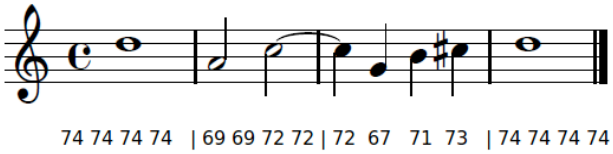


Figure 2. Counterpoint and its numerical representation of pitch.

Training a Single-Neuron Network

The first experiment involved a neural network consisting of a single neuron. It was trained to recognise the number of repeated notes in the counterpoint.

After a series of training sessions it was observed that a single neuron is capable of approximating the accurate detection of the number of repeated notes to some extent. In particular, while the number of repeated notes is always an integer number, the neuron was reporting values with fractional parts. However, those numbers were close enough to the expected value most of the time, although there were several cases of mis-approximations.

It was possible to focus the training routine on the mis-approximated cases and correct them. However, this was done at the expense of correct detections becoming less accurate. In general, it seemed impossible to train the neuron to provide completely accurate results all the time.

Training a Three-Neuron Network

The second experiment tried to replicate the entire objective assessment, which consisted of 3 parts (including repeated notes, parallel consonances and parallel motions). As the output of the neural network was expected to have 3 numbers, it needed to consist of 3 neurons.

After running a training procedure, the results were identical to the experiment involving a single neuron. In particular, focusing the training on increasing the accuracy of one particular counterpoint was done at the expense of other counterpoints.

Training a Multi-Layer Network

The third experiment involved a multi-layer architecture of a neural network. In particular a 3 layer network was tested. It contained 32 neurons in its first layer (input), 16 inside the hidden layer, and 3 neurons for the output.

During the experiments it was observed that the results were similar to those achieved by single-layer neural networks. The accuracy of detections became slightly higher. However, the training time increased dramatically.

Outcomes

All experiments demonstrated that a artificial neural network with simple architecture is hardly capable of approximating an objective aesthetic assessment accurately. Although it frequently generates results that can be considered as *close enough*, there are still cases that are too inaccurate.

When considering this result, the situation may be interpreted as a similar mechanism to a subjective aesthetic assessment. It could be perceived as the neural network exhibiting some form of subjectivity. Primarily, the results coming out from the neural network are always fractional, while the original objective values are integers. This could be described as the neural network liking something *a little bit more* or *just a bit less* than the rules suggest. Secondly, in some cases the neural network assessment differs drastically from what is expected. This can be characterised as *a strongly subjective impression* of the neural network.

Although the experiments show that the neural network demonstrates inaccuracies, there is still the possibility that 100% accurate objective response may be achieved if enough training is provided. In such a case, the neural network would loose its *subjectivity*. The next experiments focus on this situation.

ACHIEVING 100% OBJECTIVITY

This set of experiments focuses on achieving completely accurate reproduction of the objective aesthetic assessment.

Counter-Propagation Neural Network

The architecture selected for this experiment is called a *Counter-Propagation* neural network. It consists of two layers of neurons: the Kohonen layer and the Grossberg layer. The Kohonen layer has slightly modified functionality to the typical neural network. In particular, from all its neurons, the one with the highest output signal is selected as a *winner*. The value of this winning neuron is then changed to 1.0, while the responses of all the other neurons are set to 0.0. The purpose of this layer is to recognise and classify input patterns. As the training set consisted of 49 counterpoints, this layer must have at least 49 neurons. Each of them is trained to recognise one particular counterpoint. The purpose of the second layer (Grossberg) is to reconstruct the expected output signal, which is the objective aesthetic assessment itself.

Training of the Kohonen layer

The training of the first layer of *Counter-Propagation* neural network is still a tedious process. In the end, the Kohonen layer is fully capable of recognising counterpoints with 100% accuracy. However, the fact that it is capable of doing so does not guarantee success every time. A mathematical proof is a necessary ingredient here.

Taking advantage of the fact that the input to the Kohonen layer is always normalised, the content of each neuron can be mathematically calculated rather than trained. In particular, all the counterpoints from the training set can be normalised and used to populate the neurons without any further modification. Hence, it proves that it is possible to train such a layer to be 100% accurate.

Training of the Grossberg layer

The training of the second layer of *Counter-Propagation* neural network is a trivial task. As the input to the Grossberg layer will always consist of 1.0 generated by a singled neuron (and 0.0 on all others), the number of training repetition necessary is small. It can also be mathematically calculated rather than trained.

Outcomes

It has been observed that the Counter Propagation neural network is capable of accurate reproduction of the objective aesthetic assessment. However, The Kohonen layer must contain the number of neurons equal (or greater) to the number of all possible counterpoints. This number can be very high. A 12-bar melody, consisting of 48 crotchets, where each can be one of 12 pitches (assuming we put a limit of one octave), with completed disregard of the rules of counterpoint, can yield 6,319,748,715,279,270,675,921,934,218,987,893,281,199,411,530,039,296 results. This is much more than the total number of neurons present in a human brain, which is estimated at 100,000,000,000. In fact, it exceed the number of neurons in the entire human population on Earth (7 billion) by several orders of magnitude.

EXPLORING SUBJECTIVITY

The following set of experiments explores the *subjective* behaviour of the neural network's output. In particular, it focuses on creating a situation where the difference between expected and real output changes its sign. In other words, it may be interpreted that the neural network *changed its mind* and now it likes one particular counterpoint *a bit less*, while it used to *like it a bit more* than objectively prescribed.

The experiments were conducted on a two-layer neural network. It contained 16 neurons in its input layer. This number was arbitrarily selected. The number of neurons in the output layer was dictated by the size of the objective assessment vector, which is 3.

Three different training sets of counterpoints were created. All 3 sets consisted of florid counterpoints locked to one particular rhythm. The total number of counterpoints in each set was: 12, 49 and 84.



Figure 3. One counterpoint from the first set.



Figure 4. One counterpoint from the second set.



Figure 5. One counterpoint from the third set.

Experimenting with Different Sets of Counterpoints

In this experiment three identical neural networks were trained to approximate objective aesthetic assessments from each separate set. After relatively good detections were achieved, the networks swapped the training sets and it was observed that the results were too random and the number of mis-assessments was almost 100%.

Experimenting with a Pre-Conditioned Neural Network

In this experiment an already trained neural network was re-trained for detecting different counterpoint sets. However, after the training the network seemed completely to forget its prior training. The results were no different to training routines starting with random state neural network.

Changing the Order of Counterpoints within a Training Set

This experiment also was unsuccessful. After the training routing was complete, the results generated by the neural network were identical with those obtained with unchanged order.

Experimenting with Continuous Learning

In this experiment the neural network was continuously learning, while generating results at the same time. It became evident that prior expositions of counterpoints influenced the neural network's output if the learning factor was high enough. In the following example, the learning factor was set to 0.2. This result was obtained from the training set consisting of 12 counterpoints. The change of sign of the neural network's error became evident when the order of counterpoints within the training set was changed. The table below shows only the output from the second neuron, which represents the number of repeated notes within the counterpoint. This number was normalised to the range from 0.0 to 1.0, hence the fractional values of 0.625 and 0.75. The other components were omitted for clarity.

ID	Expected	Returned	Difference
0	0.625	0.634	0.009
1	0.625	0.631	0.006
2	0.625	0.631	0.006
3	0.625	0.623	-0.002
4	0.625	0.617	-0.008
5	0.625	0.616	-0.009
6	0.625	0.622	-0.003
7	0.625	0.625	0.000
8	0.625	0.623	-0.003
9	0.625	0.625	0.000
10	0.750	0.747	-0.003
11	0.750	0.748	-0.002

Table 1. The results of approximated number of repeated notes within counterpoint with original order of counterpoint.

ID	Expected	Returned	Difference
9	0.625	0.607	-0.018
6	0.625	0.633	0.008
5	0.625	0.620	-0.005
7	0.625	0.629	0.004
8	0.625	0.617	-0.008
10	0.750	0.747	-0.003
11	0.750	0.748	-0.002
4	0.625	0.628	0.003
0	0.625	0.628	0.003
1	0.625	0.629	0.004
2	0.625	0.633	0.008
3	0.625	0.623	-0.002

Table 2. The results obtained from the randomly re-ordered training set.

Interpretation of the Outcomes

In the above experiment, the difference between neural network output and the expected objective aesthetic assessment (i.e. the normalised number of repeated notes) has changed its sign in two cases. This result can be interpreted as the neural network *changed its mind* about *liking* or *disliking* a counterpoint.

According to the contrapuntal recommendation, it is desired to have a variety of different notes in the counterpoint. Therefore the number of repetitions of notes should be as small as possible. In the context of the above experiment, the lower the value of the normalised number of repeated notes, the better the aesthetic result. Hence, the positive difference represents *liking a bit less*, while the negative difference indicates *liking a bit more*. In particular, in the example above, the neural network changed its *liking* of counterpoints no. 4 and 6, after changing the order within the training set.

CONCLUSIONS

It is mathematically possible to train (or populate) a neural network to be able to provide accurate objective assessments. However, the size of such a network is too big for any practical use by many orders of magnitude.

In a typical implementation, where number of neurons and neural layers is limited to manageable amount, the neural network is able to approximate objective assessment with some small error. While this error is usually small enough, it can occasionally become too big and lead to mis-assessments. Also, the error is inversely proportional to the number of neurons.

During the experiments it also became evident that a neural network is erratic when assessing counterpoints which were not present in the original training set. This behaviour makes it difficult to user neural networks as meaningful fitness functions for evolutionary generative music [1].

However, artificial neural networks may be useful in developing some form of machine subjectivity. Their output is always *a little higher* or *lower*, which is the result of some computational error. In a typical scenario, this error becomes lower during the training process. It is nevertheless possible to keep this error continuously evolving by careful manipulation of the learning factor and maintaining the training process indefinitely. Moreover, it is possible to change the sign of this error by changing the order of expositions within the training set.

The aforementioned behaviour is to some extent similar to human subjective perception of beauty and ugliness. Our brain continuously learns. Hence, our aesthetic preferences evolve. Also, our previous experiences, even changes to our daily routine influence our judgements. While we are influenced by our human activities (such as food, entertainment, social problems, ideas, etc.), computer programs can be influenced by their previous computer activities. In this case it can be the order of counterpoint expositions.

FURTHER RESEARCH

The possible direction for further research may focus on testing more complex neural network architectures, such as those with feedback loops, including Hopfield's networks and Boltzmann machines [8]. Very large training sets could be examined as well. In particular, sets consisting of millions of possible counterpoints. It may also be interesting to investigate if it is possible to use neural networks to develop some form of machine subjectivity within bigger systems involving genetic or evolutionary algorithms.

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REVIEWED BY ABSTRACT

INTER[ACTIVE]: ARTISTS AS AGENTS OF CHANGE

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ABSTRACT

This paper explores the role of electroacoustic composers in ecological crisis and introduces the *Sonic Ecologies Framework*, a multi-platform methodology to initiate cultural change through sound. The core of this methodology pivots on a site-specific creative project embedded in a multi-layered community cultural engagement process developed in response to a specific community. This evolving model is implemented by the artist, acting as an agent of change spiraling between contextualised theory and practice. This research presents results from a series of case studies implemented with this methodology and introduces the conceptual design of a major international project highlighting the future possibilities of this model.

INTRODUCTION

In 1937, in a talk titled *The Future of Music: Credo*, John Cage said, "I believe that the use of noise to make music will continue and increase until we reach a music produced through the aid of electrical instruments which will make available for musical purposes any and all sounds that can be heard" [2].

In 2012, the centennial of John Cage, his visionary genius is clearly evident with a musical world of infinite possibilities aided by technology. The dramatic advancement of technology has truly cultivated a paradigm shift in how artists interact in both physical and virtual worlds. These changes have evolved and expanded our tools of expression but most importantly they have opened the ability to communicate at a higher level in an interdisciplinary context.

In a recent addition of *Musicworks*, Joel Chadabe stated that the current artistic practices of electroacousitc composers are rooted in the idea that new technologies, unlike traditional musical instruments, can produce sounds used to communicate core messages, including information about the state of our environment. He claims that we are all participating in the emergence of a new type of music accessible to anyone, which can be used to communicate ideas that relate more closely to life than those communicated through traditional musical forms. He believes we need to think of ourselves as "leaders in a magnificent revolution rather than the defenders of an isolate and besieged avant-garde" [3].

In a world where the catastrophic effects of climate change are rapidly becoming a bitter reality, there must be a role for sound in generating a shift in consciousness. Bill McKidden recently said; "When art both of great worth, and in great quantities, begins to cluster around an issue, it means that civilization has identified it finally as a threat" [4]. He views artists as the antibodies of the cultural bloodstream and fundamental to social change. As this social movement of creative thinking expands internationally it is worth reflecting on Attali's seminal 1985 text where he refers to music as not just simply a reflection of culture but a "harbinger of change". He states, "For twenty-five centuries, western knowledge has tried to look upon the world. It has failed to understand that the world is not for the beholding. It is for hearing. It is not legible, but audible" [1].

World leaders now looking towards the validity and possibilities of creative methodologies as tools for change, this presents both a challenge and an unprecedented opportunity for composers to gain a critical understanding of the situation, and take action in devising new processes for a sustainable future. Electroacoustic music, with the use of natural sounds, has a profound ability to ignite an awareness and connection to the environment. But is the role of the artist purely to comment on the crisis? To create awareness? Or can provocation extend beyond expression to create a behavioral shift in deeply engrained unsustainable ways of thinking?

This research explores these questions and introduces a multi-platform methodology that could provide a framework to facilitate the paradigmatic shift required to initiate cultural change. The core of this methodology pivots of a site-specific electroacoustic music project embedded in a multi-layered community cultural engagement process developed in response to the specific community. The artist, acting as a change agent spiraling between contextualised theory and practice, implements this evolving model. The results are introduced through three case studies of projects recently produced by the author. This paper concludes with the conceptual design of a major international project developed with the *Sonic Ecologies Framework* highlighting the future possibilities of this model. These projects are ultimately acting as a catalyst and represent

an unparalleled opportunity for artists taking action as agents of change in environmental emergency.

RATIONAL

The author's practice-led doctoral research involved conceiving and delivering seven original electroacoustic projects for dissemination in multi-platform environments. The divergent projects were created in cultural immersion, spanning from ambitious sonic explorations in the center of the Amazon Jungle to sounding the rivers of the world through India, Korea, China, Australia and New Zealand. The delivery and dissemination of each project was underpinned by a rich methodology that pivots on the site-specific project embedded in community cultural engagement. The concept of cultural immersion challenges the traditional notion of an isolated composer and has the ability to be profoundly influential on the composer's personal validity of their practice. For the author, it is no longer about notes on a page with a dotted bar line that finishes the project, but an entire spectrum of compositional decisions in a constantly evolving process that appears to only expand rather than dissolve. This is undeniably as a result of working directly with the communities and experiencing creative inspiration in cultural immersion.

The ambitious doctoral research began as an exploration of the sustainability of electroacoustic music and evolved into a complex web of projects harnessing electroacoustic music as a change agent. The beginning was fueled by an isolated intention, yet through the process of cultural immersion the author discovered a tool that not only provides a gratified language of creative expression, but also a voice for the communities and environments collaborating on these projects. The discoveries and observations from each individual project showed a clear trajectory towards a definitive set of tools to initiate cultural change through environmental electroacoustic music. As a result, the *Sonic Ecologies Framework* was developed as a means to create an accessible methodology for composers' interested in implementing similar projects.

SONIC ECOLOGIES FRAMEWORK

The following provides a brief overview of the five essential elements of the *Sonic Ecologies Framework*. The core of this methodology pivots on a site-specific electroacoustic music project embedded in a multi-layered community cultural engagement process developed in response to a specific community. The site specificity requires that this methodology be intrinsically flexible in order to be adaptable within a diversity of environments and communities. It is in essence a practice-led creative research process, taking an ecological approach to contextualising a project within an environment. While there is an essential degree of freedom and adaptability, the process is grounded within the theoretical contexts generated by the artists who experiment and innovate within a continual spiraling between theory and practice.

Site-specific Subject Matter

In the context of this process, the site-specific nature of the electroacoustic music project is essential. It must be pertinent to the community and grounded within a comprehensive understanding of the proposed thematic content. The sound work of Douglas Quin in Antarctica and Francisco Lopez in the Central Amazon Jungle are obvious examples.

Multi-platform Dissemination

The *Sonic Ecologies Framework* encourages collaboration and multiple outcomes where possible. While one core creative outcome is most likely the central intention, the adaptability of the project for a range of environments is essential. The sonic outcomes should be disseminated in a range of environments for maximum exposure; this includes harnessing the power of virtual platforms to facilitate global accessibility. This is exemplified in the *EcoSonus* project commissioned *Floating Land 2009*, a multi-channel sound installation with regional site-specific performances, collaborative community compositions and a interactive website streaming field recordings and alternative compositions.

Community Engagement and Education Tools

There is undeniably a strong movement associated with environmental sound art emerging internationally. This is evident through the establishment of organisations such as *Ear to the Earth*, the environmental program of the *Electronic Music Foundation*. While this profusion of research continues to evolve, the reality is the western world is still a visually dominant society and requires education to ignite the auditory perception. If the community in question is to engage and comprehend the project, they must gain a deeper understanding of their sonic environment and play a role in the process. The community engagement and education tools will always evolve depending on the nature and accessibility of the proposed community but the standard suggestions include; community sound walks, participatory field recording sessions, capacity building workshops and providing access to the appropriate technology for the community to remain engaged in the process.

Interdisciplinary Partnerships and Collaborations

In order to truly attempt to create a paradigm shift with *Sonic Ecologies*, electroacoustic music must be augmented from its traditionally isolated academic circles and expand into regional communities collaborating with environmentalists, conservationists, scientists and policy makers to expand awareness. Creating a support network around the project will be essential in its future viability and sustainability.

Long-term Strategic Vision

The artist implementing the *Sonic Ecologies Framework* is initiating a process within a community. The creative outcomes serve as significant milestones but ultimately it is the process that will continue to resonate and evolve over time. As with any form of community engagement, *Sonic Ecologies* requires time in order to facilitate change. The capacity building community engagement is designed to empower the community to continue working long after the artist has departed. It is therefore essential the artist invests critical thought into the methods in which the community will continue to engage and appropriate technology is accessible to continue working. The most obvious strategy is to leave low costs digital recorders with a key stakeholder in the community and design a web platform to enable the community to continue creating and uploading content. It should also go without saying, plans to return to the community should be instigated by the artist, whether this be a concert of future creative outcomes, workshops or simply visiting the key collaborators to maintain relationships and energy in the process.

CASE STUDY: BLUE GOLD

The first case major study is *Blue Gold*, a performance installation investigating the delicate balance between wet and dry in our natural landscape by internationally acclaimed Australian sound artist Ros Bandt. The work is composed for clarinets, cello, electronics, sculptures and digital colour field imagery. The performers navigate through the rich electroacoustic sound fields sourced from diverse areas such as the deep waters of the Southern Ocean and the Mungo desert, to the flash storms of the bush and the wetland ranges. The sculptures are designed as timers to help reflect the balance between water and earth and the role we are playing in its care and destruction. It takes its title from Maude Barlow's book of the same name referencing water as the new industrial economy, the commodity of the 21st Century. *Blue Gold* challenges us to shift our collective thinking and actions towards water at a time when water management and governance are critical issues of contemporary society.

Prior to this research, the author was inspired by the content of this work and curated two programs featuring *Blue Gold* in the context of a concert hall. The intention of this work is to shift our collective thinking and actions towards water in a live performance, but is 33 minutes really enough time? The initial data suggested that *Blue Gold* is a rich and immersive experience, but was perceived more as provocation as oppose to generating a paradigmatic shift in thinking. The thematic intention of this work was ideal to test the validity of the *Sonic Ecologies Framework* outside of the author's creative practice. Therefore in May 2011, the work was completely recontextualised as a site-specific interpretation over a Lake in one of the most pristine natural environments in Australia.

Ros Bandt participated in a dynamic ten-day program of community workshops, sound walks, forums and interactive labs designed to confront and challenge a spectrum of water issues across disciplines. The entire process was embedded in *Floating Land* 2011; one of Australia's leading art and ecology events on Lake Coorthaba in the *UNESCO* listed Biosphere of Noosa in Queensland. Over the ten days Bandt engaged in an intense process of recontexualising *Blue Gold* for this natural environment, alongside visiting national and international artists who were responding to a similar theme. *Blue Gold* became a vehicle for these conversations, ideas and actions that rippled throughout the community, a community that was changed by this process as evident in their actions and enthusiastic preparations for *Floating Land* 2013. The beginnings of a shift are without doubt present within this community, and it was clear this was as a result of implementing the five stages of the *Sonic Ecologies Framework* underpinned by the intentions of Bandt's *Blue Gold.*

CASE STUDY: REMNANT BREATH

The second case study is *Remnant Breath*, an interactive sound walk commissioned for *EcoSapiens*, New Zealand in January 2011. *EcoSapiens* is an initiative of *SCANZ;* New Zealand's premier art, technology, culture and ecology event and involves a symposium, creative residency, and public events and exhibitions. The commission itself was for the *Remnant Emergency Artlab*, which is an Australian government funded interdisciplinary research project pivoting on site-specific cultural exchange labs across the globe. As a result of embedding the project within the context of a creative residency, many aspects of the *Sonic Ecologies Framework* were already in place. This case study was therefore used as an observation process, adapting elements of the community engagement model and exploring the process on a considerably smaller scale in comparison to *Blue Gold*.

Remnant Breath was conceived and directed by the author in collaboration with Australian media artist Keith Armstrong and Maori musician/artist Jo Tito. The intention of the work was to reveal remnant sonic layers in the Tehenui rainforest walkway leading to the ocean in New Plymouth. Through a 20-minute immersive experience the work explored stories of water, breath, place and the Maori concept of interconnection and the environment. Remnants (of land, animal life or cultures) offer a warning, because they signal both loss and disconnection and the need for urgent action. So by inviting an acute sensitisation to place solely through the often ignored sense of sound, a powerful context was created for these experiences and conversations. Placing speakers along the path leading towards a live performance from Jo Tito augmented the Tehenui's

existing dense sonic environment. The audience was blindfolded for the entire performance and guided in small groups through the work. The positive responses suggested the intentions of the work were conveyed and conversations began to expand the work.

The community engagement aspects of this project involved a series of workshops designed to accompany this project, in this particular context the workshops were directed at a younger demographic, specifically 8 -15 year olds. As the future leaders, it is this generation who will be dealing with the true ramifications of humanities actions. Through this research it has become clear that encouraging young people to listen to the environment is one the most immediate and effective remedies to facilitate this shift. These workshops included sound walks and field recording sessions in addition to participating in the final event. While there wasn't a large amount of participants, the students involved left with a different perception of interacting and experiencing the environment. The visual data also clearly displays their shift in awareness before and after the workshop, simply in the way they are treading lightly on the natural environment. This project also generated a range of significant partnerships and ongoing collaborations, but the most significant outcome was clearly the critical need to engage a younger demographic in the *Sonic Ecologies* process.

CASE STUDY: SONIC BABYLON

Sonic Babylon is the creation of New York based artists Nora Farrell and Bill Duckworth, who have spent a significant amount of time in Australia developing interactive sound gardens across the country branded as *Sonic Babylon*. Riding local Wi-Fi networks, the *Sonic Babylon* sound gardens grows with music, sounds, and stories accessible on mobile devices in selected spaces within a community. The sound garden is interactive and can be both heard and manipulated by the community. As visitors move through the garden, the *Sonic Babylon* application tracks their position in the space and the 3D audio engine generates a real-time sound mix relative to the location of the planted sounds.

Sonic Babylon is built on the Tactical Sound Garden [TSG] Toolkit, an open source software platform for cultivating public "sound gardens" within contemporary cities. This is drawing on the culture of urban community gardening and is framed around creating a participatory environment for social interaction through technology. The central idea is to ride existing Wi-Fi nodes in urban cities, but the potential outcomes could be just as relevant, if not more beneficial, in regional area if the appropriate amount of Wi-Fi nodes were installed.

As a direct result of this research, the first regional version of *Sonic Babylon* was installed at the Noosa Regional Gallery on the Noosa River and in late 2010, a partnering regional sound garden was installed at the Mill Place Precinct on the Sunshine Coast with the aid of funding stemming from the urban planning sectors of state government. The project was rebranded as *{neo}sonic*, and designed to capture and connect the voices of global communities through sound based art and technology. The multi-platform nature of the project involved a virtual orchestra, indigenous soundscapes and community engagement workshops which all fed into the sound garden.

Sound Gardens have a diversity of positive outcomes for a community including the ability to repurpose existing digital content (such as oral history) and also the ability to observe a system, a virtual ecology, and hear what kind of voices and themes may arise. The key attraction is its accessibility and versatility, and its ability to grow within a community over time. In the context of *Sonic Ecologies* it can function as both the core creative work and the ongoing community engagement. It basically functions as part of the ongoing ecology of the process as oppose to an independent creative outcome. The majority of the sound materials planted in the garden initially were historic recordings, particularly revolving around the indigenous history of the region. It was extremely rewarding to see young people interacting with Sonic Babylon and gaining insight into the indigenous history of the area, learning aspects of the local Gubbi Gubbi language and clearly feeling engaged and connected.

This case study also became an observation to prove the validity and necessity of the final stages of the *Sonic Ecologies Framework*. The *{neo}sonic* process received support from local, state and federal government in the initial stages, but unfortunately there was not ongoing support and community engagement provided. It is logical that if the community engagement is not sustained, the project will gradually dissolve. It is also worthy to note that the majority of people in a regional area, particularly of the older demographic, don't have the capacity to interact with this technology without initially being guided and mentored through the process. So while the installment of *Sonic Babylon* was engaging for the community and deemed successful, this research has identified a significant area of improvement in the development of ongoing sustainable strategies. This is not necessarily about delivering a two-week workshop program along with the install, but more about developing strategic partnerships and long-term community engagement strategies that accompany a project of this nature. Although this particular case study has also produced rewarding results, it has sparked the beginning of research into interdisciplinary partnerships to deliver projects over sustained periods.

BIOSPHERE SOUNDSCAPES

The *Biosphere Soundscapes* project, currently in its first phase of research and investigation, illustrates the

conceptual design of a major international project highlighting the future possibilities of the *Sonic Ecologies Framework. Biosphere Soundscapes* pivots on a network of site-specific creative project embedded in multi-layered community engagement processes within global Biosphere Reserves. This evolving process is implemented by sound artists, acting as agents of change spiraling between contextualised theory and practice. The content generated is embedded in a virtual network of global Biosphere Reserves via google earth technology and shared though the *BioScapes* community on the *PlaceStories* system developed by *Feral Arts*. The process and creative outcomes are delivered by a core team of artists and advisors who are ultimately acting as catalysts for a global participatory environmental project accessible to anyone with an internet connection. *Biosphere Soundscapes* is designed to inspire communities across the world to listen to the environment and re-imagine the potential of Biospheres as learning laboratories for a sustainable future. This project combines art, science, technology and community to give Biosphere Reserves across the world a voice and a global audience to listen.

Composed of 580 sites in 114 countries, Biosphere Reserves are sites recognised under *UNESCO's Man and the Biosphere (MAB)* programme to promote sustainable development based on local community efforts and sound science. As places that seek to reconcile conservation of biological and cultural diversity and social development through partnerships between people and nature, they are ideal to test and demonstrate innovative approaches to sustainable development and represent a unique tool for international collaborations through sharing knowledge, exchanging experiences and building capacity for interdisciplinary partnerships. This will be the first major international sound project for the *World Network of Biosphere Reserves.*

The methodological grounding for *Biosphere Soundscapes* revolves around the creative possibilities of Soundscape Ecology, a rapidly evolving field of biology, where scientists record a given habitat and listen for patterns and changes to form an analysis of the health of the environment. The process of working with each Biosphere will modify depending on the collaborating artists and accessibility of the local community, but in any case it will be grounded in each stage of the *Sonic Ecologies Framework*. In some cases the process will involve sound labs, artist residencies and extensive community engagement, while in other cases the key community stakeholders of the Biosphere Reserve can generate content independently and engage via the website.

The first stage of delivering this project is focused on two Australian Biospheres; the Noosa Biosphere and the Great Sandy Biosphere, both situated in Queensland. The research lab for *Biosphere Soundscapes* will launch on *World Listening Day* 2012 (July 18) with an interactive sound walk and community activities in the Noosa Biosphere. The initial support, interest and committed partnerships solidify the validity of this project and highlight the potential role electroacoustic composers could play in ecological crisis.

CONCLUSION

The *Sonic Ecologies Framework* is not a complex idea, it is simple, based on logic and grounded in significant practice-led research outcomes. As a result, it is accessible for artists interested in implementing similar projects on a local and global scale. While the author will continue facilitating projects through this process, it is also hoped the wider sound community will grasp the potential of delivering work with this methodology.

Now, more than ever before there is a critical need to listen to our environment and generate a paradigm shift that engages our auditory perception. Sound, as a creative medium, is undoubtedly one of the most powerful means to stimulate this shift in consciousness. Electroacoustic music, with the use of natural sounds exposing the state of the world could be an unprecedented tool in artists taking action in ecological crisis. This research is ultimately underpinned by the realisation that artists can play a role in creating a sustainable future, and as proposed by Joel Chadabe, think of ourselves as "leaders in a magnificent revolution" [3].

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THE PROCESS OF INTEGRATING POLYMEDIA IN BLOOMS AND DEATH

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ABSTRACT

This paper will discuss the methods and meaning of the polymedia processes represented in *Blooms and Death* as a live interactive performance composition, giving audio and visual examples throughout.

Blooms and Death incorporates sounds and images based on a series of graphics created following the transformation of a freshly cut bunch of yellow roses through to their grey death of decay**.** The visuals (video samples and stills) are layers of video footage displaying blowing yellow flowers, grey pencil drawings, layered scrolls of paintings incorporating the combination of computer transparencies of images. All the visual material has been treated extensively in various computer software packages to form this series of electronic artworks and video.

Blooms and Death explores many facets of polymedia in live performance, integrating through the audio, a response to the transformation of the dying roses employing breath sounds, and digitally processed clarinet with percussive processed piano sounds (performed by David McNicol), live processed acoustic environmental sounds, fans and the sampled video projections. The process of the disintegration of the roses and the response to this is explored through manipulation of sound and visuals to another timbral plane of textural ambience, colours and exploration in the context of polymedia. The music of *Blooms and Death* is scored for Bbclarinet, fan, live audio mulching (via laptop computer), multiple video projections and piano. The opening movement, Prelude, is for solo clarinet and visual shadows, with the following movement making a clean yet rough audiovisual statement through diffused fan and air sounds with thick clusters of sound from the piano presaging the later movements of the work. This is quickly broken down with many 'peaks' and subtle layers, the piece again emerging into transformed breath sounds moving into complex paths of high overtones and repeated notes as if conversing. The live audio computer interaction in the second and third movement adds to the sonic world of layering throughout the composition, while the Interlude between the second and third movement makes a short statement combining many sounds and visuals.

1. INTRODUCTION

This paper will explore the processes of integrating sound and visual counterparts of the composition *Blooms and Death* for clarinet, live interactive audio and visuals.

It will define polymedia and other terms that correspond with evaluating the many media that are associated with creating *Blooms and Death*. In the creation of *Blooms and Death* consideration of the use of elements is discussed in individual movements of the work.

Blooms and Death is made up of a Prelude, Interlude, Movement 1, Movement 2 and Movement 3 for Bb Clarinet, piano, live and pre-recorded electronics, Fan, 2 video projections and live video mixing. The physical human presence of performers, objects, real-time interaction are key elements of exploration.

Blooms and Death will explore elements of real-time performance, microtonal exploration, electronic timbres, acoustic instrumental samples and the relationship between tuning, timbre, melody and visuals. The overall focus is to create a world of space and texture in a sonic composition.

The discussion of *Blooms and Death* will be divided into 2 sections: How was polymedia integrated in the process of the composition? How will polymedia be integrated in the process of performance?

DEFINITION OF POLYMEDIA

In the context of *Blooms and Death* polymedia is defined by the layered or different media that is performed at one moment of time in live performance. Other terms that can describe polymedia is multi-media, hybrid art, visual/music, visualized music, multi-arts and VJ-ing.

My definition of polymedia is one performer, composer and visual artist creating a whole identity. As the performer the concept of polymedia is opening new palettes for the audience to interpret in terms of form. The composition integrates acoustic sound, live audio mulching, acousmatic sound (fixed music that exists only in a recorded format (as a fixed medium), and is composed for reception via loudspeakers), live video mixing and fixed video during performance.

Visual/music is a loose term that describes a wide array of creative approaches to working with sound and image. It can also be referred to as visualized music in which the visual aspect follows the sound's amplitude, spectrum, pitch, rhythm, often in the form of light shows or computer animation, while in other instances it may refer to "image sonification" in which the audio is drawn from the image in some form. Both sound and image may be presented live, fixed, or as part of an interactive multimedia installation. [i]

Visual music has much in common with music visualization, and is used to describe electronic music visualisers and media player software. The term describes how the music generates animated imagery based on a piece of music. The imagery is usually generated and rendered in real time and is usually synchronized with the music as it is played. The changes in the music's dynamics and frequency spectrum are among the simple properties used as input to the visualization. Sometimes visual music can be described as a non-hierarchical correlation between sound and image, in which both are generated from the same algorithmic process, while in other instances, they are layered without hierarchy or correlation altogether. [ii]

VJ-ing is another term that can be loosely associated to Visual/Music as it is primarily real-time visual performance which is the creation or manipulation of imagery in real-time and synchronized music. VJ-ing is the manipulation or selection of visuals, the same way DJ-ing is a selection and manipulation of audio. One of the key elements in the practice of VJ-ing is the real-time mix of content from a "library of media", such as DVD disks, video and still image files on computer hard drives, live camera input, or from computer generated visuals. In addition to the selection of media, VJ-ing mostly implies real-time processing of the visual material. The term is also used to describe the performative use of generative software, although the word "becomes dubious (...) since no video is being mixed. [iii]

Blooms and Death
How was polymedia integrated in the process of the composition?

In the creation process of *Blooms and Death* the polymedia creator (in this case myself) makes decisive day to day notes on the theme the rose from a moody personality. My decisions within each movement both musically and visually are random, spasmodic and erratic. The overall creation especially in the sonic world is in the form of an abstract narration. Each of the movements takes you on a journey that inevitably leads to the next idea. The musical composition is written first but without the visual of the dying rose the music would have no context.

The score and acousmatic pre-recoded audio is a combination of extended clarinet and piano techniques. Clarinet techniques include: throat sounds, hums, glissando, tongue clicking, kissing sounds, micro-tonality, key clicks, screams, multiphonics, monophonic, quarter tones, over-blowing, teeth on reed and interrupted tones. [iv] The piano techniques include: transferred clusters, tone clusters, tremolo, repeated notes at changeable speeds, harmonics (keys pressed down without sounding) glissando, extreme registers, erratic rhythmic patterns with indeterminate durations and pedal effects (fluctuating at different speeds with differing pressures and improvisation on graphic notations. Many of these extended devices on the clarinet and piano such as quarter tones, multiphonics, tri-tones, microtones, transformed clusters are reorganised and work cohesively to create sound clusters by just changing how each of the devices are orchestrated. The score incorporates graphic notations especially in the 2nd and 3rd movements that can be either literally or intuitively interpreted according to the ambience and sound world the musicians are creating and visual setup of the space at the time of the performance.

Examples of electronic plug-ins I have used in the processing of the audio samples include: Stereo Delay, Delay, Transposition, Grain Duration, Pan, Granulation, Ring modulators. [v] The process continues with manipulation of files into different layers and multi channels, concentrating on microtonal interaction between the samples. [vi] A similar process is applied to the visual materials including analysis of brightness, colour, contrast, duration, speed and complexity. The images have two categories: graphic based images and film/still images. The sound and image influences the shape and analysis of each of the works. The audio in the compositions uses a real-time environment of acoustic sound and generative structures. All audio samples were pre-recorded and processed in Adobe Audition (multi-channel audio processing program.

Diagram 1 Original photographs









Diagram 2 Pencil Drawings





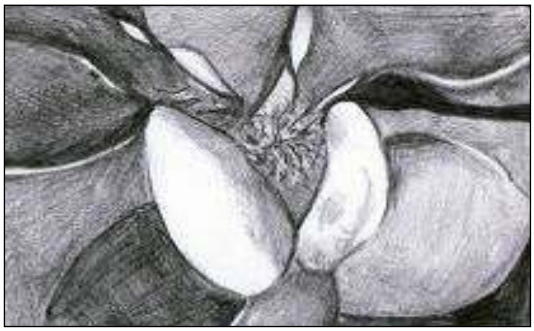


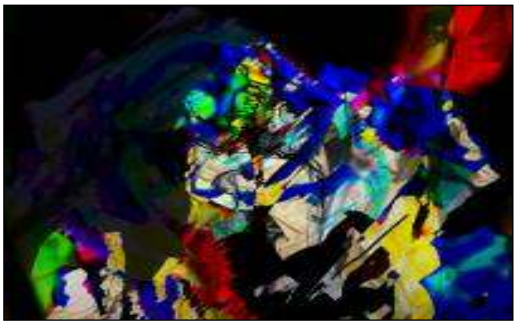


Diagram 3 Computer processed art











The imagery in *Blooms and Death* is a process of timing; every image has a fade, so it's all about the dissolve and the transformation in the image rather than in the fade. Very little manipulation is involved to fade one image to the next only the length of the fade is calculated. The layering of the transparencies with luma, chroma, speed (pulse), and cut-outs dominate many of the visual samples. The aim is to make the still images move through these effects. The use of stained glass plug-in with the treated rose images is animated.

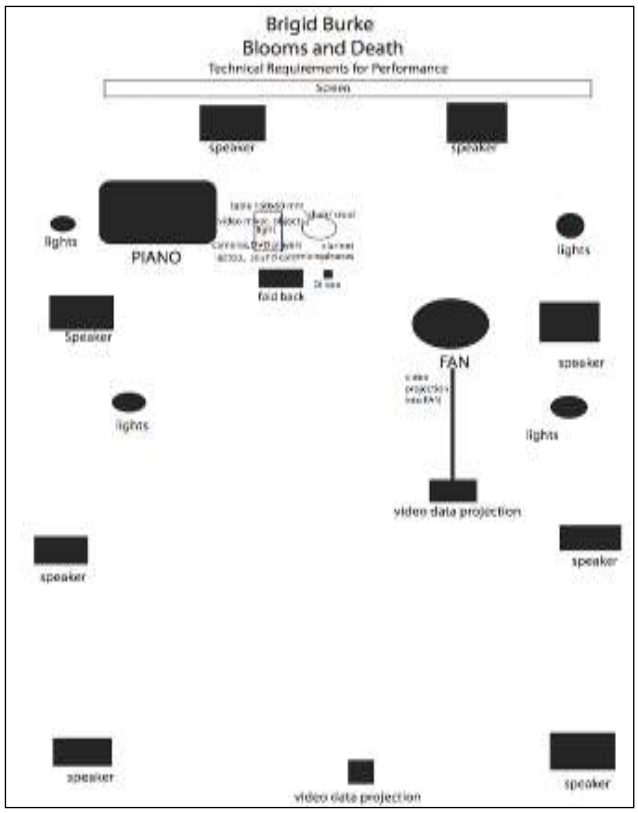
All these structures have given me freedom of process and uniquely affect the outcome of the piece. There is no correlation between the musical and video layers as the music is written at the onset and the animated video samples added afterwards so as the creation of the musical composition has created movement in a cohesive whole.

Technical Requirements for Performance

1) Eight speaker surround sound system 100+ watt stereo amplifier, speakers and mixer operated by sound engineer
2) 1 speaker (wedge) for fold back and mixer DI box for stereo sends
3) White backdrop (Screen) for visuals to be projected on to fill space
4) Two (data) Projector/Screen Composite/S video inputs
5) One 150cm x 70 cm high table
6) One chair (stool preferably)
7) One music stand
8) Lighting: two side washes and two spots (one to left and one centre) not directed on back screen
9) Two 6.35 mm plugs line-out from stage mixer into main audio mixer
10) Two Svideo/composite output connection from video mixer on stage to 2 data projectors one on Fan the other on Screen
11) Sound and lighting engineer
12) 3 microphones
13) Grand Piano
14) 30–40 inch Fan on Stand
15) Small 4to 6 channel stage mixer for on stage to be connected to main mixer operated by sound engineer

Technical Equipment performer is providing
1) Bb/Bass Clarinets
2) Lap-top computer
3) Objects
4) 1 microphone
5) Effects units, pedals and sensors
6) 2 cameras, DVD Pal player and video mixer

Diagram 4 System layout of equipment (computer/audio/visual)



Prelude

This movement for solo clarinet and visual shadows (video) starts the journey of *Blooms and Death* as a Prelude**.** It fuses the original roses as shadows into hinted layered digital images. The vivid rose is only seen in the distance as part of the many layers the movement is made up of.

Prelude is to be played in an intimate, subtle and lyrical manner, taking time over single notes with little

interruptions throughout the work. The atmosphere both visually and audibly is driven by the colour of the rose and is unique. Full of rich sonorities just waiting for the ambience of the space transform the clarinet. The aim was to try and convey moments of stillness with simplistic and guttural sound worlds while the rose with all its beauty blows in the wind.

Durational time is used to interpret Prelude. It is counting a unit for rhythmic organisation to destroy any sense of regular beat groupings. The beat or pulse, if present is an abstract concept whose function is to hold all the parts together. In durational music it is common to see ties, freely accented notes, uneven groupings and a wide variety of temporal subdivisions. The purpose of these is to destroy any feeling of the beat. Its absence makes the music non–metrical.[vii]

Diagram 5 Prelude score for acoustic Bb clarinet



MOVEMENT 1

This movement is scored for Bb Clarinet, fan, multiple video projections and piano with audiovisual statements made through diffused fan and air sounds with thick clusters of sound of extreme registers and dynamics from the piano. The clarinet adds another layer of suspense with ambient long notes in low register of the clarinet that are coloured with quarter tones and tremolos. There are many interruptions with quotations from the 1st movement which brings us back to the beauty of the rose. This is quickly broken down with many 'peaks' and subtle layers, and repeated notes and repeated rhythmic dynamically erratic clusters as if conversing.

The piano motives which were notated were recorded and divided into small samples with the blowing sounds of the fans. With many rehearsals starting out with rough ideas that soon developed into the final score. As each rehearsal was recorded the final score was now audible and the video samples of both the still images and video clips were added. The process of refining the audio samples was similar to the rehearsal refinement of the notated score and this was done in the multi-channel program Adobe Audio.

Diagram 6 MOVEMENT 1 score



MOVEMENT 2

The music of *Blooms and Death* is scored for Bb clarinet, fan, live audio mulching (via laptop computer), multiple video projections and piano with audiovisual statements made through diffused fan and air sounds again with thick clusters of sound from the piano

presaging in a processed state later in the work. This is quickly broken down with many 'peaks' and subtle layers, the piece again emerging into transformed breath sounds moving into complex paths of high overtones and repeated notes as if conversing.

Diagram 7 MOVEMENT 2 Graphic drawings in score

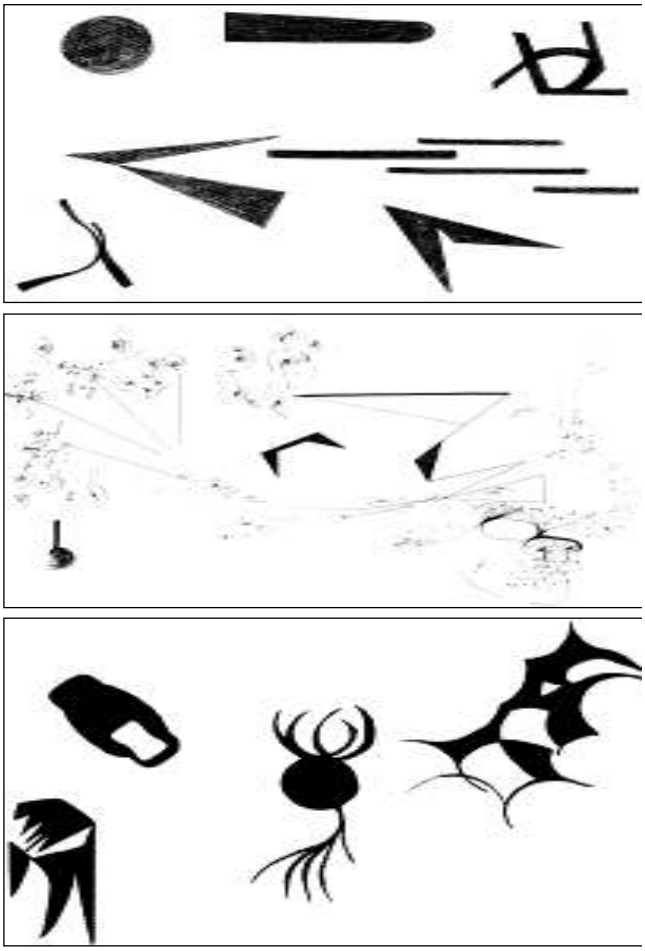
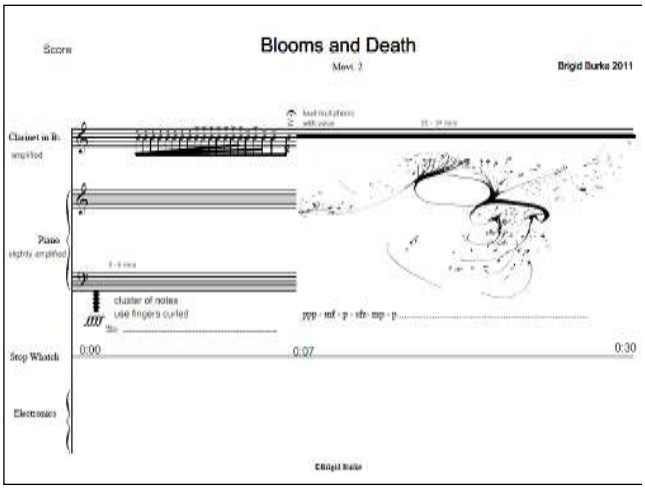
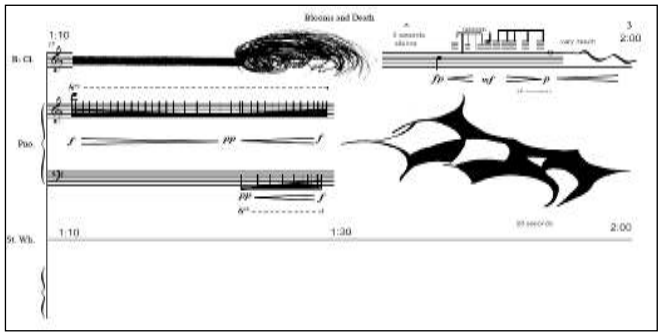
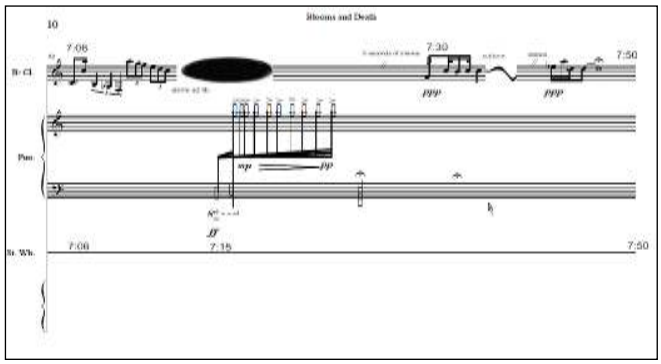


Diagram 8 MOVEMENT 2 score





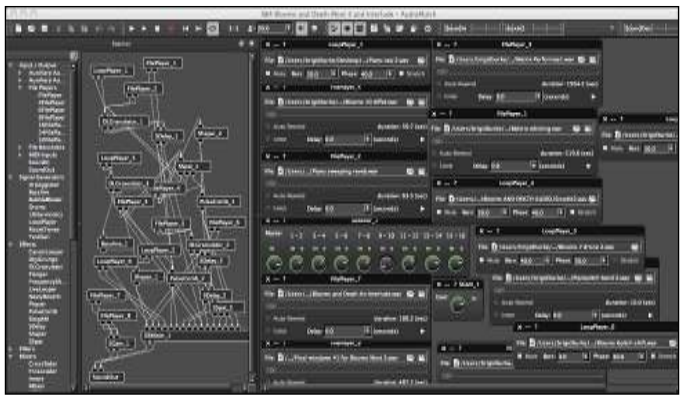


Interlude

Live Audio Mulching (laptop) and video projections.

This short Interlude brings new life to the red and blowing images in the fan of the image of red stationary rose and is fast and racy visually, accompanied by a beautiful soundscape. The Interlude also features the use of constrained random material gathered from Movt.1 and 2 interlocking with occasional raw utterances from the clarinet motives. This only adds to the freshness and sense of surprise created in the tapestry of the work. The visuals are forever moving, creating their own pulse and energy to the point where the rose fades to become unrecognizable.

Diagram 9 Audio mulch patch for Interlude Movement 3



Movement 3

The concept of the third movement takes the vivid yellow roses, breath, acoustic clarinet and percussive piano sounds to next level of textural ambience, colours

and exploration. The sounds and transformations came from images that depict grey colours and slow decay but show new life in their transformation through diffused timings and multiple layers of images and sounds. In the graphics one can see a series of yellow shades and grey pencil drawings layered to delicate tranquillity of refinement.

How will polymedia be integrated in the process of performance?

The polymedia performer in *Blooms and Death* plays Bb clarinet, controls the audio samples through the laptop and visual samples through the interactive responses on the video mixer, which is directly influenced by the sonic output of the system. In addition the first and second movement involve a pianist who is directed through a notated score and cues. During the performance one is continually creating new musical ideas that trigger fresh sonic audio improvisations that then influence different combinations of visual layering that give fluidity to the balance between the visual and audio output. Both are reacting off each at all times. In other words the performance could perhaps be conceived as creating polymedia counterpoints on player-defined fragments, effectively enabling the musician to perform a time-extended improvisation.[viii] The visual output is based on an abstract representation of how the music is progressing. The audio is affected by the live video feed samples taken which is affected by the lighting and the amount the performers interact with the footage in real-time.

The aim is to make sound and image structurally integrated. To achieve this integration in performance the audio is analysed and used directly to control the manipulation of specific aspects guided by the visuals.

Blooms and Death uses microphones and laptop to control the granular synthesis of acoustic audio material sampled during the performance, producing a musical confluence of live clarinet, piano and visual components which can only be described as polymedia.

The use of Organic Time is utilised throughout as the performers interpret the construction of sound durations not as a pulse or counting unit but using other parameters such as physiological determinants responding to the general conditions of the environment around them.

Blooms and Death explores the sonic vocabularies of extended clarinet acoustic micro-tonality techniques, keyboard, tonal and percussive techniques, interactive mapping audio devices and visual components. The interactive audio techniques used are pitch-shifters, frequency changes, room placements and granulation. All these filters and parameter modulations are controlled live using a mapping software device. The visuals go through similar processes as the audio samples but are pre recorded and include video footage of flowers blowing, photos, dead flowers and drawings with live editing through video mixing. These performance strategies open up enormous avenues for improvisation and inventiveness throughout the performance.

All audio processing is created in Audio Mulch a live audio interface for real-time audio performance, manipulated during performance.

The performance outcome aims to develop from the audio samples of the clarinet, the fan and piano audio-visual media. The element of air blowing literally from both sources symbolizes a virtual moving space. This in turn reveals that the visual images are preceded by the audio Bb clarinet, piano and audio mulching with the soft purr of the fan sounds.

The audio and visuals are all triggered manually. This is purely by choice as I treat all the components individually in a polyphonic manner. As an acoustic Bb Clarinet performer the natural state is to have no added attachments to the instrument so reaction to musical moments is spontaneous. However the addition of the laptop is another line of the control that affects/ influences the outcome resulting in a fluid and reactive performance.
A performance can be seen on youtube at:
http://www.youtube.com/watch?v=3-_xVfydqH8

Diagram 10 Photograph of set up used in performance on 6th March 2011 at UTAS



Prelude

Throughout Prelude there is always a glimpse of age, death and familiarity as a tangible sound through the clarinet with the rose being familiar and beautiful visually.

At beginning point what happens between the music and listener in the performance depends on the strategy of the composition and how familiar the audience is with the structures of the performance.

MOVEMENT 1

At the first performance of Movement 1 on 6th March 2011 the recital space was quite different from my original plan in the placement of the fan on the stage and the image it created on the big screen. So it was decided to create a double image on the screen one with the fan switched on directly in front of the projector the other with the straight image. The result was quite astonishing and created the austere atmosphere that was envisaged.

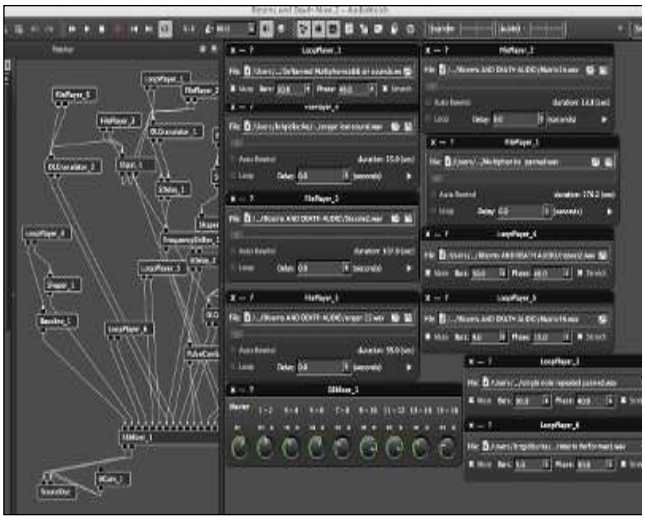
The aim in this movement was to test the material of the diffused fan, making the score; acoustic space and pulse of the visuals form a balance with the experimentation of structures and ideas.

MOVEMENT 2

The live audio mulching in this movement is controlled through the computer and adds to the sonic world of layering throughout the composition.

The aim was to take the sonic world to the next level of real-time interaction in which compositional decisions are made continually through the graphic notations in the score during the performance. There is much random computer affected samples which open up the free improvisational shaping leaving much of the decisions to the performers being intuitive with the pre-formulated material. Within these pockets of improvisation there are key meeting points. This result means no two performances are the same.[ix]

Diagram 11 Audio Mulch Patch for performance for Interlude and Movement 3



Interlude and Movement 3 for live Audio Mulching (laptop or acousmatic - fixed audio) and video projections.

These 2 movements in performance are fixed so according to the set in the performance space which should be in the form of a movie theatre with surround sound the out come is totally immersive with the integration of sound and visuals.

Conclusion

The performance work *Blooms and Death* demonstrates a creation of balance that is sensitive to environmental thought and fluidity in polymedia performance. The polymedia attributes as defined earlier in the paper achieve the use of acoustic sound in a sound environment at the present time that is of processed sound. The meaning of creating a polymedia work is each voice enhances the other (the music and visual). The aim is to create video with an emphasis on it been a moving piece with a direction even in the still image. The components of a live feed of the clarinet, piano and fans with the interactive mechanisms of video and audio mulching add a component that will captivate a moment of time. By encouraging interaction with adaptive sound as from the clarinet and visually engaging footage one can enrich the creative possibilities of polymedia processes in art, and in particular music. On a practical level, by bringing these independent processes of interactivity with computer orientated music in a traditional improvisation framework an interactive polymedia performance practice develops that offers a means of 'humanising' this age of the computer.

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ARTIST TALKS

DESIGNING INTERACTIVE PROCESSES IN RECENT ENVIRONMENTAL SOUND ART: *BLUE GOLD* AND *MEDUSA DREAMING*

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ABSTRACT

Interaction. Why and how do we want to relate to the environment? With whom and for whom? With what tools and media? What impact does our work have on the environment, its soundscape?

In two large scale environmental works, *Blue Gold* and *Medusa Dreaming*, processes of interaction will be identified, interrogating how acoustic space, sound, sculpture, film and creative musical interpretation of composed sound structures intersect. In analysing what is fixed and what is unfixed in the design process we can study different modalities of interaction and interdisciplinary collaboration at all stages of the creative process from the idea to audition. Human/environment/ human/machine, human/human interactive processes are endemic to sound art. These relationships will be identified in *Blue Gold* and *Medusa Dreaming* as they are embodied in aspects of the creative works: the site plan, the temporal and spatial design, sound recording, sound design, images, video, installation components, musical composition, to live performance and dispersion.

1. INTRODUCTION

Blue Gold and *Medusa Dreaming* are two recent works of environmental sound art, which provide excellent case studies for looking at the complex simultaneous interactive processes which may occur in interdisciplinary sound art installation and performance. Both pieces are created to focus on the most pressing environmental issue: water care. Both works interrogate our role in how we shape water care and offer time for reflection as to how we may become more attentive to our involvement. *Blue Gold* (2006-2009-12) is an Australian work commissioned by Charisma, a cello and clarinet duo. Original electro-acoustically modified soundscapes of natural and urban Australian environments are overlayed with composed western tunes representing the immigrant nation and human behaviour. Video correlates, actions with sculptures and improvisatory sound structures are required from the performers who must respond to the actual and fictional elements at the same time. *Medusa Dreaming* (2010…) Turkey, is an international cross-cultural, site-specific work celebrating the watercare of the ancients through surround sound, installation and live performance in the Yerebatan Sarayi, a palatial subterranean 6th Century water cistern in Istanbul. The new cross cultural South-North ensemble was created for this event, Erdem Helvacioglu electro-acoustic guitar and live processing, Natalia Mann Samoan harpist, and Izzet Kizzil, Turkish Percussionist and myself on Tarhu,a bowed spike fiddle, amplified slide whistle, and moving flutes. Although there is a common water theme in both works, each has very different design involving complex interactive relationships and processes in order for the work to be realised.

2. INTERACTIVITY: INTERDEPENDENT RELATIONSHIPS AND PROCESSES

Interactive relationships can be defined as those with a co-dependency on each other. "**Inter**, a prefix means 'between' 'among' 'mutually' 'reciprocally' 'together' ...**interactive**, of or pertaining to things or personas which act on each other." The narrower computer term interactivity in systems has come to mean " an immediate response to commands or data," [1] focusing on methodological techniques, often as ends in themselves. Over the years interactivity has had many applications in sound art as I have published elsewhere over the past two decades.[2] Environmental sound art embraces all types of interactive processes and relationships being firmly focused on the situational frame and environmental intention. Electro-acoustic sound is seen as an agent for this. The aesthetic design of environmental sound art requires clarity for each work and how interactivity is situated in terms of deciding

a) what is fixed, composed, not negotiable for change, e.g. a huge acoustic temple?
b) what interdependence and co-operative processes are set in place, (who and what?).

[1] Macquarie Dictionary, Second edition, 1992, p 915.
[2] See bibliography Bandt publications 1-6.

c) what is left open for adaptation and interpretation by the individual/s be it the performer or audience or swimming fish. (unpredictable, non fixed fluidities)

In *Blue Gold* (BG) and *Medusa Dreaming* (MD*)* the designs include a vast array of inter-relational correlates which determine the aesthetic pathway and journey of each work in different ways in its composition, performance and audition: These can be identified as

**environment/ human*, e.g. situational recording, dreaming (Both works)
**acoustic space/technology live sound*, e.g. hydrophone and sound mediated by the acoustic space (cistern, MD, water tank soundscape BG).
**technologically mediated acoustic space* speaker playback, amplified, filtered equalised sound during performance (BG, MD).
**sound/technology:Studio productions: Sound recordings/ mediated by technology, electroacoustic sound*, Prepared composed multi- channel surround sound files brought to the work by the composer, composer/engineer. These elaborate submixes are compositions in themselves. (BG, MD) .
**acoustic space/performer*
How does the harpist respond in the cold dank slippery water corridors at night? How do the drips and long decay time, cold and tuning problems effect performance? Can she hear and relate to the tarhu /flute player several metres and several seconds delay away from other remote corridors? The acoustic position and proximity of the speaker array (MD)?. How does the clarinettist respond playing in a very dark humid night setting on a thin dais on the lake's edge? (BG)
**soundscape/performer* responses,(BG, MD).
**notation/performer(*BG, MD).
**video/performer/audience* (double projections in sand BG, Jellies videos projected on ancient Basilica walls, MD).
**performer/performer* interactions, acoustic and live-electronics, solos, duets,(BG, MD) quartets (MD).
**speaker/audience* spatial proximity to speakers and Decibel comfort, crowds, live performance mix, (MD, BG).
**performer/audience* visibility audible proximity, (MD,BG).

These processes are conceived and overlaid in the design of each work from the outset in conceptual drawings, maps patch drawings, scores and software spectographs.

3. BLUE GOLD

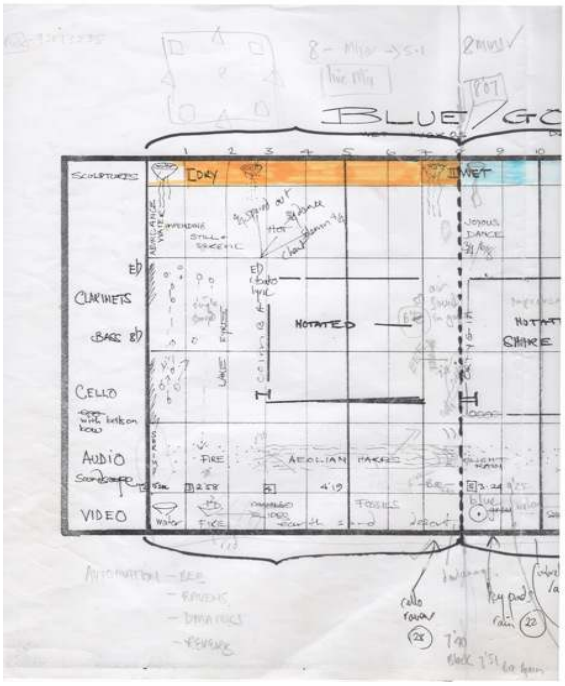


Figure 1. Blue Gold Graphic Score Excerpt.

Blue Gold has a thirty three minute electro-acoustic soundscape to situate the two instrumentalists in wet and dry environments. Their responses to the place of performance influences what and how they play from the composed musical notations and graphic scores. They pour water (cello), and sand (clarinets), into sculptures to tip the balance of the scales, a visual reminder of the state of play in the environment and how we affect the wet/dry relationships we inhabit. A colour -field video is projected behind, or on sand as the case may be, enhancing the wet and arid qualities against the actual environmental setting. In Boreen Point, for the Floating Land Festival curated by Leah Barclay, the video was set as a double projection on the night sand extending to the waters edge. This allowed visual play between the real wet, the lake and the actual dry, the sand. Projecting water on the sand changed this tango. Illumination and back lights further out to sea enhanced this depth of relationships as silver fish jumped from time to time during the performance. The durational aspect of the surround sound soundscape and the video is locked in and is not adjustable from performance to performance, but the elements the performers play can change in the graphic scored sections and in the last wet ,where there is some choice for the quasi improvisatory notation. The performers work with the site and the scored sections to their taste here which makes it very immediate. Sometimes they accompany the soundscape and other times they are submerged by it, or assert themselves as soloists.
The only time in the piece where there is composed rhythmic unison between the performers, the surround soundscape and the sound on the video, is in the cherries sections, which happen twice, representing human greed and destruction of the ecological balance by taking too much. This is fully prescriptive composition with no

room to move by the performers with all the elements fully in synch for the force of capitalism and Western greed to be fully felt. It's the loudest part of the piece. The unison dance *Mazi* (Greek for together), calls for a balance between the wet and dry through working individually and together.

These dynamic relationships may occur and change at any stage of the musical process for the composer, performer, technician, audience. It is the intensity of being in the sound in the environmental location that make the works environmentally present and alive. Not everything can be predicted. Performing *Blue Gold* in arid Wollongong in 2005 inspired Charisma to use the piece as a call for rain to break the drought. In Noosa in 2011 after the Queensland floods, the Boreen Lake performance seemed to celebrate much needed water tranquility. In each locale, the performers are working hard responding to the environments, the tools, maps, musical and graphic notations, listening and responding to the soundscape and each other. They deliver a real-time living work each time *Blue Gold* is played in each venue and particular acoustic space, with its own politic, and the overriding influence of the indoor or outdoor settings. *Blue Gold* exists as a composite of interactive opportunities for each performance. Like the environment, this flexibility of strands and interplays give the work its dynamism. A CD or DVD is merely a postage stamp freezing one of these moments.

4. MEDUSA DREAMING

4.1 Designing and Interacting with an Acoustic Space: the Yerebatan Sarnici, Istanbul. Beginning the designs.

Medusa Dreaming is coming from a very different inspiration. It is a site specific work designed for a particular acoustic space, a world heritage site, the Yerebatan Sarnici, an ancient subterranean water palace, one of the most visited world heritage sites in Turkey. Its architectural composition, acoustic identity its history and function have influenced all aspects of this work. It is 140m long, 70m wide and is a giant rectangular with 336 marble columns reaching 9 metres to support its Romanesque arched roof. 100,000 tonnes of water can be stored in it. Today the carp swim in the waters below the platforms where tourists can walk. There is usually piped music, Handel's water music or the Turkish end blown flute, the Ney coming from speakers. The first of two Saturday evening concerts, *Sounding the Basilica,* the duet with Erdem Helvacioglu, was a sonic investigation of all aspects of this chamber with an array of recordings I prepared, live electronic explosions from Erdem, quiet hydrophone recordings from the fish and improvised interactive music. We investigated how sound behaved in the air and underwater, in close proximity, playing flute through the corridors, catching the nodal spots, up down, natural and electronic, processed, unprocessed. With my hydrophone, live realtime recordings of fish swimming in the water below the platforms, were heard by the audience accompanied by the resonating condensation of drips falling from the high vaulted stone and brick Romanesque ceiling, the tears of Medusa themselves. Medusa jellyfish videos were projected on the ancient walls, the sound track having been prepared at home by playing my glass percussion sculpture, the flagong, improvising from the videos as kinetic scores.

The knowledge of the behaviour of this tricky acoustic was invaluable as sound designer and composer for *Medusa Dreaming*, although I had visited and recorded the ambience on two former occasions. Improvising on the tarhu with the harpist on the Tuesday before the second concert, taught us a lot about our duet for the sixth movement in terms of balance and possible performer interactions.The stage was quite dead and contained by comparison of working freely in the corridors.

4.2 Pre-performance Processes and Interactive Methods: composer, technician, musician interactions, Melbourne.

The surround sound files for each movement were composed and spatialised in the ABC studios beforehand together with engineer Alex Stinson. Key words inspired from the space, "water", "turned to stone", "336 marble columns" were recorded in international languages, Italian, Ancient Greek, French, Thai, Somali, Spanish, Kurdish, and positioned spatially so they would emerge from behind the columns throughout the space, as though the building itself were talking. 52 glass bells were struck representing the 52 steps everyone would climb at the end of the piece when leaving. An enormous palette of original samples was created, including pitch shifted glass bells, granulated bass slide whistle, and original water bottle sprung sculptures, bowed and played.

For the water harps track, psaltery and autoharps were performed and recorded in Melbourne together with my former sound art students, Angie Grant and Albert Mishriki. Angie treated the psaltery and autoharp improvisations with effects and created multi channeled sub-mixes as composed elements for particular movements in surround.

4.3 Composer Working with Turkish Musicians

Medusa Dreaming is a structured electroacoustic symphony in twelve movements for an international north south ensemble I arranged, Natalia Mann Samoan harpist, Erdem Helvcioglu live electroacoustic guitar and effects, Izzet kizzil, Turkish percussionist and Ros Bandt on slide whistle, bowed spike- fiddle, the tarhu, renaissance flute, and Pythagorean recorder.

Each performer was sent a page of composed notation which could be the tuning mode and modules for the improvised pieces.



Figure 2 Medusa Dreaming Notation

4.4 Medusa Dreaming's Structural Design

The programme was designed so that the performers were assigned particular roles in each movement. The lead follow, duet, exchange, support, accompany was broadly outlined as a common plan and discussed and rehearsed.

I *The Tears of Yerebetan Palace*: walking flute prelude, articulating the room, small sounds.

II *Frozen locks*: Athena's curse, (cello guitar/processing) drying airwhistle, ancient Greek "Turned to Stone " words,

III *Fish Dreaming*: fish solo enhanced hydrophone with percussion.

IV *Ode to Emperor Justinianus:* Ottoman duet, no soundscape, live electronics, Turkish Male duet.

V *Water through Glass:* Glass modules, harp water, live electronics, flute and harp.

VI *Corinthian Song* recorder percussion MD, theme introduced and built by the 4tet to rhythm 4.13

VII *Water Dreaming* water words over soundscape, water harps, flute tune and quartet accompaniment.

VIII *Doric Song* mediaeval dance, flute perussion.

IX *Medusa Dreaming* Harp takes the MD Notation with percussion, others accompany. Natalia's Medusa Solo.

X *Basilica Dreaming* electro-acoustic columns of words over aeolian harps soundscape with cello guitar.

XI *From Belgrat Forest,* woods rimu tree growing Tarhu and cello guitar.

XII *52 steps to the future of water*, all international voices, water soundscapes, improvisation by quartet.
Harp, air whistle, cello guitar /effects, percussion, apotheosis to stairs.

A CD of the electro-acoustic sound files I prepared for each movement with individual instructions was sent to all the performers beforehand. Each person was invited to make a personal response to the Medusa Myth. There would be 2 rehearsals and a long radio sound check in the space, the day of the concerts.

I made a situation which would be fun for all the professionals to enjoy the space and interaction in a public concert setting to inspire the audience to reflect on the history of the cistern in Istanbul through the ages, Doric, Corinthian, Ottomon and Modern references having been planned with the surround sound files.

4.5 Engineering Design

The ABC and TRT Radio stations agreed to provide the recording between them. I had already prepared the materials for playback. In Istanbul I made a hydrophone recording of the carp fish swimming in the cistern waters. This was mixed for a solo to entice the audience to listen just to the space with the drips near the beginning of the piece as a way in to "hearing the Cistern itself". Erdem's post-production enhanced the spatialisation of the fish from underwater to the air. This was added to the soundscape movement. Erdem would have his own live electronics set up onstage, and would oversee the soundscape files playback I had prepared as well as his own patches. This ensured integration of the material. The TRT would have a different desk which took the hydrophone line in as well. My Turkish engineering assistant Isik Yilmaz oversaw this interface, and was the translator and runner between. There was lots of essential co-operation by a team of professionals minding the sound at a number of levels in this vast and tricky space.

The multiple set ups, fold back, hydrophone live feeds, 12 microphones, and three mixing desks had to serve the peromrs, the audience and the radio documentation at the same time.

4.6 Contrapuntal processes

Experiencing the full gamut of these relationships and interactive processes in such an acoustic space was very powerful for all of us, the audience, Radio and TV, as the immediacy of having such an international event with so many contributors involved in the making of it, meant the world came into the space and was celebrating water together. The grandeur of this water palace was celestial, dense, dark and radiant, alchemical. Its overpowering strength as an acoustic modifier, a fish habitat, an elevated watery theatre, and psychological filter can not be underestimated, nor should it be. Medusa Dreaming is a celebration of the care the

ancients took with water tanks but also the references to *Dreaming* allude to the Australian aboriginal sustainable water-care, practices thousands of years before the cistern was even thought of in the Mediterranean. Water wisdom is a basic human necessity, whatever the nation or time.

5. CONCLUSIONS: COMPARISONS AND CONTRASTS

Blue Gold and Medusa Dreaming contain many similar processes, human to human, engineer to composer, performer to performer, performer audience. Both works engage in a complex array of human-machine/instrument engagements which are varied in each movement of the two works, live electronics, computerised heavily processed sound samples, derived from both natural, and original sculptural sources, live processing, improvising with surround soundscapes, shared musical performance.
The recording process of the entire work of course exists in the computer,before, during and after. In Australia, the ABC carried out post production on the TRT files I brought back, a truly international collaboration.

The engagement with the site is paramount in both works, even though in Medua Dreaming it is fixed and in BG it changes each performance. The cistern inspires, but so too does the natural environment. However Medusa Dreaming would be a very different work if it were to be performed in another venue such as the outdoor Federation Square in Melbourne. A variety of technological systemic interactions exist in the preparation of the surround sound fields both in the studio and in the live electronic performance patches.
In both works the performers have to decode instructions, read notation, improvise from modules and make decisions about intensity of self /other relationships. All the notated sections in the pieces are inspired by the Mediterranean and all the soundscapes are from Australian recordings except for the live fish recording from the waters of the cistern itself (MD).
In both works it is the contrapuntal fluidity of these changing interactions which gives each work its dynamism. Environmental water awareness is encouraged through sounding the different spaces which provide opportunities for attentive and reflective listening for all.

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SOME THOUGHTS ON 44 YEARS OF INTERACTIVE MUSIC MAKING

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ABSTRACT

In 1968, as an undergraduate student at Albany, my teacher Joel Chadabe introduced us to a new concept he'd been working on: interactive composing. Not without some questioning (how was this different from collective improvising? Or mowing the lawn?), I accepted this concept, and have been making interactive works in one form or another ever since. Today, I see the concept of interactivity as having pervaded almost every aspect of our society. In this paper, I will look at a number of pieces I've composed using this idea.

1. INTRODUCTION

Back in 1968, as an undergraduate student at Albany, my teacher Joel Chadabe introduced us to a new concept he'd been working on: interactive composing. Not without some questioning (how was this different from collective improvising? Or mowing the lawn?), I accepted this concept, and have been making interactive works in one form or another ever since. Today, I see the concept of interactivity as having pervaded almost every aspect of our society. Every time you go to an ATM, or your bank's website, you're experiencing industry's degraded concept of interactivity. And as technology has become faster and more powerful, even things which we held up as not being examples of interactivity back in the 60s – such as simple instrumental multi-track recording, or working with a text-based language such as CSound (or Music II, III or IV back then), or the Composers' Desktop Project–today have substantial elements of real-time interactivity about them. Any current DAW can be used either interactively, as well as linearly, and I'm sure someone will emerge who will be doing real-time live coding with CSound, if they aren't already doing a presentation at this conference. And aside from worrying about definitions, it seems to me that "Interactivity" is to computer music today what "I IV V I" is to the blues: the very structural basis of the form. In fact, if I wanted to be a smart-aleck (moi?) I'd quip that having "interactivity" as a theme of a computer music conference is like having "the sunrise" as a theme for a meteorology conference – interactivity is just that pervasive, as far as I can see, in computer music today.

But there are many different kinds of interactivity. And over the years, I've done many collaborative performances which used interactive processes in unique ways. I want to talk about several of these, describing how they worked and what the interactive elements were.

2. WEB (1972)

In chronological order, let's begin with *WEB (1972)*, which was a collaborative performance with Buchla synthesizers and tape delays, performed on Nov. 19, 1972, by Warren Burt, Ernie Morgan, and Bruce Rittenbach, at the University of California, San Diego, under the patient mentorship of Pauline Oliveros. This was a 4 channel performance with 3 performers. Each performer had a Buchla Box in front of them, which included a sequencer, and which fed into their own tape delay system, and in which sounds and control signals from one synthesizer was fed to the next synth in sequence, 1 to 2, 2 to 3, and 3 to 1. Each player had the option to work or not with the control and sound signals from the adjacent synth. So it was almost never a feedback loop, but did allow for some electronic interaction between the players. Each player also was playing with a patch in which the sequencer was semi-unpredictably controlling their sounds. So playing the final result was a bit like surfing an only quasi-predictable structure. Actually, though, despite this being San Diego, California, only one of us – Ernie Morgan – was actually a surfer, and at a semi-professional level, at that. Despite the potential for chaos, listening back to the performance after 40 years, I'm surprised at how gentle and sparse the performance is.

SOUND EXAMPLE – "WEB" (1972), by Warren Burt, Ernie Morgan, and Bruce Rittenbach

3. SENSUS (1988)

In 1987, I began working with Simon Veitch, inventor, with his 3DIS system. This was a system that used video cameras to detect motion and which then produced MIDI signals to control sound. The first large scale use of the system was an installation called *SENSUS (1988)* which took place at Expo 88, in fact, just a couple of hundred metres from this very spot, in the Southbank Parklands. Our pavilion was very near the Nepalese Pagoda, and in fact, when William Duckworth staged his "iOrpheus" here in 2007, he had the wedding of Orpheus and Euridice take place outside the Nepalese Pagoda, and he asked me to do a sound background for the wedding, to

be made of the sounds of the Sensus installation. So the sounds that had filled the air here in 1988 returned in 2007, to be part of an outdoor opera on the same site.

For the Sensus installation, there were about 40 places where the public could interact with it. Each interaction was very simple – someone moved near a camera, in a specific space, and a particular sound was triggered off. Some of these were inadvertent – walking through the entrance arches triggered off a choice of bird samples, and others were triggered off only by wilful gestures – such as the sound of breaking glass triggered off by sticking your hand through an empty window pane. A few sounds even allowed some very simplistic performing. Three hollow cubes, with metal frames, had pentatonic scales of samples of vibes, bass and drums, so three people could improvise a kind of pentatonic jazz, if they wanted to. As said, each interaction was very simple. What made the installation fun (and even qualify as new music), was the fact that there were so many of them, and the sound vocabulary was extremely varied. With even 3 or 4 people in the installation, things got thick very fast. And when there was a crowd around (which there usually was), the sound world was about as complex as Cage's *Williams Mix.* What kept me interested was that usually each participant was fixated on their own sound interaction, and wasn't giving attention to the sound installation as a whole, meaning the overall sound was a chance-produced result of a number of people playing at the same time.



Figure 1. Details of "SENSUS" installation, Expo88, Brisbane, by Warren Burt and Simon Veitch

SOUND EXAMPLE: "Sensus" 1988

4. HEAR THE DANCE, SEE THE MUSIC

The next year, 1989, we used 3DIS in a work for dancers. Funded by Arts Victoria and Diabetes Australia, *Hear the Dance, See the Music: Fair Exchanges (1989)* (Burt, 1989) was a collaboration between Simon Veitch (technologist), myself and Ros Bandt (composers) and Sylvia Staehli, Jane Refshauge and Shona Innes (choreographers and dancers). The program consisted of five pieces, each of which used the system in different ways. Some of the pieces were very simple in their interaction – a moving body triggered off instrumental sounds at particular places, and the dancer could play music in very traditional ways, such as Jane Refshauge's *Inside Out (for Graham)*. Other pieces such as *Free Trade Zones,* by Sylvia Staehli and me, involved elements of unpredictability in the interaction. In this piece, a political comment on the use of exploited female labour in the third world by the electronics industry, Silvia walked around an invisible figure 8 pattern on the floor. At certain points she would come to a sensitive spot. This sensitive spot had a "percentage chance" (say 40%) of triggering on a sample of "ethnic music." If she did trigger off a sound, she danced to it, and each time she passed through the area, she also had a certain chance of turning off the sound. A particular music would only happen in a particular place, but the turning on and off of the sound was probability based, and not by means of a simple on-off switch.



Figure 2. Sylvia Staehli performs "Free Trade Zones"

VIDEO EXAMPLE: "Free Trade Zones," by Burt and Staehli

In another piece, *For Percy Grainger and Burnett Cross* by myself and the three dancers, we attempted to make a dancer-controlled version of Grainger's Reed Box Tone Tool - a microtonal harmonium that used paper rolls to produce glides of closely tuned harmonium reeds. For our piece, we made a very large invisible S-shaped pattern on the floor. Low pitches were at one end, and high pitches at the other. If one walked along the length of the S, one could produce a 2 octave glide of samples of the original reedbox tuned to 36 notes per octave. If one walked across the S, one would trigger off, say, some low notes, then some middle-range notes, then some high notes. If you freely moved around the space, one could produce all manner of musical phrases, most of which (because of the layout of the floor) would have some aspect of gliding about them. This video excerpt begins with all 3 dancer on chairs in the middle of the S. After a bit of a "mad scene" in which Jane and Shona move around the chairs, then move out of the range of the cameras, Sylvia proceeds to roll back and forth along the row of chairs, producing mid-range glides. Eventually, all three dancers end up waltzing around the

space holding chairs, producing a wide ranging texture of gliding and stable notes.



Figure 3. Silvia Staehli, Jane Refshauge, Shona Innes in "For Percy Grainger and Burnett Cross"

VIDEO EXAMPLE: “For Percy Grainger and Burnett Cross” by Burt, Staehli, Refshauge and Innes.

5. HORIZON RERUN

One of my longest running artistic collaborations has been with choreographer Eva Karczag. We have been working together, off and on, since 1977. In 1994, we produced an hour-long dance and music piece called *Horizon Rerun*, which we performed in Melbourne, Sydney and Brisbane. For one section of the dance, an interesting interactive process was used. I had a Buchla Lightning – a device which used the motion of infrared sensors to produce MIDI signals to control electronic music devices. In my case, the infrared sensors were worn on my hands in the form of rings. My movements controlled a microtonal sampled electric piano. The process was this: Eva would be moving. With my hands, I would try to follow her movement. This would produce the microtonal electric piano music. She had the option of responding to the music, or not. If she did, the movement of my hands would produce different music, which she was then free to respond to or note, etc. So we had the potential for a feedback loop, mitigated by her decision whether to respond to the music or not, and my ability to follow her movements accurately with my hands. Unfortunately, no video record of this performance exists, but we do have photos from the Brisbane performance, and a sound recording.



Figure 4. Eva Karczag and Warren Burt in "Horizon Rerun" Queensland Art Gallery 1994



Figure 5. Eva Karczag in "Horizon Rerun" Queensland Art Gallery 1994

SOUND EXAMPLE: “Horizon Rerun” by Burt and Karczag

6. INSTALLATION FOR 3 LAPTOPS: REMEMBERING KENNETH’S LEMONS

The *Installation for 3 Laptops (Remembering Kenneth’s Lemons)*, from 1999, made for the “First Iterate” conference at Monash University, used a different type of interactivity. Three laptops, 2 DOS machines and one early Windows XP machine were connected in series – A to B, B to C and C to A. All machines were running a kind of chaotic generative process based on counters and shift registers, and they were all connected in a feedback loop. Two of the machines were running John Dunn's Music Box, one of the earliest algorithmic “tool kit” programs, and one was running his SoftStep, a further development of the program within a GUI environment. All three computers controlled microtonal electric piano patches. The title is a reference to Kenneth Gaburo’s early 1960s electronic composition, *Lemon Drops* which also used electronic means (Harmonic Tone Generator and splicing) to assemble an imaginary electronic piano

soundscape. The laptops were put on top of plinths, and mounted in a cross between an art-sculpture and a department store display. A panoply of small stuffed animals assisted the installation as well.



Figure 6. Installation for Three Laptops (Remembering Kenneth's Lemons) First Iteration conference, 1999

SOUND EXAMPLE: "Installation for 3 Laptops (Remembering Kenneth's Lemons)" (1999)

7. ROCK AND LIGHT TRACING, BY CATHERINE SCHIEVE

Catherine Schieve's *Rock and Light Tracing (2006)* was a piece for the reconstructed Grainger-Cross Electric Eye Tone Tool, which I constructed in 2004, and rebuilt in 2006. In this piece, the transparent plastic sheeting which constituted the score for the machine was made by placing the plastic directly onto some shale rocks in Newfoundland, (where we were performing the piece) and painting the patterns of the rocks on them. In performance, the score is not only moved along the photocells, in the manner of a traditional player piano score, but also moved in the air above and around the machine, creating highly unpredictable, but very attractive patterns of sound. By moving the score into the 3rd dimension, above the photocells, we were able to easily get beyond Grainger's 2 dimensional conception of the score, as well as having a lot of fun.



Figure 7 .Catherine Schieve: "Rock and Light Tracing" detail of score 2006



Figure 8. Warren Burt and Catherine Schieve perform "Rock and Light Tracing" Sydney 2008

VIDEO EXAMPLE: "Rock and Light Tracing" by Catherine Schieve, Warren Burt assisting.

8. ROBOJAM 2

In 2010, I participated in the second Logos international jam over the internet controlling the Logos Robot Orchestra (Logos 2012). The Robot Orchestra was built by Godfried Willem Raes at the Logos Foundation in Ghent Belgium. It consists of over 40 acoustic instruments, rigged up with solenoid control of various kinds, controlled by MIDI. For the Robo-Jams, Kristof Lauwers of Logos arranged a chat-room style interface, where international participants could type in strings of characters. These gave the central computer, at Logos, materials with which it could control the performers' selections of the Logos instruments. So each performer around the world could specify an instrument, and algorithmic instructions indicating how the instrument was to be played. In the chat-room, we could all see what was happening, and we could all hear the composite piece, which was also occurring live in a concert at Logos. In Robo-Jam 1, in 2008, internet speeds were very slow, and the improvising was difficult. By 2010, internet speeds were higher, and we could use Skype to hear the overall sound, so it was much easier. Also, none of us were specifying musical instructions that required precise rhythmic sync with the other players, so the small delays we experienced were not any significant barrier to doing a good performance. The players for this one were Kristof Lauwers, Yvan Vandersanden and Troy Rodgers at Logos Foundation, Gent, Belgium; Celio Vasconcelos at home in Aachen Germany; Scott Barton and Steve Kemper at the University of Virginia (Virginia Center for Computer Music) Charlottesville, Virginia USA; Jaime Reis in Linda-a-Velha, Portugal; Warren Burt at home in Wollongong Australia; Simon Halsberghe - in a lazy armchair in Antwerp Belgium; Brent Wetters - at home in Providence, Rhode Island, USA; and Juan Sebastian Lach Lau - at home in Morelia, Michoacan, Mexico.

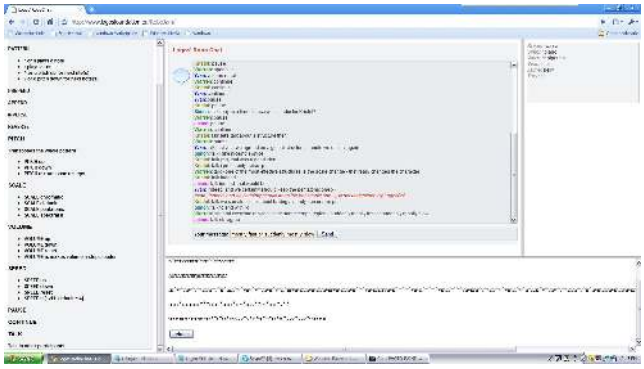


Figure 9. Example of the Chatroom like interface developed by Kristof Lauwers for RoboJam 2



Figure 10. Jaime Reis performing in RoboJam2 in Lisbon



Figure 11. QT a quarter tone organ, one of the Logos Musical Robots



Figure 12. Vibi, a computer controlled vibraphone, one of the Logos Musical Robots

SOUND EXAMPLE: “Robojam 2”

9. THREE MOVEMENTS FOR FOKKER ORGAN

My most recent international interactive performance was *3 Movements for Fokker Organ (2011)* which took place between my home in Daylesford, and the Fokker Microtonal Organ of the Huygens-Fokker Foundation in Amsterdam. The Fokker organ was designed and built by Adriaan Fokker in the early 1950s (Stichting Huygens-Fokker 2012). It is a 31-tone per octave organ, with 2 manuals and pedals. About 5 years ago it was removed from the Teyler's Museum in Haarlem, into the Muziekgebouw aan t'Ij, in Amsterdam and MIDI control was added to it. With the assistance of Ere Lievonen, who is the principal organist for the Fokker Organ and Danny De Graan and Tim Opie, who developed various versions of the Max Patch which enabled us to send midi over the internet, I played 4 algorithmic programs, written with John Dunn's ArtWonk, which generated MIDI which was then sent over to the Fokker Organ. I heard the results back via Skype. The total delay was about 1 second, which was easily good enough to control an interactive algorithmic process. My favorite of the pieces was a little encore movement, *Fokker Organ Wren*, which used DNA protein patterns from the Brazilian Uirapuru, or Organ Wren, as a basic sequencer material, which I elaborated upon in real time.



Figure 13. Warren Burt (on video screen) and Ere Livonen (at the Fokker Organ) performing Burt's "Three Movements for Fokker Organ" Muziekgebouw Amsterdam 2011



Figure 14. Warren Burt performing "Three Movements for Fokker Organ" Daylesford, Vic. 2011 - netbook is controlling the organ, laptop is used as Skype interface to hear organ and watch concert hall.

SOUND EXAMPLE: Excerpt from "Fokker Organ Wren" (2011)

10. CONCLUSION

In each of these pieces then, there is a different kind of interactivity happening. Some are algorithmic, some involve people processes, and some involve remote interaction. The variety of processes used in these pieces shows the wide range of activities that the notion of interactivity can embrace. As they used to say, it truly is a very broad church. But all of them, whether involving people or machines, involve what is an essential element of interactivity – the use of unpredictability as an essential element in musical performance.

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THE WAR ON THE CRITICAL EDITION VOLUME 2

Piano Games: Multiple compositions from a single source

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ABSTRACT

In my so called 'serious' research, (that into best practice realization of ancient and medieval music), a major theme has been the preparation of multiple realizations of a text or musical work, something that can be defined as music that has no critical or singular edition. This premise has had a profound effect on both my realized early music and new art music composition. This paper continues earlier research into the process of writing multiple compositions from a singular source. Previously I have written multiple compositions based on fragments of Hildegard of Bingen's *Ordo Virtutum* and sent a solo shakuhachi piece out to the public as a series of garageband loops, projects that stimulated discussion about the home space of composition. This part of the experiment is concerned with making new compositions from existing compositions without changing a note of 'the score'. In 2010 I wrote a series of simplistic piano preludes, recording them on a MIDI[1] instrument and treating those recordings as source data which could be exported as either a score or a MIDI file. Experimentation was then undertaken to see if new compositions can be realized wholly within a commercially available DAW.[2] Each original work has given rise to two variant compositions, which utilize some of the techniques of MIDI manipulation in screen, art and popular music styles.

1. INTRODUCTION

The process of recomposing is hardly new. It can even be argued that until the rise of the named composer that the majority of music was constantly recomposed, something that still occurs in many of the world's oral traditions. The processes of notation and transcription play a part of the solidification of music, allowing certain versions of a musical form to gain currency as the norm, whether in the preservation of historical editions or in performance practice itself. I described this in the first part of this project:

> The critical edition is at the heart of western art music. When someone asks to hear *Ave Maria* at their wedding there is an inherent cultural assumption that they will hear a particular version of the *Ave Maria*, usually composed by Schubert or Bach/Gounot. Though the Bach/Gounot is an arrangement of an earlier critical edition it has been absorbed into the cannon of western art music alongside the Schubert, and as such either composition can be attributed to a composer(s). Musicology can document the works of western composers thanks to the invention of notated music by Guido of Arrezzo in the 11th Century, and the extant works of the thousands of composers who have worked in both sacred and secular music. These thousand year old notations give us many of the best and worst aspects of our current musical life: they allow copyright and patents to flourish, intellectual capital to be recognised, but they also allow singular editions to push their multiple counterparts to the side. For example an artist may play a hit song many different ways but the critical edition is always the recorded and disseminated version of the work. (Cunio, 2009).

Recording has revolutionised the dissemination of music and also the way it is delivered. In recent times it has had a profound effect what actually constitutes a critical edition. A century later it seems impossible to imagine a world without music on demand. Though recording has signaled many changes in culture the most pertinent for this project is the singularization of the recorded work. A century ago a score was the form in which people held a critical edition, and though in itself a singular item the score was open to interpretation in reading and playing. Today for most young people music is a download, critical edition often devoid of its subtext and history. Instead of hearing many versions of a piece of music as innately equal, as with the live performances of established repertoire such as opera or jazz standards, one recording is often both the composition and dissemination of a piece of music.

It is common for there to be no formal score in a contemporary piece of music. Sometimes a score is even prepared retrospectively, particularly with popular and screen music, processes dominated by composition on an instrument or MIDI keyboard. The journal *Recent Researches in the Oral Traditions of Music* defines this point of change:

[1] Musical Instrument Digital Interface, a standard for transferring digital music related data.

[2] Digital Audio Workstation, a computer based recording, editing and sequencing program.

> Recent Researches in the Oral Traditions of Music encourages scholars to rethink the critical edition as a crucial component in the current rapprochement between ethnomusicology, historical musicology, and cultural studies. As new media make it possible to experience musics from throughout the world, as oral traditions have become essential to the globalization of local musical practices, and as popular musics give postmodern meaning to historical diasporas, so too does Recent Researches in the Oral Traditions of Music invite music scholars to conceive of editions that will contribute fundamentally to some of the most critical debates of our day. (Bohlman, 2005).

While the computer has changed the game for music making and recording, art music composers and institutions are quickly coming to terms with the ramifications. The digital transfer of music offers a potentially perfect copy of a performance that can be transcribed or learnt orally, making it a meeting place between oral and written forms. It can be argued that notation, as we historically understand it, is now one of a number of processes to preserve and record music. MIDI, the DAW and wave file composition (whereby the composition takes place after the recording of the individual parts), have replaced traditional scoring for many composers. In addition to this we now extend the term composition and composer well beyond the historical western definitions. The composer of a work does not necessarily have to know the craft of notation, nor be able to perform a work the same way twice. Reid (2007), states that a written score can range from a chord chart to a Pro Tools file, yet a standard composition degree at a tertiary institution is still mainly concerned with the authorship of singular, critical written works. Ofcourse such things are commonplace in moving image work. The composer Hans Zimmer not only developed his own sample library to replace the traditional orchestra, he has invented a hybrid sound of natural and synthetic sounds that are as much a part of the score as the notes, as these fusions sounds epitomize the thickness of the modern Hollywood sound. (Davies, 2011).

This paper therefore serves two functions: to document my developing practice based research as well as stimulating discussion into the processes and ramifications of recomposing around existing material, something that also applies to my early and intercultural music (Cunio, 2009b). Earlier in this research I recomposed around the Prologue from Hildegard of Bingen's *Ordo Virtutum*. Although new works were generated from a critical edition, it was clear that the original work still lay with the historical author and that these derivative works could be seen as some people as merely constituting an arrangement or realization. In this part of the process the notion of a home music is completely subverted, multiple compositions take place from a singular source, with the essential score data unchanged.

2. PIANO GAMES

2.1. Preamble

The genesis of *Piano Games* lies itself in technology. I was commissioned to work with the Australian author Dr Stephanie Dowrick on a CD of meditation commentaries in 2009.[3] At the initial meeting it was decided that two new pieces of music would be written in addition to an underscore for the spoken word. At this point I mentioned that I had recently updated my piano sample library and proceeded to play a short prelude that I had rendered on a virtual Fazioli piano. At this moment I had a small epiphany, realizing that for most people music technology still constituted pedals, boards and electronics, that the recent advances in synthesis and sampling have gone almost unnoticed despite their ubiquitous presence in television, film and popular music.

Although I own a 75 year old Mignon grand piano and have access to all the resources of the Conservatorium I decided at that moment to 'smuggle in' a sampled piano and cello to this commission, and pass the samples off as real players to see if anyone noticed. This was the birth of *Open Mind*, the first piece in this project. The piano part was recorded on a $200 plastic keyboard after which editing work was undertaken in a MIDI editor. The MIDI data from the performance was edited in relation to attack, decay, velocity and pedal duration, summed in the analogue domain and processed through the Bricasti M7, a newly released hardware digital reverb. After this the audio was parallel compressed to suggest the feeling of attack present in a real player. A similar process was undertaken with the cello part. A simple line was composed and played on the MIDI keyboard and different cello patches were alternated, primarily between up and down bows (the latest sample libraries now do this automatically). Extensive equalization was utilized to open the sound of the lowest notes of the cello, and provide greater bass response to the higher notes (a longstanding problem in many sample libraries). After this the part was summed in the analogue domain, limited, and again processed with the Bricasti reverb. I invented a pseudonym for the cello player,[4] and delivered the work to the company. Not a word was spoken, everyone from the company loved the work, and although it was obviously a 'fake' to me, I delighted in the realization that no-one even noticed. I then wished to extend the experiment to see if I could make new works from the one score.

[3] *Meditation Essentials,* Stephanie Dowrick and Kim Cunio: Vision Australia 2009 ISBN: 9781864826210.

[4] Eliyahu Elias, two iterations of my middle name.

2.2. Preludes

In order to proceed with the project I needed to write some source material. In keeping with the notions of subversion that constituted the birth of the project a series of structured improvisations were entered into to create new source material. Each piece would be both a prelude in the historical sense of the word, and a prelude to re-composition. The first two pieces presented themselves with ease. I exported the piano part of *Open Mind* as the first prelude and while selecting it found a sound check in the previous take, which I also selected. This piece was quickly named *Sound Check*, and the project had started. From these two works a series of parameters were developed for additional source composition:

- That each prelude be written in a particular scale.
- That modulation be kept to a minimum.
- That each prelude be capable of being notated, played acoustically or re-improvised at a later point.
- That each prelude be played into a DAW allowing the computer to store MIDI attack, decay and velocity data.
- That this project be undertaken with one commercially available DAW, in this case Digital Performer, with the Fazioli piano sample on East West's Goliath virtual instrument.5

In total six source preludes were created for which summary data is below. These works constitute CD1, with the derivative works constituting CD2 (Figure 4).

Name	Length	Key [6]	Style
Sound Check	2:55	a	Slow Persian Hijaz scale
Baby Bach improvisation	6:48	g	Contrapunctal
First Species	4:29	c	First species counterpoint
Lydia improvisation	6:16	D	Lydian mode
Ambi	6.43	e	Improvised ostinato
Open Mind	5:12	g	Filmic improvisation
Total	32:21		Improvised piano music

Figure 1. Summary data for Preludes used in Piano Games.

[5] A software based sampler loaded as an instrument track into a DAW.

[6] Capital letter denotes major key.

In a traditional music project such improvisations would probably be scored either through direct transcription or through the exporting of the MIDI data to Finale. In this project they were not altered, leaving the MIDI data looking decidedly non idiomatic on the QuickScribe window in Digital Performer.[7] The reasons for this are telling, a direct transcription would privilege the source prelude beyond its intended significance, subverting the later works. Resisting the score in this case is also tantamount to resisting the critical edition. The example below is typical of the score data in the project.



Figure 2. Sample of QuickScribe notation in *Sound Check.*

2.3. Play

The ensuing process is one that I as a researcher rarely have the luxury of entering into. I described it as a process of 'rampant experimentation' in a lecture to Music Technology students at the Queensland Conservatorium in April 2011. (Cunio, 2011). I started this phase with the objective of seeing how many fundamental parts of a composition I could manipulate without changing the score, and within a short period of time the list was extensive, as can be seen in Figure C. I was looking to manipulate this data in a mainstream DAW as opposed to a processing program such as Max MSP or Pure Data. Even though the commercial DAW is not designed for such re-composition there are certain advantages in staying within such a platform. They include time saved on writing and debugging new systems, the inherent audio quality of the commercial DAW, the integration of 3rd party products in regards to latency compensation and the availability of plug ins and the ease in which virtual instruments can be added to the initial MIDI data. There is something far more important than the method in this project however, which is the inspiration of the act of composition. Within a short period of time this project took a life of its own; processing tools replaced traditional tools and I was writing freely.

Objective	**Method**
Change instruments	Assign MIDI track to new virtual instrument
Change tempo	Change global tempo on project

[7] This is a very common issue in music for moving image where budget and timelines often do not allow for an adequate score to be prepared, leaving some session musicians with the unenviable talks of decoding a poor DAW score in a recording session. (Lee, 2009).

	Insert tempo changes into conductor track
Change key	Transpose plug in
Harmonization	Transpose plug in, chains of transpositions
Reassign scales	Humanize plug in to create new pitch maps
Change performance	Traditional MIDI dialogue tools
	Humanization plug in to control onset,
	duration on velocity, off velocity
	Time shift plug in to shift bars and beat data
Change rhythm grooves	Quantize / groove quantize, new DNA
Arpeggiation	Arpeggiator to control subdivision,
	note order, range
Delay repeats	Echo plug in to control subdivision and
Invert pitches, key	Invert pitch plug in
	Nomination of central pitch and inversions
Remap Midi data	Reassign CD plug in

Figure 3. Manipulation of initial Prelude data in Piano Games.

In this from of manipulation several things quickly became cognizant. The order of manipulation would effect the final outcome of the MIDI data just as the order of inserting plug ins to audio effects its transmission in a DAW. A number of the MIDI plug ins also had randomize functions which move the project into the realm of algorithmic composition; and the order in which data manipulation took place effected my perception of composition choices. Within a short period of time the project turned full circle. I had constructed it to a very strict brief so as to manage its scope, but it still looked too vast. Further limitation was needed in order for the project to have a unified theme and outcome. To respond to this challenge a number of parameters were developed:

- To work wholly within a portable computing environment without the aid of extensive studio equipment, analogue processing or large rack units of virtual instruments. This included bouncing final mixes from within bus tracks.
- To limit myself to four commercially available software virtual instruments, Modartt's Pianotech, a synthesis piano for piano, electric piano vibraphone, marimba, and celeste, the Wallander synthesis instrument for brass and woodwind, the Garritan instruments for Finale for strings and additional percussion and the MOTU Polysynth for synthetic pads and drones. Each of these was chosen for ease of use, price point and low CPU usage.
- To try and limit myself to two new compositions from each source Prelude.
- To not formally compose or notate a re-composition. In other words to keep each new work as an ephemeral act, undertaken in a short period of time.
- To not change a note of the initial score data or copy a plug in chain from one set of source data to another.

Once familiar with this set of parameters the task of writing music was familiar. As with any act of creation I had to find a balance between planning and intuition, or in this case prescriptive and reactive editing. Prescriptive editing in this project is defined as a deliberate manipulation to achieve a predetermined compositional outcome, whereas reactive editing is defined as a process of responding to the source data by improvising parameter changes as the work is played, in the style of live electronic and coding performance. Needless to say it is in the combination of methodologies that art takes place. In all thirteen new compositions were written. Each prelude inspired two derivative compositions though one of the source pieces *Ambi,* gave rise to a third new work. A critical packaging decision was whether to combine the Preludes with the derivative works or to present them separately. It was decided to present them separately as the derivative works comprise a full length CD in themselves, and it is anticipated that the initial preludes will be either recorded acoustically or have a new improvised realization recorded at a future point. In this project the order of the new compositions is a direct inversion of the order of the preludes. Figure 4 presents summary data for the new compositions in *Piano Games*, with a brief descriptor inserted underneath each new work.

Prelude Time/key	**New work**	
Open Mind g	*The varying nature of wind*	6:14
	Process music for piano, percussion & winds	
Bb	*Moving out*	4:22
	Ostinato led jazz groove for piano duo	
Ambi D	*Night light*	4:38
	Lullaby for solo piano	
E	*Comprovising*	6:12
	Groove variation for solo piano	
Ambi b	*Part Pärt*	5:52
	Ambient work for piano & string orchestra	
Lydia	*Case Study 1976*	3:56
	Modernist work for piano, harp, wind & brass	

D	*Tinkerbell*	4:23
	Dance for two pianos, keyboard & celeste	
First Species	*Mind games*	2:37
	Aleatoric miniature for solo piano	
	Invention for virtual prepared piano	5:09
	Aleatoric prepared piano solo	
Baby Bach g	*Driving the marimba*	3:28
	Piano, keyboard and marimba trio	
	Flight of the whole tones	3:20
	Fantasy for piano, vibes, harp & winds	
Sound Check a	*Raqam Sab'a*	4:02
	Piano duet in Arabic Hijaz scale	
F#	*Kaddish*	2:59
	Soundscape for piano, gamelan and synthesis	

Figure 4. Summary data for derivative compositions in *Piano Games.*

2.4. Manipulations

13 new pieces of music were composed from the 6 source Preludes. Each of these had a process of improvisation, formal manipulation, reflection and realization. Some pieces are recognizable with the main manipulations being related to rhythmic emphasis or subdivision. Others however are completely unrecognizable, transformed in tempo, instrumentation, pitch mapping, tempo, arpeggiation, playing emphasis and harmonization. A theme of the project has been a desire to see if the original preludes can be conclusively mapped to their derivative works. Where the original tempo, pitch or rhythm have been preserved this is straightforward, but in other cases the boundaries are not so clear. For example *Night light*, (CD2 track 3), derived from *Ambi* (CD1 track 5), appears to originate from the prelude *Lydia* (CD1 track 4), as it is in the same key with only one note changed in the scale. It also has a similar emotional quality to *Lydia,* which was an intentional part of the composition process. *Driving the Marimba* (CD2 track 10) and *The varying nature of wind* (CD2 track 1), have a great deal in common despite originating from different source preludes. They are both in the key of g, have a subtle rhythmic insistence and feature dialogue between the piano percussion and winds. Therefore it is not necessarily clear, or should it be clear where the exact boundary lies between source and derivative music. Although the pieces in CD2 have been collated according to the original preludes to allow greater reference between source and derivation this is not necessary in an artistic sense.

In order to understand this process the prelude *Lydia* and its derived compositions have been selected for study. This is because the two new works created form this prelude are dramatically different in regards to style and aesthetics. *Lydia*, the original is a response to the early 20th Century; tonal with pedalistic textures and a modal scale, in this case the Lydian mode from which its name derives. At times *Lydia* suggests the sounds of free jazz, although its limited harmonic content also imbibes a feeling of process minimalism. Its first derivative is *Case Study 1976*, a modernist work for prepared piano, harp, euphonium and tuba. *Tinkerbell*, the second derivative is a vibrant, ostinato based dance piece for two pianos, electric piano and celeste. *Tinkerbell* is immediately recognizable as it shares the same key and scale as the original, but *Case Study 1976* moves beyond the realm of rearranging the existing notes and constructs a new scale from the source data, which is combined with arpeggiation and an extended range in the piano and harp.

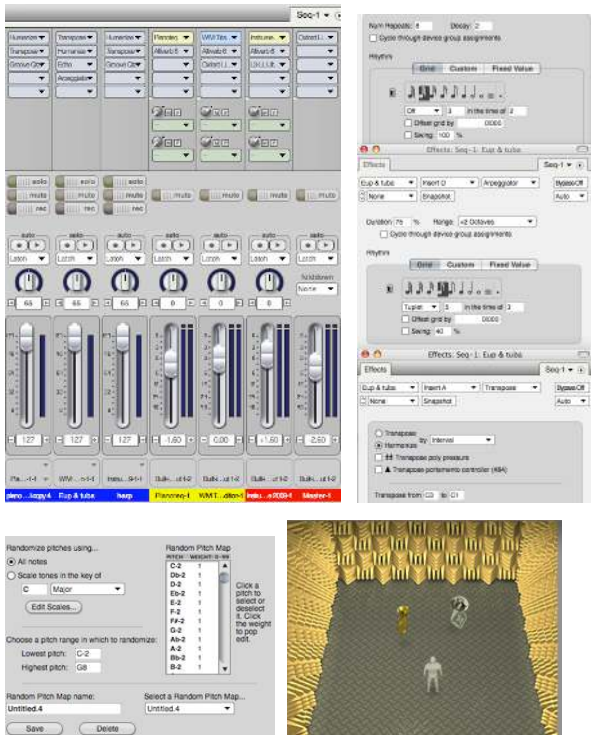


Figure 5. Example of mixing board, MIDI inserts and virtual instruments used in the production of *Case Study 1976.*

In this composition the prelude *Ambi* has been copied twice and assigned to different instruments, which in turn trigger virtual instruments. The score tracks trigger an iteration of Pianoteq for solo piano, the Garritan Personal Orchestra for harp, and the Wallander for euphonium and tuba. In this piece 10 MIDI inserts have been substantiated. Figure E also illustrates the MIDI processing used in the Wallander, which triggers both the euphonium and tuba. Seen from the top of the middle column in Figure E, they include a 32nd MIDI delay and an arpeggiator, which adds a range of two octaves to the

source sounded via a rhythmic filter set to a 5 over 3 tuplet. A harmonization of the initial data to two octaves below the source enables the Wallander to play different note combinations according to the ranges of the selected virtual instruments.

In the third column a chromatic pitch map allows all notes triggered by the Wallander, and therefore the arpeggiator and delay to be played in a randomly occurring chromatic scale. This is followed by a spatialization of the tuba and euphonium within a virtual anechoic chamber modeled in the Wallander. All the virtual instruments are then processed as audio tracks, a crude but effective way of modifying the sound of a virtual instrument. The Pianoteq is sent to a metallic drum reverb space as opposed to a concert hall, while the Wallander is sent to a small room and limited to allow greater presence and air in its 'performance'. The harp is sent to the same space as the piano with a greater tail and heavily limited to add some presence to the strings; something required to bolster the poor quality samples of the Garritan library. The tracks are then sent to a master track, which is then also limited. All the processing for this project has been undertaken on an entry-level laptop with 'in the box' plug ins available on any commercial DAW. Finally the score excerpt, shown in Figure 7 proves that the score has not been altered from the original prelude.

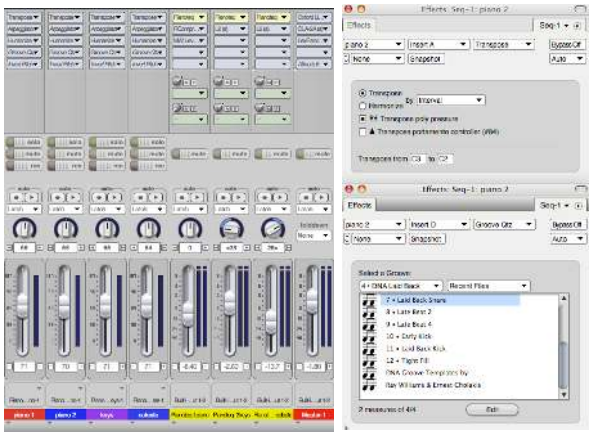


Figure 6. Example of mixing board and MIDI inserts used in the production of *Tinkerbell.*

In this composition the Prelude *Ambi* has been copied three times and assigned to different instruments, which comprise 3 different iterations of the Pianoteq plug in, one for piano, one for electric piano and one for celeste. 20 MIDI inserts are displayed in the mixer, of 14 have been substantiated, while each virtual instrument is sent to a limiter to avoid digital clipping. All the virtual instruments are also sent to a master track, which has the drafts of a limiter, compressor and two reverbs available for audition. The displayed MIDI plug ins display the Piano 2 track, transposing the data down a fourth and changing the rhythmic emphasis of the original to a programmed snare drum pattern. The score snapshot in figure 7 is taken from the Piano 1 and Piano 2 parts, which again shows that there has been no manipulation to the source data, as with *Case Study 1976.*



Figure 7. Identical score data from *Lydia*, *Case Study 1976* and *Tinkerbell*.

3. EVALUATION

Has the War been won? This is the question that beckons. There is little doubt that the process of re-composition in *Piano Games* is more radical than my previous projects, *Ordo Virtutum* by Hildegard of Bingen, and even *Namu Amida Butsu*, a solo work for shakuhachi that I exported as Garage Band loops. However, it must be asked if the resultant music is of sufficient quality to be classed as 'new composition'. A follow on question is whether this is a method of composition I might wish to use in the future, given its reliance on technology and its rigidly in regards to the initial score data. These are all subjects for ongoing debate that cannot be answered until the completion of this project. However to garner feedback I have addressed some of these issues in two lectures at the Queensland Conservatorium. In May 2011 I presented to the Music Technology Forum, a multi year undergraduate gathering of students, sessional lecturers and permanent staff. The response was mixed. While the music was broadly seen as having 'quality' the point was made that there might be a point in time in which the palette of manipulations was exhausted, where everything therefore sounded the same. Another respondent said that in order for this project to be conclusive the process should be undertaken on a number of audio platforms. (Cunio, 2011). Since that lecture a great deal of new music has been written in a number of music styles, which suggest that there is room for a significant amount of composition before the possibilities inherent in a DAW are exhausted, something I wish to explore in the future.

The research was also presented in August 2011 at the Queensland Conservatorium Higher Research Degree Colloquium, a gathering of Masters and Doctoral students that takes place once a week during the two semesters of the academic year. (Cunio, 2011b). In this presentation the response was again divided. A number of participants saw this as an arranging exercise, something akin to the tasks contained in composition technique tutorials. Other students applauded the work, seeing it as fresh, innovative and part of the melding of

popular and art music styles that is emerging in the 21st Century. It will be hard to gauge critical response until publication of this paper and dissemination of the accompanying CD, which should be completed in mid 2013. For this to happen there are still a number of decisions that must be made.

- Should the initial series of Preludes be realized on a Player Piano, played / improvised by a pianist and recorded, or presented 'as is'?
- Should the new compositions for piano be realized on a player piano, be mixed in a professional studio or presented 'as is'?
- In the case of realizations that have a significant algorithmic component should multiple realizations of the work be made before selecting a final 'new' composition?
- Should the MIDI files be made available to other composers to broaden the project?
- Is the musical material good enough to be released in a commercial environment? To be candid; does the music have artistic merit without the accompanying research?

In the meantime the results of this music will be placed in the web space of the QLD Conservatorium to enable comment, peer review and candid response, a major goal of the project.

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INTERACTING WITH SITUATIONS, INHABITED SPACES, AND TRAVELOGUES: A CASE OF A WEDDING IN THE XOCHIMILCO PARK.

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ABSTRACT

In this Artist Talk, composer Gerardo Dirié refers to an international collaboration with a Mexican virtuoso harp duo, ambience sounds from a wedding in the lake park of Xochimilco, and the expansion of artistic interactions with a city, its inhabitants, and their transit through the environment.

Gerardo Dirié, harpists Mercedes Gómez Benet and Janet Paulus collaborate with a situationalist workshop for architecture and design students at the UNAM (Universidad Nacional Autonoma de Mexico), led by Francisco Erzo and Jhavier Lloeza. Their multi experiential interfacing allows them to reach transformative levels of interpretation and knowledge, leading to a holistic educational experience that enhances the hybridization of selves with the environments in which the actions take place, all projected or published later in interrelated media.

Website: http://creaverbu.posterous.com/
www.gerardodirie.net

The presentation will include playing a CD with the studio version of “Arpegios a Xochimilco (for prerecorded sounds and improvising harps), and the projection of visual materials (quicktime movies, ppt, pdf).

THE TP-NEON: AN INFRA-INSTRUMENTAL CONTRAPTION WHICH MORPHS A MIDI TRUMPET BANDONEON-ERGONOMICS

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ABSTRACT

We describe a hardware-only improvisation and/or compositional set that merges a MIDI trumpet with a circuit bending interface that is inspired on bandoneon ergonomics by using the performer non-dominant hand - normally free in trumped playing. The instrument holistically abides by the principles of infrainstrumental design proposed by Bowens and Archers's [1]. It is termed TP-Neon and its integration within an audio and video analogue modular synthesizer, along with its playability and musicality, is discussed.

1. BACKGROUND

As a reaction to the complexities and interface setup pitfalls of laptop music improvisation, there is a growing number of composers and improvisers interested in increasingly reductionist forms of musical instruments [7]. Busy and dense possibilities of laptop improvisation are driving musicians to consider the integration of modular synthesizers and dedicated hardware equipment into their improvisational and compositional workflows. One step further is the hacking and modifying of existing musical/video devices building simple electronic contraptions. Often, the aim is to research and experiment with surprisingly fresh lo-fi hands-on approaches to sound/video generation [1].

Consequently, there has being a significant influence of lo-fi methods into pop music and academic composition and improvisation. Pop artists such as Bjork, Radio Head and Venetian Snares have readily incorporated these approaches in their recent works [4]. Concomitantly, these lo-fi methods have been featured in performance and composition work by contemporary composers such as, just ti give an example, such as, Nick Collins on his piece The Bowerbird in which several individuals each armed with a loudspeaker and battery make 'Victorian synthetic' sounds [2].

Recently, the concept of lo-fi instrument has being formalized by J. Bowers and P. Archer on his paper, Not hyper, not meta, not cyber but infra-instruments [1]. In this work the authors expand on already established concepts like 'circuit bending' [3,5], or 'hardware hacking [2,6] to define an emerging field of music and video improvisation which they term infrainstruments.

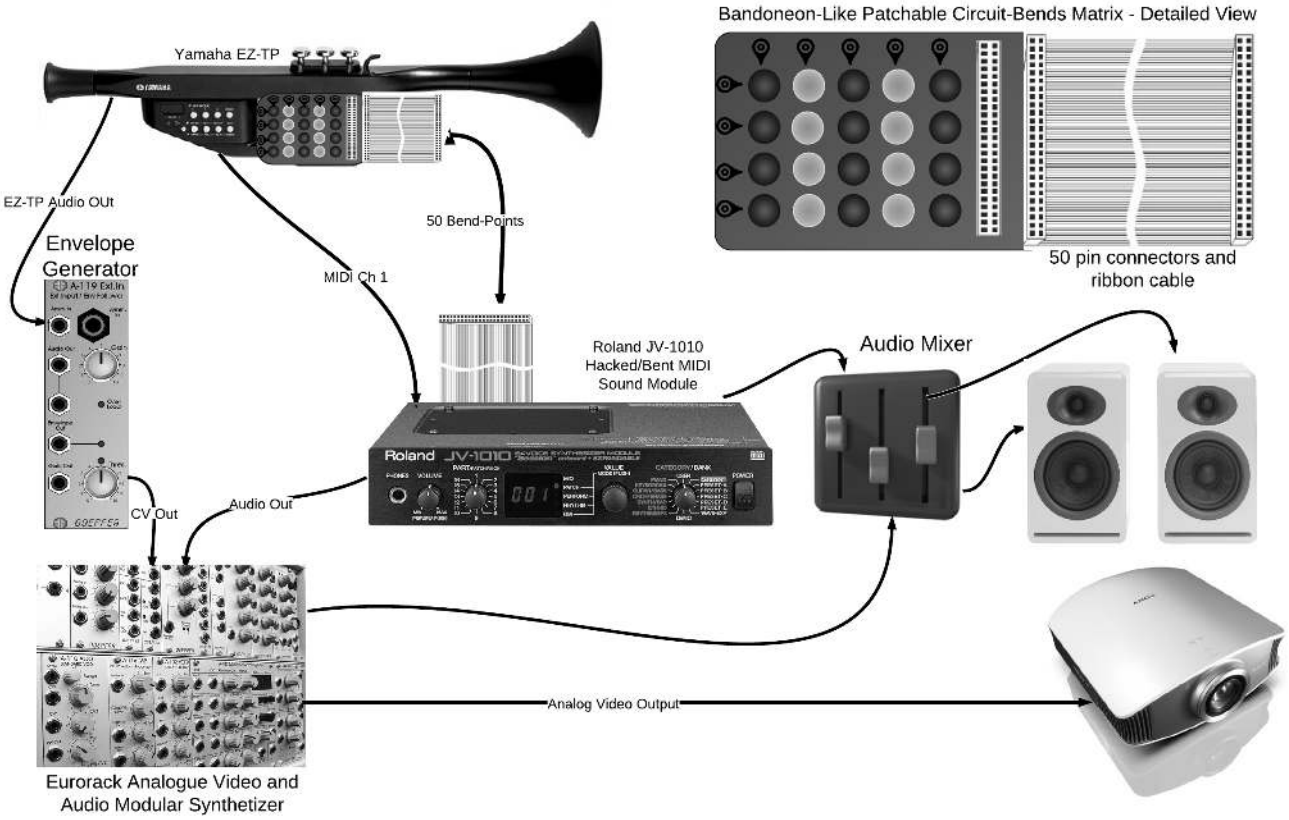
This paper describes a new improvisation set capriciously termed TP-Neon for audio and analogue video that has being inspired by Bowens and Archers's 5-axial definition of infrainstrument, for instance the TP-Neon: (1) is constrained to a limited interactive repertoire; (2) engenders relatively unpredictable music – for instance non-symphonic, non-orchestral; (3) is performed in ways that are predetermined by the architecture of the instrument; (4) is of restricted virtuosity but with a rich set of performance-driven unpredicted explorations and; (5) is ultimately engaging in terms of its visual and sonic appeal, is technically provocative and ergonomically challenging.

2. INSTRUMENT DESCRIPTION

The TP-Neon derives its name from the ergonomic morphing of a MIDI Trumpet (TP) and a Bandoneon-Like DIY control interface (Neon). The figure 1 depicts a numbered TP-Neon architecture. The TP-Neon controller based on a Yamaha EZ-TP MIDI trumpet with a DIY patchable momentary actuation matrix which can engage large number of circuit-bends' combinations with simple presses. The Matrix ergonomically resembles keys on a bandoneon so the TP-Neon instrument can be played with two hands: the non-dominant hand controlling the bend matrix and the trumpet keys played with the dominant hand. From the TP-Neon several signals emerge. The direct audio out from the EZ-TP responds to humming dynamics from the mouthpiece microphone. These are used to generate a control voltage trough an envelope generator which is used ad-hoc within the modular synthetiser. This CV is reused (multiplexed) and modified to generate several daughter CV signals that modulate analogue audio and video signals in the modular. Concomitantly, MIDI messages control a Hacked (Circuit-bent) Roland JV1010 ROM synthetiser module and the audio output is modified in real time by engaging circuit bent combinations. There is a total of 50 bending points that the authors has researched over the past 2 years and all bending points react with each other producing a very large number of bent combinations. This is helped by the DIY bend matrix which consists of momentary switches in a matrix array that can be freely patched to incoming bend points from the JV1010.

3. DISCUSSION AND CONCLUSION

Improvisation with the system results in a rich/complex audiovisual output that is generated from the dynamically controlled CV and audio from the TP-Neon audio processed trough an analogue modular synthesizer (A), EZ-TP MIDI controlled audio from the JV1010 being modulated by the circuit-bending matrix (Neon) combinations. The author has used the TP-Neon live and in his last LP [8] and finds that the technical and musical constrains of the instruments is a powerful creative driving factor. Also, the highly unpredictable character of the instrument when performing enhances the 'happening-effect' and uniqueness of every improvisation. Given that the TP-Neon architecture is reproducible, a notation system can be proposed if more prescriptive compositional works are desired. For instance trumpet notation can be augmented with matrix indexing notation to describe the bend points whilst classic notation of dynamics could be used to describe control voltages at the envelope generator stage. The number of circuit bent combinations from the bandoneon-like keypad allows for combinatorics up to the order of ten to the power of 17 (i.e. 50 bent points with a 20 key patchable switch matrix). In other words, each voice of the JV1010 can be resynthesised to produce around 1.3*1017, a great number considering the simplicity and accessibility of the interface. Also, since humming dynamics at the EZ-TP is converted to continuous control multiplexed voltages in the modular synth, it is possible to control fine nuances of analogue patches with just dynamics alone. For instance, humming softer or louder gives the performer/improviser great control over a number of CV inputs (filter cut-offs, pitches, amplitudes, timing parameters, etc) in real time. This gives the instrument a gratifying sense of nuance control but paradoxically further expends on the unpredictability of results during performance.



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DEVELOPING REAL-TIME SYSTEMS FOR CONCERT PERFORMANCE

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ABSTRACT

This artist talk describes the compositional and technical aspects of several of the author's compositional projects using the Max/MSP graphical programming environment. Musical choices and audio signal processing technologies used in the context of a real-time performance system often have a tightly interwoven relationship: sometimes technological developments are motivated by musical necessity, and conversely the construction of some of musical material is a consequence of technological considerations. This will be elucidated through examples from the author's series of preludes for flute, clarinet, and alto flute. In addition to discussing the signal processing effects used in these pieces, an important emphasis will also be made on the pitch tracking and score following techniques used.

1. INTRODUCTION

The idea behind a series of short preludes for solo instrument with real-time computer interaction was partially inspired by Mario Davidovsky's *Synchronisms* series [2]. As this collective title suggests, these pieces explore synchronization in addition to extending the timbral and gestural possibilities of the instruments used; this seemed like a useful general model for a series of short interactive pieces. In contrast to many recent compositions for instrument(s) and real-time computer interaction which sometimes overstay their welcome with audiences, the Davidovsky pieces are not particularly lengthy (a couple of them are indeed quite brief), and this mix of brevity and conciseness was also an influential factor as a musical model for the author's preludes.

2. IMPORTANT DEVELOPMENTS

The pieces which will be discussed in this talk all owe their existence to an important development: the gizmo~ object for real-time transposition of audio signals in the Max/MSP environment [4]. The gizmo~ object was developed by the author specifically with the first of these compositions in mind, and the subsequent compositions in the series also rely on it. Although by 2002 the signal processing capabilities of the Max/MSP program had been vastly expanded beyond the basic toolkit of objects that was provided with the first release of MSP [10], there were nonetheless a few necessary musical tools still missing from the environment. Some of these gaps in the tool set were addressed by diligent third-party developers who made their work available to the Max/MSP user community (for example Mille Puckette's pitch tracking objects, such as fiddle~ [8]), yet other tools were absent entirely, or of an unsatisfactory musical quality. One important musical item lacking in the environment was a way to transpose sounds in a natural, realistic manner, as the granular time-domain transposition unit provided (based on earlier analog models [1]) as an example patch with the program left a lot to be desired. Therefore the author embarked on the creation of a spectral-domain transposition object based upon published research into the phase vocoder and spectral domain effects by Laroche and Dolson [5]. The result was the development of the gizmo~ object as a standard part of the Max/MSP object set.

3. COMPOSITIONAL WORKFLOW

3.1. The Instrumental Score

The instrumental score of each of the preludes for instrument and computer was generally composed first, keeping in mind the electronic processing that would be used in different sections of the pieces. The realization of the computer part, in the form of a Max/MSP patch, was designed afterward according to indications in the instrumental score, at which point the score was often modified both for musical reasons and in order to accommodate technical limitations, such as unreliable pitch tracking. The melodic and harmonic aspects of the composition are freely composed, using no pre-determined compositional system for note choice. Generally, however, the melodic aspect of all the compositions follows diatonic sets of pitches that modulate to different pitch sets via common tones.

3.2. Audio Processing Design

Apart from some preliminary tests to try out sound processing ideas at the early stages of composition, the signal processing for these pieces was constructed based on the musical ideas and processing requirements indicated in the instrumental score. The processing was designed using Max/MSP, although the signal processing flow can be described in non-software-specific terms which could be used to re-create the real-time computer part of any of these pieces in the future using other audio signal processing software, if necessary. An example would be the block diagram for the *Prelude for Flute and Computer* (see figure 1). Each

of the preludes has its own distinct signal processing data flow, although the pieces do share some common processing algorithms, such as transposition and delay of the live input sound, recording into audio buffers and granular or algorithmic playback of the recorded sounds.

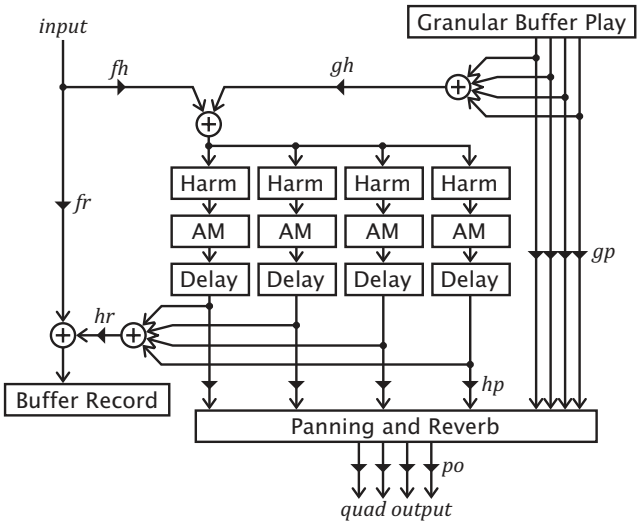


Figure 1. *Prelude for Flute and Computer* block diagram.

3.3. Compositional Choices and Necessities

A fundamental conceptual and compositional choice for the series of preludes is that the computer part begins with an "empty slate" – there is no pre-recorded material and all sounds originate from the onstage performer in concert. Some sounds from the live instrument are recorded for use later in the piece, while other sounds from the live instrument are delayed using multiple delay lines in the patch. The delay times used in the pieces are generally set using a tempo derived from the live performer.

One additional but unfortunate compositional necessity imposed on the score during the realization of the Max patches was the addition of notes in passages which had one or more repeated notes. Although human performers and accompanists are adept at recognizing discrete note events, even when the same note is repeated multiple times in a musical passage, the current computer-based score following systems are less adroit. Therefore additional notes, not harmonically related to the repeated notes, were added in some places throughout the score in order to aid the computer pitch tracking system. Although future pitch trackers may be more capable of dealing with a wider variety of musical input, these extra notes have, nonetheless, become part of the final composition.

4. EXISTING TECHNOLOGIES

There are two important pre-existing techniques without which the preludes could not exist: pitch tracking and score following.

4.1. Pitch Tracking

The first two pieces in the series, the *Prelude for Flute and Computer* and the *Prelude for Clarinet and Computer,* use a modified version of an older pitch tracker by Miller Puckette called pt~, which was originally designed for Max/FTS on the Ircam Signal Processing Workstation (ISPW) [7]. This object has proven to be much more responsive than newer pitch tracking objects for MSP such as Puckette's fiddle~ and sigmund~. Although with the flute's timbre, the pt~ object will often report erroneous pitches one octave from the actual fundamental, the incorrect octave can be corrected at the score-following stage For the clarinet, the pt~ object is rarely incorrect in its pitch and octave estimation, due to the clarinet's odd-harmonic spectrum, thereby allowing for very accurate and fast on-the-fly transpositions of individual note events.

For the *Prelude for Clarinet and Computer*, the original pt~ object needed to be modified in order to allow larger window sizes for the object's internal FFT, thereby accommodating more accurate pitch tracking of the lower notes in the clarinet's chalumeau register. Another modified version of the object, named ptg~, was also compiled specifically for this composition. It can be used inside a standard MSP pfft~ subpatch, and therefore contains only the pitch tracking code, not any code specific to the Short Term Fourier Transform, which is handled by the pfft~ object. Having the pitch tracker inside the pfft~ subpatch is useful for two reasons: 1) if a the pfft~ object is already being used in the patch for spectral-domain sound processing, a second (computationally expensive) Fourier Transform is not necessary when adding pitch tracking to the patch, and 2) it facilitates synchronization of the reported pitch estimation with any spectral-domain effects that need to use that information.

4.2. Score Following

An updated version of the score following system designed by Miller Puckette and Cort Lippe on the ISPW was used for the flute and clarinet preludes [9]. For the moment it works quite robustly with these two pieces, and makes use of objects in the standard Max/MSP distribution. The system uses the afore-mentioned pitch tracker in conjunction with the Max detonate and qlist objects. Both objects have been a regular part of the Max distribution since the mid-1990s. The three-part system works thus: the pitch estimation from the pitch tracker is sent to the detonate object, which contains a piano-roll MIDI representation of the entire score [6], in addition to cues which are output as the follower advances through the score in response to the pitch-tracked input. These cues are then sent to the cue list in the qlist object, and the events of each cue are then remotely sent to control different parameters in the patch.

4.3. Anticipatory Score Following

The *Prelude and Fantasy for Alto Flute and Computer* uses a newer pitch tracking and score following system:

Arshia Cont's antescofo~ object [3] for score following and event triggering. Although this third party object, available from IRCAM, is still in development, it is already stable enough to have been used successfully in musical productions in and outside IRCAM. The advantages of antescofo~ over the older pitch tracking, score-following and message-sending system is that it is particularly robust with fast musical material, has a built-in tempo estimation which can be used for tighter rhythmic interaction, and from a compositional point of view, merges the (text-based) score representation with the event and message-sending, allowing both the musical score and signal processing parameters to be viewed together in a single document.

5. COMPUTING TECHNIQUES

In addition to the techniques described above, the preludes share several novel signal-processing techniques. Some of these techniques are extensions of existing technologies, whereas others were designed specifically for use in the pieces.

5.1. Rhythm and Tempo Detection

A series of cues are used at several occasions in the pieces for clarinet and alto flute in order to set the tempo of various computer-processed events. In the *Prelude for Clarinet and Computer* the first of these occurs at the outset of the piece, where a series of cues sets the tempo for a sequence of transpositions of the live note. Additionally, the opening notes of the B section of that piece are used not only as a bridge from the musical material at the end of the A section, but also to allow the clarinetist to define the tempo for the following section. In the fast fantasy section of the *Prelude and Fantasy for Alto Flute and Computer* timed note sequences are adjusted according to the tempo of the regular quintuplets played by the live flautist. In both these pieces, a collection of delta-times are obtained from a series of cues, the values are sorted and the highest and lowest values are thrown out, since they could potentially be false readings, or inaccurate rhythms on the performer's part. The remaining timing values are averaged, producing an accurate tempo estimation. This tempo is used to calculate the timing and rhythm of musical events executed by the computer, including accurately setting the canon entries in the clarinet prelude, or inserting notes during the performer's rests in the alto flute piece. In all cases this allows the performer to set a comfortable tempo for the computer interaction in a natural, and musical manner.

5.2. Timbral Modification

In the clarinet and alto flute pieces some special timbral modification techniques are introduced. Since the clarinet's instrumental acoustics generate odd harmonic partials only, a principal effect used in the clarinet prelude is to fill-in the even-harmonic gaps in the clarinet's spectrum. This has the effect of transforming the clarinet sound into that resembling an alto saxophone. The effect is achieved simply by pitch-tracking the clarinet sound and using the fundamental frequency as a Hertz value to frequency-shift the audio. When the frequency-shifted audio is mixed with the original signal, it fills in the even harmonics in the clarinet spectrum. The frequency-shifted audio may be added to the original sound at any arbitrary volume level, thereby allowing smooth changes in timbre along the continuum between clarinet and saxophone. The alto flute piece also makes use of a special transposition unit which allows individual partials of the sound to be moved to alternate harmonic or inharmonic locations in the spectrum. This is used particularly to transform the spectrum of the alto flute into a bell-like spectrum.

5.3. Spectral Delay with Transposition

Although all three preludes are structured with four transpositions of the original instrumental sound which can each be delayed an arbitrary amount of time, two sections of the clarinet piece in particular required a special recursive transposition and delay effect: the end of the A section and the end of the coda. For these places an alternate spectral-domain transposition patch was made. In it, each of the four gizmo~-based transpositions is delayed a certain number of FFT frames in the spectral domain sent to the subsequent gizmo~ object. The fourth transposition is sent in a feedback loop to the input of the first. By using carefully-adjusted parameters for the transposition and volume attenuation of these transpositions, a "hazy" or "out of focus" decaying echo sound is produced, quite different from the effect that would be made if the effect were designed in the time-domain.

5.4. Glissandi

Although not a novel technique per-se, the use of an S-curve for portamento is more musically effective than a linear glissando. The standard curve~ object for MSP implements a signal-rate quasi-exponential or quasi-logarithmic curve. Two curves can be triggered successively in order to create an S-curve, which approximates well the effect of a human glissando, such as that of a finger on a violin string: starting slowly, accelerating through the center portion of the glissando and slowly easing into the destination pitch.

5.5. Pitch Grid

The most important development for the *Prelude for Clarinet and Computer* takes a small inspiration from the world of algorithmic composition, albeit performs it in the context of a digital signal processing effect. The author designed a pitch-class corrector as an event-level Max patch, whereby all incoming (MIDI) pitches are modified to fit to a pre-defined pitch grid. Each input note is compared to a given pitch-class set, and then transposed to the nearest note in the set. When used in conjunction with the robust ptg~ pitch tracker and the gizmo~ object, each note played by the clarinet may be

adjusted to a different fundamental frequency in real-time. In some ways this is the antithesis of the ubiquitous auto-tune effect in popular music, as it does not correct the fine-tuning of the note, but rather moves the entire note event to a different pitch-class level, preserving vibrato and microtonal nuances inherent in the original sound.

This effect is used heavily in the canonic B section of the composition. In this section the computer-delayed canons of the live clarinet sound are transposed to different key centers, but the individual notes of the canons are modified in order to match the current harmonic context implied by the live soloist. The effect enters subtly at first, but becomes more obvious as the piece progresses. By bar 31, the transposed canons are no longer following the original diatonic material, but rather are transposed to fit to a triadic harmonic grid.

The *Prelude and Fantasy for Alto Flute and Computer* also makes use of a similar on-the-fly transposition technique for the algorithmic playback of the short notes recorded in the opening section of the piece.

6. CONCLUSION

The technology and musical ideas behind the author's preludes for instrument and computer are at once a culmination of previous musical and technological work as well as a springboard for future compositions in the series. Especially, the algorithmic nature of the processing in the latter two of these pieces lends itself to the concept of hybridizing digital signal processing techniques with ideas taken from algorithmic composition.

7. ACKNOWLEDGEMENTS

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THE RECORDING AS ARTEFACT OF PLAY

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ABSTRACT

Zak (2010) states that sound recording provides a means for combining "musical and technological ideas and actions in a multifaceted creative process…whose resulting works expand traditional concepts of musical content, meaning and style (p.321). And yet, George Martin recalls when he first started out in the music business, a record producer's "ultimate aim" was to fashion audio facsimiles of 'live' performances with great earnestness (Badman, 2000: 256). Notwithstanding the practice of linear editing - commonplace ever since magnetic tape replaced direct-to-disc recording (Southall, Vince & Rouse, 2002) - Martin's own post-*Revolver* (1966) output increasingly constituted an illusory fabrication. Mulitrack recording (though long extant by the late sixties) facilitated an additional means of joining together many disparate musical utterances 'as if' they were one (albeit, now in a concurrent manner). Such developments in technology alone however, didn't inspire practitioners to fully exploit the interactive potential of the medium. Instead, a thoroughly *playful* frame of mind informed such fanciful 'productions' as The Beach Boys "Good Vibrations" (1966) and The Beatles' *Sgt. Pepper's Lonely Hearts Club Band* (1967); and in doing so, ushered in a new standards of practice (Miles, 1998; Leaf, 2004).

Similarly, if the proliferation in recent years of low-cost, broadcast quality Digital Audio Workstations (DAWs) is to facilitate anything more than greater convenience for practitioners, a corresponding challenging of certain tacit assumptions must be forthcoming. Brian Eno states: "'As soon as the music is on tape…it ceases to be a purely temporal art, a performance art, and becomes a plastic art.'" (quoted in Sheppard, 2009: 311). The manipulation of sound so well facilitated by DAW technology (which brings multitrack recording, non-linear editing and automated mixing to the masses), potentialises a process of *combinatorial play* (Hadamard, 1945). Such an approach can best be facilitated by a frame of mind where "implications beyond the present moment" are temporarily banished from awareness (Apter, 1991: 14). In this way, the often product-orientated concept of 'performance' can enlarged to include careful scrutiny of the phenomenological flavour of its processes. By reframing contemporary recordings as assemblages of 'artefacts of play,' rather than algorithmic (or by-rote) constructions, it will be argued that future record makers can be encouraged to more fully exploit the creative possibilities that DAW technology engenders.

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Body as Instrument – An Exploration of Gestural Interface Design

Mary Mainsbridge

ABSTRACT

As a live electronic musician and vocalist, the challenge of creating dynamic, engaging performances with digital technology has prompted an exploration of non- contact gestural systems that offer increased subtlety and intuitive control. Existing gestural controller technologies offer vocal performers with opportunities to expand their natural vocal capacity through physical manipulation of digital signal processing by tapping into the body as a natural instrument. This paper will document preliminary investigations into ways the human body can be integrated as an input into gestural systems within musical performance. The expressive potential of these technologies will be explored by tracing recurrent issues in the design of non-contact gestural controllers. Two main issues are identified – the development of suitable mapping strategies and design of feedback mechanisms. Finding simple mapping approaches that accommodate the complexity of musical performance is recognised as a primary motivation for developers. There appears to be a need among performers for interfaces that can accommodate a broad range of gestures and also communicate the relationship between gesture and sound to the audience. Vocalists in particular require a vocabulary of gestures that create flexible pathways for expression. Most gestural controllers reflect novel approaches rather than a uniform approach to gesture recognition. Whether future visions of a universal gestural language that is body-centric and not device based are possible will be discussed. Another way of utilising and increasing the range of expression available through non-contact gestural systems is ensuring that adequate feedback is present. Visual feedback is one way of delivering more information to the performer, which also establishes cues for audience comprehension of musical processes and stronger relationships between the performer's actions and sounds created. This design feature is often overlooked in the development of non-contact gestural interfaces yet can possibly deliver the finer control necessary to accommodate individual forms of performer expression.

ACOUSMATIC ART AND STORYTELLING

David Rylands

ABSTRACT

Artist Presentation - David Rylands

In this presentation I will discuss my current ideas on the interplay between multi-channel acousmatic sound art and storytelling.

I will describe my procedures for collecting sound recordings, designing loudspeaker arrays and installation approaches, and developing sound manipulation techniques for spatial sound environments.

Following this, I will talk about orality and storytelling in acousmatic composition. I will demonstrate examples in my work where I integrate the communicable voice with everyday, and abstract, sound objects and soundscapes.

I propose that the integration of storytelling with referential and abstract sound types can provide a basis for the presentation of personal contexts in which the material is unified by a clear and significant purpose.

I believe that the infusion of these materials contributes to a meaningful production process and greater communication with varied listeners.

BIOGRAPHY

David Rylands was born in Brisbane, Australia in 1979. He graduated with a Bachelor of Arts in music and visual art in 2000, followed by a Bachelor of Music Technology (Hons) at the Queensland Conservatorium in 2004. His current creative work is in experimental sound amplification and recording approaches for loudspeaker based composition. David is currently employed as the music and sound technician at the University of Auckland School of Music, New Zealand.

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THIS IS CHRISTOPHER: EXPLORING THE "SOUNDING SELF" THROUGH ELECTROACOUSTIC TREATMENT OF AN HISTORICAL AUDIO RECORDING

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ABSTRACT

In 2011 I created a fixed media electroacoustic piece, *This is Christopher,* based on a tape recording from the early 1980s. The recording was made by my grandmother and 2-year-old cousin Christopher as an "audio letter" to my family. In it, Nana told stories, chatted, and sang songs and nursery rhymes with Christopher. While listening to it I was struck by the many layers of significance in this historical audio document, in particular the process by which you can hear Christopher learning to speak and to express himself with sound and music. He listens, imitates, repeats, sings, improvises, and engages in verbal and musical play with Nana, including learning to say the phrase, "This is Christopher": the ultimate verbal declaration of self.

As one of the intended recipients of this "audio letter", turning it into a work of sonic art was an act with deep emotional significance for me. Not only was it a poignant record of a loving relationship between grandparent and grandchild, but also a personal tribute to family members. As would be expected, this informed the compositional process and treatment of the sound material. Furthermore, I've recently started teaching music to children in this age group, adding yet another layer of personal engagement.

In this talk I'll discuss the fascinating process of creating this piece. I'll focus on the compositional process, and also discuss technical issues and theoretical aspects, drawing on the work of Cathy Lane (2006).

This is Christopher (8:00) – awarded first equal in Waikato University Douglas Lilburn Composition Awards 2011.

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12 POINTERS FOR INTERNET2 REAL-TIME ELECTROACOUSTIC COMPOSITION/PERFORMANCE

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ARTIST'S TALK

Using internationally linked high-speed broadband research networks (both IPv4 and IPv6) allows multiple bi-directional audio, video, OSC data, and graphics channels for real-time electroacoustic music composition/performance synchronously. Based on extensive work over the last three years from a research, performance and composition perspective (see Fields & Whalley 2012, for example), the presentation looks at twelve useful pointers others might consider when venturing into this area (Whalley 2012), as electroacoustic music studios become increasing connected real-time globally. This covers:

1. SW/HW - How to get connected.

The use of JackAudio/JackTrip to route audio, OSC Groups for data, video conferencing units for visuals. High-speed research networks such as KAREN and AANET, static IP addresses, firewalls and IPv4/IPv6, and the need for routing and internal testing.

2. Production Space.

Creative Goals (Weinberg 2005) influence the systems used and built for a work – extremes being musical structure/composer controlled, and a processes-based approach. A structure-centred approach allows participants (agents and/or humans) to fill prescribed musical or performance outcomes. A process-centred approach explores possibilities or fulfills goals through collaboration or competition, and experiences may differ with each session.

3. Network Arrangement.

For the exchange of real-time audio two useful models are: a Peer to Peer Approach that has no central sound server - each peer streams to close peers making a cluster network, and each peer transmits many and receives many streams with no central mixing. A Star Approach provides a central server that collects individual outputs and sends out individual streams. Peers then transmit one stream and receive many, and they can alter individual streams

4. Latency and distances.

Latency cannot be avoided, and the choices are cope accept it, ignore it, or play with it (Carôt, Rebelo, Renaud, 2007). Synchronously, the two choices available they put forward are a Realistic Jam Approach (RJA) and a Latency Accepting Approach (LAA). LAA is more suited to electroacoustic music, where one learns to compose/improvise in ways that timing is not critical, and/or use strategies to compensate for it in aesthetic terms. Latency/delay is also a creative opportunity, such as using network delays as multiple musical delays.

5. Production Integration

Follmer (2005) sets out a three dimensional matrix to assess approaches to the NMP idiom that can include Internet2 practice. The matrix maps: Interplay with Network Characteristics – how the structural characteristics of the network shape the work. Interactivity and Openness or the extent that users/audiences can join in with and influence outcomes. Complexity and Flexibility – or how musically/sonically flexible the outcome can be. Follmer suggests that the production approaches best suited to the medium are hypermusic (new hybrid instruments); real-virtual/space sound installations; and algorithmic/generative installations.

6. Linear/non-liner shift and communication.

In general, linear/edited electroacoustic music has often sacrificed spontaneity and interactivity for the sake of deterministic control of increasing Parameter Space, where time is secondary to innovation and reflection toward perfect product. Interactive and generative systems often sacrifice aspects of real-time Parameter Space in favour of spontaneous outcomes that can be multiple and varied.

7. Relationships.

Roles and Motivation**.** Considering motivates people to want to invest in and take part in telemusic events in a cost/benefit sense is vital, given that in contrast to on-site music making methods it is often complicated, expensive to assemble, and more time consuming. Access to Internet2 is also limited globally to a few, and infrastructure still favours partners in closest physical proximity to each other because of time delays.

8. Motivation, net access, payoff.

Why do it? It allows the forging of new relationships that can be artistically rewarding beyond what could be produced by other means, but that these relationships take time to build through rehearsal processes. In tandem, the sense of playing simultaneously in both real and virtual environments can be rewarding in the sense of extending musical proximity through co-located action.

9. Dramaturgy models in Internet2. Relationships models (Rebelo 2009) based in traditional performance practice to get things done through Internet2: Projected Dramaturgy - one node to act as the author and other nodes to contribute or project to it; Directed Dramaturgy - authorship remains with a group or individual who takes on a director's role. Distributed Dramaturgy - where authorship is retained by each node, while contributing expertise and content to shared production.

10. Relationship: machine role considerations

Given non-trivial machines and OSC messages can be exchange between individuals/machines and between groups through Internet2, the role of intelligent machines implemented through multi-agent technology needs considering (Whalley 2009). Practitioners have two choices at or between nodes include: a multi-agent system that is self-contained (generative model), or one that can include external input (machine or human) to make a generative improvisational model using intelligent machines

11. Instrument choices.

Beyond just adding effects to acoustic instruments as new hybrids, telemusic has its own aesthetics, and needs to draw on instrument building traditions suited to this environment, and optimize performance for it. This extends the LAN centric laptop orchestra models exploring Internet2.

12. Changing roles: makers/machines/audiences.

The new space anticipates extending current interactive music paradigms, changing the traditional roles of composer, musician and producer. It also anticipates the creative role programmers are likely to play in Internet2 works, the further deployment of intelligent machines, and more participatory roles for audiences.

Extending current work using Video Conferencing Units (VCU) as one part of linked performances, our current work with international partners explores a means of distributing graphics scores that can be altered in real-time across multiple countries over a fourth network channel in addition of VCU, OSC Groups, and JackTrip Audio. This investigates what is useful from both a pragmatic performance perspective, and what is useful as a means to enhance the visual aspects of live performance layers in electroacoustic music. This is because all collated performance outcomes can also be streamed to the public Internet in real-time, extending the worldwide audience for outcomes in this new medium.

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DEMONSTRATIONS

JAM2JAM XO: AN EDUCATIONAL GENERATIVE MUSIC ACTIVITY FOR THE OLPC XO

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ABSTRACT

Jam2jam XO is a software application developed for the One Laptop Per Child (OLPC) 'XO' laptop, an educational computer designed for use by children in poor or developing countries.

Jam2jam XO is one of family of jam2jam educational multimedia software instruments. While jam2jam XO adopts techniques used by the other jam2jam's, jam2jam XO has been designed and developed to take particular advantage of the resources available on the XO laptop. The XO laptop boasts a number of innovative features, including wireless mesh networking and the 'Sugar' interface.

The jam2jam project holds a philosophy around teaching and education that shares much in common with principals held by the OLPC initiative. By developing a version of the jam2jam software for the OLPC XO laptop, jam2jam XO looks to further the objectives of both the jam2jam project and the OLPC initiative.

Having achieved over 30 000 downloads, jam2jam XO continues to be a popular activity used throughout the OLPC community.

1. ABOUT THE OLPC XO

The One Laptop Per Child (OLPC) initiative, created by Nicholas Negroponte at the Massachusetts Institute of Technology, aims to provide children in poor and developing countries with a rugged, low-cost, low-power, network-connected laptop. [1,8] To date approximately 2.5 million laptops have been distributed to children throughout the world in many places including Uruguay, Nepal, and Mongolia. Rural communities in Australia have also received laptops distributed by OLPC Australia [2].

The OLPC XO laptop is a low cost, rugged, low-powered laptop with built-in WiFi mesh networking, long lasting battery, USB ports, and a screen that is visible in daylight hours.

The Sugar Learning Platform is the 'interface' for the XO laptop. Sugar sits somewhere between a desktop and an operating system. The Sugar interface is the means by which a user interacts with applications - called Activities - saved data, and other network connected XO laptops.

Interaction with Sugar is quite different to interacting with a conventional operating system such as Windows, Mac, or even Linux: There are for example no files or folders in Sugar. Instead, snapshots of activity sessions are stored in a single location called a Journal.

Connecting to other XO laptops is done simply by 'sharing' an activity with other XO laptops in the 'neighbourhood'. The XO laptop can establish the connection using a variety of methods, either through a shared server, with the aid of a wireless router or directly with other XO laptops using peer to peer wireless mesh or ad hoc networking.

For a developer, Sugar offers an API to build a GUI, access devices and manage network connections. Sugar is built on a cut down version of Fedora Linux, and so many Linux resources are also available. The Sugar API is written in the Python programming language, and many of the additional libraries and toolkits provide Python interfaces [3]. Sugar activities are typically written in Python, although other programming languages can be used.

A Core Principal of the OLPC initiative is to be Free and Open Source [4]. As such all Sugar activities are released under an open source license. Jam2jam XO is released under the version 2 of the GNU General Public License.

2. ABOUT JAM2JAM XO

Jam2jam XO development began in 2009 to 2010 with the support of the Australasian Co-operative Research Centre for Interaction Design [5]. Jam2jam XO currently exists as a private project of the author.

The jam2jam suite of applications forms part of the Network Jamming project, led by Steve Dillon and Andrew Brown. All the jam2jam variants act as educative collaborative multi-media tools for people to create music and/or imagery. The software aims to enhance engagement and learning by being fun, and providing social interaction [6].

2.1. jam2jam AV

Jam2jam XO most closely resembles jam2jam AV. Built in the Impromptu live-coding environment, running on Mac OSX, jam2jam AV offers parameter control over generative musical transformations. The jam2jam AV user interface offers free movement of an instrument icon in a two dimensional space. The horizontal and vertical movement correspondingly adjusts two musical parameters being performed by the

instrument. Different instruments can be selected, resulting in different musical transformations for each instrument.

Jam2jam AV communicates with other jam2jam AV's on a local network or over the internet, so-as to allow virtual 'bands' to collaborate. The audible result of the collaboration is duplicated in each connected jam2jam AV, i.e. the music sounds the same on each computer [7].

Jam2jam AV also provided video manipulation, with pre-recorded video's, still images, or live video from a camera able to be processed. It is worth noting that jam2jam AV can consume a considerable amount of CPU and memory resources on a modern Apple computer.

2.2. Using jam2jam XO

The jam2jam XO 'Activity' runs on the Sugar Learning Platform for use on the XO Laptop. Jam2jam XO borrows some concepts from jam2jam AV, most notably the two dimensional parameter control. However the implementation of these concepts is significantly different. In part the differences are needed to accommodate differences in hardware, computing power and software resources between a modern Apple computer and the OLPC XO [8]. As jam2jam XO was also a complete re-write of the jam2jam software, the opportunity was taken to re-think much of the user interaction in this environment.

In keeping with OLPC's principal of collaboration, jam2jam XO is most effective as a networked activity. When another player joins a shared jam2jam XO activity, each player gains control of an instrument. i.e. one performer may operate the guitar, while the other performer operates the drums. Audio for the entire musical performance is shared across the XO laptops, such that one performer will hear only the guitar audio track from their XO laptop, while the original player operates the drums and hears all the remaining instruments from their XO. In this way, a 'band' forms with each player contributing to the audible result. The band is limited to four performers, with each player taking on a unique instrument.

Visual effects are relatively limited, as image and video processing can easily interrupt music playback on the XO laptop. The XO laptop camera is used to take pictures of the user. An accumulation of captured images is cycled through in time with the beat of the music.

3. INSIDE JAM2JAM XO

Jam2jam XO is written almost entirely in the Python programming language. There are a number of Python interfaces employed to access other tools included in the Sugar environment: The user interface makes use of a mixture of PyGTK – Python wrappers over the GTK GUI toolkit, and the 'olpcgames' library – itself a wrapper for Pygame – a 2D game and multimedia library. The olpcgames library also offers access to XO networking and the camera.

The musical styles are stored as MIDI files. These files are read, and the notes modified by algorithms, before being sent to a synthesiser.

The synthesiser in use is Csound, which is also a standard inclusion in the Sugar environment. Csound runs in real-time, with the 'notes', sent directly from the Python environment through the Python bindings of the Csound API [9]. Jam2jam XO generates appropriate Csound orchestra code at launch time, based on which MIDI files it finds in its activity bundle. This allows the addition, or change of musical styles to jam2jam XO as simple including new MIDI files into the activity bundle.

3.1. Timing

Timing of notes using the standard Python modules was found to be too inaccurate for musical purposes. The Pygame environment offers an accurate timing option, however the increase in CPU usage and the relatively slow frame rate also proved limited for musical use. The Csound API however offers a callback function which can repeatedly execute a Python function at the Csound control rate. This is fast enough for accurate musical timing. Note that Csound itself runs in its own thread, uninterrupted by Python threading or interpreter locking issues.

Jam2jam XO uses a scheduling model loosely based on the scheduling used in the Impromptu live-coding environment [7]. Events (such as notes to be sent to Csound) are sent to a scheduler queue with an execution time specified in the future. The callback function, running at the Csound control rate, polls the queue for pending events, checks the scheduled execution time with the current time and executes anything due or past due. When the event is executed it is removed from the queue.

3.2. Synchronising XO's

When a jam2jam XO activity is shared across a network to form a band, jam2jam XO attempts to play all the instruments together in time. On a fast connection a fixed latency calculation might be good enough. However on the low powered XO, with a possibly poor network connection, running a non-real-time operating system with multiple threads in a higher level interpreted programming language, the synchronisation task requires more care.

jam2jam XO attempts synchronisation using a model resembling Simple Network Time Protocol with some added sanity measures to accommodate erratic timing messages [10].

One jam2jam XO becomes the reference clock. It periodically broadcasts a message about where it is up to in the music to the other connected 'band members'. When the band members receive this message they align themselves to the same point in the music.

There is typically some delay in the time it takes the reference jam2jam XO to broadcast a message to the time the other jam2jam's receive it. This delay is calculated by the sending of a time-stamped round-trip message to tvia the reference jam2jam XO. The latency of the broadcast time message is assumed to be half the time it took the round trip message to return.

The accuracy of the latency calculation and its correspondence to the broadcast message delay can be adversely affected by numerous factors which have been mentioned above. This situation is somewhat mitigated by calculating the average of an accumulation of recorded delay calculations. The delay messages are also filtered so-as to ignore rogue results which would otherwise skew the data.

Despite these efforts, a poor or variable network connection, or timing errors in the reference jam2jam, will still cause the music to get out of time. In most cases however timing errors which do creep in eventually correct themselves.

4. MUSICAL TRANSFORMATIONS

Musical transformations occur by moving an instrument either horizontally or vertically in the main interface. When a parameter is selected for transformation along an axis, the user can control the amount of transformation that is applied to that parameter.

The transformations used in jam2jam XO are a mixture of audio modifications as well as changes in the performance of musical notes. The transformations employed are; pitch stretching/compression, rhythmic density, articulation, volume, and timbre. A brief explanation of these transformations is provided below. These methods are described in more detail in the paper 'Adaptive Music Techniques' by Brown and Kerr [11].

4.1. Pitch stretching/compression

An increase in pitch parameter has the effect of expanding the range of pitches performed in an instrumental part. Thus a melody line will have a larger pitch contour, with greater interval sizes. Chordal parts will have larger steps chosen to describe the chord.

Conversely, a decrease in the pitch parameter will reduce the pitch range, such that at the minimum level, all pitches will centre on a single note.

The drum section represents a special case. In jam2jam XO, an increase in pitch effectively filters out lower drum hits (bass drums and toms for example), while a reduction in pitch will filter out high pitched percussion. A side effect of this approach is a reduction in rhythmic density toward the extremes, with many drum ‘hits’ not being performed.

4.2. Density

The approach to modifying rhythmic density differs in jam2jam XO when compared to jam2jam AV. In the latter case three versions of a musical track (recorded as MIDI files), each representing differing levels of rhythmic density are mixed.

In jam2jam XO only a single musical track is used. At maximum density the music is played unaltered. A reduction in the rhythmic density filters notes out of the performance. The selection of which notes are removed is based on a hierarchical interpretation on rhythmic importance in a 'beat' or bar.

Reduction in rhythmic density does not lead to silent gaps in the music. Instead, the durations of the remaining notes are lengthened so-as to occupy the space left by the removed notes.

4.3. Articulation

Articulation is implemented as a duration multiplier on each note. This has the effect of increasing or reducing the duration of note lengths. Overlapping notes are allowed in jam2jam XO, with the exception of the Bass line, which is implemented as a monophonic instrument.

4.4. Volume, Timbre and Tempo

The volume and timbre parameters perform simple audio transformations. Volume adjustment literally changes the output level of an instrument. Timbre modification passes the signal through a low pass filter. Because the frequency response of the XO built in speakers is poor, and frequencies less than 200hz become inaudible, some distortion is applied to the signal as the cutoff frequency gets lower. This maintains the presence of higher partials in the sound.

At this stage tempo modification has not been implemented in jam2jam XO, although it is planned to be included in future releases. An approach to tempo change that maintains synchronisation between XO's in a network situation is being tested.

5. THE IMPACT OF JAM2JAM XO

Jam2jam has been a popular activity on the XO. At the time of writing it had been downloaded 34,600 times. In a single day on April 8th, 2010 it received 417 downloads, and for some time was the most popular 'Media' activity downloaded at Sugar labs.

Information on its actual use is harder to assess. A single download can be further distributed throughout a school from an 'XO school server', suggesting there could be many more users than downloads.

Direct feedback from end users is uncommon, as there are numerous layers of support available in the Sugar/OLPC community, including teachers, OLPC country deployments and the wider OLPC and Sugar community.

Instead, requests from 'higher up the tree' are more common. For example, there have been documentation and technical queries from the Plan Ceibal initiative in

Uruguay, responsible for managing half a million OLPC XO's.

The open source nature of Sugar and Sugar Activities also means that many technical issues are resolved without the need for the original developer to be approached. Modified versions of jam2jam XO have appeared with contributions from other developers to fix platform specific issues. A modified version of jam2jam XO was included in a recent Australian build of the 'XO server' (XS) software.

6. THE FUTURE OF JAM2JAM XO

Jam2jam aims to be an engaging, useable and creative activity for both children and teachers. In order to ascertain whether these aims are being met a form of monitoring and evaluation is required. To conduct some form of monitoring, avenues of obtaining more direct feedback from actual users are currently being explored.

Jam2jam will increasingly adhere to OLPC Core Principals, and in particular the two 'extra' principals created by OLPC Australia [12]. One of these extra OLPC Australia Core Principals, which is also a cornerstone of the jam2jam philosophy, is that of Community Engagement [4]. As such, jam2jam will be looking to expand its presence throughout Australia.

The other OLPC Australia principal is concerned with 'Empowering Teachers'. To that end, jam2jam XO will incorporate services to teachers. Jam2jam AV also provided features of this nature, such as offering time-stamped data on actual usage by students, allowing learning outcomes to be measured.

The jam2jam approach to software development is to implement features based on user feedback, and so actual development outcomes will rely on the efficacy of jam2jam XO as it is used in real world situations.

jam2jam XO has recently been recognised by the OLPC contributors program. As a result the developer has obtained two XO 1.5's to continue development.

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UNIVERSAL INTONATION SYSTEM FOR MUSIC, THE ABSOLUTE AND THE BEYOND.

Mr Svetko Lisica

This demonstration application for the ACMA 2012 Conference at the Queensland Conservatoire is for a new scientific theory and an Invention that is an auscultative method, the act of listening, that expounds upon the Natural Harmonic Series (NHS) for Music's decipherability and attunement, herein introducing Programmatic Specificity for a new, innovative and useful Musical Instrument and Composition Engine that expresses and produces Energy, Frequencies, Wavelengths, Audio Genic Music and also included is a Sonic Biodynamical Brain Entrainment Bridge that is for Binaural Beats and an active being, in the place at which an attentive commitment is at the height of it's fashion, where all of this is engineered, manufactured and irradiated from a vendible product, created by Mr Svetko Lisica's operative method, theoremic composition and hithermost creation in new wisdom, sagacity, penetration, comprehension, intelligence and apperception for the NHS that is now resolved and established into a stable unit of measure in exactitude for a tuning medium, herewith this Invention is the state or fact of existence, a practical Universal Intonation System that belongs with Music, The Absolute and The Beyond.

This auscultative method and its inventive creations liberate significant developments and substantial innovations for the NHS, formally introducing Programmatic Specificity for an innovative, new Musical Instrument, Composition Engine and a Sonic Biodynamical Brain Entrainment Bridge for Binaural Beats, expressing and producing Audio Genic Music, designed, prepared and constructed for The Human Beings Intelligible Hearing Range, the bounds of hearing and beyond it also, by this Inventions terms and conditions, is an authoritative locus, sense datum and empirical knowledge foundation stone for an all-encompassing, harmonious resolution for the NHS and by virtue of this auscultative method's array of technical features, resolves complicated equations and hithermost creating scientific certainty, quite unlike the genetically compelled imperfections and defectiveness in harmonious area under discussion from Prior Art and open to the elements by this auscultative methods inventive purport.

Prior Arts associated music and mathematical theories, in relation to Pitch Perceptions, Interval Recognitions and in so doing the basis for scale design, has long been known to be imperfect and inaccurate. It is this Invention's new Universal Intonation System For Music, The Absolute and The Beyond that establishes, crafts, fashions, produces and generates an all-embracing, sophisticated, first-rate accuracy and originality for the Pitch Perceptions, Interval Recognitions, Spatial Dimensions In Harmonic Relationships, Scale Generation, developments and constructions that are only made possible from this inventive auscultative method therein for the Programmatic Specificity, the processes and procedures in the unique artificial intelligence required by the computer software programme for the new Musical Instrument, Composition Engine and a Sonic Biodynamical Brain Entrainment Bridge for Binaural Beats.

The Author, Composer and Inventor, Mr Svetko Lisica delivers, as a consequence by this Invention, progressive development in humanities mindsets for eurhythmics, theoretical aesthetics and nature's codification system for the NHS, thereby also providing meaning and an explanation for being into other universes of human understanding and consciousness, rather than accepting the status quo of intellectual subconscious control and discipline from Prior Art, that have demonstrated inadequate, ad hoc solutions for precision and exactitude in Music's mathematical representations, herewith sets a design precedent for the NHS and a staggering scientific discovery that has significant implications for humanity.

THE PENTACLE: A WIRELESS INTERFACE FOR NEW MUSICAL EXPRESSION

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ABSTRACT

This paper describes a battery powered wireless interface focuses on free spatial movement for live musical interaction. The interface is a dodecahedron housing a specially designed 'brain' consisting of an Arduino connected to a direction-sensing sensor-fusion device offering nine degrees of freedom, twelve independently responsive contact microphones and twelve multi-colour leds. The outer surface of the dodecahedron serves as a functional physical input that allows the user to strike or press each of the faces whilst all directional movement of the device is simultaneously transmitted. Messages are converted to OpenSound Control("OSC") from a wireless receiver, and both hardware and software projects are developed continuing an inheritance of open-source licensed systems.

Introduction

The interface engages with issues of how to bridge the divide between physical and digital environments by taking established ideas about Human Computer Interaction (HCI) and extending possibilities into a three dimensional musical interaction. The central hardware of the device is two open source projects, the Funnel IO[1] (aka Arduino FIO) and FreeIMU[2]. The FIO is a device for the hardware realization of the Funnel Server tool-kit, developing a platform and libraries for interacting with hardware in any number of programming-languages and scenarios[3]. The FreeIMU is a hardware project developing decentralized sensor fusion technology and software libraries. The vibration sensing microphone inputs on each face further emphasize the physical nature of input on the device.

The core performance concept is that one can be presented with a wireless object and every gesture made will produce a "different but repeatable,"[4] outcome, with visual indicators that one can track without reference to a

For musical applications, the shaping of the ball was chosen to illustrate the applicability of geometric concepts in musical expression. The device considers the translation between forces, with vibration and movement as an input and light and sound as the output.

The casing and mounting are all made from a folded piece of cardboard. This avoids the common difficulties of realizing workable control surfaces and casings for interfaces and coexists with the ethos of an open source hardware and software solution that the user can experiment with and modify at all levels of technology development.

The core mission of this interface is to allow live performance with digital media to be more tangible, involving and accessible. Actions of movement and spatial gesture become sources of free, unimpeded, accurate information that acts as bridge between physical and digital environments.

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THE EXTEMPORE LANGUAGE: ON-THE-FLY COMPILATION FOR THE WHOLE AUDIOVISUAL STACK.

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ABSTRACT

Extempore is a systems programming language designed to support the programming of real-time systems. Extempore promotes human orchestration as a meta-model of real-time man-machine interaction in an increasingly distributed and environmentally aware computing context. Extempore is designed to support a style of programming dubbed cyberphysical programming [1].

Extempore is ideally suited for on-the-fly audiovisual software development as the language affords runtime modification of the entire audiovisual application stack. In practice this translates into a system capable of building, executing and modifying audiovisual code at both the control and signal layer in live performance. Extempore removes any divide between signal and control layers without sacrificing the performance required to build applications which remain 'close to the metal'.

Extempore is a general purpose environment and is not designed primarily as a signal processing language. This makes the environment flexible and suitable for a wide range of real-time systems programming roles. In particular Extempore is designed to support heterogeneous systems and architectures from mobile devices through to high performance clusters. This generality proves useful when building audiovisual systems designed for distributed systems and has been used extensively in QUT's new distributed 'cube' project where Extempore has formed the foundation for a 24 node real-time rendering system.

Extempore supports a wide range of audiovisual tasks including audio/visual/scientific DSP, numerical computation, algorithmic composition, 2D vector graphics, 3D model loading and animation, a 3D scene graph, gpu image processing, video processing, bullet physics integration etc..

Extempore is nearing a v1.0 release scheduled for September 2012 and is available for Linux, Windows 7, and OSX. Extempore is an open source project hosted at http://www.github.com/digego/extempore.

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POSTERS

TOWARDS IMPROVING ONSET DETECTION ACCURACY IN NON-PERCUSSIVE SOUNDS USING MULTIMODAL FUSION

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ABSTRACT

In this paper we present a novel approach for improving onset detection using multimodal fusion. Multimodal onset fusion can potentially help improve onset detection for any instrument or sound, however, the technique is particularly suited for non-percussive sounds, or other sounds (or playing techniques) with weak transients.

1. INTRODUCTION AND MOTIVATION

Across all genre and styles, music can generally be thought of as an event-based phenomenon. Whether formal pitch relationships emerge in note-based music, or timbre-spaces evolve in non-note based forms, music (in one regard) can be thought of as sequences of events happening over some length of time. Just as performers and listeners experience a piece of music through the unfolding of events, determining when events occur within a musical performance is at the core of many music information retrieval, analysis, and musical human-computer interaction scenarios. Determining the location of events in musical analysis is typically referred to as *onset detection*, and in this paper we discuss a novel approach for improving the accuracy of onset detection algorithms.

There are many established approaches to detecting note onsets [1, 4–6, 8, 12]. For percussive sounds with fast attacks and high transient changes, accurate algorithms in the time, frequency, magnitude, phase, and complex domains have been established. The task of onset detection, however, becomes much more difficult when sounds are pitched, or more complex, especially in instruments with slow or smeared attacks (such that of common stringed instruments in the orchestra).

The need for multimodal onset detection arose from collecting and analyzing data for long-term metrics tracking experiments. During initial observations of data collected from a performer improvising on a custom bowed instrument, we began to notice certain playing techniques in which onset detection algorithms could not accurately segment individual notes. As such, we began to explore other algorithms, and ultimately the multimodal approach presented in this paper.

Others have started to apply fusion techniques to the task of improving onset detection algorithms in recent years. Toh, et al. propose a machine learning based onset detection approach utilizing Gaussian Mixture Models (GMMs) to classify onset frames from non-onset frames [13]. In this work feature-level and decision-level fusion is investigated to improve classification results. Improving onset detection results using score-level fusion of peak-time and onset probability from multiple onset detection algorithms was explored by Degara, Pena, and Torres [3]. Degara and Pena have also since adapted their approach with an additional layer in which onset peaks are used to estimate rhythmic structure. The rhythmic structure is then fed-back into a second peak-fusion stage, incorporating knowledge about the rhythmic structure of the material into the final peak decisions.

While previous efforts have shown promising results, there is much room for improvement, especially when dealing with musical contexts that do not assume a fixed tempo, or that are aperiodic in music structure. Many onset detection algorithms also work well for particular sounds or instruments, but often do not generalize across the sonic spectrum of instruments easily. This is particularly true for pitched instruments, as demonstrated in the Music Information Retrieval Evaluation eXchange (MIREX) evaluations in recent years [9–11]. Added complexity also arises when trying to segment and correlate individual instruments from a single audio source or recording. These scenarios and others can be addressed by utilizing multimodal techniques, that exploit the physical dimensionalities of direct sensors on the instruments or performers.

The remainder of the paper is as follows. In section 1.1 we first provide an overview of terms used in the paper, followed by a discussion on the strengths and weaknesses of performing onset detection on acoustic and sensor signals in 1.2. An overview of our system and fusion algorithm is provided in section 2, and finally, our thoughts and conclusions in section 3.

1.1. Definition of Terms

An onset is often defined as the single point at which a musical note or sound reaches its initial transient. To further clarify what we refer to as the note onset, examine the waveform and envelope curve of a single snare drum hit shown in Figure 1. As shown in the diagram, the onset is the initial moment of the transient, whereas the attack is the interval at which the amplitude envelope of the sound increases. The transient is often thought of as the period of time at which the sound is

excited (e.g. struck with a hammer or bow), before the resonating decay. It should be noted that it is often the case that an onset detection algorithm chooses a local maxima as the onset from within the detected onset-space during a final peak-picking processing stage. This corresponds with the peak of the attack phase depicted in Figure 1 (right).

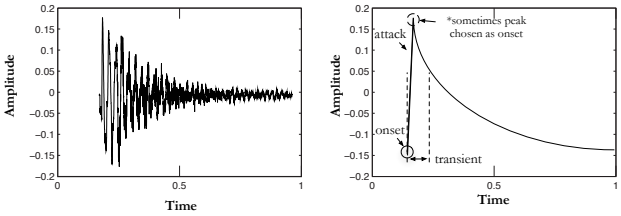


Figure 1 – Snare drum waveform (left) and envelope representation (right) of the note *onset* (circle), *attack* (bold), and *transient* (dashed)

1.2. Detection: Strengths and Weaknesses

There are many strengths and weaknesses that contribute to the overall success of audio and sensor based onset detection algorithms. The first strength of audio-based onset detection is that it is non-invasive for the performer. It is also very common to bring audio (either from a microphone or direct line input) into the computer and most modern machines provide built-in microphones and line inputs. This makes audio-based approaches applicable to a wide audience without the need of special hardware. In contrast sensors have often added wires that obstruct performance, they can alter the feel and playability of the instrument, or restrict normal body movement. For example, putting sensors on the frog of a bow could change its weight, hindering performance. In recent years however, sensors have not only become much more affordable, but also significantly smaller. Through engaging in communication with musicians during our experimental trials, we found that with careful consideration of placement and design, the invasiveness of instrumental sensor systems can be minimized such that the musicians do not notice the presence of the sensors. In fact, embeddable sensors like accelerometers and gyroscopes are already finding their way into consumer products, as demonstrated by the emerging field in wearable technology. The technologies are also beginning to appear into commercial musical instruments, and wireless sensing instrument bows already exist, such as the K-Bow from Keith McMillen instruments[1].

Another consideration between audio onsets and sensor onsets has to do with what information the onsets are actually providing. In the acoustic domain researchers have not only explored the physical onset times but the closely related perceptual onset (when the listener first hears the sound), as well as the perceptual attack time (when the sounds rhythmic emphasis is heard) [2, 14]. These distinctions are very important to make depending on the task, and when dealing with non-percussive notes, such as a slow-bowed stroke on a cello or violin (who's rhythmically perceived onset may be much later than the initial onset). This exposes a weakness in audio-based onset detection—which has trouble with non-percussive, slow, or smeared attacks. In this paper we propose a technique that uses sensor onset detection, which is capable of detecting slow, non-percussive onsets very well. This does not come without certain considerations, as described in greater detail in this section.

In the sensor domain the onset and surrounding data is often providing a trajectory of physical motion, which can sometimes be correlated with the perceptual attack time. Sometimes however, the physical onset from a sensor might not directly align with the acoustic output or perceptual attack time, and so careful co-operation between onset-fusion is necessary. In learning contexts, this trajectory can provide a highly nuanced pipe into information about the player's physical performance. The data can directly correlate with style, skill level, the physical attributes of the performance, and ultimately the acoustic sound produced.

As shown later in this section, the differences in the information provided from separate modalities can be used strengthen our beliefs in the information from either modality individually. This helps overcome weaknesses in the modalities, such as the fact that a sensor by itself may not have any musical context (e.g. gesturing a bow in-air without actually playing on the strings). Combining information from both modalities can be used to provide the missing context from one modality for the other.

Additionally, while audio onset detection has proven to work very well for non-pitched and percussive sounds, they have increased difficulty with complex and pitched sounds. This can often be addressed with sensor onset detection, as the sensor may not be affected by (and do not necessarily have any concept of) pitch.

Lastly, many musical recordings and performances are outside contain multiple instruments. This makes onset detection increasingly difficult as there is the additional task of segmenting instruments from either single stream, or from bleed in an individual stream, as well as ambient noise and interference (e.g. clapping, coughing, door shutting, etc). As there is a great deal of overlap in the typical ranges of sound produced by instruments, polyphonic sound separation is an extremely difficult task. Physical sensors, however, are naïve to other instruments and sensors other than themselves, and are normally not affected by other factors in the ambient environment. Thus they provide (in these ways) an ideally homogenous signal from which to determine, or strengthen onset predictions.

[1] http://www.keithmcmillen.com/

Strengths	Weaknesses
Audio	
• Non-invasive • Onset time can be close to perceptual attack time • No special hardware	• Algorithms have trouble with pitched and complex sounds • Algorithms have trouble with slow / smeared attacks • Ambient noise / interference • Source segmentation / non-homogenous recording
Sensor	
• Very sensitive physical measurements and trajectories • Can detect onsets from slow / smeared attacks • Not negatively affected by pitched or complex sounds • Resistant to factors in the environment / ambience • Typically mono-sources, no separation necessary	• Can sometimes be invasive • No musical context • Onsets may or may not be related to the acoustic / auditory onsets

Figure 2 – Strengths and Weaknesses of Audio and Sensor Onset Detection

2. SYSTEM DESIGN AND IMPLEMENTATION

In designing this system, a primary goal was to create a fusion algorithm that could operate independently of any one particular onset detection algorithm. In this way, the fusion algorithm was designed such that it is provided two onset streams (one for onsets detected from the acoustic or audio stream of the instrument, and one from the sensors), without bias or dependence on a particular detection algorithm. The onset algorithms can be tuned both to the task and individual modalities (perhaps one onset detection algorithm works best for a particular sensor stream vs. the audio stream). This also ensures compatibility with future onset detection algorithms that have yet to be developed. We propose multimodal onset fusion as a post-processing ("late-fusion") step. It does not replace, rather our approach improves, the robustness and accuracy of the chosen onset detection algorithm(s).

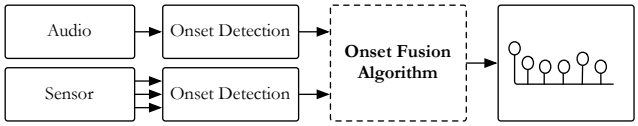


Figure 3 – General Overview of Multimodal Onset Fusion

2.1.1. Fusion Algorithm

A fusion function is provided two onset streams, one from the audio output and one for the sensor(s). First, the algorithm searches for audio onsets residing within a window (threshold) of each accelerometer onset. A typical window size is 30ms – 60ms and is an adjustable parameter called *width,* which effects the sensitivity of the algorithm. If one or more audio onsets are detected within the window of a sensor onset, our belief increases and the best (closest in time) audio onset is considered a true onset; the onset is then added to the final output fusion onset list.

If a sensor onset is detected, however, no audio onset is found within the window (width), we have a 50/50 belief that the sensor onset is an actual note onset. To give a musical context to the sensor onset, the audio window is split into multiple frames, and spectral-flux and RMS are calculated between successive frames. The max flux and RMS values are then evaluated against a threshold parameter called *burst* to determine if there is significant (relative) spectral and energy change in the framed audio-window. Because onsets are typically characterized by a sudden burst of energy, if there is enough novelty in the flux and RMS values (crosses the burst threshold), our belief in the onset increases and the sensor onset time is added to the fusion onset list. The burst threshold is a dynamic value that is calculated as a percentage above the average spectral-flux and average RMS from the audio-window. By default burst is set to equal 20% increase in the average flux value, and a 10% increase in the average RMS from the current audio selection. Increasing or decreasing the burst threshold decreases or increases the spectral flux and RMS, changing the algorithms sensitivity. Additionally, because the value looks for *relative* spectral and dynamic changes, the method works well, even for weak audio onsets.

OnsetAlgorithmPseucodeCode.cpp

OnsetAlgorithmPseucodeCode.cpp › No Selection

```
1  //  OnsetAlgorithmPseudocodeCode.cpp
2  //  2012 Jordan Hochenbaum
3
4  for each sensor onset
5  {
6      if audioOnset >= window-width and <= window+width
7      {
8          add closest audioOnset to fusion onset list;
9      }
10
11     else
12     {
13         calculate spectral flux over framed audio-window;
14         calculate rms over framed audio-window;
15
16         if max flux and rms > burst (thresholds)
17         {
18             add the onset to the fusion onset list;
19         }
20     }
21 }
22
```

Figure 4 – Multimodal Onset Detection Pseudocode

3. CONCLUSION AND FUTURE WORK

This ability to correctly differentiate musical events is at the core of all music related tasks. In this paper we have proposed a technique to improve the computers ability to detect note onsets. Our approach is a late multimodal fusion step, making it compatible with nearly any onset detection algorithm. Additionally, this allows the onset detection algorithms to be selected and tuned to individual modalities, making the technique extremely acute to the particular task.

In our preliminary trials, we were able to successfully fuse not onsets from a performer bowing a custom string hyperinstrument. When playing in a tremolo style, we were not able to achieve satisfactory results using traditional audio based onset detection alone. Performing onset detection from the gesture data provided from an accelerometer on the frog of his bow, we were able to detect most bow events, however, there were many false positives (as the bow provides a continuous stream, whether or not he was actually playing the instrument, or when he moved in between strokes). Using our fusion algorithm, we were able to

synergize the accuracy of the sensor onset detection, with the musical context provided by the audio onset detection. We are currently conducting controlled experiments to quantify the results. In the future, results could be further improved by tailoring the fusion parameters more specifically to the data being analyzed. There are many ways to do this, both by hand and dynamically, and we hope to explore these in the future. For example, dynamic range compression (DRC) during a pre-processing phase could help generalize certain parameters by reducing the amount of variance in the dynamic range of the input data, which changes from day to day and recording to recording. Additionally, there is a considerable room to experiment with the onset detection function currently used, and its various parameters.

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THE DAM(N) PROJECT: *Jobat Calling*

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ABSTRACT

Jobat Calling is a live and interactive audiovisual installation which connects Australian and Indian communities around a common concern: water security. It presents the lives of remote communities in the Narmada Valley, displaced by large-scale dam development securing hydropower for Indian cities. It presents the lives of remote communities in the Narmada Valley, displaced by large-scale dams developed to create power for Indian cities. The central idea within DAM(N) is the nature and mythology of water, its creative/destructive powers and our constant obsession to control it in the event of imminent scarcity. This presentation will showcase the long-term contribution of creative industries can make to communities in struggle.

1. BACKGROUND

The construction of large dams on the River Narmada in India and its impact on millions of people living in the river valley has become one of the most important social issues in contemporary India. The Narmada Dams Project is the biggest of its kind in the world and whilst proponents of the dams believe it will provide water security to 30 million people and provide 1.45 GW of power, the dams will internally displace up to half a million people. Global coalitions of environmental and human rights activist groups condemn the project for several reasons including corrupt and iniquitous cost benefit analyses and destruction of arable land [1,2]. Over the past 25 years, the controversy has come to symbolise the struggle for a just and equitable society in India.

2. OUR STORY

The project follows the creative team (Shakthi Sivanathan, Jehan Kanga, Leah Barclay, Meghna Nambiar, and Sylvester Mardi) through their journey into the heart of India collecting stories from affected towns. Interviews were also collected in central North-West India in the Narmada Valley from adivasi protesters and displaced farmers in the affected villages of Badwani, Jobat and Badal. In Badwani and Jobat, we interviewed key indigenous protesters such as Bhagirath Dhangar (Badwani Narmada Bachao Andolan) and Shrikanth (Jobat NBA). In Jobat, we collected stories from over 20 displaced groups who had gathered at a satyagraha (non-violent protest). They told their stories whilst preparing dinners and lunches, and singing tribal and solidarity songs.

Embedding ourselves within both the satyagraha and within remote affected communities was a vital part of capturing a complete story across the widely varied landscape. In Badwani we also interviewed Dayal Solanki, a young adivasi whose story became the common thread for our journey. He became our guide through to the extremely remote village of Badal, accessible only by fishing boat from a makeshift wharf one hour from the nearest town. The region, which is now almost completely submerged under reservoir, was formerly one of the most agriculturally productive regions in India. We stayed in Dayal's home perched on the arid crest of a mountain, and stayed with his family recording their stories of displacement and the hope they placed in their children.



Figure 1. Dayal's younger brothers Dalsingh and Kamish on the arid banks of the now submerged village of Badal.

We captured the stories and music of his whole family undertaking some digital workshops with the children to involve them in the story telling process.

Upon returning to Jobat, Meghna and Sylvester undertook dance workshops with children at a Jivanshala (NGO sponsored school), and expressed their experiences of the Narmada Valley through various site specific dance routines in Badal and on Jobat Dam. Shakthi, Leah and Jehan undertook further digital workshops at the Jobat Jivanshala.



Figure 2. A student at the Jivanshala learning about capturing sound during a digital media workshop in Jobat.

We enabled the children to collect images, video and sound to tell their story from their perspective. The incorporation of these capacity building workshops within the film will showcase the long-term contribution of creative industries can make to communities in struggle.

3. JOBAT CALLING

The presented audiovisual installation is a 20 minute is an interactive mixed media work. It is designed to work as a standalone installation or in conjunction with a choreographed dance by a member of the Attakkalari repertory. The installation also involves a solo voice interacting with the soundscape.

The installation is the first major creative outcome from a Project that will also yield a one hour mixed media dance performance, an immersive exhibition of photographs, soundscapes and short films, and finally a documentary covering the rapidly changing political landscapes around water security in India in the Narmada and in Australia in the Murray Darling as well as the creative process which becomes the lens through which this landscape unfolds.

We would like to think The DAM(IN) Project will become a model for artistic development for young artists, a model for complex collaborations between more than two major performing arts organizations which all have significant creative input, and a model for the development of new opportunities for both artists and performing arts organisations to find new audiences.



Figure 2. Co-choreographers Meghna Nambiar and Sylvester Mardi on top of Jobat Dam, December 2012.

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CREATING MOBILE COMPUTER MUSIC FOR PERCUSSIONISTS: *SNOW MUSIC*

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ABSTRACT

This paper traces the development of an *Apple iPad* and *iPhone* based musical instrument, *Snow Music,* and its use in performances by *Ensemble Evolution*, a percussion group formed by the author and two other percussionists.

This artistic research is motivated by a desire to use computer based instruments along with percussion in a group with non-programmer performers. The aims were to develop an elegant, portable, and flexible computer instrument that is accessible to the group and to find out what opportunities it enables in performance practice.

A project was undertaken to develop *Snow Music*, using *Pure Data* and *libpd*, and to collaboratively compose a musical work. The process was documented with video recordings and analysed using ethnographic methods.

The resulting setup was extremely portable and convenient for rehearsal and performance. A design that emphasised "percussive" interaction with the instrument and an improvised performance practice contributed to a collaborative development cycle. Analysis of performances gave insight into the limitations and affordances of mobile computer music devices.

1. INTRODUCTION

Starting in 2010, I co-founded a percussion group, *Ensemble Evolution*[1] in Piteå, Sweden with Jacob Remington and Maria Finkelmeier. The goal of the group was to explore projects in composition and improvisation. As a percussionist and computer musician, one of my goals was to integrate computer based instruments into this group. As the other members of the group had no experience with computer music it became clear to me that I needed new computer music tools to complement percussion instruments in my work as a music creator, collaborator and performer.

Inspired by recent developments in computer music on mobile devices, I decided to use *Apple*'s *iPad* and *iPhone*, as platforms for computer music in *Ensemble Evolution.* I thought that these devices could work better in this situation that the laptop based computer systems I had used previously.

There are several previous works that inspired this research. Schiemer and Havryliv's work, *Pocket Gamelan* [6], exploits the portability of mobile devices by having non-expert performers swing the devices on a cord in performance. Tanaka's four-hand *iPhone* performances demonstrate the affordances of sensors built into mobile devices through a collaborative improvised performance practice [7].

For *Ensemble Evolution*, it was important that the mobile devices were portable enough to easily fit into our percussion setups and to take on tour. The interesting new interfaces for musical interaction, such as touch screens, could be useful for engaging non-programmer percussionists in performance.

The research project presented in this paper was designed to study the process of producing a new *iOS*[2] based instrument to be used alongside percussion instruments. The project also studied the performance practice that emerged from collaboratively creating a musical work using this instrument. Over the course of the project, both the musical work and the *iOS* app were called *Snow Music*.

This project is part of a larger study that is the topic of my master's thesis *Mobile Computer Music for Percussionists,* completed in June 2012 at Luleå University of Technology [3]. In addition to the project described in this paper, the thesis includes a study of a mobile computer music system for vibraphone and an examination of the current state of research into mobile computer music.

2. RESEARCH QUESTIONS

The goal of the project was to address four research questions about mobile computer music:

1. *Heaviness*. Can computer music setups be made more simple, elegant and convenient using mobile devices?
2. *Shareability*. How can mobile computer music instruments be made accessible to a non-programmer percussion ensemble, and what creative processes can be used to explore them?

[1] Performance and other information about *Ensemble Evolution* can be found at http://www.ensemble-evolution.com

[2] *iOS* is the operating system that runs on *Apple*'s *iPhone, iPad* and *iPod Touch*. It is possible to produce "universal" *iOS* apps that can run on all of these devices.

3. *Playability*. How can the affordances of mobile music devices be used to create playable instruments for percussionists?
4. *Performance practice*. What new performance practices are enabled or demanded when complementing acoustic percussion instruments with mobile music devices?

3. RESEARCH METHOD

The research questions of this project were addressed qualitatively through ethnographic[3] analysis of the process of developing, rehearsing and performing the collaborative work with *Ensemble Evolution*. These processes were documented through written notes and video recordings. Analysis of the video documentation was made through a process of coding, following techniques described in Krüger's *Ethnography in the Performing Arts* [2]. In this process, key-concepts are chosen and assigned to events occurring in the video. Events can then be grouped by key-concept to make connections between the various rehearsals and performances. The process for coding verbal information is well understood, conversations can be transcribed and then coded easily in this form. Performances were coded following Östersjö's method of working directly from the video [4], using time codes to identify events.

The artistic investigation was pursued experimentally with the choice of hardware and software elements informed by current trends in computer music research.

3.1. libpd: Pure Data in *iOS*

The *iOS*-based instrument for this project was developed as a native *iOS* app, using *libpd [1],* a general project to port *Pure Data* [5] to a number of computing platforms. *libpd* is an effort to separate the audio synthesis parts of *Pure Data* from its graphical programming interface. *Pure Data* can then be used as a synthesis library for other applications.

In this project, *libpd* allowed the musical elements of the app to be composed in the familiar environment of *Pure Data* while the user interface was created in *objective-C* using *XCode*.

4. RESULTS

In order to capture the creative process in developing *Snow Music*, every rehearsal, and a series of initial public performances was video recorded. The rehearsals took place in February 2012 in Piteå, Sweden, while the performances occurred in March 2012 in Australia.

[3] Ethnography is a qualitative research method for studying cultural phenomena. The researcher conducts fieldwork to collect notes, audio and video recordings, and images relating to the phenomena, an active and subjective method. "The open-ended nature of the ethnographic approach is particularly suitable for active discovery and exploration" (Krüger [2]). Analysis of the data can be an iterative process with multiple phases of field work and analysis to refine the research question. Conclusions are drawn inductively from the data gathered.

Over the first three rehearsals, I introduced the *Snow Music* app to the other performers. The app was designed to give the performers "percussive" control of field recordings from the harsh winter in Piteå. The early rehearsals were marked by changing ideas about the musical content of the work and discovery of how to play the *iOS* devices.

In the fourth rehearsal, it was agreed that the performance would be a free improvisation, *Snow Music*, motivated by the computer sounds. Each player would use an *iOS* device and a keyboard percussion instrument. The *iOS* devices were each connected directly to a powered loudspeaker.

There were three public performances of the work in Australia, all including collaborations with performers outside of *Ensemble Evolution.* The first was at Melbourne Conservatory and included Nat Grant, a percussionist and computer musician. The second, in *Serial Space,* Sydney, took the form of a group improvisation with members of *Nomad Percussion.* The final performance was part of an *iPad* music workshop for high school students at Canberra Grammar School.

4.1. The *Snow Music* app

The final *iOS* app for *Snow Music* has two main functions. The first is a touch interface that creates snow sounds when the screen is tapped or swiped with a finger. The dynamic of tapped sounds can be controlled by the distance of a tap from the centre of the screen. The dynamics of swiped sounds corresponds to the speed of the swipe. The second function was a set of three background soundscapes, algorithmic processes that produce phrases of notes in free rhythm. The three soundscapes are of bells, cymbals, and a swooshing, wind-like, snow sound. The bells were set to use notes only from the C aeolian scale while the cymbals and snow sounds were of indeterminate but changing pitch. The overall dynamic of the sounds could be controlled using the *iOS* devices' hardware volume controls. In the first rehearsal, only a prototype of the snow sounds was available. All sounds had been developed by the fourth rehearsal.

4.2. Rehearsing and Developing *Snow Music*

4.2.1. Problems

A number of problems came up repeatedly over the rehearsal process. Most of the technical problems related to situations where an interaction with the app did not align with the player's expectation. In the first rehearsal, Maria remarked that "the interaction was throwing me off a bit". Managing the expected volume of the *iOS* devices through the loudspeakers was also a problem.

The musical problems were related to the clarity of expression that was possible with the app. Initially, it was difficult to express clear rhythms with the *iOS* devices and our early attempts to play precise, repetitive rhythms with the snow sounds were not very successful. Also, the sounds initially produced by the *iOS* devices

had a lack of timbral clarity. Jacob commented that the app felt "limited" and its available sounds just "sounded too similar" in rehearsal two.

4.2.2. Solutions

Fortunately, these problems could be addressed. In some cases, all that was needed was some more explanation from me, as my ideas about programming the interaction between the performers' gestures on the touch screen and the sound output were rapidly evolving.

The interactive concept came down to using "percussion-y technique". Common techniques for using hands to play percussion instruments – taps, finger rolls, scraping and rubbing – also work on the *iOS* devices' touch screens. Another simple explanation was to use the hardware volume control on the devices to adjust for unexpected volume levels. Finding these techniques was an important part of developing a performance practice with the *iOS* devices.

The musical problems were also addressed through updates to the app made in between rehearsals in response to suggestions and inspiration from the group. For example, the group suggested using more mimetic sounds from Piteå. In response, I added the sound of footsteps in snow and the wind that blows around the student housing.

4.3. Performing *Snow Music*

To understand the performance practice that we developed for *Snow Music,* three representative performances from different stages of the project were chosen and coded. These were: Our first[4] attempt at using the *iOS* app with three keyboard percussion instruments; a free improvisation[5] in the fourth rehearsal with a guest, Matteo Spano, playing guitar (see Figure 1); and lastly, a public concert[6] in Melbourne with a guest artist, Nat Grant, performing on laptop. The results can be categorized as concerning techniques for playing the *iOS* devices, musical affordances and musical applications of the devices in performance.



Figure 1. *Ensemble Evolution* and Matteo Spano rehearsing *Snow Music*.

[4] First *Snow Music* performance with mallet percussion instruments: http://youtu.be/W091aaKSIU4

[5] *Snow Music* performance along with Matteo Spano: http://youtu.be/MWicLWySxdI

[6] *Snow Music* performance at Melbourne Conservatory along with Nat Grant: http://youtu.be/Vz5aQxc_jlE

4.3.1. Techniques

Since all of the interactions for the *Snow Music* app occurred through the touch screen we needed to use skin contact to activate it. Although it is possible to hold mallets in our hands touch the devices with the heel of our hands, or the little finger, we generally played the *iOS* devices without simultaneously trying to hold mallets. This encouraged us to also play the keyboard percussion instruments with our fingers, an intimate, but striking sound. It also turned out to be possible to hold mallets in one hand and play the *iOS* device with the other.

One of the limitations of the *iOS* app was in expressing clear rhythm. The performers adopted several techniques to address this issue. One strategy was using "finger rolls" to express continuous rather than singular sounds. Other experiments included exaggerated swipes across the whole *iPad* screen.

4.3.2. Musical affordances

The three background sound generators (bells, cymbals and "snow-wind") were added to the app in response to the discussions in rehearsals. It is clear from the performance videos that the musical affordances of these sounds inspired their use during our improvisation. The bell sound was by far the most used. The clear tonal centre and texture cut through other sounds in our improvisation. The group responded to these affordances by using the sound to trigger sections of tonal improvisation in free rhythm.

The other background sounds, cymbals and snow wind, were not used as frequently. Where they did appear, their musical function was to create a dense texture that would support various styles of improvisation. The Melbourne performance had a much denser texture due to the fourth player on laptop and as a result, these sounds were not activated as frequently.

4.3.3. Musical application

Over the rehearsal process, the sounds we could create in the *Snow Music* app came to define the improvised piece we were developing. Even in the earliest performances, we started the piece with a "snow collage" by tapping the screens. In the later performances, this element became better defined. The sounds of footsteps in snow were strongly mimetic and gave a distinctive theme to the piece.

It is interesting to see how the *Snow Music* app was used to change the intensity of the improvisation. While the opening "snow collage" built intensity, the app was mostly used to decrease intensity. My use of the app at the end of the Melbourne performance triggered a closing "snow collage".

The ability to present a "new idea" to the ensemble is critical in free improvisation. Significantly, the "bells"

background sound was useful for driving the improvisation into a particular direction.

5. CONCLUSIONS

Overall, the *Snow Music* project was successful in introducing computer instruments to *Ensemble Evolution* and resulted in extremely satisfying performance results. The investigation uncovered conclusions to address all four research questions, which are briefly summarized here and discussed in more detail in my thesis [3].

5.1. Heaviness

The setup for *Snow Music* was extremely simple using only the three *iOS* devices and audio cables. In our rehearsal studio we used three powered speakers for performing while on tour we used a compact mixer to distribute the sound to two loudspeakers.

The light weight of the equipment and short set up time greatly contributed to the agility of our rehearsal process and performances on tour. The whole ensemble's computer music setup (not counting loudspeakers) could fit in one backpack.

5.2. Shareability

Snow Music was successful in putting computer based musical instruments in the hands of non-programmers. The performances demonstrate all the performers using the *iOS* devices confidently and musically. I received an unexpected level of input and engagement from Jacob and Maria during the rehearsals. Suggestions from the group directly connected to many of the technical and musical problems with the app. Importantly, the input from the group helped me to focus my programming in between rehearsals to address the most pressing musical issues.

5.3. Playability

The interactive focus in *Snow Music* was the touch screens of the *iOS* devices. The rehearsal process allowed us to uncover some of the problems encountered when playing these instruments. The most frustrating moments in the rehearsals were caused by a disconnect between gesture and result. These issues were able to be resolved through experience and fine tuning the app.

5.4. Performance Practice

The *Snow Music* project resulted in an investigation into free improvisation performance practice combining *iOS* devices and acoustic instruments. The group discovered that the field recording based sounds from the *Snow Music* app could define and motivate the piece and helped to give structure and meaning to our improvisation.

Analysis of the performances makes some of the constraints of the *Snow Music* app clear. Although many of the sounds are distinctive and allow a degree of precise playing, some were subtle and mainly textural. There were limits to the extent that these sounds could be used in a precise rhythmic context. While this is probably not "bad", it is important to understand such limits and decide what impact they have on the compositional process.

6. FUTURE DIRECTIONS

The rapid, collaborative development process that was achieved for *Snow Music* can serve as a model for future projects. While musical interactions with the app had limitations, it may be that techniques can be found to address these. Future projects could also benefit from an investigation into how mobile music apps could be used to contribute "new ideas" in an improvised performance.

7. ACKNOWLEDGEMENTS

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USING GENRE FUSIONS AS A TOOL FOR MUSIC COMPOSITION

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ABSTRACT

Genre is a fundamental structuring force in our musical life. The name associated with a particular music can become a particular point of reference, not only for the sound and relationship between the compositional elements, but also for the social and cultural categories to which the musical image belongs. For someone attempting to determine the distinction between these categories, the problem arises that as new music is created, new genres are conceived to create a distinction from the preceding music. How can a composer use these fluid boundaries to explore a hybrid fusion of genres as a focus for original work? How will this work be approached by audiences? How can you be sure that a true interaction between the genres has taken place?

PERFORMANCES & INSTALLATIONS

PERFORMANCES AT ACMC 2012

[LISTED BY COMPOSER IN ALPHABETICAL ORDER]

TAKOSUTBO

Takotsubo is an improvised Audio/Visual performance built around the interplay of three interfaces: traditional (guitar), the new (Kinect) and the ubiquitous (camera). These interfaces allow three different styles of interaction: the tactile and intended, the non-tactile and intended, and the non-tactile and unintended. This forms a tripartite interaction allowing varying and mutating degrees of agency and control for the participants (performers and audience).Each of these interactive systems affords different degrees of expertise and expression. In *Takotsubo,* the guitar, as traditional haptic music interface, provides fine and nuanced control to generate primary source audio material for the piece. Taking this material into a virtual spectral realm, it is then modified using live concatenative synthesis controlled by a Kinect-based gestural interaction system designed on principles of embodied music cognition. Finally, the audience, as participants in the performative moment, albeit without the same degree of agency as the performers, are included in the process via camera input into the generative audio system.

ROGER ALSOP & MARK PEDERSEN (COMPOSERS)

Roger Alsop is a sound designer, composer, musician and media artist. His activities range through working with VicHealth, La Mama, and the Royal Botanic Gardens, musical and audiovisual improvisations, live mixing, and sound and media design for performance. He also teaches Sound Design (Victorian College of the Arts: UoM), Performative Architecture (Melbourne School of Design; UoM), and Production Styles at Box Hill Institute. His interests are in developing interactive approaches for developing creative works that enhance the fundamentally hybrid nature of modern creativity.

Mark Pedersen is a Melbourne-based artist working primarily in sound and image design, and is currently doing a research Masters on gestural control systems for computer music performance at the Victorian College of the Arts: UoM. He currently co-hosts the Melbourne SeenSound: Visual/Music series with Brigid Bourke, has recorded and performed in India and Senegal with the Indian-Australian trio *Hathira Sama*, and his collaborative work with Melbourne poet and novelist Melike Ülgezer was recently published online in Cordite Poetry Review. His primary interests lie in creating environments for expressive performance which explore the connection between the physical and the virtual. Examples of his work can be found at www.chailight.com.

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RIVERS TALK
With Leah Barclay

A new collaborative radiophonic work on Australian river ecology: a joint mix of river culture from the Murray River in Mildura Victoria to the rivers leading to Wolf Rock in Queensland. Rivers talk dissolves the dense local sonic profiles from the artists' solo works Voicing the Murray and Confluence in a dynamic profile of continuing issues surrounding river care and the sensitive man/nature relationships. Through the radiophonic medium of collaborative electroacoustic sound, these issues can be heard through the authentic local voices and revisited not only in Australia but worldwide.

ROS BANDT (COMPOSER)

Ros Bandt is an award winning international sound artist from Australia who is famous for her soundings and installations of unusual acoustic spaces from wheat silos, water tanks, limestonequarries, swimming pools, ancient deserts, historic buildings, and underwater environments**.** She has pioneered the use of endangered sounds in her sound installations and radiophonic works contributing to environmental and sound culture awareness, recording and interpreting Indigenous and multi-cultural languages in change, native wild life, old machinery and lost classical languages. She has built soundplaygrounds, interactive systems, global satellite performances, made sculptures, musical instruments and is an expert sound recordist and musician. She has interpreted sites and sound cultures worldwide including the Goatherd songs of Crete, the endangered sound of the *Isobue*, Japanese Sea Whistle, and the fleeting

sounds of her own Aeolian harps and casuarinas in Lake Mungo. While living in Europe in 2010-11 she toured her international collaborative sound art performance with hair dye paintings with Johannes S Sistermanns Germany, *Tracings* CD Wergo, and composing and performing on Tarhu the *Black Flacon* CD with Erdem Helvacioglu, Pozitif Yappim, CD of the week in Istanbul. Her site specific electro-acoustic symphony*, Medusa Dreaming* was composed for the Yerebatan Sarayi underwater cistern in Istanbul, performing two concerts there with the Medusa ensemble she created for this event. She is a founding member of six ensembles incuding the early music ensemble La Romanesca with whom she is currently touring.

She has been awarded Australia's highest honours, the Don Banks Composers Award, The ABC sound artist in residence, the Sound Art Australia prize, and in the USA, she became the Inaugural Benjamin Cohen Peace fellow for her installation *Altars of Power and Desire*. Her commissions include the Paris Autumn Festival, Radio ORF Vienna, Transit, the Studio of Akustische Kunst WDR KOln, ABC, Melbourne International Festival, Ball State University USA, ReMap Athens and the Ionian University Kerkyra, Greece. The ensemble Charisma commissioned *Blue Gold*. She is published by Wergo, New Albion, EMI, Move Records, Sonic Gallery, Au Courant, L'agent des refuses, Cambridge Scholars Press, Fine Arts Press, The Australian Music Centre, Composers Autograph Limited edition and distributed by CDE Music and Indie Records.

She founded the first online gallery of soundart in public space in Australia, www.sounddesign.unimelb.edu.au and has written several definitive books on sound art and culture, winning research awards for excellent scholarship at the University of Melbourne. Her PhD on performance practice of repetitive music was awarded from Monash University and she has since had three ARC grants. She lives from her art practice.

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THE DAM(N) PROJECT: JOBAT CALLING

Jobat Calling is a live and interactive audiovisual installation that connects Australian and Indian communities around a common concern: water security. The DAM(N) Project is a collaboration between director Jehan Kanga, vocal ensemble The Song Company, Bangalore based contemporary dance company Attakkalari, new/digital media company CuriousWorks and sound artist Leah Barclay. It presents the lives of remote communities in the Narmada Valley, displaced by large-scale dams developed to create power for Indian cities. The Project does not seek to engage in the heavily politicised issues around the Narmada, rather it tells stories taken from the many viewpoints and the living culture in the affected areas. The central idea within DAM(N) is the nature and mythology of water, its creative and destructive powers and our constant obsession to control it in the event of imminent scarcity. This is an issue that resonates with both the Indian and Australian psyche.

LEAH BARCLAY AND JEHAN KANGA (COMPOSERS)

Jehan is an emerging artistic director/producer - currently the assistant artistic director under Roland Peelman of The Song Company. His study as a baroque violin and as a director has taken him to Switzerland, Spain, Italy and USA. He has also directed several projects Australian Baroque Brass, The Pocket Score Company, Seven Teares, The Madrigal Society, Ensemble Con Brio and The Renaissance Players. He has produced and directed several collaborative productions – the most recent being *Sweet Dreams* with shadow puppeteer Stephen Mushin and The Song Company.

Leah Barclay is an award-winning composer and interdisciplinary artist recognised internationally for her distinctive sonic language. She creates immersive performance and multi-disciplinary installations at the intersection of art, science, technology and the environment. Leah's recent major projects include Uravum (2011) with Attakkalari, Floating Land Festival as guest curator, Dhakan (ACMC11), Transvergence (2011) an installation on Cockatoo Island, Econtro Das Aguas commission for The Australian Voices, Mamori Sound Project in the Amazon, and lead artist within the Remnant/Emergency Artlab.

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SWITCHED ON **(2011)**

This piece began by exploring the sounds of switches, dials and buttons collected from my home and my place of work. A particularly interesting sound was sourced from turning on an old TV, which ignited a series of high frequency pitches and crackling static flutters before eventually powering on. Lower transpositions of this sound revealed otherwise inaudible collection of electronic-like frequencies that feature throughout the work. Together with this the

switch sounds –short in duration, are clustered into intricate groupings, cascades and explosive flourishes. In addition, I was particularly drawn to the concepts of inanimate object powering up from moments of inactivity, and surging electricity running and humming through circuitry.

Thanks go to Lynn Holland and David Lewis for their help and extensive switch hunting in the art department, and Andrew Hall for his valued sound contribution. This work was created and completed at Liverpool Hope University, UK and the Elektronmusikstudion (EMS) in Stockholm, Sweden.

MANUELLA BLACKBURN (COMPOSER)

(b. London in 1984). Dr. Blackburn did music studies at The University of Manchester. She completed her doctorate at this university with Dr Ricardo Climent's supervision. Manuella is now a a lecturer in music technology at Liverpool Hope University. Successes include First Prize for her work Vista Points in the 10th Musica Viva Electroacoustic Music Competition, 2009, Portugal, and Grand Prize in the Digital Arts Awards, Japan for Kitchen Alchemy. She has received Honorary Mentions in the CMMAS competition in Mexico and in the VII CIMESP (Concurso Internacional de Musica Eletroacustica de Sao Paulo 2007. Other awards include First Prize for her acousmatic work, Causal Impacts, in the 7th Musica Viva Electroacoustic Competition 2006, Portugal 2nd Prize in the Diffusion Competition, Limerick, Selection in the 2006 Bourges International Electroacoustic Music and Sonic Arts Competition, and the Public Prize in the CEMJKO competition in Brazil. Manuella is also a member of *The Splice Girls* live laptop improvisation duo. Highlights include a 'sonic ferry' at the Sonic Arts Network in Plymouth and Florida at the Atlantic Centre for the Arts (2008). [Source: http://www.manuella.co.uk/about.htm].

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BRISBANE NOCTURNE REVISITED

“In 2000, I wrote piece for the ACMC, which that year was held in Brisbane. It was a microtonal, algorithmic, real-time improvisational computer piece. The control part of the program was written using John Dunn’s SoftStep. The synthesis was done with Martin Fay’s Vaz Modular. For the conference, I also wrote a paper, describing in detail the processes used in the piece. For this 2012 conference, I am reconstructing the piece, but not referring to the original SoftStep patch or the recording of the original piece. Instead, I am reading the paper and reconstructing the piece from my descriptions, using the original (now obsolete) SoftStep. After the conference, I will listen to the original, and post both versions on my website, seeing if, after a lapse of 12 years, and based on my descriptions in the original paper, I came close to reconstructing the original.” (WB).

WARREN BURT (COMPOSER)

Notes from the composer’s website: “I'm a composer, performer, video artist, sound poet, writer, instrument maker (both in hardware and software, electronic and acoustic) and a few other things. Currently, I'm living in Daylesford, Victoria, Australia, and teaching at Box Hill Institute, Melbourne, and Bendigo Regional Institute of TAFE, Bendigo. This website has an introduction to me and my work. On this page, I'm including a kind of official summary I put in grant applications, job searches, etc. It gives some idea of what I've been up to....

CAREER HIGHLIGHTS 1984 – present: Born 1949 Baltimore, Md. USA; lived in Waterford NY, 1950-1971; San Diego CA 1971-75; Melbourne, Australia 1975- 2004; Wollongong 2004-present. Education; BA 1971 State U. of NY at Albany; MA 1975 University of California, San Diego, PhD 2007 University of Wollongong

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IN A WINTER LANDSCAPE
Peter Sheridan, *Bass Flute*

This piece is an introspective composition intended to convey an austere Winter landscape, after a powerful storm. The storm, having cleared out the old growth, has made room for new life. It is inspired by the work of Arvo Pärt and John Cage. *In A Winter Landscape* was commissioned by Peter Sheridan.

MADELYN BYRNE (COMPOSER)

Madelyn Byrne is an active composer whose work frequently combines acoustic instruments with computer-generated sounds. Some compositions in this area include In A Winter Landscape (commissioned by Peter Sheridan), For ANWR

(commissioned by Yoon Jeong Heo), Rain, Sea, and Sky (commissioned by the NOISE Ensemble), and Dream Tableaux (commissioned by Colin McAllister). Past honors include ASCAPlus Awards, honorable mention in the "Suoni di Legno" competition, winner of the Friends and Enemies of New Music Composition Competition, and recordings on Innova, Everglade, and New World/CRI Records. Madelyn's music has also been selected for a wide range of new music festivals including several International Computer Music Conferences, Electronic Music Midwest, WIU's New Music Festival, Imagine II, and Merging Voices among others. Her music has received additional national and international performances and broadcasts on television, radio, and the internet. Madelyn completed her DMA in Composition at The Graduate Center in 1999 and joined the Palomar College Faculty in the Fall 2000 semester. Madelyn has also been a guest composer at Columbia University's Computer Music Center.

PETER SHERIDAN (Flutes)

Peter Sheridan, was raised in New York City, and studied at the Mannes College of Music. He has lived in London, Arizona, Los Angeles and now takes up residence in Melbourne Australia. Mr. Sheridan currently maintains an active schedule performing, teaching, creating and touring. Peter is director of the Monash University Flute Ensemble and a lecturer of flute for the University of Melbourne. As a recording artist he has appeared on various Hollywood soundtracks, recorded two chamber music discs and has completed two solo albums featuring the unique tones of the low flutes .

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_DERIVATIONS
Alana Blackburn, *Recorder*

This composition presents an interactive performance system, designed to facilitate possible modes of interactivity between an instrumentalist and the computer in improvised performance. The system derives all of its sonic responses directly from the improvising instrumentalist; listening to, comparing and transforming analysed musical phrases stored in an ever expanding memory of past performer's gestures. Part of the recent development of the _derivations software has been in facilitating the use of multiple 'rehearsals' sessions with the system in addition to information recorded and analysis during performance. The premise is that a rich database of cumulative interactions between human and machine will deepen and complicate any eventual live performance. For today's performance, neither system nor performer will begin with a 'blank memory', as both system and recorder player will have been rehearsing for some time previously.

BENJAMIN CAREY (COMPOSER)

Ben is a saxophonist/composer/technologist living in Sydney, and he is involved in the area of contemporary music performance of 20th and 21st century solo and chamber repertoire. He also works regularly with young composers in premiering new works for the saxophone in a variety of contexts. After his graduation from the Sydney Conservatorium in 2005, he spent two years in France studying saxophone the Conservatoire de Bordeaux. He is currently pursuing a PhD degree at the University of Technology, Sydney, where he is researching and developing systems for live interaction between instrumentalists and computer.

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PRELUDE FOR CLARINET AND COMPUTER

This Prelude is the second in a series of interactive pieces for solo wind instrument with live computer processing. The piece explores (and blurs) the fine line between the sound of the live clarinet and the computer processed clarinet sound by the use of multiphonics in the clarinet part. The computer is also used to fill in the "missing" even harmonic partials in the clarinet's odd-harmonic sound spectrum, morphing its sound into one resembling a saxophone. It is worth noting that all of the sounds in the electronic component of the piece come from the live performer on stage: sometimes the computer processes the live sound of the clarinet, and sometimes records some specific sounds for playback and processing later in the piece.

RICHARD DUDAS (COMPOSER)

Richard Dudas holds degrees in Music Composition from The Peabody Conservatory of Music of the Johns Hopkins University, and from The University of California, Berkeley. He additionally studied at the Franz Liszt Academy of

Music in Budapest, Hungary and the National Regional Conservatory of Nice, France. In addition to composing music for acoustic instruments, he has been actively involved with music technology since the late 1980s. As a computer musician, he has taught courses at IRCAM, and developed musical tools for Cycling '74. Since 2007 he has been teaching music composition and computer music at Hanyang University in Seoul, Korea.

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PATOISE II: SOW

Nine months of the year I nurture a healthy obsession with water footage I've taken in the past while being land logged here in Indiana. The footage mostly comes from Australia where for the other three months of the year I'm surrounded by it constantly. In this piece I've taken my experimentation with film a little further and tried to parallel an abstract sound environment with worlds of water...

> *The cure for anything is salt water - sweat, tears, or the sea*. (Isak Dinesen)

MELODY EÖTVÖS (COMPOSER)

Melody Eötvös is an Australian composer currently completing a Doctorate of Music at the Jacobs School of Music, Indiana University, Bloomington, USA. As well as composing instrumental music, Eötvös has also strongly applied herself to electronic sound composition. Both styles of composing are strongly influenced by philosophical reflections and extra-musical elements, which were enhanced by completing an MA in philosophy, and extensive international travel experience. She has had works performed internationally, including the London Sinfonietta & BBC Singers, and been the recipient of several awards both in Australia and overseas. Past composition professors include Stephen Leek (AUS), Gerardo Dirié (AUS), Simon Bainbridge (UK), P.Q. Phan (USA), Aaron Travers (USA), and David Dzubay (USA). www.melodyeotvos.com.au

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***AERIAL VAPOURS* (2011)**

This piece explores notions of lightness and transcendence within spectral space. Many of its defining morphologies suggest skyward trajectories that, on occasion, one might feel are achieved. The twilight flicker of soft, filtered grains occurring towards the end of the piece, which become extended and lift to reveal a momentarily suspended field of effervescing pulses, is perhaps a good example of this sense of empyreal gain. In other cases, however, these vertical tendencies are thwarted by sudden hostile announcements, or else they seem to lack the necessary physical properties to escape the burden of gravity. The title is taken from a line in John Milton's *Paradise Lost*, describing the flight of unfortunate souls whose self-profiting ways in life will earn them a place in the "Limbo of Vanity" within "the bare convex of this World's outermost orb". It reflects a situation which is both fixed and transitory, and provides an analogy with attributes of this work.

LEE DANIEL FRASER (COMPOSER)

Lee Fraser (UK, 1981) is a composer specialising in electroacoustic music. He studied composition with Frank Denyer and David Prior at Dartington College of Arts, before taking a degree in sonic arts at Middlesex University, and in 2009 completed an MA in electroacoustic composition with Denis Smalley at City University, London, for which he was awarded a distinction. Lee is currently pursuing a PhD in electroacoustic composition under the supervision of David Berezan at the University of Manchester. His research, funded by the AHRC, is concerned with the aesthetics of acousmatic music. In recent years, he has been involved in a number of collaborative and performance-based projects, including the reworking of material for sound artist and composer Mikhail Karikis, released on the Sub Rosa imprint in 2009. His music has been performed and broadcast internationally.

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ONE

"In 2003 I began developing works for the eMic –a new musical interface I have designed. The eMic consists of a modified microphone stand that allows the performer to manipulate her/his voice in real time by capturing the physical gestures via an array of sensing devices. My most recent composition *Idol* (funded by the Australia Council and presented at a range of venues internationally in 2010/11) represented a new direction for working with the interface.

The compositional process was commenced with choreographed gesture rather than sound. Working with movement as a starting point uses choreographic gestures as the basis for musical structures. Thus, the compositional process must commence with gestural data sets. In *One*, this approach is expanded and applies to new sound materials. *One* uses only voice as a sound source for generation of the musical materials and draws upon vocal utterances that accompany choreographed movement." [The composer]

DONNA HEWITT (COMPOSER)

Dr Donna Hewitt is a vocalist, electronic music composer and instrument designer. Her primary interest in recent years has been investigating new ways of interfacing the voice with electronic media. She is the inventor of the eMic, a sensor enhanced microphone stand for electronic music performance. The eMic brings together her work as a vocal performer with her compositional work using microphone-captured audio. Much of Donna's composition work involves re-processing audio via various digital processes to produce complex textures. Donna's work has attracted funding from the Australian council for the Arts, most recently her collaboration in 2010 with dance artist Avril Huddy to develop the work 'Idol' for eMic and dancers. In 2010 Donna was also awarded funding to undertake residency with Julian Knowles at STEIM (Amsterdam) – an internationally renowned centre for research and development of instruments & tools for performers in the electronic performance arts.

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TINTS OF JULY (for flute, guitar, and live electronics)
Kathleen Gallagher, *Flute*
Paul Ballam-Cross, *Guitar*
Haruka Hirayama, *Live Electronics*

The timbres of Guitar and Flute sound best the tones of my July.
Dappled sunlight, shimmering, pleasure, transience, quiet rain and a feeling of openness.....
This is the first composition for flute, guitar and live electronics, and various extended techniques were explored in order to extract as much unique desirable timbres as possible. As a main compositional concern, creating organic sound relationship between musical instruments and computer was sought so that audience can be engaged. The electronic sound was created in the Max/ MSP programming environment, and has been renewed with each performance: as a primary change, three pre-recorded and modified sound clips were also included after an original performance, where all of the electronic sound was generated in real-time.

HARUKA HIRAYAMA (COMPOSER)

Haruka studied composition and computer music with Cort Lippe and Takayuki Rai at Kunitachi College of Music in Tokyo. She was a winner of the Residence Prize at the 32nd International Competition of Electroacoustic Music and Sonic Art, and invited to the Institute for Electroacoustic Music in Sweden (EMS) as a composer-in-residence. Her interactive pieces for musical instruments and computers have also been performed at the International Computer Music Conferences of 2005, 2006 and 2009, and various works have been performed in many countries. Currently she lives in the UK, and has been a PhD student in Electroacoustic composition at the NOVARS Research Centre, The University of Manchester under the supervision of Dr. Ricardo Climent.

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THREE STUDIES FOR BOWL AND COMPUTER
Study 101 Thunderbowl: 3' 30 "
Study 102 Scrabbling: 4' 29"
Study 103 Reson: 4' 18"

Three Studies for Bowl and Computer uses a number of sounds from a set of stainless steel kitchen bowls, and several thunderstorms that I recorded at the beachside town of Torquay where we holiday every summer. In *Study 101: Thunderbowl*, I used the basic two classes of sound, the bowl and the thunderstorm, literally at first, then they begin to interact and influence each other as the piece progresses. Then the thunderstorm raindrops are actually playing the bowl. Finally, the thunderstorm and the bowl sounds separate and we hear them in their natural form at the end. *Study 102: Scrabbling* uses scraping bowl sounds with movement and processing. *Study 103: Reson* is probably the most 'electronic' sounding of the three studies, but it still uses the bowls as the source sounds.

DAVID HIRST (COMPOSER)

Hirst has lectured on electronic music at the Tasmanian Conservatorium of Music and La Trobe University, where he served as Chairperson of the Music Department from 1993 to 1995. In 1997, Hirst studied composition with Jonty Harrison at the University of Birmingham. He has published extensively on electroacoustic music, in journals such as Organised Sound, Array, Chroma, Sounds Australian, New Music Articles, Press Press Magazine, and he was the founding editor of Mikropolyphonie, the online contemporary music journal. Hirst was a Senior Lecturer in Educational Design at the University of Melbourne from 2000 to 2007. At the University of Melbourne, he completed a PhD with a music composition folio and a music psychology thesis entitled The Development of a Cognitive Framework for the Analysis of Acousmatic Music, and subsequently became a Senior Research Fellow in the Music and Neurocognition Laboratory of the School of Behavioural Sciences. Since 2008, he has been Senior Lecturer in Curriculum and Academic Development at La Trobe University. Hirst's works have been released by Frog Peak Music, USA, PeP Recordings, Canada, the Australian Computer Music Association, and NMA Tapes, Australia

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VISCOSITY

Everything tangible has mass, everything fluid has viscosity (a fluid's internal resistance to *flow*). Psychological *Flow* (Csíkszentmihályi) relates to the ability to lose oneself in-the-moment, to deep focus, where time and space distort. Psychological *flow* (for me) has viscosity. This piece is inspired by two of my favourite places of flow – the studio and the Great Barrier Reef. This piece attempts to describe some of the ways the Great Barrier Reef makes me feel, and documents my attempts to cultivate these same feelings of flow while creating this work in the studio. On the Barrier Reef, my sense of scale is intensified; discernment of contrast is amplified - the smallest drops of water in the largest drop of water (the ocean); coral spawn and biolumniscent plankton surrounding me on a night dive at the same time as the expanse of the milky way, no separation between the sky and water; the sense of the tiniest particle in as intense focus as the massive, time and space distorted. This piece attempts to describe these multiple moments, places and spaces, flowing between and juxtaposing them (musically) to reflect the way memory can distort time and space by merging multiple locations or events, not tied to any simple structure or direction as our minds wander - the peaceful, dramatic, calm, turbulent, tiny and massive all coexisting and then merging into a single moment – the moment when flow happens.

MATT HITCHCOCK (COMPOSER)

Matt has worked extensively (as an independent as well as for multi-national corporations) as a professional musician, recording engineer, studio owner, music producer, programmer, sound designer and composer. Currently working as a Senior Lecturer in Music Technologies & e-Media and head of Music Technology programs at Queensland Conservatorium, Matt's current research projects include: surround sound composition (7.1); sound production for music and linear and non-linear moving image contexts (film, animation, gaming, performance and installation); research on higher education, learning communities, curriculum and learning environments; and ongoing recording and multimedia authoring projects within the IMERSD (Intermedia, Music Education & Research Design) recording studios. He has also had many years experience in digital technologies including working as multimedia author, web designer and software programmer. More recently Matt's national leadership in these areas has been recognised by national and local teaching awards.

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III

'Cut away the damaged material using a saw. Remove all damaged material. No fibers of the base material should be exposed. File the edge to create a smooth flat surface. To clearly see that all damaged material has been removed, flood the area with alcohol or solvent.'

JORDAN HOCHENBAUM, JASON ERSKINE & BLAKE JOHNSTON (COMPOSERS)

Jordan Hochenbaum is a musician and multimedia artist concerned with finding meaningful connections between music, art, and technology. At a young age, Jordan developed a keen interest in the composition, and decomposition of both audio and electronics. From a list spanning custom-made tube Theremins to the large-scale musical multi-touch interface Bricktable, Jordan's work is informed by an eclectic background and interests in sound and technology.

Jason Erskine is a graduate of the New Zealand School of Music with an honours degree in Composition. Under the guidance of Ajay Kapur, his output includes instrumental composition with live electronics, installation work, and custom user interface design. With a strong background in popular music he currently works as a freelance mix engineer on a wide array of musical genres. Jason is also a recipient of the judge's prize at the 2011 Australasian Computer Music Conference Young Composers competition.

Blake Johnston is currently in his fourth year of study at the New Zealand School of Music and Victoria University studying music composition and philosophy. He has placed in Jeu de temps 2010 and had his electroacoustic work played around Europe and America. He currently works mostly with designing electronic interfaces and writing pieces for small ensembles.

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THE FLASHCARD SEQUENCES – PART 1: AMSTERDAM 14.12.10

This work comprises a series of short semi-improvisatory audio-visual pieces produced with handheld mobile phone video and portable laptop production technologies which seek to investigate new compositional relationships between sound and image. The production process is intentionally swift and the works are produced (often in hotel rooms) as quickly as possible following the image capture.

Amsterdam 14.12.10 is the first piece in the series, and was created during a residency at STEIM, The Netherlands, in December 2010. The harsh European winter of 2010/11 gave rise to heavy snowfalls, closing roads and bringing many essential transport services to a halt. The visual material consists of a series of static video shots containing patterns of snow falling from the sky. In this piece, the video source is digitally analysed and used to affect processing parameters in the audio domain. Such an approach attempts to find, and compositionally engage with, the direct structural relationships between the visual and sound materials whilst stopping short of direct and literal sonification. Such an approach aims to take transductive visual/sound mapping processes and bring them into creative dialogue with freely interpretive compositional processes. This work was created with funding assistance from Arts Queensland, the Australia Council with residency support from STEIM, Holland.

JULIAN KNOWLES (COMPOSER)

Professor Knowles is a composer, performer and media artist. He has worked in a range of solo and group contexts since the late 1980s and is a member of the Australian electro-environmental music group *Social Interiors* who have released several CDs on Extreme Records. In recent years, Julian's music and audio/visual work has been presented at events and venues such as the San Francisco Museum of Modern Art, Experimental Intermedia in New York City, What is Music?, Australian Perspecta, Melbourne International Film Festival, Liquid Architecture, Brisbane Festival and the Sydney Opera House. Julian is a Professor of Music and Research Leader: Creative Arts and Writing at Queensland University of Technology.

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REVOICING THE STRIATED LANDSCAPE

Revoicing the Striated Soundscape is an 8-channel interpretation of a sound-artwork recently commissioned by the City of Melbourne Public Art Commission to be installed in a CBD laneway. Using onsite sounds the artwork repatterns a laneway soundscape creating an acoustic space in which air-conditioners speak, chant and sing to passers-by. It is the intention of the artwork to bring attention to the homogenising effects of air-conditioners and ventilation units on city soundscapes, and to indicate creative responses that attempt to diversify the listening experiences of city dwellers. The installation uses the software program WasP, an interactive performance environment for multichannel spatial soundscape composition, developed at SIAL Sound Studios.

JORDAN LACEY (COMPOSER)

Jordan is a PhD candidate and sessional lecturer at SIAL Sound Studios in the School of Architecture & Design at RMIT University. He is researching Urban Soundscape Design, Biophilic Sound Design, Rhythmanalysis and Soundscape Composition. He is currently engaged in the creation of sonic philosophies and soundscape compositions in a variety of contexts. His work has a particular focus on spatial analysis of the everyday, and the potential for its

transformation through soundscape design intervention and composition. Jordan has been published in a number of journals and is a practicing sound-artist and musician.

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FAITH

"Plenty out of pain, life out of death,"- Anonymous (In the Hour of Silence)

XIAO LI (COMPOSER)

Eva Li was originally from China. She is currently studying her Master degree at the University of Auckland majoring in composition, where she has studied with Associate Professor John Elmsly, Dr. Eve de Castro-Robinson and Dr. John Coulter.From the beginning of 2009, Eva has developed a strong interest in composing electronic music. Her goal in composition is to write music that has the potential to speak to audiences and enable self-understanding.

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ALL ALONG THE BELL TOWER

This piece investigates the illusion of the internal and external sound world of a church/cathedral space and the sound-world within those sounds. The church building is treated as a deconstructed sound object exposing the multiple layers within. Space is explored both as intimate/internal and as distant. *"The most salient sound signal in the Christian community is the church bell"*. (R.M Schafer). This sound can dominate a busy city landscape such as Oxford Road, in Manchester where even peak time traffic and the volume created by 40,000 students cannot drown it out. Therefore the church bell introduces the listener into this new electroacoustic soundscape before focusing on the micro sounds within this environment. These include the smallest movements within the church such as the movement of a person against the wood, people murmuring their prayers and lighting candles and how they are amplified by this space. The aim of the piece is to create an illusion that connects the listener to an environment that one is perhaps very familiar with. However it is created to make the listener aware of the acoustic properties contained within this large reverberant space that has dominated our landscape for centuries.

BRONA MARTIN (COMPOSER)

Brona Martin is an electroacoustic composer and sound artist from Banagher, Co. Offaly, Ireland. Following a BA in History and Geography at the University of Dublin, Brona studied music at the University of Cork where she had her first encounter with computer music, noise and experimental music. At the University of Limerick she completed a Masters in Music Technology with Jürgen Simpson and Dr. Kerry Hagan. After some teaching and traveling Brona started her PhD in September 2010 under the supervision of Dr. David Berezan. Research interests include soundscape composition, augmented auralities, acoustic ecology and narrative within Electroacoustic music.

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ECHOING SPACES **(2009)**

This is a nine-minute work for single-channel video and stereo audio. The work explores a number of virtual environments in which the primary elements recur (echo) both in immediate succession and at different times throughout the piece, always in varied form. The visual imagery employs a number of similar shaped elements that appear in overlapping, morphing configurations, and the restricted color palette helps maintain a focus on the primary objects. The music of Echoing Spaces was created using a variety of modern techniques including convolution and physical modeling. The quasi-tonal context adds an additional layer of continuity to the work.

DENNIS MILLER (COMPOSER)

Dennis Miller received his Doctorate in Music Composition from Columbia University and is currently on the Music faculty of Northeastern University in Boston where he heads the Music Technology program. Miller is also the founder and artistic director of the Visual Music Marathon (www.2009vmm.neu.edu), an ongoing program that includes screenings of works by filmmakers from around the world. His mixed media works have been presented at numerous

venues throughout the world, most recently the DeCordova Museum, the New York Digital Salon Traveling Exhibit, the 2005 Art in Motion screenings, Images du Nouveau Monde, CynetArts, Sonic Circuits, the Cuban International Festival of Music, and the 2004 New England Film and Video Festival. His work was also presented at the gala opening of the new Disney Hall in Los Angeles and at SIGGRAPH 2006 in the both the Animation Theatre and the Art Gallery. Recent exhibits of his 3D still images include the Boston Computer Museum and the Biannual Conference on Art and Technology, as well as publication in *Sonic Graphics: Seeing Sound* (Rizzoli Books) and *Art in the Digital Age* (Thames and Hudson). Miller's music and artworks are available at www.dennismiller.neu.edu.

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OBJECTS

This composition was generated entirely through digital synthesis using Csound.

BEN MURPHY (COMPOSER)

He is a composer and performer of acoustic and electroacoustic music, currently living in Missoula, Montana, US. His music has been performed in the United States and Europe at such events as the Society for Electroacoustic Music in the United States (SEAMUS) National Conference, Mountain Computer Music Festival, International Computer Music Conference, and soundSCAPE International Composition and Performance Exchange in Maccagno, Italy. This year, he won the Music Teachers National Association (MTNA) Young Artist Composition Competition. He recently completed a Bachelor's degree in Composition and Music Technology at the University of Montana, where he was also an active member of the University of Montana Percussion Ensemble and Symphonic Wind Ensemble, and co-composed and performed as a percussionist in an electroacoustic duo. In 2011, he was awarded the Presser Foundation Undergraduate Scholar Award, the highest honor for a music student at the University of Montana. He will continue his study in composition and music technology in autumn 2012 at Bowling Green State University in Ohio, US.

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APHÂR

'Aphâr is a Hebrew word that means "dust". This piece is inspired on the dream of Jacob, described in the Old Testament (Genesis Chapter 28):

Jacob had a dream: He saw a stairway erected on the earth with its top reaching to the heavens. The angels of God were going up and coming down it and the Lord stood at its top. He said, I am the Lord, the God of your grandfather Abraham and the God of your father Isaac. I will give you and your descendants the ground you are lying on. Your descendants will be like the dust of the earth.

To climb Jacob's ladder is a very difficult process. Each step you take is harder than the previous one and takes more time and effort to accomplish. These steps are heavy, and the dust in your feet is spread around, and finally disappears. If you reach the top of the stair, everything else is gone, and a door closes to the rest of the world, you have reached infinity.

'Aphâr was composed at the composer's personal studio and at the Electroacoustic Music Studio of the University of Aveiro. It received the first prize in Yamaha-Visiones Sonoras (Mexico) competition in 2007.

JOÃO PEDRO OLIVEIRA (COMPOSER)

Oliveira is one of the most prominent Portuguese composers of his generation. He studied organ performance at the Gregorian Institute of Lisbon. As a Fulbright and Gulbekian fellow, he completed a PhD in Music at the University of New York at Stony Brook. He has composed numerous instrumental and electroacoustic works, including the exploration of interaction between instrumental and electroacoustic sounds, and other media. He has received numerous prizes and awards, including three Prizes at Bourges Electroacoustic Music Competition, its prestigious Magisterium Prize, the Giga-Hertz Special Award, and many others. He teaches at the Federal University of Minas Gerais (Brazil) and Aveiro University (Portugal) .He has published several articles in journals, and is the author of a book about analysis and 20th century music theory.

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CALLILPHORA

Calliphora is a beautifully misleading word which refers, not to an idyllic fantasy island set on crystal blue waters, but the genus of the common Blowfly. Using the numerology of the Flies life cycle and computer generated sounds which aim to mimic the sonifications of these insectoid organisms, *Calliphora* is a segmented work of alien sounds in a familiar environment. As if under the microscope, studied from various angles, this work attempts to find beauty and interest in a subject most find abhorrent. *Calliphora* was written for another striking and misunderstood phenomenon, the bass recorder. Flies are still gross, though.
Alicia Crossley, Bass Recorder

MARK OLIVEIRO (COMPOSER)

Born in Sydney, Australia in 1983. His music has been performed by the Darwin Symphony Orchestra and Orchestra Victoria, by the Sydney Eclectic Composer Society, BIT20, The Juventas Ensemble, Nexas Sax, the Song Company, Pictures on Silence, and Chronology Arts, and international appearances at the workshop for Young Composers in Latvia (2006) and the International Festival of Stresa, Italy (2006). He has studied at the University of Sydney (Australia), Indiana University (USA) and is currently in pursuing a PhD degree at the University of North Texas. His teachers include Michael Smetanin, Mary Finsterer, Luca rancesconi, Klas Torstensson, John Gibson, Don Freund, Andew May, Joseph Klein and Jon Nelson.

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MUSIC FOR GUITAR AND COMPUTER

This improvised ten-minute piece deliberately sets out to destabilize the performer by changing the sound of the guitar aperiodically, challenging the player to adjust things to make the inevitable transitions sound as if they were on purpose. The player is only trying to play some blues but this will prove difficult under the circumstances.

MILLER PUCKETTE (COMPOSER)

Miller Puckette obtained a B.S. in Mathematics from MIT (1980) and Ph. D. in Mathematics from Harvard (1986). He was a member of MIT's Media Lab from its inception until 1987, and then a researcher at IRCAM (l'Institut de Recherche et de Coordination Musique/Acoustique), founded by composer and conductor Pierre Boulez. There he wrote the Max program for MacIntosh computers, which was first distributed commercially by Opcode Systems in 1990 and is now available from Cycling74.com. In 1989 Puckette joined IRCAM's musical workstation team and wrote an extended version of Max, called Max/FTS, for the Ircam Signal Processing Workstation.

Puckette joined the Music department of the University of California, San Diego in 1994, where he is now chairman. From 2000 to 2011 he was Associate Director of Center for Research in Computing and the Arts (CRCA) at UCSD.

He is currently working on Pure Data ("Pd"), a leading open-source real-time multimedia arts programming environment, with software contributions by many others worldwide. Puckette has collaborated with many artists and musicians, including Philipe Manoury (whose Sonus ex Machina cycle was thie first major work to use Max), and Rand Steiger, and Vibeke Sorensen (as part of the Global Visual Music project). Since 2004 he has performed with the Convolution Brothers.

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ELECTROACOUSTIC WORKS BY KAIJA SAARIAHO

SIX JAPANESE GARDENS (1994)
Vanessa Tomlinson, *Percussion*
Andrew Brown, *Live Electronics Processing*

Six Japanese Gardens is a collection of impressions of the gardens I saw in Kyoto during my stay in Japan in the summer of 1993 and my reflection on rhythm at that time. As the title indicates, the piece is divided into six parts. All these parts give specific look at a rhythmic material, starting from the simplistic first part, in which the main instrumentation is introduced, going to complex polyrhythmic or ostinato figures, or alternation of rhythmic and purely coloristic material. The selection of instruments played by the percussionist is voluntarily reduced to give space for the

perception of rhythmic evolutions. Also, the reduced colours are extended with the addition of an electronics part, in which we hear nature's sounds, ritual singing, and percussion instruments recorded in the Kuntachi College of Music with Shinti Ueno. The ready-mixed sections are triggered by the percussionist during the piece, from a Macintosh computer. All the work for processing and mixing the pre-recorded material was done with a Macintosh computer in my home studio. Some transformations are made with the resonant filters in the CHANT program, and with the SVP Phaser Vocoder. This work was made with Jean-Baptiste Barrière. The final mixing was made with the Protools program with the assistance of Hanspeter Stubbe Teglbjaerg. The piece is commissioned by the Kunitachi College of Music and written for Shinti Ueno. [Kaija Saariaho. From: www.chesternovello.com]

NOA NOA (1992)
Kathleen Gallagher, *Flute*
Andrew Brown, *Live Electronics Processing*

NoaNoa ('Fragrant' 1992) was born from the ideas I had for flute while writing my ballet music Maa. I wanted to write down, exaggerate, even abuse certain flute mannerisms that had been haunting me for some years, and thus force myself to move onto something new. Formally I experimented with an idea of developing several elements simultaneously, first sequentially, then superimposed on each other. The title refers to a wood cut by Paul Gauguin called *NoaNoa*. It also refers to a travel diary of the same name, written by Gauguin during his visit to Tahiti in 1891-93. The fragments of phrases selected for the voice part in the piece come from this book. *NoaNoa* is also a team work. Many details in the flute part were worked out with Camilla Hoitenga. The electronic part was developed under the supervision of Jean-Baptiste Barrière and programmed by Xavier Chabot.
(Kaija Saariaho) [From: www.chesternovello.com]

TROIS RIVIÈRES: DELTA (2001)
Angus Wilson*, Percussion*
Andrew Brown, *Live Electronics Processing*

Trois rivières is divided into three separate sections. The first part introduces all the instrumental colours used in the piece. The rhythmic aspect is here nearly eliminated, giving room for the timbral scales, colours, resonances, attacks and textures to come to the fore. The instruments come from all members of the percussion family, but unpitched instruments are given a primary role.

KAIJA SAARIAHO (COMPOSER)

Kaija Saariaho studied composition in Helsinki, Freiburg and Paris, where she has lived since 1982. Her studies and research at IRCAM have had a major influence on her music and her characteristically luxuriant and mysterious textures are often created by combining live music and electronics. Although much of her catalogue comprises chamber works, from the mid-nineties she has turned increasingly to larger forces and broader structures, such as the operas L'Amour de loin and Adriana Mater and the oratorio La Passion de Simone.

PERFORMERS

Andrew Brown is an active computational artist working in music and visual domains. He is Professor of Digital Arts at the Queensland Conservatorium of Music, Griffith University, in Brisbane, Australia where his work explores the aesthetics of process and often involves programming of software as part of the creative process. In addition to a history of computer-assisted composition and rendered animations, Andrew has in recent years focused on real-time art works using generative processes and musical live-coding where the software to generate a work is written as part of the performance. He has performed live coding around Australia and internationally including in London, Copenhagen, and Boston. His digital media art work has been shown in galleries in Australia and China. For more visit http://andrewrbrown.net.au

Angus Wilson is a Queensland based percussionist, he is currently completing a Master of Music (research) at Queensland Conservatorium Griffith University under Vanessa Tomlinson. Angus moved to Brisbane in 2011 after completing a Bachelor of Music in performance at the Victorian College of the Arts. He is an active member of the music scene performing in a wide range of styles genres, largely with a focus on contemporary music. Angus has performed with ensembles and soloists such as Elision ensemble, eighth blackbird, Kroumata percussion, Lisa Moore, Annea Lockwood and Queensland Symphony Orchestra as well as at festivals such as Crossbows and the Brisbane Festival.

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FEEDBACK SUITE

In several brief sessions (recorded at the Queensland Conservatorium's IMERSD studio), Stefulj free improvised, making sonic beds with a suite of wind and percussion, then handed these artifacts to Cunio for re-composition, processing and manipulation. In some sessions, Cunio suggested motifs or fragments to play, which Stefulj interpreted in real time. These fragments were then taken away and manipulated extensively. Finally the artists returned to the IMERSD studio to listen to and finalise the music, taking stock of underlying aesthetics and the flow of the project as a whole.

In this conference performance, there are four stereo pairs of processed audio playing alongside the live performance. The performance is primarily a re-imagination of the original project, utilising many of the devices of the recording in real time, with the option of radical changes in the improvised field depending on the nature of the performance. In some instances, harmonic creations (made from time-stretched single drone notes) are left intact for counter-melody to resurface as a live component.

JOHN STEFULJ & KIM CUNIO (COMPOSERS)

Dr Kim Cunio has studied with a number of Australia's finest musicians including Australian composer Nigel Butterly, conductor and producer Eric Clapham, and Jazz guitar legend Ike Isaacs. His work with the ABC has seen him compose and produce music projects for CD, radio and television over the last decade. He is one of Australia's most accomplished researching composers and was awarded an ABC Golden Manuscript Award in 2004 in recognition of his work with traditional and Islamic music.

Kim works in new art music, music research, traditional music, acousmatics and screen. He plays a large number of traditional instruments and his music is expansive, beautiful and multi layered. He has a state of the art studio and lectures at the Queensland Conservatorium.

John Stefulj has composed, arranged, performed, recorded, researched and taught with artists throughout Australia as a professional multi-instrumentalist and educator since the 1980s. His work continues within and across a variety of contemporary, traditional and indigenous music as part of his lifelong commitment to cultural enhancement throughout communities, including activities in ensembles, recordings, clinics and festivals. Through primary, secondary and tertiary level teaching, John has shared his experiences from a broad range of music and associated environments, and similarly, he finds teaching to be integral to his sense of creativity and contribution in performance. John realises his enthusiasm towards composed and improvised music by blending both approaches comfortably, which has led him to experiment in many acoustic and electronic mediums, permitting him the enjoyment of a broadening media base through involvement in a range of educational, theatre, film, television, radio, street performance, multimedia environments and other art forms.

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_PROSTHESIS

MICHAEL YOUNG (COMPOSER)

Dr.Young is a composer with interests in real-time computer music, interdiscipinary practice and performance. He was born in South Shields in 1968. He studied music at Oxford University and composition with John Casken & Peter Manning at Durham University, completing a PhD in 1995. He is a Senior Lecturer in Music at Goldsmiths, University of London in the Electronic Music Studios (EMS). Previously, he lectured at Oxford Brookes University and University of Wales, Bangor. His recent music has focussed on interactive/generative systems, exploring the interface between composed and improvised music-making. He is co-founder of the Live Algorithms for Music Network. He has worked with jazz musicians and improvisers as pianist, laptop musician & composer.
[source: http://homepage.mac.com/oobop/myounginfo/about.html]

INSTALLATIONS

HIBIKI 1.0

Hibiki 1.0 is part an interactive sound installation and part a musical piece. Hibiki 1.0 will implement real-time sonification of data streams obtained from the internet. Such data may be collected using specialised online tools such as Nimbits which makes available hundreds of sensor data streams. The focus of Hibiki is to sonify environmental data and the impact of human consumption of resources; this represents a sonification layer. Other sonification layers may also be implemented, for example: Internet traffic, the weather, stock exchange data etc.

Another important aspect of Hibiki is interactivity: using their web-enabled mobile phones, the public will be able to modify parameters of the sonification mapping and the current sonification layer. We implement interactivity using an HTML5 web page and custom server-side software which makes our technique cross-platform and no native app installation is required. Hibiki will be implemented as a permanent sound installation in the Nishi building which is a modern and 6 green star rated building complex currently being developed in Canberra. Hibiki was commissioned as an experiment in architectural audio, data sonification and user interactivity.

Hibiki will be shown for the first time to the public at ACMC 2012.

Guillaume Potard is a Sydney-based sound researcher and artist. In 2006 he obtained a PhD on '3-D audio object oriented coding' from the University of Wollongong. Some of his research interests are: acoustics, digital signal processing, perception of sound (psychoacoustics), design of novel interactive multimedia systems, data sonification and installation art. After working 6 years at Dolby Labs in Sydney, Guillaume started Lofty Consulting in 2011, a consultancy firm which provides R&D and engineering services in Digital Signal Processing, sound and multimedia systems. Guillaume is the main author on two patents.

Stephen Barrass is an academic, researcher and artist with interests in data sonification, sound art, digital design and fabrication, interactive storytelling, social media in creative collaboration, physical computing, immersive and wearable interfaces. He holds a Ph.D. in Information Technology from the Australian National University on the topic of Auditory Information Design (1998) and a Bachelor of Electrical Engineering from the University of NSW with an honours dissertation on Digital Holograms (1987). His creative works include the Welcome Space at the National Museum of Australia 2001~, OpShop at Experimenta 2003, Listening to the Mind Listening at the Sydney Opera House Studio 2004, Baroque EEG at Sound and Musci Computing in Barcelona 2010, and Zizi the Affectionate Couch at MONA 2011~.

Olivier Zeller was born and raised in Paris. He has lived and worked in Paris, London and Sydney over the last 15 years. As an avid student of music and all things technology, he has completed higher education in a range of disciplines, including a diploma in sound engineering at SAE Paris, a Master in computer science at Paris V and INSA Toulouse, as well as studies in music composition at the Australian Institute of Music (Sydney). His professional life has been just as varied, ranging from software developer for IRCAM in Paris and a DSP design engineer for Dolby in Sydney. In 2008 he became an independent music producer and DJ, specialising in electronic music, with a focus on dubstep.

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Blue Gold - a new album of environmental sounds works by Ros Bandt and Leah Barclay
Ros Bandt with Leah Barclay

Blue gold is an electroacoustic work for cello and clarinets, sculptures and digital colour field and moving imagery. It investigates the delicate balance between wet and dry, hugros, xeros, symbolised in the colours of the title BLUE GOLD. The performers, cello (wet) and the clarinets (dry), navigate their way through the electroacoustic sound fields sourced from the Southern Ocean underwater recordings, aeolian harps in the arid Mungo desert, flash storms in the bush and the wetlands of the Otway Ranges. A range of ecstatic dances composed in the Peloponnese, Greece, remind us that European settlement has reformed and changed the Australian landscape irrevocably. The performers respond to soundscape recordings, the imagery, the graphic and notated scores, to bring this alive in each place they perform, inside or outside, desert or flood. They pour earth and water into the glass timer sculptures or the large bamboo beam

balance showing the changing balances between the water and the earth, and the role we are playing in its care and destruction. These resonate at the metaphorical levels of civilization/nature, and european/indigenous perspectives. Water, land care and global warming are surely the greatest challenges facing us and demand attentiveness. BLUE GOLD is a plea to become more mindful and not take so much.

BLUE GOLD was commissioned, premiered and toured by charisma, ros dunlop, clarinets and julia ryder, cello.
thanks to canadian activist maude barlow for her encouragement and permission to use the title inspired by her book *blue gold: the fight to stop the corporate theft of the world's water,* 2003.

Ros Bandt is an award winning international sound artist from Australia who is famous for her soundings and installations of unusual acoustic spaces from wheat silos, water tanks, limestonequarries, swimming pools, ancient deserts, historic buildings, and underwater environments**.** She has pioneered the use of endangered sounds in her sound installations and radiophonic works contributing to environmental and sound culture awareness, recording and interpreting Indigenous and multi-cultural languages in change, native wild life, old machinery and lost classical languages. She has built soundplaygrounds, interactive systems, global satellite performances, made sculptures, musical instruments and is an expert sound recordist and musician. She has interpreted sites and sound cultures worldwide including the Goatherd songs of Crete, the endangered sound of the *Isobue*, Japanese Sea Whistle, and the fleeting sounds of her own Aeolian harps and casuarinas in Lake Mungo. While living in Europe in 2010-11 she toured her international collaborative sound art performance with hair dye paintings with Johannes S Sistermanns Germany, *Tracings* CD Wergo, and composing and performing on Tarhu the *Black Flacon* CD with Erdem Helvacioglu, Pozitif Yappim, CD of the week in Istanbul. Her site specific electro-acoustic symphony*, Medusa Dreaming* was composed for the Yerebatan Sarayi underwater cistern in Istanbul, performing two concerts there with the Medusa ensemble she created for this event. She is a founding member of six ensembles incuding the early music ensemble La Romanesca with whom she is currently touring.

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SKIDIBIDOU [ZAK]

Skibidibow is every crazy cool party you've rocked, rolled into one song and wrapped in disco balls. Comprised of the dynamic disco duo Dr Offig and DJ Teh, Skibidibow are masters of mashtiche, blending classic party songs, dirty dubstep, and live audio manipulation. For this performance Skibidibow will throw caution to the wind and embark on a full-scale assault of live audio destruction.

POP GOES THE WEASEL

Pop Goes the Weasel is a proposed live performance of selected students works from the Queensland Conservatorium Popular Music Program's third year course in Computer Music. The course revolves around the creation of a creative musical work, with a focus on interactivity. The best (and most interactive) student works will be selected for the performance, curated by Toby Gifford.

Dr Offig (Toby Gifford) is a music technologist, DJ and acoustic musician. Recent (2011) performances have included a DJ residency at the Brisbane Festival, live electroacoustic improvisation at the ElectroFringe festival, and hybrid acoustic/electronic residency at the GoMA Cinemateque performing music to accompany silent Hitchcock films.

Dr Offig's interactive music software, the Jambot, has received national media attention, appearing on the ABC New Inventors program.

DJ Teh honed his skills and tricks deep in the UK underground under the tutelage of anarchist production genius Chopper of the STONKA crew in East London. Whilst deep in the genre-bending studio dungeon making hardcore industrial music, Teh took his electro, drum'n'bass roots and exploded his technical production knowledge. While in the UK, Teh produced and performed with the UK underground groups, Gatchiman, Sanyasa and MEM. After moving back to Australia TEH formed the electronic dance act Edge of Colour with Michelle Xen. Edge of Colour released two EPs and a remix EP and toured Australia and Japan playing at festivals and venues including MONA FOMA Festival, Woodford Folk Festival and the Hi-Fi Brisbane.

EVENTS PROGRAM

Thursday, July 12 2012
11:30 am, Basil Jones Orchestral Hall

Switched On (2011) — Manuella Blackburn (Liverpool, UK)

Objects (2011) — Ben Murphy (Montana, USA)

for computer-generated sounds

Mint Cascade — Andrew Dolphin (Leeds, UK)

Prelude for Clarinet and Computer (2006) — Richard Dudas (Seoul, South Korea)

John Stefulj, *Clarinet*

Feedback Suite (2012) — John Stefulj and Kim Cunio (Brisbane – Sydney, Australia)

John Stefulj, *Miscellaneous Instruments*
Kim Cunio, *DSP*

The DAM(N) Project: *Jobat Calling* (2011) — Leah Barclay (Queensland, Australia)

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Thursday, July 12 2012
5:30 pm, Basil Jones Orchestral Hall

Aphâr (2007) — João Pedro Oliveira
electroacoustic (8-channels) — (Minas Gerais, Brazil; Aveiro, Portugal)

Brisbane Nocturne Revisited (2000. Rev.2012) — Warren Burt
(Wollongong, Australia)

Warren Burt, *Laptop and DSP*

Aerial Vapours (2011) — Lee Daniel Fraser
acousmatic work for fixed media, stereo — (Manchester, UK)

One (2012) — Donna Hewitt
(Brisbane, Australia)

Donna Hewitt, *Voice and eMic Interface*

_prosthesis — Michael Young
(London, UK)

Brigid Burke, *Bass Clarinet*
Michael Young, *Computer Improviser*

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Friday, July 13 2012
12 noon, Basil Jones Orchestral Hall

All Along The Bell Tower
for 5.1 surround

Brona Martin
(Offaly, Ireland; Manchester, UK)

Revoicing The Striated Landscape
for 8-channel system

Jordan Lacey
(Melbourne, Australia)

Rivers Talk (2011)
radiophonic work for enhanced stereo

Ros Bandt and Leah Barclay
(Melbourne – Sundshine Coast, Australia)

Faith (2012)
for 8-channel system

Xiao Li
(Auckland, New Zealand)

Calliphora

Mark Oliveiro
(Sydney, Australia; Texas, USA)

Alicia Crossley, *Bass Recorder*
Mark Oliveiro*, DSP*

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Friday, July 13 2012
5:30pm, Basil Jones Orchestral Hall

Three Studies For Bowl and Computer
stereo

David Hirst
(Melbourne, Australia)

The Flashcard Sequence - Part 1: Amsterdam 14.12.10
electronics and video activated digital audio processing

Julian Knowles

Julian Knowles, *DSP*

Tints of July (2011)

Haruka Hirayama
(Tokyo, Japan; Manchester, UK)

Kathleen Gallagher, *Flute*
Paul Ballam-Cross, *Guitar*
Haruka Hirayama, *Steiner Controller*

Patoise II: SoW (2012)
audiovisual

Melody Eötvös
(Gold Coast, Australia; Indiana, USA)

Takotsubo (2012)
improvised audiovisual performance

Mark Pedersen and Roger Alsop
(Melbourne, Australia)

Viscosity (2012)
for 7.1 system

Matt Hitchcock
(Brisbane, Australia)

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Saturday, July 14 2012
8:30pm, Basil Jones Orchestral Hall

Noa Noa (1992) — Kaija Saariaho (Finland; France)

Kathleen Gallagher, *Flute*
Andrew Brown, *DSP*

Trois rivières: Delta (1994)

Angus Wilson, *Percussion*
Andrew Brown, *DSP*

_derivations — Benjamin Carey (Melbourne, Australia)

Alana Blackburn, *Bass Recorder*
Benjamin Carey, *Computer Improviser*

In A Winter Landscape — Madelyn Byrne (California, USA)
I. In A Winter Landscape
II. First Flight

Peter Sheridan, *Bass Flute*

Echoing Spaces (2009) — Dennis Miller (Boston, USA)
audiovisual

Improvisations — Miller Puckette, Eric Griswold, Vanessa Tomlinson (California, USA; Brisbane, Australia)

Eric Griswold, *Piano*
Vanessa Tomlinson, *Percussion*
Miller Puckette, *Computer Improviser*

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Installations

July 12-14 2012
IMERSD, Queensland Conservatorium

HIBIKI 1.0 — Stephen Barras

for 4 or 9 channels system — (Sydney, Australia)

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July 12-14 2012
Foyer, Queensland Conservatorium

Selection of works — Various artists

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Thursday 13 and Friday 14 2012
Outside Venues

Skidibidou — Tobby Gifford

(Brisbane, Australia)

Pop Goes The Wisel

Tobby Gifford, *Computer Improviser*

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WORKSHOPS

Michael Young Workshop: Interact and Generate
This workshop will explore the fundamental elements necessary in live algorithms theory: machine listening and algorithmic process for synthesis, which can be summarized as the "PfQ" functionality of a live algorithmic system.

In collaboration with fellow performer Brigid Burke, the workshop will include a demonstration of elements of the Max/MSP systems used in Michael Young's recent compositions *piano_prosthesis* and *Trio for Two Players*.

Participants are welcome to bring along their own work where relevant to this topic, and as time permits, a hacking and/or discussion session will be included. Prior notice of interest in contributing would be appreciated: please contact Michael Young directly.

Miller Puckette Workshop: Using audio signals as controllers

The most natural source of control for a real-time computer process is a microphone pointed at a person or instrument. It's easy enough to measure things like pitch, amplitude, or spectral envelope, but then many interesting considerations come up when we try to use these to control a sound-generating process in a computer. For example, how accurately can one or another parameter be controlled by an instrument or voice? How quickly can it change? Can continuously changing values be parsed into yes-or-no events, and with what time lag, time accuracy, and robustness? This workshop will provide an exploration of these issues using some specific examples. We'll use Pd for the demonstrations, although participants can probably work along in Max or Supercollider as they prefer, and should be able to use their own laptops and built-in microphones.